

```

/**
 * @license
 * Copyright 2010-2021 Three.js Authors
 * SPDX-License-Identifier: MIT
 */
!function(t,e){"object"==typeof exports&&"undefined"!=typeof module?
e(exports):"function"==typeof define&&define.amd?
define(["exports"],e):e((t="undefined"!=typeof globalThis?
globalThis:t||self).THREE={})}(this,(function(t){{"use strict";const
e="128",n=100,i=300,r=301,s=302,a=303,o=304,l=306,c=307,h=1e3,u=1001,d=1002,p=1003,
m=1004,f=1005,g=1006,v=1007,y=1008,x=1009,_=1012,w=1014,b=1015,M=1016,S=1020,T=1022
,E=1023,A=1026,L=1027,R=33776,C=33777,P=33778,D=33779,I=35840,N=35841,B=35842,z=358
43,F=37492,O=37496,H=2300,G=2301,U=2302,k=2400,V=2401,W=2402,j=2500,q=2501,X=3e3,Y=
3001,Z=3007,J=3002,Q=3004,K=3005,$=3006,tt=7680,et=35044,nt=35048,it="300 es";class
rt{addEventListener(t,e){void 0===this._listeners&&(this._listeners={});const
n=this._listeners;void 0===n[t]&&(n[t]=
[]),-1===n[t].indexOf(e)&&n[t].push(e)}hasEventListener(t,e){if(void
0===this._listeners)return!1;const n=this._listeners;return void
0!==(n[t]&&-1!==(n[t].indexOf(e)}removeEventListener(t,e){if(void
0===this._listeners)return;const n=this._listeners[t];if(void 0!==(n){const
t=n.indexOf(e);-1!==t&&n.splice(t,1)}}dispatchEvent(t){if(void
0===this._listeners)return;const e=this._listeners[t.type];if(void 0!==(e)
{t.target=this;const n=e.slice(0);for(let
e=0,i=n.length;e<i;e++)n[e].call(this,t);t.target=null}}}const st=[];for(let
t=0;t<256;t++)st[t]=(t<16?"0":"")+t.toString(16);let at=1234567;const
ot=Math.PI/180,lt=180/Math.PI;function ct(){const
t=4294967295*Math.random()|0,e=4294967295*Math.random()|0,n=4294967295*Math.random(
)|0,i=4294967295*Math.random()|0;return(st[255&t]+st[t>>8&255]+st[t>>16&255]+st[t>>
24&255]+"-"+st[255&e]+st[e>>8&255]+"-"+st[e>>16&15|64]+st[e>>24&255]+"-
"+st[63&n|128]+st[n>>8&255]+"-
"+st[n>>16&255]+st[n>>24&255]+st[255&i]+st[i>>8&255]+st[i>>16&255]+st[i>>24&255])).t
oUpperCase()}function ht(t,e,n){return Math.max(e,Math.min(n,t))}function ut(t,e)
{return(t%e)%e}function dt(t,e,n){return(1-n)*t+n*e}function pt(t){return 0==
(t&t-1)&&0!==(t)}function mt(t){return
Math.pow(2,Math.ceil(Math.log(t)/Math.LN2))}function ft(t){return
Math.pow(2,Math.floor(Math.log(t)/Math.LN2))}var
gt=Object.freeze({__proto__:null,DEG2RAD:ot,RAD2DEG:lt,generateUUID:ct,clamp:ht,euc
lideanModulo:ut,mapLinear:function(t,e,n,i,r){return i+(t-e)*(r-i)/(n-
e)},inverseLerp:function(t,e,n){return t!==(e?(n-t)/(e-
t):0)},lerp:dt,damp:function(t,e,n,i){return dt(t,e,1-Math.exp(-
n*i))},pingpong:function(t,e=1){return e-Math.abs(ut(t,2*e)-
e)},smoothstep:function(t,e,n){return t<=e?0:t>=n?1:(t=(t-e)/(n-e))*t*(3-
2*t)},smootherstep:function(t,e,n){return t<=e?0:t>=n?1:(t=(t-e)/(n-e))*t*t*(t*
(6*t-15)+10)},randInt:function(t,e){return t+Math.floor(Math.random()*(e-
t+1))},randFloat:function(t,e){return t+Math.random()*(e-
t)},randFloatSpread:function(t){return t*(.5-
Math.random())},seededRandom:function(t){return void 0!==(t&&
(at=t%2147483647),at=16807*at%2147483647,(at-1)/2147483646},degToRad:function(t)
{return t*ot},radToDeg:function(t){return
t*lt},isPowerOfTwo:pt,ceilPowerOfTwo:mt,floorPowerOfTwo:ft,setQuaternionFromProperE
uler:function(t,e,n,i,r){const
s=Math.cos,a=Math.sin,o=s(n/2),l=a(n/2),c=s((e+i)/2),h=a((e+i)/2),u=s((e-
i)/2),d=a((e-i)/2),p=s((i-e)/2),m=a((i-e)/2);switch(r)
{case"XYX":t.set(o*h,l*u,l*d,o*c);break;case"YZY":t.set(l*d,o*h,l*u,o*c);break;case
"XZX":t.set(l*u,l*d,o*h,o*c);break;case"XZY":t.set(o*h,l*m,l*p,o*c);break;case"YXY"
:t.set(l*p,o*h,l*m,o*c);break;case"YZX":t.set(l*m,l*p,o*h,o*c);break;default:consol
e.warn("THREE.MathUtils: .setQuaternionFromProperEuler() encountered an unknown

```

```

order: "+r)}}});class vt{constructor(t=0,e=0){this.x=t,this.y=e}get width(){return
this.x}set width(t){this.x=t}get height(){return this.y}set height(t)
{this.y=t}set(t,e){return this.x=t,this.y=e,this}setScalar(t){return
this.x=t,this.y=t,this}setX(t){return this.x=t,this}setY(t){return
this.y=t,this}setComponent(t,e){switch(t){case 0:this.x=e;break;case
1:this.y=e;break;default:throw new Error("index is out of range: "+t)}return
this}getComponent(t){switch(t){case 0:return this.x;case 1:return
this.y;default:throw new Error("index is out of range: "+t)}}clone(){return new
this.constructor(this.x,this.y)}copy(t){return this.x=t.x,this.y=t.y,this}add(t,e)
{return void 0!==e?(console.warn("THREE.Vector2: .add() now only accepts one
argument. Use .addVectors( a, b ) instead."),this.addVectors(t,e)):
(this.x+=t.x,this.y+=t.y,this)}addScalar(t){return
this.x+=t,this.y+=t,this}addVectors(t,e){return
this.x=t.x+e.x,this.y=t.y+e.y,this}addScaledVector(t,e){return
this.x+=t.x*e,this.y+=t.y*e,this}sub(t,e){return void 0!==e?
(console.warn("THREE.Vector2: .sub() now only accepts one argument. Use
.subVectors( a, b ) instead."),this.subVectors(t,e)):(this.x-=t.x,this.y-=
t.y,this)}subScalar(t){return this.x-=t,this.y-=t,this}subVectors(t,e){return
this.x=t.x-e.x,this.y=t.y-e.y,this}multiply(t){return
this.x*=t.x,this.y*=t.y,this}multiplyScalar(t){return
this.x*=t,this.y*=t,this}divide(t){return
this.x/=t.x,this.y/=t.y,this}divideScalar(t){return
this.multiplyScalar(1/t)}applyMatrix3(t){const
e=this.x,n=this.y,i=t.elements;return
this.x=i[0]*e+i[3]*n+i[6],this.y=i[1]*e+i[4]*n+i[7],this}min(t){return
this.x=Math.min(this.x,t.x),this.y=Math.min(this.y,t.y),this}max(t){return
this.x=Math.max(this.x,t.x),this.y=Math.max(this.y,t.y),this}clamp(t,e){return
this.x=Math.max(t.x,Math.min(e.x,this.x)),this.y=Math.max(t.y,Math.min(e.y,this.y))
,this}clampScalar(t,e){return
this.x=Math.max(t,Math.min(e,this.x)),this.y=Math.max(t,Math.min(e,this.y)),this}cl
ampLength(t,e){const n=this.length();return
this.divideScalar(n||1).multiplyScalar(Math.max(t,Math.min(e,n)))}floor(){return
this.x=Math.floor(this.x),this.y=Math.floor(this.y),this}ceil(){return
this.x=Math.ceil(this.x),this.y=Math.ceil(this.y),this}round(){return
this.x=Math.round(this.x),this.y=Math.round(this.y),this}roundToZero(){return
this.x=this.x<0?Math.ceil(this.x):Math.floor(this.x),this.y=this.y<0?
Math.ceil(this.y):Math.floor(this.y),this}negate(){return this.x=-this.x,this.y=-
this.y,this}dot(t){return this.x*t.x+this.y*t.y}cross(t){return this.x*t.y-
this.y*t.x}lengthSq(){return this.x*this.x+this.y*this.y}length(){return
Math.sqrt(this.x*this.x+this.y*this.y)}manhattanLength(){return
Math.abs(this.x)+Math.abs(this.y)}normalize(){return
this.divideScalar(this.length()||1)}angle(){return Math.atan2(-this.y,-
this.x)+Math.PI}distanceTo(t){return
Math.sqrt(this.distanceToSquared(t))}distanceToSquared(t){const e=this.x-
t.x,n=this.y-t.y;return e*e+n*n}manhattanDistanceTo(t){return Math.abs(this.x-
t.x)+Math.abs(this.y-t.y)}setLength(t){return
this.normalize().multiplyScalar(t)}lerp(t,e){return this.x+=(t.x-this.x)*e,this.y+=
(t.y-this.y)*e,this}lerpVectors(t,e,n){return this.x=t.x+(e.x-t.x)*n,this.y=t.y+
(e.y-t.y)*n,this>equals(t){return t.x===this.x&&this.y===this.y}fromArray(t,e=0)
{return this.x=t[e],this.y=t[e+1],this}toArray(t=[],e=0){return
t[e]=this.x,t[e+1]=this.y,t}fromBufferAttribute(t,e,n){return void
0!==n&&console.warn("THREE.Vector2: offset has been removed from
.fromBufferAttribute()."),this.x=t.getX(e),this.y=t.getY(e),this}rotateAround(t,e)
{const n=Math.cos(e),i=Math.sin(e),r=this.x-t.x,s=this.y-t.y;return this.x=r*n-
s*i+t.x,this.y=r*i+s*n+t.y,this}random(){return
this.x=Math.random(),this.y=Math.random(),this}}vt.prototype.isVector2=!0;class
yt{constructor(){this.elements=
[1,0,0,0,1,0,0,0,1],arguments.length>0&&console.error("THREE.Matrix3: the

```

```

constructor no longer reads arguments. use .set() instead.")]set(t,e,n,i,r,s,a,o,l)
{const c=this.elements;return
c[0]=t,c[1]=i,c[2]=a,c[3]=e,c[4]=r,c[5]=o,c[6]=n,c[7]=s,c[8]=l,this}identity()
{return this.set(1,0,0,0,1,0,0,0,1),this}copy(t){const
e=this.elements,n=t.elements;return
e[0]=n[0],e[1]=n[1],e[2]=n[2],e[3]=n[3],e[4]=n[4],e[5]=n[5],e[6]=n[6],e[7]=n[7],e[8]
]=n[8],this}extractBasis(t,e,n){return
t.setFromMatrix3Column(this,0),e.setFromMatrix3Column(this,1),n.setFromMatrix3Column
n(this,2),this}setFromMatrix4(t){const e=t.elements;return
this.set(e[0],e[4],e[8],e[1],e[5],e[9],e[2],e[6],e[10]),this}multiply(t){return
this.multiplyMatrices(this,t)}premultiply(t){return
this.multiplyMatrices(t,this)}multiplyMatrices(t,e){const
n=t.elements,i=e.elements,r=this.elements,s=n[0],a=n[3],o=n[6],l=n[1],c=n[4],h=n[7]
,u=n[2],d=n[5],p=n[8],m=i[0],f=i[3],g=i[6],v=i[1],y=i[4],x=i[7],_=i[2],w=i[5],b=i[8]
];return
r[0]=s*m+a*v+o*_ ,r[3]=s*f+a*y+o*w,r[6]=s*g+a*x+o*b,r[1]=l*m+c*v+h*_ ,r[4]=l*f+c*y+h*
w,r[7]=l*g+c*x+h*b,r[2]=u*m+d*v+p*_ ,r[5]=u*f+d*y+p*w,r[8]=u*g+d*x+p*b,this}multiply
Scalar(t){const e=this.elements;return
e[0]*=t,e[3]*=t,e[6]*=t,e[1]*=t,e[4]*=t,e[7]*=t,e[2]*=t,e[5]*=t,e[8]*=t,this}determ
inant(){const
t=this.elements,e=t[0],n=t[1],i=t[2],r=t[3],s=t[4],a=t[5],o=t[6],l=t[7],c=t[8];retu
rn e*s*c-e*a*l-n*r*c+n*a*o+i*r*l-i*s*o}invert(){const
t=this.elements,e=t[0],n=t[1],i=t[2],r=t[3],s=t[4],a=t[5],o=t[6],l=t[7],c=t[8],h=c*
s-a*l,u=a*o-c*r,d=l*r-s*o,p=e*h+n*u+i*d;if(0===p)return
this.set(0,0,0,0,0,0,0,0,0);const m=1/p;return t[0]=h*m,t[1]=(i*l-c*n)*m,t[2]=(a*n-
i*s)*m,t[3]=u*m,t[4]=(c*e-i*o)*m,t[5]=(i*r-a*e)*m,t[6]=d*m,t[7]=(n*o-l*e)*m,t[8]=
(s*e-n*r)*m,this}transpose(){let t;const e=this.elements;return
t=e[1],e[1]=e[3],e[3]=t,t=e[2],e[2]=e[6],e[6]=t,t=e[5],e[5]=e[7],e[7]=t,this}getNor
malMatrix(t){return
this.setFromMatrix4(t).invert().transpose()}transposeIntoArray(t){const
e=this.elements;return
t[0]=e[0],t[1]=e[3],t[2]=e[6],t[3]=e[1],t[4]=e[4],t[5]=e[7],t[6]=e[2],t[7]=e[5],t[8]
]=e[8],this}setUvTransform(t,e,n,i,r,s,a){const o=Math.cos(r),l=Math.sin(r);return
this.set(n*o,n*l,-n*(o*s+l*a)+s+t,-i*l,i*o,-i*(-l*s+o*a)+a+e,0,0,1),this}scale(t,e)
{const n=this.elements;return
n[0]*=t,n[3]*=t,n[6]*=t,n[1]*=e,n[4]*=e,n[7]*=e,this}rotate(t){const
e=Math.cos(t),n=Math.sin(t),i=this.elements,r=i[0],s=i[3],a=i[6],o=i[1],l=i[4],c=i[
7];return i[0]=e*r+n*o,i[3]=e*s+n*l,i[6]=e*a+n*c,i[1]=-n*r+e*o,i[4]=-n*s+e*l,i[7]=-
n*a+e*c,this}translate(t,e){const n=this.elements;return
n[0]+=t*n[2],n[3]+=t*n[5],n[6]+=t*n[8],n[1]+=e*n[2],n[4]+=e*n[5],n[7]+=e*n[8],this}
equals(t){const e=this.elements,n=t.elements;for(let
t=0;t<9;t++)if(e[t]!=n[t])return!1;return!0}fromArray(t,e=0){for(let
n=0;n<9;n++)this.elements[n]=t[n+e];return this}toArray(t=[],e=0){const
n=this.elements;return
t[e]=n[0],t[e+1]=n[1],t[e+2]=n[2],t[e+3]=n[3],t[e+4]=n[4],t[e+5]=n[5],t[e+6]=n[6],t
[e+7]=n[7],t[e+8]=n[8],t}clone(){return(new
this.constructor).fromArray(this.elements)}}let xt,yt.prototype.isMatrix3=!0;class
_t{static getDataURL(t){if(/^data:/i.test(t.src))return
t.src;if("undefined"===typeof HTMLCanvasElement)return t.src;let e;if(t instanceof
HTMLCanvasElement)e=t;else{void 0===xt&&
(xt=document.createElementNS("http://www.w3.org/1999/xhtml","canvas")),xt.width=t.w
idth,xt.height=t.height;const n=xt.getContext("2d");t instanceof ImageData?
n.putImageData(t,0,0):n.drawImage(t,0,0,t.width,t.height),e=xt}return
e.width>2048||e.height>2048?(console.warn("THREE.ImageUtils.getDataURL: Image
converted to jpg for performance
reasons",t),e.toDataURL("image/jpeg",.6)):e.toDataURL("image/png")}}let wt=0;class
bt extends
rt{constructor(t=bt.DEFAULT_IMAGE,e=bt.DEFAULT_MAPPING,n=1001,i=1001,r=1006,s=1008,

```

```

a=1023,o=1009,l=1,c=3e3){super(),Object.defineProperty(this,"id",
{value:wt++}),this.uuid=ct(),this.name="",this.image=t,this.mipmaps=
[],this.mapping=e,this.wrapS=n,this.wrapT=i,this.magFilter=r,this.minFilter=s,this.
anisotropy=1,this.format=a,this.internalFormat=null,this.type=o,this.offset=new
vt(0,0),this.repeat=new vt(1,1),this.center=new
vt(0,0),this.rotation=0,this.matrixAutoUpdate=!0,this.matrix=new
yt,this.generateMipmaps=!0,this.premultiplyAlpha=!1,this.flipY=!0,this.unpackAlignm
ent=4,this.encoding=c,this.version=0,this.onUpdate=null}updateMatrix()
{this.matrix.setUvTransform(this.offset.x,this.offset.y,this.repeat.x,this.repeat.y
,this.rotation,this.center.x,this.center.y)}clone(){return(new
this.constructor).copy(this)}copy(t){return
this.name=t.name,this.image=t.image,this.mipmaps=t.mipmaps.slice(0),this.mapping=t.
mapping,this.wrapS=t.wrapS,this.wrapT=t.wrapT,this.magFilter=t.magFilter,this.minFi
lter=t.minFilter,this.anisotropy=t.anisotropy,this.format=t.format,this.internalFor
mat=t.internalFormat,this.type=t.type,this.offset.copy(t.offset),this.repeat.copy(t
.repeat),this.center.copy(t.center),this.rotation=t.rotation,this.matrixAutoUpdate=
t.matrixAutoUpdate,this.matrix.copy(t.matrix),this.generateMipmaps=t.generateMipmap
s,this.premultiplyAlpha=t.premultiplyAlpha,this.flipY=t.flipY,this.unpackAlignment=
t.unpackAlignment,this.encoding=t.encoding,this}toJSON(t){const e=void
0===t||"string"==typeof t;if(!e&&void 0!==t.textures[this.uuid])return
t.textures[this.uuid];const n={metadata:
{version:4.5,type:"Texture",generator:"Texture.toJSON"},uuid:this.uuid,name:this.na
me,mapping:this.mapping,repeat:[this.repeat.x,this.repeat.y],offset:
[this.offset.x,this.offset.y],center:
[this.center.x,this.center.y],rotation:this.rotation,wrap:
[this.wrapS,this.wrapT],format:this.format,type:this.type,encoding:this.encoding,mi
nFilter:this.minFilter,magFilter:this.magFilter,anisotropy:this.anisotropy,flipY:th
is.flipY,premultiplyAlpha:this.premultiplyAlpha,unpackAlignment:this.unpackAlignmen
t};if(void 0!==this.image){const i=this.image;if(void 0===i.uuid&&
(i.uuid=ct()),!e&&void 0===t.images[i.uuid]){let e;if(Array.isArray(i)){e=
[];for(let t=0,n=i.length;t<n;t++)i[t].isDataTexture?
e.push(Mt(i[t].image)):e.push(Mt(i[t]))}else e=Mt(i);t.images[i.uuid]=
{uuid:i.uuid,url:e}}n.image=i.uuid}return e||(t.textures[this.uuid]=n,n).dispose()
{this.dispatchEvent({type:"dispose"})}transformUv(t){if(this.mapping!==i)return
t;if(t.applyMatrix3(this.matrix),t.x<0||t.x>1)switch(this.wrapS){case h:t.x=t.x-
Math.floor(t.x);break;case u:t.x=t.x<0?0:1;break;case
d:l===Math.abs(Math.floor(t.x)%2)?t.x=Math.ceil(t.x)-t.x:t.x=t.x-
Math.floor(t.x)}if(t.y<0||t.y>1)switch(this.wrapT){case h:t.y=t.y-
Math.floor(t.y);break;case u:t.y=t.y<0?0:1;break;case
d:l===Math.abs(Math.floor(t.y)%2)?t.y=Math.ceil(t.y)-t.y:t.y=t.y-
Math.floor(t.y)}return this.flipY&&(t.y=1-t.y),t}set needsUpdate(t)
{!0===t&&this.version++}function Mt(t){return"undefined"!==typeof
HTMLImageElement&&t instanceof HTMLImageElement||"undefined"!==typeof
HTMLCanvasElement&&t instanceof HTMLCanvasElement||"undefined"!==typeof
ImageBitmap&&t instanceof ImageBitmap?t.getDataURL(t):t.data?
{data:Array.prototype.slice.call(t.data),width:t.width,height:t.height,type:t.data.
constructor.name):(console.warn("THREE.Texture: Unable to serialize Texture."),
{}})}bt.DEFAULT_IMAGE=void 0,bt.DEFAULT_MAPPING=i,bt.prototype.isTexture=!0;class
St{constructor(t=0,e=0,n=0,i=1){this.x=t,this.y=e,this.z=n,this.w=i}get width()
{return this.z}set width(t){this.z=t}get height(){return this.w}set height(t)
{this.w=t}set(t,e,n,i){return this.x=t,this.y=e,this.z=n,this.w=i,this}setScalar(t)
{return this.x=t,this.y=t,this.z=t,this.w=t,this}setX(t){return
this.x=t,this}setY(t){return this.y=t,this}setZ(t){return this.z=t,this}setW(t)
{return this.w=t,this}setComponent(t,e){switch(t){case 0:this.x=e;break;case
1:this.y=e;break;case 2:this.z=e;break;case 3:this.w=e;break;default:throw new
Error("index is out of range: "+t)}return this}getComponent(t){switch(t){case
0:return this.x;case 1:return this.y;case 2:return this.z;case 3:return
this.w;default:throw new Error("index is out of range: "+t)}}clone(){return new

```

```

this.constructor(this.x,this.y,this.z,this.w)}copy(t){return
this.x=t.x,this.y=t.y,this.z=t.z,this.w=void 0!==t.w?t.w:1,this}add(t,e){return
void 0!==e?(console.warn("THREE.Vector4: .add() now only accepts one argument. Use
.addVectors( a, b ) instead."),this.addVectors(t,e)):
(this.x+=t.x,this.y+=t.y,this.z+=t.z,this.w+=t.w,this)}addScalar(t){return
this.x+=t,this.y+=t,this.z+=t,this.w+=t,this}addVectors(t,e){return
this.x=t.x+e.x,this.y=t.y+e.y,this.z=t.z+e.z,this.w=t.w+e.w,this}addScaledVector(t,
e){return this.x+=t.x*e,this.y+=t.y*e,this.z+=t.z*e,this.w+=t.w*e,this}sub(t,e)
{return void 0!==e?(console.warn("THREE.Vector4: .sub() now only accepts one
argument. Use .subVectors( a, b ) instead."),this.subVectors(t,e)):(this.x-
=t.x,this.y-=t.y,this.z-=t.z,this.w-=t.w,this)}subScalar(t){return this.x-
=t,this.y-=t,this.z-=t,this.w-=t,this}subVectors(t,e){return this.x=t.x-
e.x,this.y=t.y-e.y,this.z=t.z-e.z,this.w=t.w-e.w,this}multiply(t){return
this.x*=t.x,this.y*=t.y,this.z*=t.z,this.w*=t.w,this}multiplyScalar(t){return
this.x*=t,this.y*=t,this.z*=t,this.w*=t,this}applyMatrix4(t){const
e=this.x,n=this.y,i=this.z,r=this.w,s=t.elements;return
this.x=s[0]*e+s[4]*n+s[8]*i+s[12]*r,this.y=s[1]*e+s[5]*n+s[9]*i+s[13]*r,this.z=s[2]
*e+s[6]*n+s[10]*i+s[14]*r,this.w=s[3]*e+s[7]*n+s[11]*i+s[15]*r,this}divideScalar(t)
{return this.multiplyScalar(1/t)}setAxisAngleFromQuaternion(t)
{this.w=2*Math.acos(t.w);const e=Math.sqrt(1-t.w*t.w);return e<1e-4?
(this.x=1,this.y=0,this.z=0):
(this.x=t.x/e,this.y=t.y/e,this.z=t.z/e),this}setAxisAngleFromRotationMatrix(t){let
e,n,i,r;const
s=.01,a=.1,o=t.elements,l=o[0],c=o[4],h=o[8],u=o[1],d=o[5],p=o[9],m=o[2],f=o[6],g=o
[10];if(Math.abs(c-u)<s&&Math.abs(h-m)<s&&Math.abs(p-f)<s){if(Math.abs(c+u)
<a&&Math.abs(h+m)<a&&Math.abs(p+f)<a&&Math.abs(l+d+g-3)<a)return
this.set(1,0,0,0),this;e=Math.PI;const t=(l+1)/2,o=(d+1)/2,v=(g+1)/2,y=(c+u)/4,x=
(h+m)/4,_=(p+f)/4;return t>o&&t>v?t<s?(n=0,i=.707106781,r=.707106781):
(n=Math.sqrt(t),i=y/n,r=x/n):o>v?o<s?(n=.707106781,i=0,r=.707106781):
(i=Math.sqrt(o),n=y/i,r=_/i):v<s?(n=.707106781,i=.707106781,r=0):
(r=Math.sqrt(v),n=x/r,i=_/r),this.set(n,i,r,e),this}let v=Math.sqrt((f-p)*(f-p)+(h-
m)*(h-m)+(u-c)*(u-c));return Math.abs(v)<.001&&(v=1),this.x=(f-p)/v,this.y=(h-
m)/v,this.z=(u-c)/v,this.w=Math.acos((l+d+g-1)/2),this}min(t){return
this.x=Math.min(this.x,t.x),this.y=Math.min(this.y,t.y),this.z=Math.min(this.z,t.z)
,this.w=Math.min(this.w,t.w),this}max(t){return
this.x=Math.max(this.x,t.x),this.y=Math.max(this.y,t.y),this.z=Math.max(this.z,t.z)
,this.w=Math.max(this.w,t.w),this}clamp(t,e){return
this.x=Math.max(t.x,Math.min(e.x,this.x)),this.y=Math.max(t.y,Math.min(e.y,this.y))
,this.z=Math.max(t.z,Math.min(e.z,this.z)),this.w=Math.max(t.w,Math.min(e.w,this.w)
),this}clampScalar(t,e){return
this.x=Math.max(t,Math.min(e,this.x)),this.y=Math.max(t,Math.min(e,this.y)),this.z=
Math.max(t,Math.min(e,this.z)),this.w=Math.max(t,Math.min(e,this.w)),this}clampLeng
th(t,e){const n=this.length();return
this.divideScalar(n||1).multiplyScalar(Math.max(t,Math.min(e,n)))}floor(){return
this.x=Math.floor(this.x),this.y=Math.floor(this.y),this.z=Math.floor(this.z),this.
w=Math.floor(this.w),this}ceil(){return
this.x=Math.ceil(this.x),this.y=Math.ceil(this.y),this.z=Math.ceil(this.z),this.w=M
ath.ceil(this.w),this}round(){return
this.x=Math.round(this.x),this.y=Math.round(this.y),this.z=Math.round(this.z),this.
w=Math.round(this.w),this}roundToZero(){return this.x=this.x<0?
Math.ceil(this.x):Math.floor(this.x),this.y=this.y<0?
Math.ceil(this.y):Math.floor(this.y),this.z=this.z<0?
Math.ceil(this.z):Math.floor(this.z),this.w=this.w<0?
Math.ceil(this.w):Math.floor(this.w),this}negate(){return this.x=-this.x,this.y=-
this.y,this.z=-this.z,this.w=-this.w,this}dot(t){return
this.x*t.x+this.y*t.y+this.z*t.z+this.w*t.w}lengthSq(){return
this.x*this.x+this.y*this.y+this.z*this.z+this.w*this.w}length(){return
Math.sqrt(this.x*this.x+this.y*this.y+this.z*this.z+this.w*this.w)}manhattanLength(

```

```

){return
Math.abs(this.x)+Math.abs(this.y)+Math.abs(this.z)+Math.abs(this.w)}normalize()
{return this.divideScalar(this.length()||1)}setLength(t){return
this.normalize().multiplyScalar(t)}lerp(t,e){return this.x+=(t.x-this.x)*e,this.y+=(
t.y-this.y)*e,this.z+=(t.z-this.z)*e,this.w+=(t.w-
this.w)*e,this}lerpVectors(t,e,n){return this.x=t.x+(e.x-t.x)*n,this.y=t.y+(e.y-
t.y)*n,this.z=t.z+(e.z-t.z)*n,this.w=t.w+(e.w-t.w)*n,this}equals(t){return
t.x===this.x&&t.y===this.y&&t.z===this.z&&t.w===this.w}fromArray(t,e=0){return
this.x=t[e],this.y=t[e+1],this.z=t[e+2],this.w=t[e+3],this}toArray(t=[],e=0){return
t[e]=this.x,t[e+1]=this.y,t[e+2]=this.z,t[e+3]=this.w,t}fromBufferAttribute(t,e,n)
{return void 0!==n&&console.warn("THREE.Vector4: offset has been removed from
.fromBufferAttribute()."),this.x=t.getX(e),this.y=t.getY(e),this.z=t.getZ(e),this.w
=t.getW(e),this}random(){return
this.x=Math.random(),this.y=Math.random(),this.z=Math.random(),this.w=Math.random()
,this}}St.prototype.isVector4=!0;class Tt extends rt{constructor(t,e,n)
{super(),this.width=t,this.height=e,this.depth=1,this.scissor=new
St(0,0,t,e),this.scissorTest=!1,this.viewport=new St(0,0,t,e),n=n||
{}},this.texture=new bt(void
0,n.mapping,n.wrapS,n.wrapT,n.magFilter,n.minFilter,n.format,n.type,n.anisotropy,n.
encoding),this.texture.image=
{}},this.texture.image.width=t,this.texture.image.height=e,this.texture.image.depth=
1,this.texture.generateMipmaps=void
0!==n.generateMipmaps&&n.generateMipmaps,this.texture.minFilter=void
0!==n.minFilter?n.minFilter:g,this.depthBuffer=void
0===n.depthBuffer||n.depthBuffer,this.stencilBuffer=void
0!==n.stencilBuffer&&n.stencilBuffer,this.depthTexture=void 0!==n.depthTexture?
n.depthTexture:null}setTexture(t){t.image=
{width:this.width,height:this.height,depth:this.depth},this.texture=t}setSize(t,e,n
=1){this.width===t&&this.height===e&&this.depth===n||
(this.width=t,this.height=e,this.depth=n,this.texture.image.width=t,this.texture.im
age.height=e,this.texture.image.depth=n,this.dispose()),this.viewport.set(0,0,t,e),
this.scissor.set(0,0,t,e)}clone(){return(new this.constructor).copy(this)}copy(t)
{return
this.width=t.width,this.height=t.height,this.depth=t.depth,this.viewport.copy(t.vie
wport),this.texture=t.texture.clone(),this.depthBuffer=t.depthBuffer,this.stencilBu
ffer=t.stencilBuffer,this.depthTexture=t.depthTexture,this}dispose()
{this.dispatchEvent({type:"dispose"})}Tt.prototype.isWebGLRenderTarget=!0;class Et
extends Tt{constructor(t,e,n){super(t,e,n),this.samples=4}copy(t){return
super.copy.call(this,t),this.samples=t.samples,this}Et.prototype.isWebGLMultisampl
eRenderTarget=!0;class At{constructor(t=0,e=0,n=0,i=1)
{this._x=t,this._y=e,this._z=n,this._w=i}static slerp(t,e,n,i){return
console.warn("THREE.Quaternion: Static .slerp() has been deprecated. Use
qm.slerpQuaternions( qa, qb, t ) instead."),n.slerpQuaternions(t,e,i)}static
slerpFlat(t,e,n,i,r,s,a){let o=n[i+0],l=n[i+1],c=n[i+2],h=n[i+3];const
u=r[s+0],d=r[s+1],p=r[s+2],m=r[s+3];if(0===a)return
t[e+0]=o,t[e+1]=l,t[e+2]=c,void(t[e+3]=h);if(1===a)return
t[e+0]=u,t[e+1]=d,t[e+2]=p,void(t[e+3]=m);if(h!==m||o!==u||l!==d||c!==p){let t=1-
a;const e=o*u+l*d+c*p+h*m,n=e>=0?1:-1,i=1-e*e;if(i>Number.EPSILON){const
r=Math.sqrt(i),s=Math.atan2(r,e*n);t=Math.sin(t*s)/r,a=Math.sin(a*s)/r}const
r=a*n;if(o=o*t+u*r,l=l*t+d*r,c=c*t+p*r,h=h*t+m*r,t===1-a){const
t=1/Math.sqrt(o*o+l*l+c*c+h*h);o*=t,l*=t,c*=t,h*=t}}t[e]=o,t[e+1]=l,t[e+2]=c,t[e+3]
=h}static multiplyQuaternionsFlat(t,e,n,i,r,s){const
a=n[i],o=n[i+1],l=n[i+2],c=n[i+3],h=r[s],u=r[s+1],d=r[s+2],p=r[s+3];return
t[e]=a*p+c*h+o*d-l*u,t[e+1]=o*p+c*u+l*h-a*d,t[e+2]=l*p+c*d+a*u-o*h,t[e+3]=c*p-a*h-
o*u-l*d,t}get x(){return this._x}set x(t){this._x=t,this._onChangeCallback()}get
y(){return this._y}set y(t){this._y=t,this._onChangeCallback()}get z(){return
this._z}set z(t){this._z=t,this._onChangeCallback()}get w(){return this._w}set w(t)
{this._w=t,this._onChangeCallback()}set(t,e,n,i){return

```

```

this._x=t,this._y=e,this._z=n,this._w=i,this._onChangeCallback(),this.clone()
{return new this.constructor(this._x,this._y,this._z,this._w)}copy(t){return
this._x=t.x,this._y=t.y,this._z=t.z,this._w=t.w,this._onChangeCallback(),this}setFr
omEuler(t,e){if(!t||!t.isEuler)throw new Error("THREE.Quaternion: .setFromEuler()
now expects an Euler rotation rather than a Vector3 and order.");const
n=t._x,i=t._y,r=t._z,s=t._order,a=Math.cos,o=Math.sin,l=a(n/2),c=a(i/2),h=a(r/2),u=
o(n/2),d=o(i/2),p=o(r/2);switch(s){case"XYZ":this._x=u*c*h+l*d*p,this._y=l*d*h-
u*c*p,this._z=l*c*p+u*d*h,this._w=l*c*h-
u*d*p;break;case"YXZ":this._x=u*c*h+l*d*p,this._y=l*d*h-u*c*p,this._z=l*c*p-
u*d*h,this._w=l*c*h+u*d*p;break;case"ZXY":this._x=u*c*h-
l*d*p,this._y=l*d*h+u*c*p,this._z=l*c*p+u*d*h,this._w=l*c*h-
u*d*p;break;case"ZYX":this._x=u*c*h-l*d*p,this._y=l*d*h+u*c*p,this._z=l*c*p-
u*d*h,this._w=l*c*h+u*d*p;break;case"YZX":this._x=u*c*h+l*d*p,this._y=l*d*h+u*c*p,t
his._z=l*c*p-u*d*h,this._w=l*c*h-u*d*p;break;case"XZY":this._x=u*c*h-
l*d*p,this._y=l*d*h-
u*c*p,this._z=l*c*p+u*d*h,this._w=l*c*h+u*d*p;break;default:console.warn("THREE.Qua
ternion: .setFromEuler() encountered an unknown order:
"+s)}return!!e&&this._onChangeCallback(),this}setFromAxisAngle(t,e){const
n=e/2,i=Math.sin(n);return
this._x=t.x*i,this._y=t.y*i,this._z=t.z*i,this._w=Math.cos(n),this._onChangeCallbac
k(),this}setFromRotationMatrix(t){const
e=t.elements,n=e[0],i=e[4],r=e[8],s=e[1],a=e[5],o=e[9],l=e[2],c=e[6],h=e[10],u=n+a+
h;if(u>0){const t=.5/Math.sqrt(u+1);this._w=.25/t,this._x=(c-o)*t,this._y=(r-
l)*t,this._z=(s-i)*t}else if(n>a&&n>h){const t=2*Math.sqrt(1+n-a-h);this._w=(c-
o)/t,this._x=.25*t,this._y=(i+s)/t,this._z=(r+l)/t}else if(a>h){const
t=2*Math.sqrt(1+a-n-h);this._w=(r-l)/t,this._x=(i+s)/t,this._y=.25*t,this._z=
(o+c)/t}else{const t=2*Math.sqrt(1+h-n-a);this._w=(s-i)/t,this._x=(r+l)/t,this._y=
(o+c)/t,this._z=.25*t}return this._onChangeCallback(),this}setFromUnitVectors(t,e)
{let n=t.dot(e)+1;return n<Number.EPSILON?(n=0,Math.abs(t.x)>Math.abs(t.z)?
(this._x=-t.y,this._y=t.x,this._z=0,this._w=n):(this._x=0,this._y=-
t.z,this._z=t.y,this._w=n)):(this._x=t.y*e.z-t.z*e.y,this._y=t.z*e.x-
t.x*e.z,this._z=t.x*e.y-t.y*e.x,this._w=n),this.normalize()}angleTo(t){return
2*Math.acos(Math.abs(ht(this.dot(t),-1,1)))}rotateTowards(t,e){const
n=this.angleTo(t);if(0===n)return this;const i=Math.min(1,e/n);return
this.slerp(t,i),this}identity(){return this.set(0,0,0,1)}invert(){return
this.conjugate()}conjugate(){return
this._x*=-1,this._y*=-1,this._z*=-1,this._onChangeCallback(),this}dot(t){return
this._x*t._x+this._y*t._y+this._z*t._z+this._w*t._w}lengthSq(){return
this._x*this._x+this._y*this._y+this._z*this._z+this._w*this._w}length(){return
Math.sqrt(this._x*this._x+this._y*this._y+this._z*this._z+this._w*this._w)}normaliz
e(){let t=this.length();return 0===t?(this._x=0,this._y=0,this._z=0,this._w=1):
(t=1/t,this._x=this._x*t,this._y=this._y*t,this._z=this._z*t,this._w=this._w*t),thi
s._onChangeCallback(),this}multiply(t,e){return void 0!==e?
(console.warn("THREE.Quaternion: .multiply() now only accepts one argument. Use
.multiplyQuaternions( a, b )
instead."),this.multiplyQuaternions(t,e)):this.multiplyQuaternions(this,t)}premulti
ply(t){return this.multiplyQuaternions(t,this)}multiplyQuaternions(t,e){const
n=t._x,i=t._y,r=t._z,s=t._w,a=e._x,o=e._y,l=e._z,c=e._w;return this._x=n*c+s*a+i*l-
r*o,this._y=i*c+s*o+r*a-n*l,this._z=r*c+s*l+n*o-i*a,this._w=s*c-n*a-i*o-
r*l,this._onChangeCallback(),this}slerp(t,e){if(0===e)return this;if(1===e)return
this.copy(t);const n=this._x,i=this._y,r=this._z,s=this._w;let
a=s*t._w+n*t._x+i*t._y+r*t._z;if(a<0?(this._w=-t._w,this._x=-t._x,this._y=-
t._y,this._z=-t._z,a=-a):this.copy(t),a>1)return
this._w=s,this._x=n,this._y=i,this._z=r,this;const o=1-a*a;if(o<=Number.EPSILON)
{const t=1-e;return
this._w=t*s+e*this._w,this._x=t*n+e*this._x,this._y=t*i+e*this._y,this._z=t*r+e*thi
s._z,this.normalize(),this._onChangeCallback(),this}const
l=Math.sqrt(o),c=Math.atan2(l,a),h=Math.sin((1-e)*c)/l,u=Math.sin(e*c)/l;return

```

```

this._w=s*h+this._w*u,this._x=n*h+this._x*u,this._y=i*h+this._y*u,this._z=r*h+this._z*u,this._onChangeCallback(),this}slerpQuaternions(t,e,n)
{this.copy(t).slerp(e,n)}equals(t){return
t._x===this._x&&this._y===this._y&&this._z===this._z&&this._w===this._w}fromArray(t,e=0)
{return
this._x=t[e],this._y=t[e+1],this._z=t[e+2],this._w=t[e+3],this._onChangeCallback(),
this}toArray(t=[],e=0){return
t[e]=this._x,t[e+1]=this._y,t[e+2]=this._z,t[e+3]=this._w,t}fromBufferAttribute(t,e)
){return
this._x=t.getX(e),this._y=t.getY(e),this._z=t.getZ(e),this._w=t.getW(e),this}_onChange(t){return this._onChangeCallback=t,this}_onChangeCallback()
{}}At.prototype.isQuaternion=!0;class Lt{constructor(t=0,e=0,n=0)
{this.x=t,this.y=e,this.z=n}set(t,e,n){return void 0===n&&
(n=this.z),this.x=t,this.y=e,this.z=n,this}setScalar(t){return
this.x=t,this.y=t,this.z=t,this}setX(t){return this.x=t,this}setY(t){return
this.y=t,this}setZ(t){return this.z=t,this}setComponent(t,e){switch(t){case
0:this.x=e;break;case 1:this.y=e;break;case 2:this.z=e;break;default:throw new
Error("index is out of range: "+t)}return this}getComponent(t){switch(t){case
0:return this.x;case 1:return this.y;case 2:return this.z;default:throw new
Error("index is out of range: "+t)}}clone(){return new
this.constructor(this.x,this.y,this.z)}copy(t){return
this.x=t.x,this.y=t.y,this.z=t.z,this}add(t,e){return void 0!==e?
(console.warn("THREE.Vector3: .add() now only accepts one argument. Use
.addVectors( a, b ) instead."),this.addVectors(t,e)):
(this.x+=t.x,this.y+=t.y,this.z+=t.z,this)}addScalar(t){return
this.x+=t,this.y+=t,this.z+=t,this}addVectors(t,e){return
this.x=t.x+e.x,this.y=t.y+e.y,this.z=t.z+e.z,this}addScaledVector(t,e){return
this.x+=t.x*e,this.y+=t.y*e,this.z+=t.z*e,this}sub(t,e){return void 0!==e?
(console.warn("THREE.Vector3: .sub() now only accepts one argument. Use
.subVectors( a, b ) instead."),this.subVectors(t,e)):(this.x-=t.x,this.y-=
t.y,this.z-=t.z,this)}subScalar(t){return this.x-=t,this.y-=t,this.z-=
t,this}subVectors(t,e){return this.x=t.x-e.x,this.y=t.y-e.y,this.z=t.z-
e.z,this}multiply(t,e){return void 0!==e?(console.warn("THREE.Vector3: .multiply()
now only accepts one argument. Use .multiplyVectors( a, b )
instead."),this.multiplyVectors(t,e)):
(this.x*=t.x,this.y*=t.y,this.z*=t.z,this)}multiplyScalar(t){return
this.x*=t,this.y*=t,this.z*=t,this}multiplyVectors(t,e){return
this.x=t.x*e.x,this.y=t.y*e.y,this.z=t.z*e.z,this}applyEuler(t){return
t&&t.isEuler||console.error("THREE.Vector3: .applyEuler() now expects an Euler
rotation rather than a Vector3 and
order."),this.applyQuaternion(Ct.setFromEuler(t))}applyAxisAngle(t,e){return
this.applyQuaternion(Ct.setFromAxisAngle(t,e))}applyMatrix3(t){const
e=this.x,n=this.y,i=this.z,r=t.elements;return
this.x=r[0]*e+r[3]*n+r[6]*i,this.y=r[1]*e+r[4]*n+r[7]*i,this.z=r[2]*e+r[5]*n+r[8]*i
,this}applyNormalMatrix(t){return this.applyMatrix3(t).normalize()}applyMatrix4(t)
{const
e=this.x,n=this.y,i=this.z,r=t.elements,s=1/(r[3]*e+r[7]*n+r[11]*i+r[15]);return
this.x=(r[0]*e+r[4]*n+r[8]*i+r[12])*s,this.y=(r[1]*e+r[5]*n+r[9]*i+r[13])*s,this.z=
(r[2]*e+r[6]*n+r[10]*i+r[14])*s,this}applyQuaternion(t){const
e=this.x,n=this.y,i=this.z,r=t.x,s=t.y,a=t.z,o=t.w,l=o*e+s*i-a*n,c=o*n+a*e-
r*i,h=o*i+r*n-s*e,u=-r*e-s*n-a*i;return this.x=l*o+u*-r+c*-a-h*-s,this.y=c*o+u*-
s+h*-r-l*-a,this.z=h*o+u*-a+l*-s-c*-r,this}project(t){return
this.applyMatrix4(t.matrixWorldInverse).applyMatrix4(t.projectionMatrix)}unproject(
t){return
this.applyMatrix4(t.projectionMatrixInverse).applyMatrix4(t.matrixWorld)}transformD
irection(t){const e=this.x,n=this.y,i=this.z,r=t.elements;return
this.x=r[0]*e+r[4]*n+r[8]*i,this.y=r[1]*e+r[5]*n+r[9]*i,this.z=r[2]*e+r[6]*n+r[10]*
i,this.normalize()}divide(t){return

```



```

this.x/=t.x,this.y/=t.y,this.z/=t.z,this}divideScalar(t){return
this.multiplyScalar(1/t)}min(t){return
this.x=Math.min(this.x,t.x),this.y=Math.min(this.y,t.y),this.z=Math.min(this.z,t.z)
,this}max(t){return
this.x=Math.max(this.x,t.x),this.y=Math.max(this.y,t.y),this.z=Math.max(this.z,t.z)
,this}clamp(t,e){return
this.x=Math.max(t.x,Math.min(e.x,this.x)),this.y=Math.max(t.y,Math.min(e.y,this.y))
,this.z=Math.max(t.z,Math.min(e.z,this.z)),this}clampScalar(t,e){return
this.x=Math.max(t,Math.min(e,this.x)),this.y=Math.max(t,Math.min(e,this.y)),this.z=
Math.max(t,Math.min(e,this.z)),this}clampLength(t,e){const n=this.length();return
this.divideScalar(n||1).multiplyScalar(Math.max(t,Math.min(e,n)))}floor(){return
this.x=Math.floor(this.x),this.y=Math.floor(this.y),this.z=Math.floor(this.z),this}
ceil(){return
this.x=Math.ceil(this.x),this.y=Math.ceil(this.y),this.z=Math.ceil(this.z),this}rou
nd(){return
this.x=Math.round(this.x),this.y=Math.round(this.y),this.z=Math.round(this.z),this}
roundToZero(){return this.x=this.x<0?
Math.ceil(this.x):Math.floor(this.x),this.y=this.y<0?
Math.ceil(this.y):Math.floor(this.y),this.z=this.z<0?
Math.ceil(this.z):Math.floor(this.z),this}negate(){return this.x=-this.x,this.y=-
this.y,this.z=-this.z,this}dot(t){return
this.x*t.x+this.y*t.y+this.z*t.z}lengthSq(){return
this.x*this.x+this.y*this.y+this.z*this.z}length(){return
Math.sqrt(this.x*this.x+this.y*this.y+this.z*this.z)}manhattanLength(){return
Math.abs(this.x)+Math.abs(this.y)+Math.abs(this.z)}normalize(){return
this.divideScalar(this.length()||1)}setLength(t){return
this.normalize().multiplyScalar(t)}lerp(t,e){return this.x+=(t.x-this.x)*e,this.y+=
(t.y-this.y)*e,this.z+=(t.z-this.z)*e,this}lerpVectors(t,e,n){return this.x=t.x+
(e.x-t.x)*n,this.y=t.y+(e.y-t.y)*n,this.z=t.z+(e.z-t.z)*n,this}cross(t,e){return
void 0!==e?(console.warn("THREE.Vector3: .cross() now only accepts one argument.
Use .crossVectors( a, b )
instead."),this.crossVectors(t,e)):this.crossVectors(this,t)}crossVectors(t,e)
{const n=t.x,i=t.y,r=t.z,s=e.x,a=e.y,o=e.z;return this.x=i*o-r*s,this.y=r*s-
n*o,this.z=n*a-i*s,this}projectOnVector(t){const e=t.lengthSq();if(0===e)return
this.set(0,0,0);const n=t.dot(this)/e;return
this.copy(t).multiplyScalar(n)}projectOnPlane(t){return
Rt.copy(this).projectOnVector(t),this.sub(Rt)}reflect(t){return
this.sub(Rt.copy(t).multiplyScalar(2*this.dot(t)))}angleTo(t){const
e=Math.sqrt(this.lengthSq()*t.lengthSq());if(0===e)return Math.PI/2;const
n=this.dot(t)/e;return Math.acos(ht(n,-1,1))}distanceTo(t){return
Math.sqrt(this.distanceToSquared(t))}distanceToSquared(t){const e=this.x-
t.x,n=this.y-t.y,i=this.z-t.z;return e*e+n*n+i*i}manhattanDistanceTo(t){return
Math.abs(this.x-t.x)+Math.abs(this.y-t.y)+Math.abs(this.z-t.z)}setFromSpherical(t)
{return
this.setFromSphericalCoords(t.radius,t.phi,t.theta)}setFromSphericalCoords(t,e,n)
{const i=Math.sin(e)*t;return
this.x=i*Math.sin(n),this.y=Math.cos(e)*t,this.z=i*Math.cos(n),this}setFromCylindri
cal(t){return
this.setFromCylindricalCoords(t.radius,t.theta,t.y)}setFromCylindricalCoords(t,e,n)
{return
this.x=t*Math.sin(e),this.y=n,this.z=t*Math.cos(e),this}setFromMatrixPosition(t)
{const e=t.elements;return
this.x=e[12],this.y=e[13],this.z=e[14],this}setFromMatrixScale(t){const
e=this.setFromMatrixColumn(t,0).length(),n=this.setFromMatrixColumn(t,1).length(),i
=this.setFromMatrixColumn(t,2).length();return
this.x=e,this.y=n,this.z=i,this}setFromMatrixColumn(t,e){return
this.fromArray(t.elements,4*e)}setFromMatrix3Column(t,e){return
this.fromArray(t.elements,3*e)}equals(t){return

```

```

t.x===this.x&&t.y===this.y&&t.z===this.z}fromArray(t,e=0){return
this.x=t[e],this.y=t[e+1],this.z=t[e+2],this}toArray(t=[],e=0){return
t[e]=this.x,t[e+1]=this.y,t[e+2]=this.z,t}fromBufferAttribute(t,e,n){return void
0!==n&&console.warn("THREE.Vector3: offset has been removed from
.fromBufferAttribute()."),this.x=t.getX(e),this.y=t.getY(e),this.z=t.getZ(e),this}r
andom(){return
this.x=Math.random(),this.y=Math.random(),this.z=Math.random(),this}}Lt.prototype.i
sVector3=!0;const Rt=new Lt,Ct=new At;class Pt{constructor(t=new
Lt(1/0,1/0,1/0),e=new Lt(-1/0,-1/0,-1/0)){this.min=t,this.max=e}set(t,e){return
this.min.copy(t),this.max.copy(e),this}setFromArray(t){let
e=1/0,n=1/0,i=1/0,r=-1/0,s=-1/0,a=-1/0;for(let o=0,l=t.length;o<l;o+=3){const
l=t[o],c=t[o+1],h=t[o+2];l<e&&(e=l),c<n&&(n=c),h<i&&(i=h),l>r&&(r=l),c>s&&
(s=c),h>a&&(a=h)}return
this.min.set(e,n,i),this.max.set(r,s,a),this}setFromBufferAttribute(t){let
e=1/0,n=1/0,i=1/0,r=-1/0,s=-1/0,a=-1/0;for(let o=0,l=t.count;o<l;o++){const
l=t.getX(o),c=t.getY(o),h=t.getZ(o);l<e&&(e=l),c<n&&(n=c),h<i&&(i=h),l>r&&
(r=l),c>s&&(s=c),h>a&&(a=h)}return
this.min.set(e,n,i),this.max.set(r,s,a),this}setFromPoints(t)
{this.makeEmpty();for(let e=0,n=t.length;e<n;e++)this.expandByPoint(t[e]);return
this}setFromCenterAndSize(t,e){const n=It.copy(e).multiplyScalar(.5);return
this.min.copy(t).sub(n),this.max.copy(t).add(n),this}setFromObject(t){return
this.makeEmpty(),this.expandByObject(t)}clone(){return(new
this.constructor).copy(this)}copy(t){return
this.min.copy(t.min),this.max.copy(t.max),this}makeEmpty(){return
this.min.x=this.min.y=this.min.z=1/0,this.max.x=this.max.y=this.max.z=-1/0,this}isE
mpty(){return
this.max.x<this.min.x||this.max.y<this.min.y||this.max.z<this.min.z}getCenter(t)
{return void 0===t&&(console.warn("THREE.Box3: .getCenter() target is now
required"),t=new Lt),this.isEmpty()?
t.set(0,0,0):t.addVectors(this.min,this.max).multiplyScalar(.5)}getSize(t){return
void 0===t&&(console.warn("THREE.Box3: .getSize() target is now required"),t=new
Lt),this.isEmpty()?t.set(0,0,0):t.subVectors(this.max,this.min)}expandByPoint(t)
{return this.min.min(t),this.max.max(t),this}expandByVector(t){return
this.min.sub(t),this.max.add(t),this}expandByScalar(t){return this.min.addScalar(-
t),this.max.addScalar(t),this}expandByObject(t){t.updateWorldMatrix(!1,!1);const
e=t.geometry;void 0!==e&&
(null===e.boundingBox&&e.computeBoundingBox(),Nt.copy(e.boundingBox),Nt.applyMatrix
4(t.matrixWorld),this.union(Nt));const n=t.children;for(let
t=0,e=n.length;t<e;t++)this.expandByObject(n[t]);return this}containsPoint(t)
{return!
(t.x<this.min.x||t.x>this.max.x||t.y<this.min.y||t.y>this.max.y||t.z<this.min.z||t.
z>this.max.z)}containsBox(t){return
this.min.x<=t.min.x&&t.max.x<=this.max.x&&this.min.y<=t.min.y&&t.max.y<=this.max.y&
&this.min.z<=t.min.z&&t.max.z<=this.max.z}getParameter(t,e){return void 0===e&&
(console.warn("THREE.Box3: .getParameter() target is now required"),e=new
Lt),e.set((t.x-this.min.x)/(this.max.x-this.min.x),(t.y-this.min.y)/(this.max.y-
this.min.y),(t.z-this.min.z)/(this.max.z-this.min.z))}intersectsBox(t){return!
(t.max.x<this.min.x||t.min.x>this.max.x||t.max.y<this.min.y||t.min.y>this.max.y||t.
max.z<this.min.z||t.min.z>this.max.z)}intersectsSphere(t){return
this.clampPoint(t.center,It),It.distanceToSquared(t.center)
<=t.radius*t.radius}intersectsPlane(t){let e,n;return t.normal.x>0?
(e=t.normal.x*this.min.x,n=t.normal.x*this.max.x):
(e=t.normal.x*this.max.x,n=t.normal.x*this.min.x),t.normal.y>0?
(e+=t.normal.y*this.min.y,n+=t.normal.y*this.max.y):
(e+=t.normal.y*this.max.y,n+=t.normal.y*this.min.y),t.normal.z>0?
(e+=t.normal.z*this.min.z,n+=t.normal.z*this.max.z):
(e+=t.normal.z*this.max.z,n+=t.normal.z*this.min.z),e<=-t.constant&&n>=-
t.constant}intersectsTriangle(t)

```

```

{if(this.isEmpty())return!1;this.getCenter(Ut),kt.subVectors(this.max,Ut),Bt.subVec
tors(t.a,Ut),zt.subVectors(t.b,Ut),Ft.subVectors(t.c,Ut),Ot.subVectors(zt,Bt),Ht.su
bVectors(Ft,zt),Gt.subVectors(Bt,Ft);let e=[0,-Ot.z,Ot.y,0,-Ht.z,Ht.y,0,-
Gt.z,Gt.y,Ot.z,0,-Ot.x,Ht.z,0,-Ht.x,Gt.z,0,-Gt.x,-Ot.y,Ot.x,0,-Ht.y,Ht.x,0,-
Gt.y,Gt.x,0];return!!jt(e,Bt,zt,Ft,kt)&&(e=
[1,0,0,0,1,0,0,0,1],!!jt(e,Bt,zt,Ft,kt)&&(Vt.crossVectors(Ot,Ht),e=
[Vt.x,Vt.y,Vt.z],jt(e,Bt,zt,Ft,kt)))}clampPoint(t,e){return void 0===e&&
(console.warn("THREE.Box3: .clampPoint() target is now required"),e=new
Lt),e.copy(t).clamp(this.min,this.max)}distanceToPoint(t){return
It.copy(t).clamp(this.min,this.max).sub(t).length()}getBoundingSphere(t){return
void 0===t&&console.error("THREE.Box3: .getBoundingSphere() target is now
required"),this.getCenter(t.center),t.radius=.5*this.getSize(It).length(),t}interse
ct(t){return
this.min.max(t.min),this.max.min(t.max),this.isEmpty()&&this.makeEmpty(),this}union
(t){return this.min.min(t.min),this.max.max(t.max),this}applyMatrix4(t){return
this.isEmpty()||
(Dt[0].set(this.min.x,this.min.y,this.min.z).applyMatrix4(t),Dt[1].set(this.min.x,t
his.min.y,this.max.z).applyMatrix4(t),Dt[2].set(this.min.x,this.max.y,this.min.z).a
pplyMatrix4(t),Dt[3].set(this.min.x,this.max.y,this.max.z).applyMatrix4(t),Dt[4].se
t(this.max.x,this.min.y,this.min.z).applyMatrix4(t),Dt[5].set(this.max.x,this.min.y
,this.max.z).applyMatrix4(t),Dt[6].set(this.max.x,this.max.y,this.min.z).applyMatri
x4(t),Dt[7].set(this.max.x,this.max.y,this.max.z).applyMatrix4(t),this.setFromPoint
s(Dt)),this}translate(t){return this.min.add(t),this.max.add(t),this}equals(t)
{return
t.min.equals(this.min)&&t.max.equals(this.max)}}Pt.prototype.isBox3=!0;const Dt=
[new Lt,new Lt,new Lt,new Lt,new Lt,new Lt,new Lt,new Lt],It=new Lt,Nt=new
Pt,Bt=new Lt,zt=new Lt,Ft=new Lt,Ot=new Lt,Ht=new Lt,Gt=new Lt,Ut=new Lt,kt=new
Lt,Vt=new Lt,Wt=new Lt;function jt(t,e,n,i,r){for(let s=0,a=t.length-3;s<=a;s+=3)
{Wt.fromArray(t,s);const
a=r.x*Math.abs(Wt.x)+r.y*Math.abs(Wt.y)+r.z*Math.abs(Wt.z),o=e.dot(Wt),l=n.dot(Wt),
c=i.dot(Wt);if(Math.max(-Math.max(o,l,c),Math.min(o,l,c))>a)return!1}return!0}const
qt=new Pt,Xt=new Lt,Yt=new Lt,Zt=new Lt;class Jt{constructor(t=new Lt,e=-1)
{this.center=t,this.radius=e}set(t,e){return
this.center.copy(t),this.radius=e,this}setFromPoints(t,e){const n=this.center;void
0!==e?n.copy(e):qt.setFromPoints(t).getCenter(n);let i=0;for(let
e=0,r=t.length;e<r;e++)i=Math.max(i,n.distanceToSquared(t[e]));return
this.radius=Math.sqrt(i),this}copy(t){return
this.center.copy(t.center),this.radius=t.radius,this}isEmpty(){return
this.radius<0}makeEmpty(){return
this.center.set(0,0,0),this.radius=-1,this}containsPoint(t){return
t.distanceToSquared(this.center)<=this.radius*this.radius}distanceToPoint(t){return
t.distanceTo(this.center)-this.radius}intersectsSphere(t){const
e=this.radius+t.radius;return t.center.distanceToSquared(this.center)
<=e*e}intersectsBox(t){return t.intersectsSphere(this)}intersectsPlane(t){return
Math.abs(t.distanceToPoint(this.center))<=this.radius}clampPoint(t,e){const
n=this.center.distanceToSquared(t);return void 0===e&&(console.warn("THREE.Sphere:
.clampPoint() target is now required"),e=new
Lt),e.copy(t),n>this.radius*this.radius&&
(e.sub(this.center).normalize(),e.multiplyScalar(this.radius).add(this.center)),e}g
etBoundingBox(t){return void 0===t&&(console.warn("THREE.Sphere: .getBoundingBox()
target is now required"),t=new Pt),this.isEmpty()?(t.makeEmpty(),t):
(t.set(this.center,this.center),t.expandByScalar(this.radius),t)}applyMatrix4(t)
{return
this.center.applyMatrix4(t),this.radius=this.radius*t.getMaxScaleOnAxis(),this}tran
slate(t){return this.center.add(t),this}expandByPoint(t)
{Zt.subVectors(t,this.center);const e=Zt.lengthSq();if(e>this.radius*this.radius)
{const t=Math.sqrt(e),n=.5*(t-
this.radius);this.center.add(Zt.multiplyScalar(n/t)),this.radius+=n}return

```

```

this}union(t){return
Yt.subVectors(t.center,this.center).normalize().multiplyScalar(t.radius),this.expandByPoint(Xt.copy(t.center).add(Yt)),this.expandByPoint(Xt.copy(t.center).sub(Yt)),this}equals(t){return t.center.equals(this.center)&&t.radius===this.radius}clone(){return(new this.constructor).copy(this)}const Qt=new Lt,Kt=new Lt,$t=new Lt,te=new Lt,ee=new Lt,ne=new Lt,ie=new Lt;class re{constructor(t=new Lt,e=new Lt(0,0,-1)){this.origin=t,this.direction=e}set(t,e){return this.origin.copy(t),this.direction.copy(e),this}copy(t){return this.origin.copy(t.origin),this.direction.copy(t.direction),this}at(t,e){return void 0===e&&(console.warn("THREE.Ray: .at() target is now required"),e=new Lt),e.copy(this.direction).multiplyScalar(t.add(this.origin))}lookAt(t){return this.direction.copy(t.sub(this.origin).normalize(),this}recast(t){return this.origin.copy(this.at(t,Qt)),this}closestPointToPoint(t,e){void 0===e&&(console.warn("THREE.Ray: .closestPointToPoint() target is now required"),e=new Lt),e.subVectors(t,this.origin);const n=e.dot(this.direction);return n<0?e.copy(this.origin):e.copy(this.direction).multiplyScalar(n.add(this.origin))}distanceToPoint(t){return Math.sqrt(this.distanceSqToPoint(t))}distanceSqToPoint(t){const e=Qt.subVectors(t,this.origin).dot(this.direction);return e<0?this.origin.distanceToSquared(t):(Qt.copy(this.direction).multiplyScalar(e).add(this.origin),Qt.distanceToSquared(t))}distanceSqToSegment(t,e,n,i){Kt.copy(t).add(e).multiplyScalar(.5),$t.copy(e).sub(t).normalize(),te.copy(this.origin).sub(Kt);const r=.5*t.distanceTo(e),s=-this.direction.dot($t),a=te.dot(this.direction),o=-te.dot($t),l=te.lengthSq(),c=Math.abs(1-s*s);let h,u,d,p;if(c>0){if(h=s*o-a,u=s*a-o,p=r*c,h>=0){if(u>=-p){if(u<=p){const t=l/c,h*=t,u*=t,d=h*(h+s*u+2*a)+u*(s*h+u+2*o)+1}else u=r,h=Math.max(0,-(s*u+a)),d=-h*h+u*(u+2*o)+1}else u=-r,h=Math.max(0,-(s*u+a)),d=-h*h+u*(u+2*o)+1}else u<=-p?(h=Math.max(0,-(s*r+a)),u=h>0?-r:Math.min(Math.max(-r,-o),r),d=-h*h+u*(u+2*o)+1):u<=p?(h=0,u=Math.min(Math.max(-r,-o),r),d=u*(u+2*o)+1):(h=Math.max(0,-(s*r+a)),u=h>0?r:Math.min(Math.max(-r,-o),r),d=-h*h+u*(u+2*o)+1}else u=s>0?-r:r,h=Math.max(0,-(s*u+a)),d=-h*h+u*(u+2*o)+1;return n&&n.copy(this.direction).multiplyScalar(h).add(this.origin),i&&i.copy($t).multiplyScalar(u).add(Kt),d}intersectSphere(t,e){Qt.subVectors(t.center,this.origin);const n=Qt.dot(this.direction),i=Qt.dot(Qt)-n*n,r=t.radius*t.radius;if(i>r)return null;const s=Math.sqrt(r-i),a=n-s,o=n+s;return a<0&&o<0?null:a<0?this.at(o,e):this.at(a,e)}intersectsSphere(t){return this.distanceSqToPoint(t.center)<=t.radius*t.radius}distanceToPlane(t){const e=t.normal.dot(this.direction);if(0===e)return 0===t.distanceToPoint(this.origin)?0:null;const n=-(this.origin.dot(t.normal)+t.constant)/e;return n>=0?n:null}intersectPlane(t,e){const n=this.distanceToPlane(t);return null===n?null:this.at(n,e)}intersectsPlane(t){const e=t.distanceToPoint(this.origin);if(0===e)return!0;return t.normal.dot(this.direction)*e<0}intersectBox(t,e){let n,i,r,s,a,o;const l=1/this.direction.x,c=1/this.direction.y,h=1/this.direction.z,u=this.origin;return l>=0?(n=(t.min.x-u.x)*l,i=(t.max.x-u.x)*l):(n=(t.max.x-u.x)*l,i=(t.min.x-u.x)*l),c>=0?(r=(t.min.y-u.y)*c,s=(t.max.y-u.y)*c):(r=(t.max.y-u.y)*c,s=(t.min.y-u.y)*c),n>s||r>i?null:((r>n||n!=n)&&(n=r),(s<i||i!=i)&&(i=s),h>=0?(a=(t.min.z-u.z)*h,o=(t.max.z-u.z)*h):(a=(t.max.z-u.z)*h,o=(t.min.z-u.z)*h),n>o||a>i?null:((a>n||n!=n)&&(n=a),(o<i||i!=i)&&(i=o),i<0?null:this.at(n>=0?n:i,e)))}intersectsBox(t){return null!==this.intersectBox(t,Qt)}intersectTriangle(t,e,n,i,r){{ee.subVectors(e,t),ne.subVectors(n,t),ie.crossVectors(ee,ne);let s,a=this.direction.dot(ie);if(a>0){if(i)return null;s=1}else{if(!(a<0))return null;s=-1,a=-a}te.subVectors(this.origin,t);const o=s*this.direction.dot(ne.crossVectors(te,ne));if(o<0)return null;const l=s*this.direction.dot(ee.cross(te));if(l<0)return null;if(o+l>a)return null;const c=-s*te.dot(ie);return c<0?null:this.at(c/a,r)}applyMatrix4(t){return

```

```

this.origin.applyMatrix4(t),this.direction.transformDirection(t),this.equals(t)
{return t.origin.equals(this.origin)&&t.direction.equals(this.direction)}clone()
{return(new this.constructor).copy(this)}}class se{constructor(){this.elements=
[1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1],arguments.length>0&&console.error("THREE.Matrix4:
the constructor no longer reads arguments. use .set()
instead.")}set(t,e,n,i,r,s,a,o,l,c,h,u,d,p,m,f){const g=this.elements;return
g[0]=t,g[4]=e,g[8]=n,g[12]=i,g[1]=r,g[5]=s,g[9]=a,g[13]=o,g[2]=l,g[6]=c,g[10]=h,g[14]=u,g[3]=d,g[7]=p,g[11]=m,g[15]=f,this}identity(){return
this.set(1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1),this}clone(){return(new
se).fromArray(this.elements)}copy(t){const e=this.elements,n=t.elements;return
e[0]=n[0],e[1]=n[1],e[2]=n[2],e[3]=n[3],e[4]=n[4],e[5]=n[5],e[6]=n[6],e[7]=n[7],e[8]=n[8],e[9]=n[9],e[10]=n[10],e[11]=n[11],e[12]=n[12],e[13]=n[13],e[14]=n[14],e[15]=n[15],this}copyPosition(t){const e=this.elements,n=t.elements;return
e[12]=n[12],e[13]=n[13],e[14]=n[14],this}setFromMatrix3(t){const
e=t.elements;return
this.set(e[0],e[3],e[6],0,e[1],e[4],e[7],0,e[2],e[5],e[8],0,0,0,0,1),this}extractBasis(t,e,n){return
t.setFromMatrixColumn(this,0),e.setFromMatrixColumn(this,1),n.setFromMatrixColumn(this,2),this}makeBasis(t,e,n){return
this.set(t.x,e.x,n.x,0,t.y,e.y,n.y,0,t.z,e.z,n.z,0,0,0,0,1),this}extractRotation(t){const
e=this.elements,n=t.elements,i=1/ae.setFromMatrixColumn(t,0).length(),r=1/ae.setFromMatrixColumn(t,1).length(),s=1/ae.setFromMatrixColumn(t,2).length();return
e[0]=n[0]*i,e[1]=n[1]*i,e[2]=n[2]*i,e[3]=0,e[4]=n[4]*r,e[5]=n[5]*r,e[6]=n[6]*r,e[7]=0,e[8]=n[8]*s,e[9]=n[9]*s,e[10]=n[10]*s,e[11]=0,e[12]=0,e[13]=0,e[14]=0,e[15]=1,this}makeRotationFromEuler(t){t&&t.isEuler||console.error("THREE.Matrix4:
.makeRotationFromEuler() now expects a Euler rotation rather than a Vector3 and
order.");const
e=this.elements,n=t.x,i=t.y,r=t.z,s=Math.cos(n),a=Math.sin(n),o=Math.cos(i),l=Math.sin(i),c=Math.cos(r),h=Math.sin(r);if("XYZ"===t.order){const
t=s*c,n=s*h,i=a*c,r=a*h;e[0]=o*c,e[4]=-o*h,e[8]=l,e[1]=n+i*l,e[5]=t-r*l,e[9]=-a*o,e[2]=r-t*l,e[6]=i+n*l,e[10]=s*o}else if("YXZ"===t.order){const
t=o*c,n=o*h,i=l*c,r=l*h;e[0]=t+r*a,e[4]=i*a-n,e[8]=s*l,e[1]=s*h,e[5]=s*c,e[9]=-a,e[2]=n*a-i,e[6]=r+t*a,e[10]=s*o}else if("ZXY"===t.order){const
t=o*c,n=o*h,i=l*c,r=l*h;e[0]=t-r*a,e[4]=-s*h,e[8]=i+n*a,e[1]=n+i*a,e[5]=s*c,e[9]=r-t*a,e[2]=-s*l,e[6]=a,e[10]=s*o}else if("ZYX"===t.order){const
t=s*c,n=s*h,i=a*c,r=a*h;e[0]=o*c,e[4]=i*l-n,e[8]=t*l+r,e[1]=o*h,e[5]=r*l+t,e[9]=n*l-i,e[2]=-l,e[6]=a*o,e[10]=s*o}else
if("YZX"===t.order){const t=s*o,n=s*l,i=a*o,r=a*l;e[0]=o*c,e[4]=r-t*h,e[8]=i*h+n,e[1]=h,e[5]=s*c,e[9]=-a*c,e[2]=-l*c,e[6]=n*h+i,e[10]=t-r*h}else
if("XZY"===t.order){const t=s*o,n=s*l,i=a*o,r=a*l;e[0]=o*c,e[4]=-h,e[8]=l*c,e[1]=t*h+r,e[5]=s*c,e[9]=n*h-i,e[2]=i*h-n,e[6]=a*c,e[10]=r*h+t}return
e[3]=0,e[7]=0,e[11]=0,e[12]=0,e[13]=0,e[14]=0,e[15]=1,this}makeRotationFromQuaternion(t){return this.compose(1e,t,ce)}lookAt(t,e,n){const i=this.elements;return
de.subVectors(t,e),0===de.lengthSq()&&
(de.z=1),de.normalize(),he.crossVectors(n,de),0===he.lengthSq()&&
(1===Math.abs(n.z)?de.x+=1e-4:de.z+=1e-4,de.normalize(),he.crossVectors(n,de)),he.normalize(),ue.crossVectors(de,he),i[0]=he.x,i[4]=ue.x,i[8]=de.x,i[1]=he.y,i[5]=ue.y,i[9]=de.y,i[2]=he.z,i[6]=ue.z,i[10]=de.z,this}multiply(t,e){return void 0!==e?(console.warn("THREE.Matrix4: .multiply()
now only accepts one argument. Use .multiplyMatrices( a, b )
instead."),this.multiplyMatrices(t,e)):this.multiplyMatrices(this,t)}premultiply(t){return this.multiplyMatrices(t,this)}multiplyMatrices(t,e){const
n=t.elements,i=e.elements,r=this.elements,s=n[0],a=n[4],o=n[8],l=n[12],c=n[1],h=n[5],u=n[9],d=n[13],p=n[2],m=n[6],f=n[10],g=n[14],v=n[3],y=n[7],x=n[11],_=n[15],w=i[0],b=i[4],M=i[8],S=i[12],T=i[1],E=i[5],A=i[9],L=i[13],R=i[2],C=i[6],P=i[10],D=i[14],I=i[3],N=i[7],B=i[11],z=i[15];return
r[0]=s*w+a*T+o*R+l*I,r[4]=s*b+a*E+o*C+l*N,r[8]=s*M+a*A+o*P+l*B,r[12]=s*S+a*L+o*D+l*

```

```

z,r[1]=c*w+h*T+u*R+d*I,r[5]=c*b+h*E+u*C+d*N,r[9]=c*M+h*A+u*P+d*B,r[13]=c*S+h*L+u*D+
d*z,r[2]=p*w+m*T+f*R+g*I,r[6]=p*b+m*E+f*C+g*N,r[10]=p*M+m*A+f*P+g*B,r[14]=p*S+m*L+f
*D+g*z,r[3]=v*w+y*T+x*R+_I,r[7]=v*b+y*E+x*C+_N,r[11]=v*M+y*A+x*P+_B,r[15]=v*S+y*
L+x*D+_z,this}multiplyScalar(t){const e=this.elements;return
e[0]*=t,e[4]*=t,e[8]*=t,e[12]*=t,e[1]*=t,e[5]*=t,e[9]*=t,e[13]*=t,e[2]*=t,e[6]*=t,e
[10]*=t,e[14]*=t,e[3]*=t,e[7]*=t,e[11]*=t,e[15]*=t,this}determinant(){const
t=this.elements,e=t[0],n=t[4],i=t[8],r=t[12],s=t[1],a=t[5],o=t[9],l=t[13],c=t[2],h=
t[6],u=t[10],d=t[14];return t[3]*(+r*o*h-i*l*h-r*a*u+n*l*u+i*a*d-n*o*d)+t[7]*
(+e*o*d-e*l*u+r*s*u-i*s*d+i*l*c-r*o*c)+t[11]*(+e*l*h-e*a*d-r*s*h+n*s*d+r*a*c-
n*l*c)+t[15]*(-i*a*c-e*o*h+e*a*u+i*s*h-n*s*u+n*o*c)}transpose(){const
t=this.elements;let e;return
e=t[1],t[1]=t[4],t[4]=e,e=t[2],t[2]=t[8],t[8]=e,e=t[6],t[6]=t[9],t[9]=e,e=t[3],t[3]
=t[12],t[12]=e,e=t[7],t[7]=t[13],t[13]=e,e=t[11],t[11]=t[14],t[14]=e,this}setPositi
on(t,e,n){const i=this.elements;return t.isVector3?(i[12]=t.x,i[13]=t.y,i[14]=t.z):
(i[12]=t,i[13]=e,i[14]=n),this}invert(){const
t=this.elements,e=t[0],n=t[1],i=t[2],r=t[3],s=t[4],a=t[5],o=t[6],l=t[7],c=t[8],h=t[
9],u=t[10],d=t[11],p=t[12],m=t[13],f=t[14],g=t[15],v=h*f*l-m*u*l+m*o*d-a*f*d-
h*o*g+a*u*g,y=p*u*l-c*f*l-p*o*d+s*f*d+c*o*g-s*u*g,x=c*m*l-p*h*l+p*a*d-s*m*d-
c*a*g+s*h*g,_=p*h*o-c*m*o-p*a*u+s*m*u+c*a*f-s*h*f,w=e*v+n*y+i*x+r*_;if(0===w)return
this.set(0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0);const b=1/w;return t[0]=v*b,t[1]=(m*u*r-
h*f*r-m*i*d+n*f*d+h*i*g-n*u*g)*b,t[2]=(a*f*r-m*o*r+m*i*l-n*f*l-a*i*g+n*o*g)*b,t[3]=
(h*o*r-a*u*r-h*i*l+n*u*l+a*i*d-n*o*d)*b,t[4]=y*b,t[5]=(c*f*r-p*u*r+p*i*d-e*f*d-
c*i*g+e*u*g)*b,t[6]=(p*o*r-s*f*r-p*i*l+e*f*l+s*i*g-e*o*g)*b,t[7]=(s*u*r-
c*o*r+c*i*l-e*u*l-s*i*d+e*o*d)*b,t[8]=x*b,t[9]=(p*h*r-c*m*r-p*n*d+e*m*d+c*n*g-
e*h*g)*b,t[10]=(s*m*r-p*a*r+p*n*l-e*m*l-s*n*g+e*a*g)*b,t[11]=(c*a*r-s*h*r-
c*n*l+e*h*l+s*n*d-e*a*d)*b,t[12]=_b,t[13]=(c*m*i-p*h*i+p*n*u-e*m*u-
c*n*f+e*h*f)*b,t[14]=(p*a*i-s*m*i-p*n*o+e*m*o+s*n*f-e*a*f)*b,t[15]=(s*h*i-
c*a*i+c*n*o-e*h*o-s*n*u+e*a*u)*b,this}scale(t){const
e=this.elements,n=t.x,i=t.y,r=t.z;return
e[0]*=n,e[4]*=i,e[8]*=r,e[1]*=n,e[5]*=i,e[9]*=r,e[2]*=n,e[6]*=i,e[10]*=r,e[3]*=n,e[
7]*=i,e[11]*=r,this}getMaxScaleOnAxis(){const
t=this.elements,e=t[0]*t[0]+t[1]*t[1]+t[2]*t[2],n=t[4]*t[4]+t[5]*t[5]+t[6]*t[6],i=t
[8]*t[8]+t[9]*t[9]+t[10]*t[10];return
Math.sqrt(Math.max(e,n,i))}makeTranslation(t,e,n){return
this.set(1,0,0,t,0,1,0,e,0,0,1,n,0,0,0,1),this}makeRotationX(t){const
e=Math.cos(t),n=Math.sin(t);return this.set(1,0,0,0,0,e,-
n,0,0,n,e,0,0,0,0,1),this}makeRotationY(t){const e=Math.cos(t),n=Math.sin(t);return
this.set(e,0,n,0,0,1,0,0,-n,0,e,0,0,0,0,1),this}makeRotationZ(t){const
e=Math.cos(t),n=Math.sin(t);return this.set(e,-
n,0,0,n,e,0,0,0,0,1,0,0,0,0,1),this}makeRotationAxis(t,e){const
n=Math.cos(e),i=Math.sin(e),r=1-n,s=t.x,a=t.y,o=t.z,l=r*s,c=r*a;return
this.set(l*s+n,l*a-i*o,l*o+i*a,0,l*a+i*o,c*a+n,c*o-i*s,0,l*o-
i*a,c*o+i*s,r*o*o+n,0,0,0,0,1),this}makeScale(t,e,n){return
this.set(t,0,0,0,0,e,0,0,0,0,n,0,0,0,0,1),this}makeShear(t,e,n){return
this.set(1,e,n,0,t,1,n,0,t,e,1,0,0,0,0,1),this}compose(t,e,n){const
i=this.elements,r=e._x,s=e._y,a=e._z,o=e._w,l=r+r,c=s+s,h=a+a,u=r*l,d=r*c,p=r*h,m=s
*c,f=s*h,g=a*h,v=o*l,y=o*c,x=o*h,_=n.x,w=n.y,b=n.z;return i[0]=(1-(m+g))*_,i[1]=
(d+x)*_,i[2]=(p-y)*_,i[3]=0,i[4]=(d-x)*w,i[5]=(1-(u+g))*w,i[6]=(f+v)*w,i[7]=0,i[8]=
(p+y)*b,i[9]=(f-v)*b,i[10]=(1-
(u+m))*b,i[11]=0,i[12]=t.x,i[13]=t.y,i[14]=t.z,i[15]=1,this}decompose(t,e,n){const
i=this.elements;let r=ae.set(i[0],i[1],i[2]).length();const
s=ae.set(i[4],i[5],i[6]).length(),a=ae.set(i[8],i[9],i[10]).length();this.determina
nt()<0&&(r=-r),t.x=i[12],t.y=i[13],t.z=i[14],oe.copy(this);const
o=1/r,l=1/s,c=1/a;return
oe.elements[0]*=o,oe.elements[1]*=o,oe.elements[2]*=o,oe.elements[4]*=l,oe.elements
[5]*=l,oe.elements[6]*=l,oe.elements[8]*=c,oe.elements[9]*=c,oe.elements[10]*=c,e.s
etFromRotationMatrix(oe),n.x=r,n.y=s,n.z=a,this}makePerspective(t,e,n,i,r,s){void
0===s&&console.warn("THREE.Matrix4: .makePerspective() has been redefined and has a

```

```

new signature. Please check the docs.");const a=this.elements,o=2*r/(e-t),l=2*r/(n-i),c=(e+t)/(e-t),h=(n+i)/(n-i),u=-(s+r)/(s-r),d=-2*s*r/(s-r);return
a[0]=o,a[4]=0,a[8]=c,a[12]=0,a[1]=0,a[5]=l,a[9]=h,a[13]=0,a[2]=0,a[6]=0,a[10]=u,a[14]=d,a[3]=0,a[7]=0,a[11]=-1,a[15]=0,this}makeOrthographic(t,e,n,i,r,s){const
a=this.elements,o=1/(e-t),l=1/(n-i),c=1/(s-r),h=(e+t)*o,u=(n+i)*l,d=(s+r)*c;return
a[0]=2*o,a[4]=0,a[8]=0,a[12]=-h,a[1]=0,a[5]=2*l,a[9]=0,a[13]=-
u,a[2]=0,a[6]=0,a[10]=-2*c,a[14]=-d,a[3]=0,a[7]=0,a[11]=0,a[15]=1,this}equals(t)
{const e=this.elements,n=t.elements;for(let
t=0;t<16;t++)if(e[t]!==n[t])return!1;return!0}fromArray(t,e=0){for(let
n=0;n<16;n++)this.elements[n]=t[n+e];return this}toArray(t=[],e=0){const
n=this.elements;return
t[e]=n[0],t[e+1]=n[1],t[e+2]=n[2],t[e+3]=n[3],t[e+4]=n[4],t[e+5]=n[5],t[e+6]=n[6],t
[e+7]=n[7],t[e+8]=n[8],t[e+9]=n[9],t[e+10]=n[10],t[e+11]=n[11],t[e+12]=n[12],t[e+13
]=n[13],t[e+14]=n[14],t[e+15]=n[15],t}}se.prototype.isMatrix4=!0;const ae=new
Lt,oe=new se,le=new Lt(0,0,0),ce=new Lt(1,1,1),he=new Lt,ue=new Lt,de=new Lt,pe=new
se,me=new At;class fe{constructor(t=0,e=0,n=0,i=fe.DefaultOrder)
{this._x=t,this._y=e,this._z=n,this._order=i}get x(){return this._x}set x(t)
{this._x=t,this._onChangeCallback()}get y(){return this._y}set y(t)
{this._y=t,this._onChangeCallback()}get z(){return this._z}set z(t)
{this._z=t,this._onChangeCallback()}get order(){return this._order}set order(t)
{this._order=t,this._onChangeCallback()}set(t,e,n,i){return
this._x=t,this._y=e,this._z=n,this._order=i||this._order,this._onChangeCallback(),t
his}clone(){return new
this.constructor(this._x,this._y,this._z,this._order)}copy(t){return
this._x=t._x,this._y=t._y,this._z=t._z,this._order=t._order,this._onChangeCallback(
),this}setFromRotationMatrix(t,e,n){const
i=t.elements,r=i[0],s=i[4],a=i[8],o=i[1],l=i[5],c=i[9],h=i[2],u=i[6],d=i[10];switch
(e=e||this._order){case"XYZ":this._y=Math.asin(ht(a,-1,1)),Math.abs(a)<.9999999?
(this._x=Math.atan2(-c,d),this._z=Math.atan2(-s,r)):
(this._x=Math.atan2(u,l),this._z=0);break;case"YXZ":this._x=Math.asin(-
ht(c,-1,1)),Math.abs(c)<.9999999?(this._y=Math.atan2(a,d),this._z=Math.atan2(o,l)):
(this._y=Math.atan2(-
h,r),this._z=0);break;case"ZXY":this._x=Math.asin(ht(u,-1,1)),Math.abs(u)<.9999999?
(this._y=Math.atan2(-h,d),this._z=Math.atan2(-s,l)):
(this._y=0,this._z=Math.atan2(o,r));break;case"ZYX":this._y=Math.asin(-
ht(h,-1,1)),Math.abs(h)<.9999999?(this._x=Math.atan2(u,d),this._z=Math.atan2(o,r)):
(this._x=0,this._z=Math.atan2(-
s,l));break;case"YZX":this._z=Math.asin(ht(o,-1,1)),Math.abs(o)<.9999999?
(this._x=Math.atan2(-c,l),this._y=Math.atan2(-h,r)):
(this._x=0,this._y=Math.atan2(a,d));break;case"XZY":this._z=Math.asin(-
ht(s,-1,1)),Math.abs(s)<.9999999?(this._x=Math.atan2(u,l),this._y=Math.atan2(a,r)):
(this._x=Math.atan2(-c,d),this._y=0);break;default:console.warn("THREE.Euler:
.setFromRotationMatrix() encountered an unknown order: "+e)}return
this._order=e,!1!==n&&this._onChangeCallback(),this}setFromQuaternion(t,e,n){return
pe.makeRotationFromQuaternion(t),this.setFromRotationMatrix(pe,e,n)}setFromVector3(
t,e){return this.set(t.x,t.y,t.z,e||this._order)}reorder(t){return
me.setFromEuler(this),this.setFromQuaternion(me,t)}equals(t){return
t._x===this._x&&t._y===this._y&&t._z===this._z&&t._order===this._order}fromArray(t)
{return this._x=t[0],this._y=t[1],this._z=t[2],void 0!==t[3]&&
(this._order=t[3]),this._onChangeCallback(),this}toArray(t=[],e=0){return
t[e]=this._x,t[e+1]=this._y,t[e+2]=this._z,t[e+3]=this._order,t}toVector3(t){return
t?t.set(this._x,this._y,this._z):new Lt(this._x,this._y,this._z)}_onChange(t)
{return this._onChangeCallback=t,this}_onChangeCallback()
{}}fe.prototype.isEuler=!0,fe.DefaultOrder="XYZ",fe.RotationOrders=
["XYZ","YZX","ZXY","XZY","YXZ","ZYX"];class ge{constructor(){this.mask=1}set(t)
{this.mask=1<<t|0}enable(t){this.mask|=1<<t|0}enableAll(){this.mask=-1}toggle(t)
{this.mask^=1<<t|0}disable(t){this.mask&=~(1<<t|0)}disableAll(){this.mask=0}test(t)
{return 0!==(this.mask&t.mask)}}let ve=0;const ye=new Lt,xe=new At,_e=new se,we=new

```

```

Lt,be=new Lt,Me=new Lt,Se=new At,Te=new Lt(1,0,0),Ee=new Lt(0,1,0),Ae=new
Lt(0,0,1),Le={type:"added"},Re={type:"removed"};class Ce extends rt{constructor()
{super(),Object.defineProperty(this,"id",
{value:ve++}),this.uuid=ct(),this.name="",this.type="Object3D",this.parent=null,thi
s.children=[],this.up=Ce.DefaultUp.clone();const t=new Lt,e=new fe,n=new At,i=new
Lt(1,1,1);e._onChange((function(){n.setFromEuler(e,!1)})),n._onChange((function()
{e.setFromQuaternion(n,void 0,!1)})),Object.defineProperties(this,{position:
{configurable:!0,enumerable:!0,value:t},rotation:
{configurable:!0,enumerable:!0,value:e},quaternion:
{configurable:!0,enumerable:!0,value:n},scale:
{configurable:!0,enumerable:!0,value:i},modelViewMatrix:{value:new
se},normalMatrix:{value:new yt}}),this.matrix=new se,this.matrixWorld=new
se,this.matrixAutoUpdate=Ce.DefaultMatrixAutoUpdate,this.matrixWorldNeedsUpdate=!1,
this.layers=new
ge,this.visible=!0,this.castShadow=!1,this.receiveShadow=!1,this.frustumCulled=!0,t
his.renderOrder=0,this.animations=[],this.userData={}}onBeforeRender()
{}onAfterRender(){}applyMatrix4(t)
{this.matrixAutoUpdate&&this.updateMatrix(),this.matrix.premultiply(t),this.matrix.
decompose(this.position,this.quaternion,this.scale)}applyQuaternion(t){return
this.quaternion.premultiply(t),this.setRotationFromAxisAngle(t,e)
{this.quaternion.setFromAxisAngle(t,e)}setRotationFromEuler(t)
{this.quaternion.setFromEuler(t,!0)}setRotationFromMatrix(t)
{this.quaternion.setFromRotationMatrix(t)}setRotationFromQuaternion(t)
{this.quaternion.copy(t)}rotateOnAxis(t,e){return
xe.setFromAxisAngle(t,e),this.quaternion.multiply(xe),this}rotateOnWorldAxis(t,e)
{return xe.setFromAxisAngle(t,e),this.quaternion.premultiply(xe),this}rotateX(t)
{return this.rotateOnAxis(Te,t)}rotateY(t){return
this.rotateOnAxis(Ee,t)}rotateZ(t){return
this.rotateOnAxis(Ae,t)}translateOnAxis(t,e){return
ye.copy(t).applyQuaternion(this.quaternion),this.position.add(ye.multiplyScalar(e))
,this}translateX(t){return this.translateOnAxis(Te,t)}translateY(t){return
this.translateOnAxis(Ee,t)}translateZ(t){return
this.translateOnAxis(Ae,t)}localToWorld(t){return
t.applyMatrix4(this.matrixWorld)}worldToLocal(t){return
t.applyMatrix4(_e.copy(this.matrixWorld).invert())}lookAt(t,e,n){t.isVector3?
we.copy(t):we.set(t,e,n);const
i=this.parent;this.updateWorldMatrix(!0,!1),be.setFromMatrixPosition(this.matrixWor
ld),this.isCamera||this.isLight?
_e.lookAt(be,we,this.up):_e.lookAt(we,be,this.up),this.quaternion.setFromRotationMa
trix(_e),i&&
(_e.extractRotation(i.matrixWorld),xe.setFromRotationMatrix(_e),this.quaternion.pre
multiply(xe.invert()))}add(t){if(arguments.length>1){for(let
t=0;t<arguments.length;t++)this.add(arguments[t]);return this}return t===this?
(console.error("THREE.Object3D.add: object can't be added as a child of
itself.",t),this):(t&&t.isObject3D?
(null!==t.parent&&t.parent.remove(t),t.parent=this,this.children.push(t),t.dispatch
Event(Le)):console.error("THREE.Object3D.add: object not an instance of
THREE.Object3D.",t),this)}remove(t){if(arguments.length>1){for(let
t=0;t<arguments.length;t++)this.remove(arguments[t]);return this}const
e=this.children.indexOf(t);return-1!==e&&
(t.parent=null,this.children.splice(e,1),t.dispatchEvent(Re)),this}clear(){for(let
t=0;t<this.children.length;t++){const
e=this.children[t];e.parent=null,e.dispatchEvent(Re)}return
this.children.length=0,this}attach(t){return
this.updateWorldMatrix(!0,!1),_e.copy(this.matrixWorld).invert(),null!==t.parent&&
(t.parent.updateWorldMatrix(!0,!1),_e.multiply(t.parent.matrixWorld)),t.applyMatrix
4(_e),this.add(t),t.updateWorldMatrix(!1,!0),this}getObjectById(t){return
this.getObjectByProperty("id",t)}getObjectByName(t){return

```



```

this.getObjectByProperty("name",t)}getObjectByProperty(t,e){if(this[t]===e)return
this;for(let n=0,i=this.children.length;n<i;n++){const
i=this.children[n].getObjectByProperty(t,e);if(void 0!==i)return
i}}getWorldPosition(t){return void 0===t&&(console.warn("THREE.Object3D:
.getWorldPosition() target is now required"),t=new
Lt),this.updateWorldMatrix(!0,!1),t.setFromMatrixPosition(this.matrixWorld)}getWorldQuaternion(t){return void 0===t&&(console.warn("THREE.Object3D:
.getWorldQuaternion() target is now required"),t=new
At),this.updateWorldMatrix(!0,!1),this.matrixWorld.decompose(be,t,Me),t}getWorldScale(t){return void 0===t&&(console.warn("THREE.Object3D: .getWorldScale() target is
now required"),t=new
Lt),this.updateWorldMatrix(!0,!1),this.matrixWorld.decompose(be,Se,t),t}getWorldDirection(t){void 0===t&&(console.warn("THREE.Object3D: .getWorldDirection() target is
now required"),t=new Lt),this.updateWorldMatrix(!0,!1);const
e=this.matrixWorld.elements;return t.set(e[8],e[9],e[10]).normalize()}raycast()
{}traverse(t){t(this);const e=this.children;for(let
n=0,i=e.length;n<i;n++)e[n].traverse(t)}traverseVisible(t)
{if(!1===this.visible)return;t(this);const e=this.children;for(let
n=0,i=e.length;n<i;n++)e[n].traverseVisible(t)}traverseAncestors(t){const
e=this.parent;null!==e&&(t(e),e.traverseAncestors(t))}updateMatrix()
{this.matrix.compose(this.position,this.quaternion,this.scale),this.matrixWorldNeedsUpdate=!0}updateMatrixWorld(t){this.matrixAutoUpdate&&this.updateMatrix(),
(this.matrixWorldNeedsUpdate|t)&&(null===this.parent?
this.matrixWorld.copy(this.matrix):this.matrixWorld.multiplyMatrices(this.parent.matrixWorld,this.matrix),this.matrixWorldNeedsUpdate=!1,t=!0);const
e=this.children;for(let
n=0,i=e.length;n<i;n++)e[n].updateMatrixWorld(t)}updateWorldMatrix(t,e){const
n=this.parent;if(!0===t&&null!==n&&n.updateWorldMatrix(!0,!1),this.matrixAutoUpdate&&this.updateMatrix(),null===this.parent?
this.matrixWorld.copy(this.matrix):this.matrixWorld.multiplyMatrices(this.parent.matrixWorld,this.matrix),!0===e){const t=this.children;for(let
e=0,n=t.length;e<n;e++)t[e].updateWorldMatrix(!1,!0)}}toJSON(t){const e=void
0===t||"string"===typeof t,n={};e&&(t={geometries:{},materials:{},textures:
{},images:{},shapes:{},skeletons:{},animations:{},n.metadata=
{version:4.5,type:"Object",generator:"Object3D.toJSON"}});const i={};function r(e,n)
{return void 0===e[n.uuid]&&
(e[n.uuid]=n.toJSON(t)),n.uuid}if(i.uuid=this.uuid,i.type=this.type,""!==this.name&
&(i.name=this.name),!0===this.castShadow&&
(i.castShadow=!0),!0===this.receiveShadow&&(i.receiveShadow=!0),!1===this.visible&&
(i.visible=!1),!1===this.frustumCulled&&(i.frustumCulled=!1),0!==this.renderOrder&&
(i.renderOrder=this.renderOrder),"{}"!==JSON.stringify(this.userData)&&
(i.userData=this.userData),i.layers=this.layers.mask,i.matrix=this.matrix.toArray(),
!1===this.matrixAutoUpdate&&(i.matrixAutoUpdate=!1),this.isInstancedMesh&&
(i.type="InstancedMesh",i.count=this.count,i.instanceMatrix=this.instanceMatrix.toJSON(),null!==this.instanceColor&&
(i.instanceColor=this.instanceColor.toJSON()),this.isMesh||this.isLine||this.isPoints){i.geometry=r(t.geometries,this.geometry);const
e=this.geometry.parameters;if(void 0!==e&&void 0!==e.shapes){const
n=e.shapes;if(Array.isArray(n))for(let e=0,i=n.length;e<i;e++){const
i=n[e];r(t.shapes,i)}else r(t.shapes,n)}}if(this.isSkinnedMesh&&
(i.bindMode=this.bindMode,i.bindMatrix=this.bindMatrix.toArray(),void
0!==this.skeleton&&
(r(t.skeletons,this.skeleton),i.skeleton=this.skeleton.uuid)),void
0!==this.material)if(Array.isArray(this.material)){const e=[];for(let
n=0,i=this.material.length;n<i;n++)e.push(r(t.materials,this.material[n]));i.material=e}else i.material=r(t.materials,this.material);if(this.children.length>0)
{i.children=[];for(let
e=0;e<this.children.length;e++)i.children.push(this.children[e].toJSON(t).object)}i

```

```

f(this.animations.length>0){i.animations=[];for(let
e=0;e<this.animations.length;e++){const
n=this.animations[e];i.animations.push(r(t.animations,n))}}if(e){const
e=s(t.geometries),i=s(t.materials),r=s(t.textures),a=s(t.images),o=s(t.shapes),l=s(
t.skeletons),c=s(t.animations);e.length>0&&(n.geometries=e),i.length>0&&
(n.materials=i),r.length>0&&(n.textures=r),a.length>0&&(n.images=a),o.length>0&&
(n.shapes=o),l.length>0&&(n.skeletons=l),c.length>0&&(n.animations=c)}return
n.object=i,n;function s(t){const e=[];for(const n in t){const i=t[n];delete
i.metadata,e.push(i)}return e}}clone(t){return(new
this.constructor).copy(this,t)}copy(t,e=!0)
{if(this.name=t.name,this.up.copy(t.up),this.position.copy(t.position),this.rotatio
n.order=t.rotation.order,this.quaternion.copy(t.quaternion),this.scale.copy(t.scale
),this.matrix.copy(t.matrix),this.matrixWorld.copy(t.matrixWorld),this.matrixAutoUp
date=t.matrixAutoUpdate,this.matrixWorldNeedsUpdate=t.matrixWorldNeedsUpdate,this.l
ayers.mask=t.layers.mask,this.visible=t.visible,this.castShadow=t.castShadow,this.r
eceiveShadow=t.receiveShadow,this.frustumCulled=t.frustumCulled,this.renderOrder=t.
renderOrder,this.userData=JSON.parse(JSON.stringify(t.userData)),!0===e)for(let
e=0;e<t.children.length;e++){const n=t.children[e];this.add(n.clone())}return
this}}Ce.DefaultUp=new
Lt(0,1,0),Ce.DefaultMatrixAutoUpdate=!0,Ce.prototype.isObject3D=!0;const Pe=new
Lt,De=new Lt,Ie=new yt;class Ne{constructor(t=new Lt(1,0,0),e=0)
{this.normal=t,this.constant=e}set(t,e){return
this.normal.copy(t),this.constant=e,this}setComponents(t,e,n,i){return
this.normal.set(t,e,n),this.constant=i,this}setFromNormalAndCoplanarPoint(t,e)
{return this.normal.copy(t),this.constant=-
e.dot(this.normal),this}setFromCoplanarPoints(t,e,n){const
i=Pe.subVectors(n,e).cross(De.subVectors(t,e)).normalize();return
this.setFromNormalAndCoplanarPoint(i,t),this}copy(t){return
this.normal.copy(t.normal),this.constant=t.constant,this}normalize(){const
t=1/this.normal.length();return
this.normal.multiplyScalar(t),this.constant*=t,this}negate(){return
this.constant*=-1,this.normal.negate(),this}distanceToPoint(t){return
this.normal.dot(t)+this.constant}distanceToSphere(t){return
this.distanceToPoint(t.center)-t.radius}projectPoint(t,e){return void 0===e&&
(console.warn("THREE.Plane: .projectPoint() target is now required"),e=new
Lt),e.copy(this.normal).multiplyScalar(-
this.distanceToPoint(t)).add(t)}intersectLine(t,e){void 0===e&&
(console.warn("THREE.Plane: .intersectLine() target is now required"),e=new
Lt);const n=t.delta(Pe),i=this.normal.dot(n);if(0===i)return
0===this.distanceToPoint(t.start)?e.copy(t.start):null;const r=-
(t.start.dot(this.normal)+this.constant)/i;return r<0||r>1?
null:e.copy(n).multiplyScalar(r).add(t.start)}intersectsLine(t){const
e=this.distanceToPoint(t.start),n=this.distanceToPoint(t.end);return
e<0&&n>0||n<0&&e>0}intersectsBox(t){return
t.intersectsPlane(this)}intersectsSphere(t){return
t.intersectsPlane(this)}coplanarPoint(t){return void 0===t&&
(console.warn("THREE.Plane: .coplanarPoint() target is now required"),t=new
Lt),t.copy(this.normal).multiplyScalar(-this.constant)}applyMatrix4(t,e){const
n=e||Ie.getNormalMatrix(t),i=this.coplanarPoint(Pe).applyMatrix4(t),r=this.normal.a
pplyMatrix3(n).normalize();return this.constant=-i.dot(r),this}translate(t){return
this.constant-=t.dot(this.normal),this}equals(t){return
t.normal.equals(this.normal)&&t.constant===this.constant}clone(){return(new
this.constructor).copy(this)}}Ne.prototype.isPlane=!0;const Be=new Lt,ze=new
Lt,Fe=new Lt,Oe=new Lt,He=new Lt,Ge=new Lt,Ue=new Lt,ke=new Lt,Ve=new Lt,We=new
Lt;class je{constructor(t=new Lt,e=new Lt,n=new Lt)
{this.a=t,this.b=e,this.c=n}static getNormal(t,e,n,i){void 0===i&&
(console.warn("THREE.Triangle: .getNormal() target is now required"),i=new
Lt),i.subVectors(n,e),Be.subVectors(t,e),i.cross(Be);const r=i.lengthSq();return

```

```

r>0?i.multiplyScalar(1/Math.sqrt(r)):i.set(0,0,0)}static getBarycoord(t,e,n,i,r)
{Be.subVectors(i,e),ze.subVectors(n,e),Fe.subVectors(t,e);const
s=Be.dot(Be),a=Be.dot(ze),o=Be.dot(Fe),l=ze.dot(ze),c=ze.dot(Fe),h=s*l-a*a;if(void
0===r&&(console.warn("THREE.Triangle: .getBarycoord() target is now
required"),r=new Lt),0===h)return r.set(-2,-1,-1);const u=1/h,d=(l*o-a*c)*u,p=(s*c-
a*o)*u;return r.set(1-d-p,p,d)}static containsPoint(t,e,n,i){return
this.getBarycoord(t,e,n,i,Oe),Oe.x>=0&&Oe.y>=0&&Oe.x+Oe.y<=1}static
getUV(t,e,n,i,r,s,a,o){return
this.getBarycoord(t,e,n,i,Oe),o.set(0,0),o.addScaledVector(r,Oe.x),o.addScaledVecto
r(s,Oe.y),o.addScaledVector(a,Oe.z),o}static isFrontFacing(t,e,n,i){return
Be.subVectors(n,e),ze.subVectors(t,e),Be.cross(ze).dot(i)<0}set(t,e,n){return
this.a.copy(t),this.b.copy(e),this.c.copy(n),this}setFromPointsAndIndices(t,e,n,i)
{return this.a.copy(t[e]),this.b.copy(t[n]),this.c.copy(t[i]),this}clone()
{return(new this.constructor).copy(this)}copy(t){return
this.a.copy(t.a),this.b.copy(t.b),this.c.copy(t.c),this}getArea(){return
Be.subVectors(this.c,this.b),ze.subVectors(this.a,this.b),.5*Be.cross(ze).length()}
getMidpoint(t){return void 0===t&&(console.warn("THREE.Triangle: .getMidpoint()
target is now required"),t=new
Lt),t.addVectors(this.a,this.b).add(this.c).multiplyScalar(1/3)}getNormal(t){return
je.getNormal(this.a,this.b,this.c,t)}getPlane(t){return void 0===t&&
(console.warn("THREE.Triangle: .getPlane() target is now required"),t=new
Ne),t.setFromCoplanarPoints(this.a,this.b,this.c)}getBarycoord(t,e){return
je.getBarycoord(t,this.a,this.b,this.c,e)}getUV(t,e,n,i,r){return
je.getUV(t,this.a,this.b,this.c,e,n,i,r)}containsPoint(t){return
je.containsPoint(t,this.a,this.b,this.c)}isFrontFacing(t){return
je.isFrontFacing(this.a,this.b,this.c,t)}intersectsBox(t){return
t.intersectsTriangle(this)}closestPointToPoint(t,e){void 0===e&&
(console.warn("THREE.Triangle: .closestPointToPoint() target is now
required"),e=new Lt);const n=this.a,i=this.b,r=this.c;let
s,a;He.subVectors(i,n),Ge.subVectors(r,n),ke.subVectors(t,n);const
o=He.dot(ke),l=Ge.dot(ke);if(o<=0&&l<=0)return e.copy(n);Ve.subVectors(t,i);const
c=He.dot(Ve),h=Ge.dot(Ve);if(c>=0&&h<=c)return e.copy(i);const u=o*h-
c*l;if(u<=0&&o>=0&&c<=0)return s=o/(o-
c),e.copy(n).addScaledVector(He,s);We.subVectors(t,r);const
d=He.dot(We),p=Ge.dot(We);if(p>=0&&d<=p)return e.copy(r);const m=d*l-
o*p;if(m<=0&&l>=0&&p<=0)return a=1/(1-p),e.copy(n).addScaledVector(Ge,a);const
f=c*p-d*h;if(f<=0&&h-c>=0&&d-p>=0)return Ue.subVectors(r,i),a=(h-c)/(h-c+(d-
p)),e.copy(i).addScaledVector(Ue,a);const g=1/(f+m+u);return
s=m*g,a=u*g,e.copy(n).addScaledVector(He,s).addScaledVector(Ge,a)}equals(t){return
t.a.equals(this.a)&&t.b.equals(this.b)&&t.c.equals(this.c)}let qe=0;function Xe()
{Object.defineProperty(this,"id",
{value:qe++}),this.uuid=ct(),this.name="",this.type="Material",this.fog=!0,this.ble
nding=1,this.side=0,this.vertexColors=!1,this.opacity=1,this.transparent=!1,this.bl
endSrc=204,this.blendDst=205,this.blendEquation=n,this.blendSrcAlpha=null,this.blend
dDstAlpha=null,this.blendEquationAlpha=null,this.depthFunc=3,this.depthTest=!0,this
.depthWrite=!0,this.stencilWriteMask=255,this.stencilFunc=519,this.stencilRef=0,thi
s.stencilFuncMask=255,this.stencilFail=tt,this.stencilZFail=tt,this.stencilZPass=tt
,this.stencilWrite=!1,this.clippingPlanes=null,this.clipIntersection=!1,this.clipSh
adows=!1,this.shadowSide=null,this.colorWrite=!0,this.precision=null,this.polygonOf
fset=!1,this.polygonOffsetFactor=0,this.polygonOffsetUnits=0,this.dithering=!1,this
.alphaTest=0,this.alphaToCoverage=!1,this.premultipliedAlpha=!1,this.visible=!0,thi
s.toneMapped=!0,this.userData=
{}},this.version=0}Xe.prototype=Object.assign(Object.create(rt.prototype),
{constructor:Xe,isMaterial:!0,onBuild:function(){},onBeforeCompile:function()
{}},customProgramCacheKey:function(){return
this.onBeforeCompile.toString()},setValues:function(t){if(void 0!==t)for(const e in
t){const n=t[e];if(void 0===n){console.warn("THREE.Material: '"+e+"' parameter is
undefined.");continue}if("shading"===e){console.warn("THREE."+this.type+": .shading

```

```

has been removed. Use the boolean .flatShading
instead."),this.flatShading=1===n;continue}const i=this[e];void 0!==i?i&&i.isColor?
i.set(n):i&&i.isVector3&&n&&n.isVector3?
i.copy(n):this[e]=n:console.warn("THREE."+this.type+": '"+e+"' is not a property of
this material.")]},toJSON:function(t){const e=void 0===t||"string"===typeof t;e&&(t=
{textures:{},images:{}});const n={metadata:
{version:4.5,type:"Material",generator:"Material.toJSON"}};function i(t){const e=
[];for(const n in t){const i=t[n];delete i.metadata,e.push(i)}return
e}if(n.uuid=this.uuid,n.type=this.type,"!"!==this.name&&
(n.name=this.name),this.color&&this.color.isColor&&
(n.color=this.color.getHex()),void 0!==this.roughness&&
(n.roughness=this.roughness),void 0!==this.metalness&&
(n.metalness=this.metalness),this.sheen&&this.sheen.isColor&&
(n.sheen=this.sheen.getHex()),this.emissive&&this.emissive.isColor&&
(n.emissive=this.emissive.getHex()),this.emissiveIntensity&&1!==this.emissiveIntens
ity&&
(n.emissiveIntensity=this.emissiveIntensity),this.specular&&this.specular.isColor&&
(n.specular=this.specular.getHex()),void 0!==this.shininess&&
(n.shininess=this.shininess),void 0!==this.clearcoat&&
(n.clearcoat=this.clearcoat),void 0!==this.clearcoatRoughness&&
(n.clearcoatRoughness=this.clearcoatRoughness),this.clearcoatMap&&this.clearcoatMap
.isTexture&&
(n.clearcoatMap=this.clearcoatMap.toJSON(t).uuid),this.clearcoatRoughnessMap&&this.
clearcoatRoughnessMap.isTexture&&
(n.clearcoatRoughnessMap=this.clearcoatRoughnessMap.toJSON(t).uuid),this.clearcoatN
ormalMap&&this.clearcoatNormalMap.isTexture&&
(n.clearcoatNormalMap=this.clearcoatNormalMap.toJSON(t).uuid,n.clearcoatNormalScale
=this.clearcoatNormalScale.toArray()),this.map&&this.map.isTexture&&
(n.map=this.map.toJSON(t).uuid),this.matcap&&this.matcap.isTexture&&
(n.matcap=this.matcap.toJSON(t).uuid),this.alphaMap&&this.alphaMap.isTexture&&
(n.alphaMap=this.alphaMap.toJSON(t).uuid),this.lightMap&&this.lightMap.isTexture&&
(n.lightMap=this.lightMap.toJSON(t).uuid,n.lightMapIntensity=this.lightMapIntensity
),this.aoMap&&this.aoMap.isTexture&&
(n.aoMap=this.aoMap.toJSON(t).uuid,n.aoMapIntensity=this.aoMapIntensity),this.bumpM
ap&&this.bumpMap.isTexture&&
(n.bumpMap=this.bumpMap.toJSON(t).uuid,n.bumpScale=this.bumpScale),this.normalMap&&
this.normalMap.isTexture&&
(n.normalMap=this.normalMap.toJSON(t).uuid,n.normalMapType=this.normalMapType,n.nor
malScale=this.normalScale.toArray()),this.displacementMap&&this.displacementMap.isT
exture&&
(n.displacementMap=this.displacementMap.toJSON(t).uuid,n.displacementScale=this.dis
placementScale,n.displacementBias=this.displacementBias),this.roughnessMap&&this.ro
ughnessMap.isTexture&&
(n.roughnessMap=this.roughnessMap.toJSON(t).uuid),this.metalnessMap&&this.metalness
Map.isTexture&&
(n.metalnessMap=this.metalnessMap.toJSON(t).uuid),this.emissiveMap&&this.emissiveMa
p.isTexture&&
(n.emissiveMap=this.emissiveMap.toJSON(t).uuid),this.specularMap&&this.specularMap.
isTexture&&
(n.specularMap=this.specularMap.toJSON(t).uuid),this.envMap&&this.envMap.isTexture&
&(n.envMap=this.envMap.toJSON(t).uuid,void 0!==this.combine&&
(n.combine=this.combine)),void 0!==this.envMapIntensity&&
(n.envMapIntensity=this.envMapIntensity),void 0!==this.reflectivity&&
(n.reflectivity=this.reflectivity),void 0!==this.refractionRatio&&
(n.refractionRatio=this.refractionRatio),this.gradientMap&&this.gradientMap.isTextu
re&&(n.gradientMap=this.gradientMap.toJSON(t).uuid),void 0!==this.size&&
(n.size=this.size),null!==this.shadowSide&&(n.shadowSide=this.shadowSide),void
0!==this.sizeAttenuation&&

```

```

(n.sizeAttenuation=this.sizeAttenuation),!0!==this.blending&&
(n.blending=this.blending),0!==this.side&&(n.side=this.side),this.vertexColors&&
(n.vertexColors=!0),this.opacity<1&&
(n.opacity=this.opacity),!0===this.transparent&&
(n.transparent=this.transparent),n.depthFunc=this.depthFunc,n.depthTest=this.depthT
est,n.depthWrite=this.depthWrite,n.colorWrite=this.colorWrite,n.stencilWrite=this.s
tencilWrite,n.stencilWriteMask=this.stencilWriteMask,n.stencilFunc=this.stencilFunc
,n.stencilRef=this.stencilRef,n.stencilFuncMask=this.stencilFuncMask,n.stencilFail=
this.stencilFail,n.stencilZFail=this.stencilZFail,n.stencilZPass=this.stencilZPass,
this.rotation&&0!==this.rotation&&
(n.rotation=this.rotation),!0===this.polygonOffset&&
(n.polygonOffset=!0),0!==this.polygonOffsetFactor&&
(n.polygonOffsetFactor=this.polygonOffsetFactor),0!==this.polygonOffsetUnits&&
(n.polygonOffsetUnits=this.polygonOffsetUnits),this.linewidth&&1!==this.linewidth&&
(n.linewidth=this.linewidth),void 0!==this.dashSize&&
(n.dashSize=this.dashSize),void 0!==this.gapSize&&(n.gapSize=this.gapSize),void
0!==this.scale&&(n.scale=this.scale),!0===this.dithering&&
(n.dithering=!0),this.alphaTest>0&&
(n.alphaTest=this.alphaTest),!0===this.alphaToCoverage&&
(n.alphaToCoverage=this.alphaToCoverage),!0===this.premultipliedAlpha&&
(n.premultipliedAlpha=this.premultipliedAlpha),!0===this.wireframe&&
(n.wireframe=this.wireframe),this.wireframeLinewidth>1&&
(n.wireframeLinewidth=this.wireframeLinewidth),"round"!==this.wireframeLinecap&&
(n.wireframeLinecap=this.wireframeLinecap),"round"!==this.wireframeLinejoin&&
(n.wireframeLinejoin=this.wireframeLinejoin),!0===this.morphTargets&&
(n.morphTargets=!0),!0===this.morphNormals&&
(n.morphNormals=!0),!0===this.skinning&&(n.skinning=!0),!0===this.flatShading&&
(n.flatShading=this.flatShading),!1===this.visible&&
(n.visible=!1),!1===this.toneMapped&&(n.toneMapped=!1),"
{}"!==JSON.stringify(this.userData)&&(n.userData=this.userData),e){const
e=i(t.textures),r=i(t.images);e.length>0&&(n.textures=e),r.length>0&&
(n.images=r)}return n},clone:function(){return(new
this.constructor).copy(this)},copy:function(t)
{this.name=t.name,this.fog=t.fog,this.blending=t.blending,this.side=t.side,this.ver
texColors=t.vertexColors,this.opacity=t.opacity,this.transparent=t.transparent,this
.blendSrc=t.blendSrc,this.blendDst=t.blendDst,this.blendEquation=t.blendEquation,th
is.blendSrcAlpha=t.blendSrcAlpha,this.blendDstAlpha=t.blendDstAlpha,this.blendEquat
ionAlpha=t.blendEquationAlpha,this.depthFunc=t.depthFunc,this.depthTest=t.depthTest
,this.depthWrite=t.depthWrite,this.stencilWriteMask=t.stencilWriteMask,this.stencil
Func=t.stencilFunc,this.stencilRef=t.stencilRef,this.stencilFuncMask=t.stencilFuncM
ask,this.stencilFail=t.stencilFail,this.stencilZFail=t.stencilZFail,this.stencilZPa
ss=t.stencilZPass,this.stencilWrite=t.stencilWrite;const e=t.clippingPlanes;let
n=null;if(null!==e){const t=e.length;n=new Array(t);for(let
i=0;i!==t;++i)n[i]=e[i].clone()}return
this.clippingPlanes=n,this.clipIntersection=t.clipIntersection,this.clipShadows=t.c
lipShadows,this.shadowSide=t.shadowSide,this.colorWrite=t.colorWrite,this.precision
=t.precision,this.polygonOffset=t.polygonOffset,this.polygonOffsetFactor=t.polygonO
ffsetFactor,this.polygonOffsetUnits=t.polygonOffsetUnits,this.dithering=t.dithering
,this.alphaTest=t.alphaTest,this.alphaToCoverage=t.alphaToCoverage,this.premultipli
edAlpha=t.premultipliedAlpha,this.visible=t.visible,this.toneMapped=t.toneMapped,th
is.userData=JSON.parse(JSON.stringify(t.userData)),this},dispose:function()
{this.dispatchEvent({type:"dispose"})},Object.defineProperty(Xe.prototype,"needsU
pdate",{set:function(t){!0===t&&this.version++}});const Ye=
{aliceblue:15792383,antiquewhite:16444375,aqua:65535,aquamarine:8388564,azure:15794
175,beige:16119260,bisque:16770244,black:0,blanchedalmond:16772045,blue:255,bluevio
let:9055202,brown:10824234,burlywood:14596231,cadetblue:6266528,chartreuse:8388352,
chocolate:13789470,coral:16744272,cornflowerblue:6591981,cornsilk:16775388,crimson:
14423100,cyan:65535,darkblue:139,darkcyan:35723,darkgoldenrod:12092939,darkgray:111

```

```

19017,darkgreen:25600,darkgrey:11119017,darkkhaki:12433259,darkmagenta:9109643,dark
olivegreen:5597999,darkorange:16747520,darkorchid:10040012,darkred:9109504,darksalm
on:15308410,darkseagreen:9419919,darkslateblue:4734347,darkslategray:3100495,darksl
ategrey:3100495,darkturquoise:52945,darkviolet:9699539,deeppink:16716947,deepskyblu
e:49151,dimgray:6908265,dimgrey:6908265,dodgerblue:2003199,firebrick:11674146,flora
lwhite:16775920,forestgreen:2263842,fuchsia:16711935,gainsboro:14474460,ghostwhite:
16316671,gold:16766720,goldenrod:14329120,gray:8421504,green:32768,greenyellow:1140
3055,greyscale:8421504,honeydew:15794160,hotpink:16738740,indianred:13458524,indigo:4915
330,ivory:16777200,khaki:15787660,lavender:15132410,lavenderblush:16773365,lawngree
n:8190976,lemonchiffon:16775885,lightblue:11393254,lightcoral:15761536,lightcyan:14
745599,lightgoldenrodyellow:16448210,lightgray:13882323,lightgreen:9498256,lightgre
y:13882323,lightpink:16758465,lightsalmon:16752762,lightseagreen:2142890,lightskybl
ue:8900346,lightslategray:7833753,lightslategrey:7833753,lightsteelblue:11584734,li
ghtyellow:16777184,lime:65280,limegreen:3329330,linen:16445670,magenta:16711935,mar
oon:8388608,mediumaquamarine:6737322,mediumblue:205,mediumorchid:12211667,mediumpur
ple:9662683,mediumseagreen:3978097,mediumslateblue:8087790,mediumspringgreen:64154,
mediumturquoise:4772300,mediumvioletred:13047173,midnightblue:1644912,mintcream:161
21850,mistyrose:16770273,moccasin:16770229,navajowhite:16768685,navy:128,oldlace:16
643558,olive:8421376,olivedrab:7048739,orange:16753920,orangered:16729344,orchid:14
315734,palegoldenrod:15657130,palegreen:10025880,paleturquoise:11529966,palevioletr
ed:14381203,papayawhip:16773077,peachpuff:16767673,peru:13468991,pink:16761035,plum
:14524637,powderblue:11591910,purple:8388736,rebeccapurple:6697881,red:16711680,ros
ybrown:12357519,royalblue:4286945,saddlebrown:9127187,salmon:16416882,sandybrown:16
032864,seagreen:3050327,seashell:16774638,sienna:10506797,silver:12632256,skyblue:8
900331,slateblue:6970061,slategray:7372944,slategrey:7372944,snow:16775930,springgr
een:65407,steelblue:4620980,tan:13808780,teal:32896,thistle:14204888,tomato:1673709
5,turquoise:4251856,violet:15631086,wheat:16113331,white:16777215,whitesmoke:161192
85,yellow:16776960,yellowgreen:10145074},Ze={h:0,s:0,l:0},Je={h:0,s:0,l:0};function
Qe(t,e,n){return n<0&&(n+=1),n>1&&(n-=1),n<1/6?t+6*(e-t)*n:n<.5?e:n<2/3?t+6*(e-t)*
(2/3-n):t}function Ke(t){return
t<.04045?.0773993808*t:Math.pow(.9478672986*t+.0521327014,2.4)}function $e(t)
{return t<.0031308?12.92*t:1.055*Math.pow(t,.41666)-.055}class
tn{constructor(t,e,n){return void 0===e&&void 0===n?
this.set(t):this.setRGB(t,e,n)}set(t){return t&&t.isColor?
this.copy(t):"number"===typeof t?this.setHex(t):"string"===typeof
t&&this.setStyle(t),this}setScalar(t){return
this.r=t,this.g=t,this.b=t,this}setHex(t){return t=Math.floor(t),this.r=
(t>>16&255)/255,this.g=(t>>8&255)/255,this.b=(255&t)/255,this}setRGB(t,e,n){return
this.r=t,this.g=e,this.b=n,this}setHSL(t,e,n)
{if(t=ut(t,1),e=ht(e,0,1),n=ht(n,0,1),0===e)this.r=this.g=this.b=n;else{const
i=n<=.5?n*(1+e):n+e-n*e,r=2*n-
i;this.r=Qe(r,i,t+1/3),this.g=Qe(r,i,t),this.b=Qe(r,i,t-1/3)}return
this}setStyle(t){function e(e){void 0!==e&&parseFloat(e)
<1&&console.warn("THREE.Color: Alpha component of "+t+" will be ignored.")}let
n;if(n=/^((?:rgb|hsl)a?)\(([\^\s]*)\)$/i.exec(t)){let t;const i=n[1],r=n[2];switch(i)
{case"rgb":case"rgba":if(t=/^\s*(\d+)\s*,\s*(\d+)\s*,\s*(\d+)\s*(?:,\s*(\d*\.\d+
?\d+)\s*)?$/i.exec(r))return
this.r=Math.min(255,parseInt(t[1],10))/255,this.g=Math.min(255,parseInt(t[2],10))/2
55,this.b=Math.min(255,parseInt(t[3],10))/255,e(t[4]),this;if(t=/^\s*(\d+)\s*\%\s*,\s*
(\d+)\s*\%\s*,\s*(\d+)\s*\%\s*(?:,\s*(\d*\.\d+?\d+)\s*)?$/i.exec(r))return
this.r=Math.min(100,parseInt(t[1],10))/100,this.g=Math.min(100,parseInt(t[2],10))/1
00,this.b=Math.min(100,parseInt(t[3],10))/100,e(t[4]),this;break;case"hsl":case"hsl
a":if(t=/^\s*(\d*\.\d+?\d+)\s*,\s*(\d+)\s*\%\s*,\s*(\d+)\s*\%\s*(?:,\s*(\d*\.\d+
?\d+)\s*)?$/i.exec(r)){const
n=parseFloat(t[1])/360,i=parseInt(t[2],10)/100,r=parseInt(t[3],10)/100;return
e(t[4]),this.setHSL(n,i,r)}}else if(n=/^#([A-Fa-f\d]+)$/i.exec(t)){const
t=n[1],e=t.length;if(3===e)return
this.r=parseInt(t.charAt(0)+t.charAt(0),16)/255,this.g=parseInt(t.charAt(1)+t.charAt

```

```

t(1),16)/255,this.b=parseInt(t.charAt(2)+t.charAt(2),16)/255,this;if(6===e)return
this.r=parseInt(t.charAt(0)+t.charAt(1),16)/255,this.g=parseInt(t.charAt(2)+t.charA
t(3),16)/255,this.b=parseInt(t.charAt(4)+t.charAt(5),16)/255,this}return
t&& t.length>0?this.setColorName(t):this}setColorName(t){const
e=Ye[t.toLowerCase()];return void 0!==e?this.setHex(e):console.warn("THREE.Color:
Unknown color "+t),this}clone(){return new
this.constructor(this.r,this.g,this.b)}copy(t){return
this.r=t.r,this.g=t.g,this.b=t.b,this}copyGammaToLinear(t,e=2){return
this.r=Math.pow(t.r,e),this.g=Math.pow(t.g,e),this.b=Math.pow(t.b,e),this}copyLinea
rToGamma(t,e=2){const n=e>0?1/e:1;return
this.r=Math.pow(t.r,n),this.g=Math.pow(t.g,n),this.b=Math.pow(t.b,n),this}convertGa
mmaToLinear(t){return this.copyGammaToLinear(this,t),this}convertLinearToGamma(t)
{return this.copyLinearToGamma(this,t),this}copySRGBToLinear(t){return
this.r=Ke(t.r),this.g=Ke(t.g),this.b=Ke(t.b),this}copyLinearToSRGB(t){return
this.r=$e(t.r),this.g=$e(t.g),this.b=$e(t.b),this}convertSRGBToLinear(){return
this.copySRGBToLinear(this),this}convertLinearToSRGB(){return
this.copyLinearToSRGB(this),this}getHex(){return
255*this.r<<16^255*this.g<<8^255*this.b<<0}getHexString()
{return("000000"+this.getHex().toString(16)).slice(-6)}getHSL(t){void 0===t&&
(console.warn("THREE.Color: .getHSL() target is now required"),t=
{h:0,s:0,l:0});const
e=this.r,n=this.g,i=this.b,r=Math.max(e,n,i),s=Math.min(e,n,i);let a,o;const l=
(s+r)/2;if(s===r)a=0,o=0;else{const t=r-s;switch(o=l<=.5?t/(r+s):t/(2-r-s),r){case
e:a=(n-i)/t+(n<i?6:0);break;case n:a=(i-e)/t+2;break;case i:a=(e-n)/t+4;a/=6}return
t.h=a,t.s=o,t.l=l,t}getStyle(){return"rgb("+255*this.r|0+", "+255*this.g|0+", "+
255*this.b|0+")"}offsetHSL(t,e,n){return
this.getHSL(Ze),Ze.h+=t,Ze.s+=e,Ze.l+=n,this.setHSL(Ze.h,Ze.s,Ze.l),this}add(t)
{return this.r+=t.r,this.g+=t.g,this.b+=t.b,this}addColors(t,e){return
this.r=t.r+e.r,this.g=t.g+e.g,this.b=t.b+e.b,this}addScalar(t){return
this.r+=t,this.g+=t,this.b+=t,this}sub(t){return this.r=Math.max(0,this.r-
t.r),this.g=Math.max(0,this.g-t.g),this.b=Math.max(0,this.b-t.b),this}multiply(t)
{return this.r*=t.r,this.g*=t.g,this.b*=t.b,this}multiplyScalar(t){return
this.r*=t,this.g*=t,this.b*=t,this}lerp(t,e){return this.r+=(t.r-this.r)*e,this.g+=
(t.g-this.g)*e,this.b+=(t.b-this.b)*e,this}lerpColors(t,e,n){return this.r=t.r+
(e.r-t.r)*n,this.g=t.g+(e.g-t.g)*n,this.b=t.b+(e.b-t.b)*n,this}lerpHSL(t,e)
{this.getHSL(Ze),t.getHSL(Je);const
n=dt(Ze.h,Je.h,e),i=dt(Ze.s,Je.s,e),r=dt(Ze.l,Je.l,e);return
this.setHSL(n,i,r),this}equals(t){return
t.r===this.r&&t.g===this.g&&t.b===this.b}fromArray(t,e=0){return
this.r=t[e],this.g=t[e+1],this.b=t[e+2],this}toArray(t=[],e=0){return
t[e]=this.r,t[e+1]=this.g,t[e+2]=this.b,t}fromBufferAttribute(t,e){return
this.r=t.getX(e),this.g=t.getY(e),this.b=t.getZ(e),!0===t.normalized&&
(this.r/=255,this.g/=255,this.b/=255),this}toJSON(){return
this.getHex()}}tn.NAMES=Ye,tn.prototype.isColor=!0,tn.prototype.r=1,tn.prototype.g=
1,tn.prototype.b=1;class en extends Xe{constructor(t)
{super(),this.type="MeshBasicMaterial",this.color=new
tn(16777215),this.map=null,this.lightMap=null,this.lightMapIntensity=1,this.aoMap=n
ull,this.aoMapIntensity=1,this.specularMap=null,this.alphaMap=null,this.envMap=null
,this.combine=0,this.reflectivity=1,this.refractionRatio=.98,this.wireframe=!1,this
.wireframeLinewidth=1,this.wireframeLinecap="round",this.wireframeLinejoin="round",
this.skinning=!1,this.morphTargets=!1,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.map=t.map,this.lightMap=t.lightMap,this
.lightMapIntensity=t.lightMapIntensity,this.aoMap=t.aoMap,this.aoMapIntensity=t.aoM
apIntensity,this.specularMap=t.specularMap,this.alphaMap=t.alphaMap,this.envMap=t.e
nvMap,this.combine=t.combine,this.reflectivity=t.reflectivity,this.refractionRatio=
t.refractionRatio,this.wireframe=t.wireframe,this.wireframeLinewidth=t.wireframeLin
ewidth,this.wireframeLinecap=t.wireframeLinecap,this.wireframeLinejoin=t.wireframeL
inejoin,this.skinning=t.skinning,this.morphTargets=t.morphTargets,this}}en.prototype

```

```

e.isMeshBasicMaterial=!0;const nn=new Lt,rn=new vt;class sn{constructor(t,e,n)
{if(Array.isArray(t))throw new TypeError("THREE.BufferAttribute: array should be a
Typed Array.");this.name="",this.array=t,this.itemSize=e,this.count=void 0!==t?
t.length/e:0,this.normalized=!0===n,this.usage=et,this.updateRange=
{offset:0,count:-1},this.version=0,this.onUploadCallback=function(){}set
needsUpdate(t){!0===t&&this.version++}setUsage(t){return this.usage=t,this}copy(t)
{return this.name=t.name,this.array=new
t.array.constructor(t.array),this.itemSize=t.itemSize,this.count=t.count,this.norma
lized=t.normalized,this.usage=t.usage,this}copyAt(t,e,n)
{t*=this.itemSize,n*=e.itemSize;for(let
i=0,r=this.itemSize;i<r;i++)this.array[t+i]=e.array[n+i];return this}copyArray(t)
{return this.array.set(t),this}copyColorsArray(t){const e=this.array;let
n=0;for(let i=0,r=t.length;i<r;i++){let r=t[i];void 0===r&&
(console.warn("THREE.BufferAttribute.copyColorsArray(): color is
undefined",i),r=new tn),e[n++]=r.r,e[n++]=r.g,e[n++]=r.b)return
this}copyVector2sArray(t){const e=this.array;let n=0;for(let
i=0,r=t.length;i<r;i++){let r=t[i];void 0===r&&
(console.warn("THREE.BufferAttribute.copyVector2sArray(): vector is
undefined",i),r=new vt),e[n++]=r.x,e[n++]=r.y}return this}copyVector3sArray(t)
{const e=this.array;let n=0;for(let i=0,r=t.length;i<r;i++){let r=t[i];void 0===r&&
(console.warn("THREE.BufferAttribute.copyVector3sArray(): vector is
undefined",i),r=new Lt),e[n++]=r.x,e[n++]=r.y,e[n++]=r.z}return
this}copyVector4sArray(t){const e=this.array;let n=0;for(let
i=0,r=t.length;i<r;i++){let r=t[i];void 0===r&&
(console.warn("THREE.BufferAttribute.copyVector4sArray(): vector is
undefined",i),r=new St),e[n++]=r.x,e[n++]=r.y,e[n++]=r.z,e[n++]=r.w}return
this}applyMatrix3(t){if(2===this.itemSize)for(let
e=0,n=this.count;e<n;e++)rn.fromBufferAttribute(this,e),rn.applyMatrix3(t),this.set
XY(e,rn.x,rn.y);else if(3===this.itemSize)for(let
e=0,n=this.count;e<n;e++)nn.fromBufferAttribute(this,e),nn.applyMatrix3(t),this.set
XYZ(e,nn.x,nn.y,nn.z);return this}applyMatrix4(t){for(let
e=0,n=this.count;e<n;e++)nn.x=this.getX(e),nn.y=this.getY(e),nn.z=this.getZ(e),nn.a
pplyMatrix4(t),this.setXYZ(e,nn.x,nn.y,nn.z);return this}applyNormalMatrix(t)
{for(let
e=0,n=this.count;e<n;e++)nn.x=this.getX(e),nn.y=this.getY(e),nn.z=this.getZ(e),nn.a
pplyNormalMatrix(t),this.setXYZ(e,nn.x,nn.y,nn.z);return this}transformDirection(t)
{for(let
e=0,n=this.count;e<n;e++)nn.x=this.getX(e),nn.y=this.getY(e),nn.z=this.getZ(e),nn.t
ransformDirection(t),this.setXYZ(e,nn.x,nn.y,nn.z);return this}set(t,e=0){return
this.array.set(t,e),this}getX(t){return this.array[t*this.itemSize]}setX(t,e)
{return this.array[t*this.itemSize]=e,this}getY(t){return
this.array[t*this.itemSize+1]}setY(t,e){return
this.array[t*this.itemSize+1]=e,this}getZ(t){return
this.array[t*this.itemSize+2]}setZ(t,e){return
this.array[t*this.itemSize+2]=e,this}getW(t){return
this.array[t*this.itemSize+3]}setW(t,e){return
this.array[t*this.itemSize+3]=e,this}setXY(t,e,n){return
t*=this.itemSize,this.array[t+0]=e,this.array[t+1]=n,this}setXYZ(t,e,n,i){return
t*=this.itemSize,this.array[t+0]=e,this.array[t+1]=n,this.array[t+2]=i,this}setXYZW
(t,e,n,i,r){return
t*=this.itemSize,this.array[t+0]=e,this.array[t+1]=n,this.array[t+2]=i,this.array[t
+3]=r,this}onUpload(t){return this.onUploadCallback=t,this}clone(){return new
this.constructor(this.array,this.itemSize).copy(this)}toJSON(){const t=
{itemSize:this.itemSize,type:this.array.constructor.name,array:Array.prototype.slic
e.call(this.array),normalized:this.normalized};return""!==this.name&&
(t.name=this.name),this.usage!==et&&
(t.usage=this.usage),0===this.updateRange.offset&&-1===this.updateRange.count||
(t.updateRange=this.updateRange),t}}sn.prototype.isBufferAttribute=!0;class an

```



```

extends sn{constructor(t,e,n){super(new Int8Array(t),e,n)}}class on extends
sn{constructor(t,e,n){super(new Uint8Array(t),e,n)}}class ln extends
sn{constructor(t,e,n){super(new Uint8ClampedArray(t),e,n)}}class cn extends
sn{constructor(t,e,n){super(new Int16Array(t),e,n)}}class hn extends
sn{constructor(t,e,n){super(new Uint16Array(t),e,n)}}class un extends
sn{constructor(t,e,n){super(new Int32Array(t),e,n)}}class dn extends
sn{constructor(t,e,n){super(new Uint32Array(t),e,n)}}class pn extends
sn{constructor(t,e,n){super(new
Uint16Array(t),e,n)}}pn.prototype.isFloat16BufferAttribute=!0;class mn extends
sn{constructor(t,e,n){super(new Float32Array(t),e,n)}}class fn extends
sn{constructor(t,e,n){super(new Float64Array(t),e,n)}}function gn(t)
{if(0===t.length)return-1/0;let e=t[0];for(let n=1,i=t.length;n<i;++n)t[n]>e&&
(e=t[n]);return e}const vn=
{Int8Array:Int8Array,Uint8Array:Uint8Array,Uint8ClampedArray:Uint8ClampedArray,Int1
6Array:Int16Array,Uint16Array:Uint16Array,Int32Array:Int32Array,Uint32Array:Uint32A
rray,Float32Array:Float32Array,Float64Array:Float64Array};function yn(t,e){return
new vn[t](e)}let xn=0;const _n=new se,wn=new Ce,bn=new Lt,Mn=new Pt,Sn=new
Pt,Tn=new Lt;class En extends rt{constructor()
{super(),Object.defineProperty(this,"id",
{value:xn++}),this.uuid=ct(),this.name="",this.type="BufferGeometry",this.index=nul
l,this.attributes={},this.morphAttributes=
{},this.morphTargetsRelative=!1,this.groups=
[],this.boundingBox=null,this.boundingSphere=null,this.drawRange=
{start:0,count:1/0},this.userData={}}getIndex(){return this.index}setIndex(t)
{return Array.isArray(t)?this.index=new(gn(t)>65535?dn:hn)
(t,1):this.index=t,this}getAttribute(t){return this.attributes[t]}setAttribute(t,e)
{return this.attributes[t]=e,this}deleteAttribute(t){return delete
this.attributes[t],this}hasAttribute(t){return void
0!==this.attributes[t]}addGroup(t,e,n=0)
{this.groups.push({start:t,count:e,materialIndex:n})}clearGroups(){this.groups=
[]}setDrawRange(t,e){this.drawRange.start=t,this.drawRange.count=e}applyMatrix4(t)
{const e=this.attributes.position;void 0!==e&&
(e.applyMatrix4(t),e.needsUpdate=!0);const n=this.attributes.normal;if(void 0!==n)
{const e=(new yt).getNormalMatrix(t);n.applyNormalMatrix(e),n.needsUpdate=!0}const
i=this.attributes.tangent;return void 0!==i&&
(i.transformDirection(t),i.needsUpdate=!0),null!==this.boundingBox&&this.computeBou
ndingBox(),null!==this.boundingSphere&&this.computeBoundingSphere(),this}rotateX(t)
{return _n.makeRotationX(t),this.applyMatrix4(_n),this}rotateY(t){return
_n.makeRotationY(t),this.applyMatrix4(_n),this}rotateZ(t){return
_n.makeRotationZ(t),this.applyMatrix4(_n),this}translate(t,e,n){return
_n.makeTranslation(t,e,n),this.applyMatrix4(_n),this}scale(t,e,n){return
_n.makeScale(t,e,n),this.applyMatrix4(_n),this}lookAt(t){return
wn.lookAt(t),wn.updateMatrix(),this.applyMatrix4(wn.matrix),this}center(){return
this.computeBoundingBox(),this.boundingBox.getCenter(bn).negate(),this.translate(bn
.x,bn.y,bn.z),this}setFromPoints(t){const e=[];for(let n=0,i=t.length;n<i;n++)
{const i=t[n];e.push(i.x,i.y,i.z||0)}return this.setAttribute("position",new
mn(e,3)),this}computeBoundingBox(){null===this.boundingBox&&(this.boundingBox=new
Pt);const
t=this.attributes.position,e=this.morphAttributes.position;if(t&&t.isGLBufferAttrib
ute)return console.error('THREE.BufferGeometry.computeBoundingBox():
GLBufferAttribute requires a manual bounding box. Alternatively set
"mesh.frustumCulled" to "false".',this),void this.boundingBox.set(new
Lt(-1/0,-1/0,-1/0),new Lt(1/0,1/0,1/0));if(void 0!==t)
{if(this.boundingBox.setFromBufferAttribute(t),e)for(let t=0,n=e.length;t<n;t++)
{const n=e[t];Mn.setFromBufferAttribute(n),this.morphTargetsRelative?
(Tn.addVectors(this.boundingBox.min,Mn.min),this.boundingBox.expandByPoint(Tn),Tn.a
ddVectors(this.boundingBox.max,Mn.max),this.boundingBox.expandByPoint(Tn)):
(this.boundingBox.expandByPoint(Mn.min),this.boundingBox.expandByPoint(Mn.max))}}el

```

```

se this.boundingBox.makeEmpty();
(isNaN(this.boundingBox.min.x)||isNaN(this.boundingBox.min.y)||isNaN(this.boundingBox.min.z))&&console.error('THREE.BufferGeometry.computeBoundingBox(): Computed min/max have NaN values. The "position" attribute is likely to have NaN values.',this)}computeBoundingSphere(){null===this.boundingSphere&&(this.boundingSphere=new Jt);const t=this.attributes.position,e=this.morphAttributes.position;if(t&&t.isGLBufferAttribute)return console.error('THREE.BufferGeometry.computeBoundingSphere(): GLBufferAttribute requires a manual bounding sphere. Alternatively set "mesh.frustumCulled" to "false".',this),void this.boundingSphere.set(new Lt,1/0);if(t){const n=this.boundingSphere.center;if(Mn.setFromBufferAttribute(t,e)for(let t=0,n=e.length;t<n;t++){const n=e[t];Sn.setFromBufferAttribute(n,this.morphTargetsRelative?(Tn.addVectors(Mn.min,Sn.min),Mn.expandByPoint(Tn),Tn.addVectors(Mn.max,Sn.max),Mn.expandByPoint(Tn)):(Mn.expandByPoint(Sn.min),Mn.expandByPoint(Sn.max)))Mn.getCenter(n);let i=0;for(let e=0,r=t.count;e<r;e++)Tn.fromBufferAttribute(t,e),i=Math.max(i,n.distanceToSquared(Tn));if(e)for(let r=0,s=e.length;r<s;r++){const s=e[r],a=this.morphTargetsRelative;for(let e=0,r=s.count;e<r;e++)Tn.fromBufferAttribute(s,e),a&&(bn.fromBufferAttribute(t,e),Tn.add(bn)),i=Math.max(i,n.distanceToSquared(Tn))}this.boundingSphere.radius=Math.sqrt(i),isNaN(this.boundingSphere.radius)&&console.error('THREE.BufferGeometry.computeBoundingSphere(): Computed radius is NaN. The "position" attribute is likely to have NaN values.',this)}computeFaceNormals(){}computeTangents(){const t=this.index,e=this.attributes;if(null===t||void 0===e.position||void 0===e.normal||void 0===e.uv)return void console.error("THREE.BufferGeometry: .computeTangents() failed. Missing required attributes (index, position, normal or uv)");const n=t.array,i=e.position.array,r=e.normal.array,s=e.uv.array,a=i.length/3;void 0===e.tangent&&this.setAttribute("tangent",new sn(new Float32Array(4*a),4));const o=e.tangent.array,l=[],c=[];for(let t=0;t<a;t++){l[t]=new Lt,c[t]=new Lt;const h=new Lt,u=new Lt,d=new Lt,p=new vt,m=new vt,f=new vt,g=new Lt,v=new Lt;function y(t,e,n){h.fromArray(i,3*t),u.fromArray(i,3*e),d.fromArray(i,3*n),p.fromArray(s,2*t),m.fromArray(s,2*e),f.fromArray(s,2*n),u.sub(h),d.sub(h),m.sub(p),f.sub(p);const r=1/(m.x*f.y-f.x*m.y);isFinite(r)&&(g.copy(u).multiplyScalar(f.y).addScaledVector(d,-m.y).multiplyScalar(r),v.copy(d).multiplyScalar(m.x).addScaledVector(u,-f.x).multiplyScalar(r),l[t].add(g),l[e].add(g),l[n].add(g),c[t].add(v),c[e].add(v),c[n].add(v))}let x=this.groups;0===x.length&&(x=[{start:0,count:n.length}]);for(let t=0,e=x.length;t<e;t++){const e=x[t],i=e.start;for(let t=i,r=i+e.count;t<r;t+=3)y(n[t+0],n[t+1],n[t+2])}const _=new Lt,w=new Lt,b=new Lt,M=new Lt;function S(t){b.fromArray(r,3*t),M.copy(b);const e=l[t];_.copy(e),_.sub(b.multiplyScalar(b.dot(e))).normalize(),w.crossVectors(M,e);const n=w.dot(c[t])<0?-1:1;o[4*t]=_.x,o[4*t+1]=_.y,o[4*t+2]=_.z,o[4*t+3]=n}for(let t=0,e=x.length;t<e;t++){const e=x[t],i=e.start;for(let t=i,r=i+e.count;t<r;t+=3)S(n[t+0]),S(n[t+1]),S(n[t+2])}computeVertexNormals(){const t=this.index,e=this.getAttribute("position");if(void 0!==e){let n=this.getAttribute("normal");if(void 0===n)n=new sn(new Float32Array(3*e.count),3),this.setAttribute("normal",n);else for(let t=0,e=n.count;t<e;t++){n.setXYZ(t,0,0,0);const i=new Lt,r=new Lt,s=new Lt,a=new Lt,o=new Lt,l=new Lt,c=new Lt,h=new Lt;if(t)for(let u=0,d=t.count;u<d;u+=3){const d=t.getX(u+0),p=t.getX(u+1),m=t.getX(u+2);i.fromBufferAttribute(e,d),r.fromBufferAttribute(e,p),s.fromBufferAttribute(e,m),c.subVectors(s,r),h.subVectors(i,r),c.cross(h),a.fromBufferAttribute(n,d),o.fromBufferAttribute(n,p),l.fromBufferAttribute(n,m),a.add(c),o.add(c),l.add(c),n.setXYZ(d,a.x,a.y,a.z),n.setXYZ(p,o.x,o.y,o.z),n.setXYZ(m,l.x,l.y,l.z)}else for(let t=0,a=e.count;t<a;t+=3)i.fromBufferAttribute(e,t+0),r.fromBufferAttribute(e,t+1),s.

```

```

fromBufferAttribute(e,t+2),c.subVectors(s,r),h.subVectors(i,r),c.cross(h),n.setXYZ(
t+0,c.x,c.y,c.z),n.setXYZ(t+1,c.x,c.y,c.z),n.setXYZ(t+2,c.x,c.y,c.z);this.normalize
Normals(),n.needsUpdate=!0}}merge(t,e){if(!t||!t.isBufferGeometry)return void
console.error("THREE.BufferGeometry.merge(): geometry not an instance of
THREE.BufferGeometry.",t);void 0===e&&
(e=0,console.warn("THREE.BufferGeometry.merge(): Overwriting original geometry,
starting at offset=0. Use BufferGeometryUtils.mergeBufferGeometries() for lossless
merge."));const n=this.attributes;for(const i in n){if(void
0===t.attributes[i])continue;const
r=n[i].array,s=t.attributes[i],a=s.array,o=s.itemSize*e,l=Math.min(a.length,r.lengt
h-o);for(let t=0,e=o;t<l;t++,e++)r[e]=a[t]}return this}normalizeNormals(){const
t=this.attributes.normal;for(let
e=0,n=t.count;e<n;e++)Tn.fromBufferAttribute(t,e),Tn.normalize(),t.setXYZ(e,Tn.x,Tn
.y,Tn.z)}toNonIndexed(){function t(t,e){const
n=t.array,i=t.itemSize,r=t.normalized,s=new n.constructor(e.length*i);let
a=0,o=0;for(let t=0,r=e.length;t<r;t++){a=e[t]*i;for(let
t=0;t<i;t++)s[o++]=n[a++]}}return new sn(s,i,r)}if(null===this.index)return
console.warn("THREE.BufferGeometry.toNonIndexed(): BufferGeometry is already non-
indexed."),this;const e=new En,n=this.index.array,i=this.attributes;for(const r in
i){const s=t(i[r],n);e.setAttribute(r,s)}const r=this.morphAttributes;for(const i
in r){const s=[],a=r[i];for(let e=0,i=a.length;e<i;e++){const
i=t(a[e],n);s.push(i)}e.morphAttributes[i]=s}e.morphTargetsRelative=this.morphTarge
tsRelative;const s=this.groups;for(let t=0,n=s.length;t<n;t++){const
n=s[t];e.addGroup(n.start,n.count,n.materialIndex)}return e}toJSON(){const t=
{metadata:
{version:4.5,type:"BufferGeometry",generator:"BufferGeometry.toJSON"}};if(t.uuid=th
is.uuid,t.type=this.type,"!"!==this.name&&
(t.name=this.name),Object.keys(this.userData).length>0&&
(t.userData=this.userData),void 0!==this.parameters){const
e=this.parameters;for(const n in e)void 0!==e[n]&&(t[n]=e[n]);return t}t.data=
{attributes:{}};const e=this.index,null!==e&&(t.data.index=
{type:e.array.constructor.name,array:Array.prototype.slice.call(e.array)});const
n=this.attributes;for(const e in n){const
i=n[e];t.data.attributes[e]=i.toJSON(t.data)}const i={};let r=!1;for(const e in
this.morphAttributes){const n=this.morphAttributes[e],s=[];for(let
e=0,i=n.length;e<i;e++){const i=n[e];s.push(i.toJSON(t.data))}s.length>0&&
(i[e]=s,r=!0)}r&&
(t.data.morphAttributes=i,t.data.morphTargetsRelative=this.morphTargetsRelative);co
nst s=this.groups;s.length>0&&(t.data.groups=JSON.parse(JSON.stringify(s)));const
a=this.boundingBox;return null!==a&&(t.data.boundingBox=
{center:a.center.toArray(),radius:a.radius}),t.clone(){return(new
En).copy(this)}copy(t){this.index=null,this.attributes={},this.morphAttributes=
{},this.groups=[],this.boundingBox=null,this.boundingBox=null;const e=
{};this.name=t.name;const n=t.index,null!==n&&this.setIndex(n.clone(e));const
i=t.attributes;for(const t in i){const n=i[t];this.setAttribute(t,n.clone(e))}const
r=t.morphAttributes;for(const t in r){const n=[],i=r[t];for(let
t=0,r=i.length;t<r;t++)n.push(i[t].clone(e));this.morphAttributes[t]=n}this.morphTa
rgetsRelative=t.morphTargetsRelative;const s=t.groups;for(let
t=0,e=s.length;t<e;t++){const
e=s[t];this.addGroup(e.start,e.count,e.materialIndex)}const
a=t.boundingBox,null!==a&&(this.boundingBox=a.clone());const
o=t.boundingBox;return null!==o&&
(this.boundingBox=o.clone()),this.drawRange.start=t.drawRange.start,this.drawRan
ge.count=t.drawRange.count,this.userData=t.userData,this}dispose()
{this.dispatchEvent({type:"dispose"})}}En.prototype.isBufferGeometry=!0;const
An=new se,Ln=new re,Rn=new Jt,Cn=new Lt,Pn=new Lt,Dn=new Lt,In=new Lt,Nn=new
Lt,Bn=new Lt,zn=new Lt,Fn=new Lt,On=new Lt,Hn=new vt,Gn=new vt,Un=new vt,kn=new
Lt,Vn=new Lt;class Wn extends Ce{constructor(t=new En,e=new en)

```

```

{super(),this.type="Mesh",this.geometry=t,this.material=e,this.updateMorphTargets()
}copy(t){return super.copy(t),void 0!==t.morphTargetInfluences&&
(this.morphTargetInfluences=t.morphTargetInfluences.slice()),void
0!==t.morphTargetDictionary&&
(this.morphTargetDictionary=Object.assign({},t.morphTargetDictionary)),this.materia
l=t.material,this.geometry=t.geometry,this.updateMorphTargets(){const
t=this.geometry;if(t.isBufferGeometry){const
e=t.morphAttributes,n=Object.keys(e);if(n.length>0){const t=e[n[0]];if(void 0!==t)
{this.morphTargetInfluences=[],this.morphTargetDictionary={};for(let
e=0,n=t.length;e<n;e++){const
n=t[e].name||String(e);this.morphTargetInfluences.push(0),this.morphTargetDictionar
y[n]=e}}}}else{const e=t.morphTargets;void
0!==e&&e.length>0&&console.error("THREE.Mesh.updateMorphTargets() no longer
supports THREE.Geometry. Use THREE.BufferGeometry instead.")}}raycast(t,e){const
n=this.geometry,i=this.material,r=this.matrixWorld;if(void
0===i)return;if(null===n.boundingSphere&&n.computeBoundingSphere(),Rn.copy(n.boundi
ngSphere),Rn.applyMatrix4(r),!l===t.ray.intersectsSphere(Rn))return;if(An.copy(r).i
nvert(),Ln.copy(t.ray).applyMatrix4(An),null!==n.boundingBox&&!l===Ln.intersectsBox
(n.boundingBox))return;let s;if(n.isBufferGeometry){const
r=n.index,a=n.attributes.position,o=n.morphAttributes.position,l=n.morphTargetsRela
tive,c=n.attributes.uv,h=n.attributes.uv2,u=n.groups,d=n.drawRange;if(null!==r)if(A
rray.isArray(i))for(let n=0,p=u.length;n<p;n++){const
p=u[n],m=i[p.materialIndex];for(let
n=Math.max(p.start,d.start),i=Math.min(p.start+p.count,d.start+d.count);n<i;n+=3)
{const
i=r.getX(n),u=r.getX(n+1),d=r.getX(n+2);s=jn(this,m,t,Ln,a,o,l,c,h,i,u,d),s&&
(s.faceIndex=Math.floor(n/3),s.face.materialIndex=p.materialIndex,e.push(s))}}else{
for(let n=Math.max(0,d.start),u=Math.min(r.count,d.start+d.count);n<u;n+=3){const
u=r.getX(n),d=r.getX(n+1),p=r.getX(n+2);s=jn(this,i,t,Ln,a,o,l,c,h,u,d,p),s&&
(s.faceIndex=Math.floor(n/3),e.push(s))}}else if(void
0!==a)if(Array.isArray(i))for(let n=0,r=u.length;n<r;n++){const
r=u[n],p=i[r.materialIndex];for(let
n=Math.max(r.start,d.start),i=Math.min(r.start+r.count,d.start+d.count);n<i;n+=3)
{s=jn(this,p,t,Ln,a,o,l,c,h,n,n+1,n+2),s&&
(s.faceIndex=Math.floor(n/3),s.face.materialIndex=r.materialIndex,e.push(s))}}else{
for(let n=Math.max(0,d.start),r=Math.min(a.count,d.start+d.count);n<r;n+=3)
{s=jn(this,i,t,Ln,a,o,l,c,h,n,n+1,n+2),s&&
(s.faceIndex=Math.floor(n/3),e.push(s))}}}}else
n.isGeometry&&console.error("THREE.Mesh.raycast() no longer supports
THREE.Geometry. Use THREE.BufferGeometry instead.")}}function
jn(t,e,n,i,r,s,a,o,l,c,h,u)
{Cn.fromBufferAttribute(r,c),Pn.fromBufferAttribute(r,h),Dn.fromBufferAttribute(r,u
);const d=t.morphTargetInfluences;if(e.morphTargets&&s&&d)
{zn.set(0,0,0),Fn.set(0,0,0),On.set(0,0,0);for(let t=0,e=s.length;t<e;t++){const
e=d[t],n=s[t];0!==e&&
(In.fromBufferAttribute(n,c),Nn.fromBufferAttribute(n,h),Bn.fromBufferAttribute(n,u
),a?(zn.addScaledVector(In,e),Fn.addScaledVector(Nn,e),On.addScaledVector(Bn,e)):
(zn.addScaledVector(In.sub(Cn),e),Fn.addScaledVector(Nn.sub(Pn),e),On.addScaledVect
or(Bn.sub(Dn),e)))Cn.add(zn),Pn.add(Fn),Dn.add(On)}t.isSkinnedMesh&&e.skinning&&
(t.boneTransform(c,Cn),t.boneTransform(h,Pn),t.boneTransform(u,Dn));const
p=function(t,e,n,i,r,s,a,o){let l;if(l=1===e.side?
i.intersectTriangle(a,s,r,!0,o):i.intersectTriangle(r,s,a,!0,o),null===l)re
turn null;Vn.copy(o),Vn.applyMatrix4(t.matrixWorld);const
c=n.ray.origin.distanceTo(Vn);return c<n.near||c>n.far?null:
{distance:c,point:Vn.clone(),object:t}}(t,e,n,i,Cn,Pn,Dn,kn);if(p){o&&
(Hn.fromBufferAttribute(o,c),Gn.fromBufferAttribute(o,h),Un.fromBufferAttribute(o,u
),p.uv=je.getUV(kn,Cn,Pn,Dn,Hn,Gn,Un,new vt)),l&&
(Hn.fromBufferAttribute(l,c),Gn.fromBufferAttribute(l,h),Un.fromBufferAttribute(l,u

```

```

),p.uv2=je.getUV(kn,Cn,Pn,Dn,Hn,Gn,Un,new vt));const t={a:c,b:h,c:u,normals:new
Lt,materialIndex:0};je.getNormal(Cn,Pn,Dn,t.normal),p.face=t}return
p}Wn.prototype.isMesh=!0;class qn extends En{constructor(t=1,e=1,n=1,i=1,r=1,s=1)
{super(),this.type="BoxGeometry",this.parameters=
{width:t,height:e,depth:n,widthSegments:i,heightSegments:r,depthSegments:s};const
a=this,i=Math.floor(i),r=Math.floor(r),s=Math.floor(s);const o=[],l=[],c=[],h=
[];let u=0,d=0;function p(t,e,n,i,r,s,p,m,f,g,v){const
y=s/f,x=p/g,_=s/2,w=p/2,b=m/2,M=f+1,S=g+1;let T=0,E=0;const A=new Lt;for(let
s=0;s<S;s++){const a=s*x-w;for(let o=0;o<M;o++){const u=o*y-
_;A[t]=u*i,A[e]=a*r,A[n]=b,l.push(A.x,A.y,A.z),A[t]=0,A[e]=0,A[n]=m>0?
l:-1,c.push(A.x,A.y,A.z),h.push(o/f),h.push(1-s/g),T+=1}}for(let
t=0;t<g;t++)for(let e=0;e<f;e++){const n=u+e*M*t,i=u+e*M*(t+1),r=u+(e+1)+M*
(t+1),s=u+
(e+1)+M*t;o.push(n,i,s),o.push(i,r,s),E+=6)a.addGroup(d,E,v),d+=E,u+=T}p("z","y","x
",-1,-1,n,e,t,s,r,0),p("z","y","x",1,-1,n,e,-
t,s,r,1),p("x","z","y",1,1,t,n,e,i,s,2),p("x","z","y",1,-1,t,n,-
e,i,s,3),p("x","y","z",1,-1,t,e,n,i,r,4),p("x","y","z",-1,-1,t,e,-
n,i,r,5),this.setIndex(o),this.setAttribute("position",new
mn(1,3)),this.setAttribute("normal",new mn(c,3)),this.setAttribute("uv",new
mn(h,2))}}function Xn(t){const e={};for(const n in t){e[n]={};for(const i in t[n])
{const r=t[n][i];r&&
(r.isColor||r.isMatrix3||r.isMatrix4||r.isVector2||r.isVector3||r.isVector4||r.isTe
xture||r.isQuaternion)?e[n][i]=r.clone():Array.isArray(r)?e[n][i]=r.slice():e[n]
[i]=r}}return e}function Yn(t){const e={};for(let n=0;n<t.length;n++){const
i=Xn(t[n]);for(const t in i)e[t]=i[t]}return e}const Zn={clone:Xn,merge:Yn};class
Jn extends Xe{constructor(t){super(),this.type="ShaderMaterial",this.defines=
{},this.uniforms={},this.vertexShader="void main() {\n\tgl_Position =
projectionMatrix * modelViewMatrix * vec4( position, 1.0
);\n}",this.fragmentShader="void main() {\n\tgl_FragColor = vec4( 1.0, 0.0, 0.0,
1.0
);\n}",this.linewidth=1,this.wireframe=!1,this.wireframeLinewidth=1,this.fog=!1,thi
s.lights=!1,this.clipping=!1,this.skinning=!1,this.morphTargets=!1,this.morphNormal
s=!1,this.extensions=
{derivatives:!1,fragDepth:!1,drawBuffers:!1,shaderTextureLOD:!1},this.defaultAttrib
uteValues={color:[1,1,1],uv:[0,0],uv2:[0,0]},this.index0AttributeName=void
0,this.uniformsNeedUpdate=!1,this.glslVersion=null,void 0!==t&&(void
0!==t.attributes&&console.error("THREE.ShaderMaterial: attributes should now be
defined in THREE.BufferGeometry instead.")),this.setValues(t))}copy(t){return
super.copy(t),this.fragmentShader=t.fragmentShader,this.vertexShader=t.vertexShader
,this.uniforms=Xn(t.uniforms),this.defines=Object.assign({},t.defines),this.wirefra
me=t.wireframe,this.wireframeLinewidth=t.wireframeLinewidth,this.lights=t.lights,th
is.clipping=t.clipping,this.skinning=t.skinning,this.morphTargets=t.morphTargets,th
is.morphNormals=t.morphNormals,this.extensions=Object.assign({},t.extensions),this.
glslVersion=t.glslVersion,this}toJSON(t){const
e=super.toJSON(t);e.glslVersion=this.glslVersion,e.uniforms={};for(const n in
this.uniforms){const i=this.uniforms[n].value;i&&i.isTexture?e.uniforms[n]=
{type:"t",value:i.toJSON(t).uuid}:i&&i.isColor?e.uniforms[n]=
{type:"c",value:i.getHex()}:i&&i.isVector2?e.uniforms[n]=
{type:"v2",value:i.toArray()}:i&&i.isVector3?e.uniforms[n]=
{type:"v3",value:i.toArray()}:i&&i.isVector4?e.uniforms[n]=
{type:"v4",value:i.toArray()}:i&&i.isMatrix3?e.uniforms[n]=
{type:"m3",value:i.toArray()}:i&&i.isMatrix4?e.uniforms[n]=
{type:"m4",value:i.toArray()}:e.uniforms[n]=
{value:i}}Object.keys(this.defines).length>0&&
(e.defines=this.defines),e.vertexShader=this.vertexShader,e.fragmentShader=this.fra
gmentShader;const n={};for(const t in this.extensions)!0===this.extensions[t]&&
(n[t]=!0);return Object.keys(n).length>0&&
(e.extensions=n),e}}Jn.prototype.isShaderMaterial=!0;class Qn extends

```

```

Ce{constructor(){super(),this.type="Camera",this.matrixWorldInverse=new
se,this.projectionMatrix=new se,this.projectionMatrixInverse=new se}copy(t,e)
{return
super.copy(t,e),this.matrixWorldInverse.copy(t.matrixWorldInverse),this.projectionM
atrix.copy(t.projectionMatrix),this.projectionMatrixInverse.copy(t.projectionMatrix
Inverse),this}getWorldDirection(t){void 0===t&&(console.warn("THREE.Camera:
.getWorldDirection() target is now required")),t=new
Lt),this.updateWorldMatrix(!0,!1);const e=this.matrixWorld.elements;return t.set(-
e[8],-e[9],-e[10]).normalize()}updateMatrixWorld(t)
{super.updateMatrixWorld(t),this.matrixWorldInverse.copy(this.matrixWorld).invert()
}updateWorldMatrix(t,e)
{super.updateWorldMatrix(t,e),this.matrixWorldInverse.copy(this.matrixWorld).invert
()}clone(){return(new this.constructor).copy(this)}}Qn.prototype.isCamera=!0;class
Kn extends Qn{constructor(t=50,e=1,n=.1,i=2e3)
{super(),this.type="PerspectiveCamera",this.fov=t,this.zoom=1,this.near=n,this.far=
i,this.focus=10,this.aspect=e,this.view=null,this.filmGauge=35,this.filmOffset=0,th
is.updateProjectionMatrix()}copy(t,e){return
super.copy(t,e),this.fov=t.fov,this.zoom=t.zoom,this.near=t.near,this.far=t.far,thi
s.focus=t.focus,this.aspect=t.aspect,this.view=null===t.view?
null:Object.assign({},t.view),this.filmGauge=t.filmGauge,this.filmOffset=t.filmOffs
et,this}setFocalLength(t){const
e=.5*this.getFilmHeight()/t;this.fov=2*Math.atan(e),this.updateProjectionMatrix(
)}getFocalLength(){const
t=Math.tan(.5*Math.PI*this.fov);return.5*this.getFilmHeight()/t}getEffectiveFOV(){return
2*Math.PI*Math.atan(Math.tan(.5*Math.PI*this.fov)/this.zoom)}getFilmWidth(){return
this.filmGauge*Math.min(this.aspect,1)}getFilmHeight(){return
this.filmGauge/Math.max(this.aspect,1)}setViewOffset(t,e,n,i,r,s)
{this.aspect=t/e,null===this.view&&(this.view=
{enabled:!0,fullWidth:1,fullHeight:1,offsetX:0,offsetY:0,width:1,height:1}),this.vi
ew.enabled=!0,this.view.fullWidth=t,this.view.fullHeight=e,this.view.offsetX=n,this
.view.offsetY=i,this.view.width=r,this.view.height=s,this.updateProjectionMatrix())
clearViewOffset(){null!==this.view&&
(this.view.enabled=!1),this.updateProjectionMatrix()}updateProjectionMatrix(){const
t=this.near;let
e=t*Math.tan(.5*Math.PI*this.fov)/this.zoom,n=2*e,i=this.aspect*n,r=-.5*i;const
s=this.view;if(null!==this.view&&this.view.enabled){const
t=s.fullWidth,a=s.fullHeight;r+=s.offsetX*i/t,e-
=s.offsetY*n/a,i*=s.width/t,n*=s.height/a}const a=this.filmOffset;0!==a&&
(r+=t*a/this.getFilmWidth()),this.projectionMatrix.makePerspective(r,r+i,e,e-
n,t,this.far),this.projectionMatrixInverse.copy(this.projectionMatrix).invert()}toJ
SON(t){const e=super.toJSON(t);return
e.object.fov=this.fov,e.object.zoom=this.zoom,e.object.near=this.near,e.object.far=
this.far,e.object.focus=this.focus,e.object.aspect=this.aspect,null!==this.view&&
(e.object.view=Object.assign({},this.view)),e.object.filmGauge=this.filmGauge,e.obj
ect.filmOffset=this.filmOffset,e}}Kn.prototype.isPerspectiveCamera=!0;const
$n=90;class ti extends Ce{constructor(t,e,n)
{if(super(),this.type="CubeCamera",!0!==n.isWebGLCubeRenderTarget)return void
console.error("THREE.CubeCamera: The constructor now expects an instance of
WebGLCubeRenderTarget as third parameter.");this.renderTarget=n;const i=new
Kn($n,1,t,e);i.layers=this.layers,i.up.set(0,-1,0),i.lookAt(new
Lt(1,0,0)),this.add(i);const r=new
Kn($n,1,t,e);r.layers=this.layers,r.up.set(0,-1,0),r.lookAt(new
Lt(-1,0,0)),this.add(r);const s=new
Kn($n,1,t,e);s.layers=this.layers,s.up.set(0,0,1),s.lookAt(new
Lt(0,1,0)),this.add(s);const a=new
Kn($n,1,t,e);a.layers=this.layers,a.up.set(0,0,-1),a.lookAt(new
Lt(0,-1,0)),this.add(a);const o=new
Kn($n,1,t,e);o.layers=this.layers,o.up.set(0,-1,0),o.lookAt(new

```

[illegible]

```

f).normalize(),e[1].setComponents(a+i,h+o,m+u,y+f).normalize(),e[2].setComponents(a
+r,h+l,m+d,y+g).normalize(),e[3].setComponents(a-r,h-l,m-d,y-
g).normalize(),e[4].setComponents(a-s,h-c,m-p,y-
v).normalize(),e[5].setComponents(a+s,h+c,m+p,y+v).normalize(),this}intersectsObjec
t(t){const e=t.geometry;return
null===e.boundingSphere&&e.computeBoundingSphere(),ri.copy(e.boundingSphere).applyM
atrix4(t.matrixWorld),this.intersectsSphere(ri)}intersectsSprite(t){return
ri.center.set(0,0,0),ri.radius=.7071067811865476,ri.applyMatrix4(t.matrixWorld),thi
s.intersectsSphere(ri)}intersectsSphere(t){const e=this.planes,n=t.center,i=-
t.radius;for(let t=0;t<6;t++){if(e[t].distanceToPoint(n)
<i)return!1}return!0}intersectsBox(t){const e=this.planes;for(let n=0;n<6;n++)
{const i=e[n];if(si.x=i.normal.x>0?t.max.x:t.min.x,si.y=i.normal.y>0?
t.max.y:t.min.y,si.z=i.normal.z>0?t.max.z:t.min.z,i.distanceToPoint(si)
<0)return!1}return!0}containsPoint(t){const e=this.planes;for(let
n=0;n<6;n++)if(e[n].distanceToPoint(t)<0)return!1;return!0}clone(){return(new
this.constructor).copy(this)}}function oi(){let t=null,e=!1,n=null,i=null;function
r(e,s){n(e,s),i=t.requestAnimationFrame(r)}return{start:function()
{!0!==e&&null!==n&&(i=t.requestAnimationFrame(r),e=!0)},stop:function()
{t.cancelAnimationFrame(i),e=!1},setAnimationLoop:function(t)
{n=t},setContext:function(e){t=e}}}function li(t,e){const n=e.isWebGL2,i=new
WeakMap;return{get:function(t){return t.isInterleavedBufferAttribute&&
(t=t.data),i.get(t)},remove:function(e){e.isInterleavedBufferAttribute&&
(e=e.data);const n=i.get(e);n&&
(t.deleteBuffer(n.buffer),i.delete(e))},update:function(e,r)
{if(e.isGLBufferAttribute){const t=i.get(e);return
void(!t||t.version<e.version)&&i.set(e,
{buffer:e.buffer,type:e.type,bytesPerElement:e.elementSize,version:e.version})}}e.i
sInterleavedBufferAttribute&&(e=e.data);const s=i.get(e);void 0===s?
i.set(e,function(e,i){const
r=e.array,s=e.usage,a=t.createBuffer();t.bindBuffer(i,a),t.bufferData(i,r,s),e.onUp
loadCallback();let o=5126;return r instanceof Float32Array?o=5126:r instanceof
Float64Array?console.warn("THREE.WebGLAttributes: Unsupported data buffer format:
Float64Array."):r instanceof Uint16Array?e.isFloat16BufferAttribute?n?
o=5131:console.warn("THREE.WebGLAttributes: Usage of Float16BufferAttribute
requires WebGL2."):o=5123:r instanceof Int16Array?o=5122:r instanceof Uint32Array?
o=5125:r instanceof Int32Array?o=5124:r instanceof Int8Array?o=5120:r instanceof
Uint8Array&&(o=5121),
{buffer:a,type:o,bytesPerElement:r.BYTES_PER_ELEMENT,version:e.version}}
(e,r)):s.version<e.version&&(!function(e,i,r){const
s=i.array,a=i.updateRange;t.bindBuffer(r,e),-1===a.count?t.bufferSubData(r,0,s):(n?
t.bufferSubData(r,a.offset*s.BYTES_PER_ELEMENT,s,a.offset,a.count):t.bufferSubData(
r,a.offset*s.BYTES_PER_ELEMENT,s.subarray(a.offset,a.offset+a.count)),a.count=-1)}
(s.buffer,e,r),s.version=e.version)}}}class ci extends
En{constructor(t=1,e=1,n=1,i=1){super(),this.type="PlaneGeometry",this.parameters=
{width:t,height:e,widthSegments:n,heightSegments:i};const
r=t/2,s=e/2,a=Math.floor(n),o=Math.floor(i),l=a+1,c=o+1,h=t/a,u=e/o,d=[],p=[],m=
[],f=[];for(let t=0;t<c;t++){const e=t*u-s;for(let n=0;n<l;n++){const i=n*h-
r;p.push(i,-e,0),m.push(0,0,1),f.push(n/a),f.push(1-t/o)}}for(let
t=0;t<o;t++)for(let e=0;e<a;e++){const n=e+1*t,i=e+1*(t+1),r=e+1+l*
(t+1),s=e+1+l*t;d.push(n,i,s),d.push(i,r,s)}this.setIndex(d),this.setAttribute("pos
ition",new mn(p,3)),this.setAttribute("normal",new
mn(m,3)),this.setAttribute("uv",new mn(f,2))}const hi={alphanap_fragment:"#ifdef
USE_ALPHAMAP\n\t diffuseColor.a *= texture2D( alphaMap, vUv
).g;\n#endif",alphanap_pars_fragment:"#ifdef USE_ALPHAMAP\n\t uniform sampler2D
alphaMap;\n#endif",alphatest_fragment:"#ifdef ALPHATEST\n\t if ( diffuseColor.a <
ALPHATEST ) discard;\n#endif",aomap_fragment:"#ifdef USE_AOMAP\n\t float
ambientOcclusion = ( texture2D( aoMap, vUv2 ).r - 1.0 ) * aoMapIntensity +
1.0;\n\t reflectedLight.indirectDiffuse *= ambientOcclusion;\n\t #if defined(

```



```

USE_ENVMAP ) && defined( STANDARD )\n\t\tfloat dotNV = saturate( dot(
geometry.normal, geometry.viewDir ) );\n\t\treflectedLight.indirectSpecular *=
computeSpecularOcclusion( dotNV, ambientOcclusion, material.specularRoughness
);\n\t#endif\n#endif",aomap_pars_fragment:"#ifdef USE_AOMAP\n\t\tuniform sampler2D
aoMap;\n\t\tuniform float aoMapIntensity;\n#endif",begin_vertex:"vec3 transformed =
vec3( position );",beginnormal_vertex:"vec3 objectNormal = vec3( normal );\n#ifdef
USE_TANGENT\n\t\tvec3 objectTangent = vec3( tangent.xyz );\n#endif",bsdfs:"vec2
integrateSpecularBRDF( const in float dotNV, const in float roughness ) {\n\tconst
vec4 c0 = vec4( - 1, - 0.0275, - 0.572, 0.022 );\n\tconst vec4 c1 = vec4( 1,
0.0425, 1.04, - 0.04 );\n\tvec4 r = roughness * c0 + c1;\n\tfloat a004 = min( r.x *
r.x, exp2( - 9.28 * dotNV ) ) * r.x + r.y;\n\treturn vec2( -1.04, 1.04 ) * a004 +
r.zw;\n}\nfloat punctualLightIntensityToIrradianceFactor( const in float
lightDistance, const in float cutoffDistance, const in float decayExponent ) {\n#if
defined( PHYSICALLY_CORRECT_LIGHTS )\n\tfloat distanceFalloff = 1.0 / max( pow(
lightDistance, decayExponent ), 0.01 );\n\tif( cutoffDistance > 0.0 )
{\n\t\tfloat distanceFalloff *= pow2( saturate( 1.0 - pow4( lightDistance /
cutoffDistance ) ) );\n\t}\n\treturn distanceFalloff;\n#else\n\tif( cutoffDistance
> 0.0 && decayExponent > 0.0 ) {\n\t\treturn pow( saturate( -lightDistance /
cutoffDistance + 1.0 ), decayExponent );\n\t}\n\treturn 1.0;\n#endif\n}\nvec3
BRDF_Diffuse_Lambert( const in vec3 diffuseColor ) {\n\treturn RECIPROCAL_PI *
diffuseColor;\n}\nvec3 F_Schlick( const in vec3 specularColor, const in float dotLH
) {\n\tfloat fresnel = exp2( ( -5.55473 * dotLH - 6.98316 ) * dotLH );\n\treturn (
1.0 - specularColor ) * fresnel + specularColor;\n}\nvec3
F_Schlick_RoughnessDependent( const in vec3 F0, const in float dotNV, const in
float roughness ) {\n\tfloat fresnel = exp2( ( -5.55473 * dotNV - 6.98316 ) * dotNV
);\n\tvec3 Fr = max( vec3( 1.0 - roughness ), F0 ) - F0;\n\treturn Fr * fresnel +
F0;\n}\nfloat G_GGX_Smith( const in float alpha, const in float dotNL, const in
float dotNV ) {\n\tfloat a2 = pow2( alpha );\n\tfloat gl = dotNL + sqrt( a2 + ( 1.0
- a2 ) * pow2( dotNL ) );\n\tfloat gv = dotNV + sqrt( a2 + ( 1.0 - a2 ) * pow2(
dotNV ) );\n\treturn 1.0 / ( gl * gv );\n}\nfloat G_GGX_SmithCorrelated( const in
float alpha, const in float dotNL, const in float dotNV ) {\n\tfloat a2 = pow2(
alpha );\n\tfloat gv = dotNL * sqrt( a2 + ( 1.0 - a2 ) * pow2( dotNV ) );\n\tfloat
gl = dotNV * sqrt( a2 + ( 1.0 - a2 ) * pow2( dotNL ) );\n\treturn 0.5 / max( gv +
gl, EPSILON );\n}\nfloat D_GGX( const in float alpha, const in float dotNH )
{\n\tfloat a2 = pow2( alpha );\n\tfloat denom = pow2( dotNH ) * ( a2 - 1.0 ) +
1.0;\n\treturn RECIPROCAL_PI * a2 / pow2( denom );\n}\nvec3 BRDF_Specular_GGX(
const in IncidentLight incidentLight, const in vec3 viewDir, const in vec3 normal,
const in vec3 specularColor, const in float roughness ) {\n\tfloat alpha = pow2(
roughness );\n\tvec3 halfDir = normalize( incidentLight.direction + viewDir
);\n\tfloat dotNL = saturate( dot( normal, incidentLight.direction ) );\n\tfloat
dotNV = saturate( dot( normal, viewDir ) );\n\tfloat dotNH = saturate( dot( normal,
halfDir ) );\n\tfloat dotLH = saturate( dot( incidentLight.direction, halfDir )
);\n\tvec3 F = F_Schlick( specularColor, dotLH );\n\tfloat G =
G_GGX_SmithCorrelated( alpha, dotNL, dotNV );\n\tfloat D = D_GGX( alpha, dotNH
);\n\treturn F * ( G * D );\n}\nvec2 LTC_Uv( const in vec3 N, const in vec3 V,
const in float roughness ) {\n\tconst float LUT_SIZE = 64.0;\n\tconst float
LUT_SCALE = ( LUT_SIZE - 1.0 ) / LUT_SIZE;\n\tconst float LUT_BIAS = 0.5 /
LUT_SIZE;\n\tfloat dotNV = saturate( dot( N, V ) );\n\tvec2 uv = vec2( roughness,
sqrt( 1.0 - dotNV ) );\n\tuv = uv * LUT_SCALE + LUT_BIAS;\n\treturn uv;\n}\nfloat
LTC_ClipppedSphereFormFactor( const in vec3 f ) {\n\tfloat l = length( f
);\n\treturn max( ( l * l + f.z ) / ( l + 1.0 ), 0.0 );\n}\nvec3
LTC_EdgeVectorFormFactor( const in vec3 v1, const in vec3 v2 ) {\n\tfloat x = dot(
v1, v2 );\n\tfloat y = abs( x );\n\tfloat a = 0.8543985 + ( 0.4965155 + 0.0145206 *
y ) * y;\n\tfloat b = 3.4175940 + ( 4.1616724 + y ) * y;\n\tfloat v = a /
b;\n\tfloat theta_sintheta = ( x > 0.0 ) ? v : 0.5 * inversesqrt( max( 1.0 - x * x,
1e-7 ) ) - v;\n\treturn cross( v1, v2 ) * theta_sintheta;\n}\nvec3 LTC_Evaluate(
const in vec3 N, const in vec3 V, const in vec3 P, const in mat3 mInv, const in
vec3 rectCoords[ 4 ] ) {\n\tvec3 v1 = rectCoords[ 1 ] - rectCoords[ 0 ];\n\tvec3 v2

```

```

= rectCoords[ 3 ] - rectCoords[ 0 ];\n\tvec3 lightNormal = cross( v1, v2 );\n\tif(
dot( lightNormal, P - rectCoords[ 0 ] ) < 0.0 ) return vec3( 0.0 );\n\tvec3 T1,
T2;\n\tT1 = normalize( V - N * dot( V, N ) );\n\tT2 = - cross( N, T1 );\n\tmat3 mat
= mInv * transposeMat3( mat3( T1, T2, N ) );\n\tvec3 coords[ 4 ];\n\tcoords[ 0 ] =
mat * ( rectCoords[ 0 ] - P );\n\tcoords[ 1 ] = mat * ( rectCoords[ 1 ] - P
);\n\tcoords[ 2 ] = mat * ( rectCoords[ 2 ] - P );\n\tcoords[ 3 ] = mat * (
rectCoords[ 3 ] - P );\n\tcoords[ 0 ] = normalize( coords[ 0 ] );\n\tcoords[ 1 ] =
normalize( coords[ 1 ] );\n\tcoords[ 2 ] = normalize( coords[ 2 ] );\n\tcoords[ 3 ]
= normalize( coords[ 3 ] );\n\tvec3 vectorFormFactor = vec3( 0.0
);\n\tvectorFormFactor += LTC_EdgeVectorFormFactor( coords[ 0 ], coords[ 1 ]
);\n\tvectorFormFactor += LTC_EdgeVectorFormFactor( coords[ 1 ], coords[ 2 ]
);\n\tvectorFormFactor += LTC_EdgeVectorFormFactor( coords[ 2 ], coords[ 3 ]
);\n\tvectorFormFactor += LTC_EdgeVectorFormFactor( coords[ 3 ], coords[ 0 ]
);\n\tfloat result = LTC_ClipppedSphereFormFactor( vectorFormFactor );\n\treturn
vec3( result );\n}\n\nvec3 BRDF_Specular_GGX_Environment( const in vec3 viewDir,
const in vec3 normal, const in vec3 specularColor, const in float roughness )
{\n\tfloat dotNV = saturate( dot( normal, viewDir ) );\n\tvec2 brdf =
integrateSpecularBRDF( dotNV, roughness );\n\treturn specularColor * brdf.x +
brdf.y;\n}\n\nvoid BRDF_Specular_Multiscattering_Environment( const in
GeometricContext geometry, const in vec3 specularColor, const in float roughness,
inout vec3 singleScatter, inout vec3 multiScatter ) {\n\tfloat dotNV = saturate(
dot( geometry.normal, geometry.viewDir ) );\n\tvec3 F =
F_Schlick_RoughnessDependent( specularColor, dotNV, roughness );\n\tvec2 brdf =
integrateSpecularBRDF( dotNV, roughness );\n\tvec3 FssEss = F * brdf.x +
brdf.y;\n\tfloat Ess = brdf.x + brdf.y;\n\tfloat Ems = 1.0 - Ess;\n\tvec3 Favg =
specularColor + ( 1.0 - specularColor ) * 0.047619;\n\tvec3 Fms = FssEss * Favg / (
1.0 - Ems * Favg );\n\tsingleScatter += FssEss;\n\tmultiScatter += Fms *
Ems;\n}\n\nfloat G_BlinnPhong_Implicit( ) {\n\treturn 0.25;\n}\n\nfloat D_BlinnPhong(
const in float shininess, const in float dotNH ) {\n\treturn RECIPROCAL_PI * (
shininess * 0.5 + 1.0 ) * pow( dotNH, shininess );\n}\n\nvec3
BRDF_Specular_BlinnPhong( const in IncidentLight incidentLight, const in
GeometricContext geometry, const in vec3 specularColor, const in float shininess )
{\n\tvec3 halfDir = normalize( incidentLight.direction + geometry.viewDir
);\n\tfloat dotNH = saturate( dot( geometry.normal, halfDir ) );\n\tfloat dotLH =
saturate( dot( incidentLight.direction, halfDir ) );\n\tvec3 F = F_Schlick(
specularColor, dotLH );\n\tfloat G = G_BlinnPhong_Implicit( );\n\tfloat D =
D_BlinnPhong( shininess, dotNH );\n\treturn F * ( G * D );\n}\n\nfloat
GGXRoughnessToBlinnExponent( const in float ggxRoughness ) {\n\treturn ( 2.0 /
pow2( ggxRoughness + 0.0001 ) - 2.0 );\n}\n\nfloat BlinnExponentToGGXRoughness( const
in float blinnExponent ) {\n\treturn sqrt( 2.0 / ( blinnExponent + 2.0 ) );\n}\n\n#if
defined( USE_SHEEN )\n\nfloat D_Charlie(float roughness, float NoH) {\n\tfloat
invAlpha = 1.0 / roughness;\n\tfloat cos2h = NoH * NoH;\n\tfloat sin2h = max(1.0 -
cos2h, 0.0078125);\n\treturn (2.0 + invAlpha) * pow(sin2h, invAlpha * 0.5) / (2.0 *
PI);\n}\n\nfloat V_Neubelt(float NoV, float NoL) {\n\treturn saturate(1.0 / (4.0 *
(NoL + NoV - NoL * NoV)));\n}\n\nvec3 BRDF_Specular_Sheen( const in float roughness,
const in vec3 L, const in GeometricContext geometry, vec3 specularColor ) {\n\tvec3
N = geometry.normal;\n\tvec3 V = geometry.viewDir;\n\tvec3 H = normalize( V + L
);\n\tfloat dotNH = saturate( dot( N, H ) );\n\treturn specularColor * D_Charlie(
roughness, dotNH ) * V_Neubelt( dot(N, V), dot(N, L)
);\n}\n\n#endif",bumpmap_pars_fragment:"#ifdef USE_BUMPMAP\n\tuniform sampler2D
bumpMap;\n\tuniform float bumpScale;\n\tvec2 dHdxy_fwd() {\n\t\tvec2 dSTdx = dFdx(
vUv );\n\t\tvec2 dSTdy = dFdy( vUv );\n\t\tfloat Hl1 = bumpScale * texture2D(
bumpMap, vUv ).x;\n\t\tfloat dBx = bumpScale * texture2D( bumpMap, vUv + dSTdx ).x
- Hl1;\n\t\tfloat dBy = bumpScale * texture2D( bumpMap, vUv + dSTdy ).x -
Hl1;\n\t\treturn vec2( dBx, dBy );\n\t}\n\tvec3 perturbNormalArb( vec3 surf_pos,
vec3 surf_norm, vec2 dHdxy, float faceDirection ) {\n\t\tvec3 vSigmaX = vec3( dFdx(
surf_pos.x ), dFdx( surf_pos.y ), dFdx( surf_pos.z ) );\n\t\tvec3 vSigmaY = vec3(
dFdy( surf_pos.x ), dFdy( surf_pos.y ), dFdy( surf_pos.z ) );\n\t\tvec3 vN =

```

```

surf_norm;\n\t\tvec3 R1 = cross( vSigmaY, vN );\n\t\tvec3 R2 = cross( vN, vSigmaX
);\n\t\tfloat fDet = dot( vSigmaX, R1 ) * faceDirection;\n\t\tvec3 vGrad = sign(
fDet ) * ( dHdxy.x * R1 + dHdxy.y * R2 );\n\t\treturn normalize( abs( fDet ) *
surf_norm - vGrad );\n\t}\n#endif",clipping_planes_fragment:"#if
NUM_CLIPPING_PLANES > 0\n\tvec4 plane;\n\t#pragma unroll_loop_start\n\tfor ( int i
= 0; i < UNION_CLIPPING_PLANES; i ++ ) {\n\t\tplane = clippingPlanes[ i ];\n\t\tif
( dot( vClipPosition, plane.xyz ) > plane.w ) discard;\n\t}\n\t#pragma
unroll_loop_end\n\tif UNION_CLIPPING_PLANES < NUM_CLIPPING_PLANES\n\t\tbool
clipped = true;\n\t\t#pragma unroll_loop_start\n\t\tfor ( int i =
UNION_CLIPPING_PLANES; i < NUM_CLIPPING_PLANES; i ++ ) {\n\t\t\tplane =
clippingPlanes[ i ];\n\t\t\tclipped = ( dot( vClipPosition, plane.xyz ) > plane.w )
&& clipped;\n\t\t}\n\t\t#pragma unroll_loop_end\n\t\tif ( clipped )
discard;\n\t}\n#endif",clipping_planes_pars_fragment:"#if NUM_CLIPPING_PLANES
> 0\n\tvarying vec3 vClipPosition;\n\tuniform vec4 clippingPlanes[
NUM_CLIPPING_PLANES ];\n#endif",clipping_planes_pars_vertex:"#if
NUM_CLIPPING_PLANES > 0\n\tvarying vec3
vClipPosition;\n#endif",clipping_planes_vertex:"#if NUM_CLIPPING_PLANES >
0\n\tvClipPosition = - mvPosition.xyz;\n#endif",color_fragment:"#if defined(
USE_COLOR_ALPHA )\n\t diffuseColor *= vColor;\n#elif defined( USE_COLOR
)\n\t diffuseColor.rgb *= vColor;\n#endif",color_pars_fragment:"#if defined(
USE_COLOR_ALPHA )\n\t varying vec4 vColor;\n#elif defined( USE_COLOR )\n\t varying
vec3 vColor;\n#endif",color_pars_vertex:"#if defined( USE_COLOR_ALPHA )\n\t varying
vec4 vColor;\n#elif defined( USE_COLOR ) || defined( USE_INSTANCING_COLOR
)\n\t varying vec3 vColor;\n#endif",color_vertex:"#if defined( USE_COLOR_ALPHA
)\n\t vColor = vec4( 1.0 );\n#elif defined( USE_COLOR ) || defined(
USE_INSTANCING_COLOR )\n\t vColor = vec3( 1.0 );\n#endif\n#ifdef USE_COLOR\n\t vColor
*= color;\n#endif\n#ifdef USE_INSTANCING_COLOR\n\t vColor.xyz *=
instanceColor.xyz;\n#endif",common:"#define PI 3.141592653589793\n#define PI2
6.283185307179586\n#define PI_HALF 1.5707963267948966\n#define RECIPROCAL_PI
0.3183098861837907\n#define RECIPROCAL_PI2 0.15915494309189535\n#define EPSILON 1e-
6\n#ifdef saturate\n#define saturate(a) clamp( a, 0.0, 1.0 )\n#endif\n#define
whiteComplement(a) ( 1.0 - saturate( a ) )\nfloat pow2( const in float x ) { return
x*x; }\nfloat pow3( const in float x ) { return x*x*x; }\nfloat pow4( const in
float x ) { float x2 = x*x; return x2*x2; }\nfloat average( const in vec3 color ) {
return dot( color, vec3( 0.3333 ) ); }\nhighp float rand( const in vec2 uv )
{\n\tconst highp float a = 12.9898, b = 78.233, c = 43758.5453;\n\t highp float dt =
dot( uv.xy, vec2( a,b ) ), sn = mod( dt, PI );\n\treturn fract(sin(sn) *
c);\n}\n#ifdef HIGH_PRECISION\n\tfloat precisionSafeLength( vec3 v ) { return
length( v ); }\n#else\n\tfloat max3( vec3 v ) { return max( max( v.x, v.y ), v.z );
}\n\tfloat precisionSafeLength( vec3 v ) {\n\t\tfloat maxComponent = max3( abs( v )
);\n\t\treturn length( v / maxComponent ) * maxComponent;\n\t}\n#endif\nstruct
IncidentLight {\n\tvec3 color;\n\tvec3 direction;\n\tbool visible;\n};\nstruct
ReflectedLight {\n\tvec3 directDiffuse;\n\tvec3 directSpecular;\n\tvec3
indirectDiffuse;\n\tvec3 indirectSpecular;\n};\nstruct GeometricContext {\n\tvec3
position;\n\tvec3 normal;\n\tvec3 viewDir;\n#ifdef CLEARCOAT\n\tvec3
clearcoatNormal;\n#endif\n}\nvec3 transformDirection( in vec3 dir, in mat4 matrix
) {\n\treturn normalize( ( matrix * vec4( dir, 0.0 ) ).xyz );\n}\nvec3
inverseTransformDirection( in vec3 dir, in mat4 matrix ) {\n\treturn normalize( (
vec4( dir, 0.0 ) * matrix ).xyz );\n}\nvec3 projectOnPlane(in vec3 point, in vec3
pointOnPlane, in vec3 planeNormal ) {\n\tfloat distance = dot( planeNormal, point -
pointOnPlane );\n\treturn - distance * planeNormal + point;\n}\nfloat sideOfPlane(
in vec3 point, in vec3 pointOnPlane, in vec3 planeNormal ) {\n\treturn sign( dot(
point - pointOnPlane, planeNormal ) );\n}\nvec3 linePlaneIntersect( in vec3
pointOnLine, in vec3 lineDirection, in vec3 pointOnPlane, in vec3 planeNormal )
{\n\treturn lineDirection * ( dot( planeNormal, pointOnPlane - pointOnLine ) / dot(
planeNormal, lineDirection ) ) + pointOnLine;\n}\nmat3 transposeMat3( const in mat3
m ) {\n\tmat3 tmp;\n\ttmp[ 0 ] = vec3( m[ 0 ].x, m[ 1 ].x, m[ 2 ].x );\n\ttmp[ 1 ]
= vec3( m[ 0 ].y, m[ 1 ].y, m[ 2 ].y );\n\ttmp[ 2 ] = vec3( m[ 0 ].z, m[ 1 ].z, m[

```

```

2 ].z );\n\treturn tmp;\n}\n\nfloat linearToRelativeLuminance( const in vec3 color )
{\n\tvec3 weights = vec3( 0.2126, 0.7152, 0.0722 );\n\treturn dot( weights,
color.rgb );\n}\n\nbool isPerspectiveMatrix( mat4 m ) {\n\treturn m[ 2 ][ 3 ] == -
1.0;\n}\n\nvec2 equirectUv( in vec3 dir ) {\n\tfloat u = atan( dir.z, dir.x ) *
RECIPROCAL_PI2 + 0.5;\n\tfloat v = asin( clamp( dir.y, - 1.0, 1.0 ) ) *
RECIPROCAL_PI + 0.5;\n\treturn vec2( u, v
);\n}" ,cube_uv_reflection_fragment:"#ifndef ENVMAP_TYPE_CUBE_UV\n\t#define
cubeUV_maxMipLevel 8.0\n\t#define cubeUV_minMipLevel 4.0\n\t#define
cubeUV_maxTileSize 256.0\n\t#define cubeUV_minTileSize 16.0\n\tfloat getFace( vec3
direction ) {\n\t\tvec3 absDirection = abs( direction );\n\t\tfloat face = -
1.0;\n\t\tif ( absDirection.x > absDirection.z ) {\n\t\t\tif ( absDirection.x >
absDirection.y )\n\t\t\t\tface = direction.x > 0.0 ? 0.0 :
3.0;\n\t\t\telse\n\t\t\t\tface = direction.y > 0.0 ? 1.0 : 4.0;\n\t\t} else
{\n\t\t\tif ( absDirection.z > absDirection.y )\n\t\t\t\tface = direction.z > 0.0 ?
2.0 : 5.0;\n\t\t\telse\n\t\t\t\tface = direction.y > 0.0 ? 1.0 :
4.0;\n\t\t}\n\t\treturn face;\n\t}\n\tvec2 getUV( vec3 direction, float face )
{\n\t\tvec2 uv;\n\t\tif ( face == 0.0 ) {\n\t\t\tuv = vec2( direction.z,
direction.y ) / abs( direction.x );\n\t\t} else if ( face == 1.0 ) {\n\t\t\tuv =
vec2( - direction.x, - direction.z ) / abs( direction.y );\n\t\t} else if ( face ==
2.0 ) {\n\t\t\tuv = vec2( - direction.x, direction.y ) / abs( direction.z );\n\t\t}
else if ( face == 3.0 ) {\n\t\t\tuv = vec2( - direction.z, direction.y ) / abs(
direction.x );\n\t\t} else if ( face == 4.0 ) {\n\t\t\tuv = vec2( - direction.x,
direction.z ) / abs( direction.y );\n\t\t} else {\n\t\t\tuv = vec2( direction.x,
direction.y ) / abs( direction.z );\n\t\t}\n\t\treturn 0.5 * ( uv + 1.0
);\n\t}\n\tvec3 bilinearCubeUV( sampler2D envMap, vec3 direction, float mipInt )
{\n\t\tfloat face = getFace( direction );\n\t\tfloat filterInt = max(
cubeUV_minMipLevel - mipInt, 0.0 );\n\t\tint mipInt = max( mipInt, cubeUV_minMipLevel
);\n\t\tfloat faceSize = exp2( mipInt );\n\t\tfloat texelSize = 1.0 / ( 3.0 *
cubeUV_maxTileSize );\n\t\tvec2 uv = getUV( direction, face ) * ( faceSize - 1.0
);\n\t\tvec2 f = fract( uv );\n\t\tuv += 0.5 - f;\n\t\tif ( face > 2.0 )
{\n\t\t\tuv.y += faceSize;\n\t\t\tface -= 3.0;\n\t\t}\n\t\tuv.x += face *
faceSize;\n\t\tif ( mipInt < cubeUV_maxMipLevel ) {\n\t\t\tuv.y += 2.0 *
cubeUV_maxTileSize;\n\t\t}\n\t\tuv.y += filterInt * 2.0 *
cubeUV_minTileSize;\n\t\tuv.x += 3.0 * max( 0.0, cubeUV_maxTileSize - 2.0 *
faceSize );\n\t\tuv *= texelSize;\n\t\tvec3 tl = envMapTexelToLinear( texture2D(
envMap, uv ) ).rgb;\n\t\tuv.x += texelSize;\n\t\tvec3 tr = envMapTexelToLinear(
texture2D( envMap, uv ) ).rgb;\n\t\tuv.y += texelSize;\n\t\tvec3 br =
envMapTexelToLinear( texture2D( envMap, uv ) ).rgb;\n\t\tuv.x -=
texelSize;\n\t\tvec3 bl = envMapTexelToLinear( texture2D( envMap, uv )
).rgb;\n\t\tvec3 tm = mix( tl, tr, f.x );\n\t\tvec3 bm = mix( bl, br, f.x
);\n\t\treturn mix( tm, bm, f.y );\n\t}\n\t#define r0 1.0\n\t#define v0
0.339\n\t#define m0 - 2.0\n\t#define r1 0.8\n\t#define v1 0.276\n\t#define m1 -
1.0\n\t#define r4 0.4\n\t#define v4 0.046\n\t#define m4 2.0\n\t#define r5
0.305\n\t#define v5 0.016\n\t#define m5 3.0\n\t#define r6 0.21\n\t#define v6
0.0038\n\t#define m6 4.0\n\tfloat roughnessToMip( float roughness ) {\n\t\tfloat
mip = 0.0;\n\t\tif ( roughness >= r1 ) {\n\t\t\tmip = ( r0 - roughness ) * ( m1 -
m0 ) / ( r0 - r1 ) + m0;\n\t\t} else if ( roughness >= r4 ) {\n\t\t\tmip = ( r1 -
roughness ) * ( m4 - m1 ) / ( r1 - r4 ) + m1;\n\t\t} else if ( roughness >= r5 )
{\n\t\t\tmip = ( r4 - roughness ) * ( m5 - m4 ) / ( r4 - r5 ) + m4;\n\t\t} else if
( roughness >= r6 ) {\n\t\t\tmip = ( r5 - roughness ) * ( m6 - m5 ) / ( r5 - r6 ) +
m5;\n\t\t} else {\n\t\t\tmip = - 2.0 * log2( 1.16 * roughness );\n\t\t}\n\t\treturn
mip;\n\t}\n\tvec4 textureCubeUV( sampler2D envMap, vec3 sampleDir, float roughness
) {\n\t\tfloat mip = clamp( roughnessToMip( roughness ), m0, cubeUV_maxMipLevel
);\n\t\tfloat mipF = fract( mip );\n\t\tfloat mipInt = floor( mip );\n\t\tvec3
color0 = bilinearCubeUV( envMap, sampleDir, mipInt );\n\t\tif ( mipF == 0.0 )
{\n\t\t\treturn vec4( color0, 1.0 );\n\t\t} else {\n\t\t\tvec3 color1 =
bilinearCubeUV( envMap, sampleDir, mipInt + 1.0 );\n\t\t\treturn vec4( mix( color0,
color1, mipF ), 1.0 );\n\t\t}\n\t}\n#endif",defaultNormal_vertex:"vec3

```

```

transformedNormal = objectNormal;\n#ifdef USE_INSTANCING\n\tmat3 m = mat3(
instanceMatrix );\n\ttransformedNormal /= vec3( dot( m[ 0 ], m[ 0 ] ), dot( m[ 1 ],
m[ 1 ] ), dot( m[ 2 ], m[ 2 ] ) );\n\ttransformedNormal = m *
transformedNormal;\n#endif\ntransformedNormal = normalMatrix *
transformedNormal;\n#ifdef FLIP_SIDED\n\ttransformedNormal = -
transformedNormal;\n#endif\n#ifdef USE_TANGENT\n\tvec3 transformedTangent = (
modelViewMatrix * vec4( objectTangent, 0.0 ) ).xyz;\n\t#ifdef
FLIP_SIDED\n\t\ttransformedTangent = -
transformedTangent;\n\t#endif\n#endif",displacementmap_pars_vertex:"#ifdef
USE_DISPLACEMENTMAP\n\tuniform sampler2D displacementMap;\n\tuniform float
displacementScale;\n\tuniform float
displacementBias;\n#endif",displacementmap_vertex:"#ifdef
USE_DISPLACEMENTMAP\n\ttransformed += normalize( objectNormal ) * ( texture2D(
displacementMap, vUv ).x * displacementScale + displacementBias
);\n#endif",emissivemap_fragment:"#ifdef USE_EMISSIVEMAP\n\tvec4 emissiveColor =
texture2D( emissiveMap, vUv );\n\tvec3 emissiveColor.rgb = emissiveMapTexelToLinear(
emissiveColor ).rgb;\n\tfloat totalEmissiveRadiance *=
emissiveColor.rgb;\n#endif",emissivemap_pars_fragment:"#ifdef
USE_EMISSIVEMAP\n\tuniform sampler2D
emissiveMap;\n#endif",encodings_fragment:"gl_FragColor = linearToOutputTexel(
gl_FragColor );",encodings_pars_fragment:"\nvec4 LinearToLinear( in vec4 value )
{\n\treturn value;\n}\nvec4 GammaToLinear( in vec4 value, in float gammaFactor )
{\n\treturn vec4( pow( value.rgb, vec3( gammaFactor ) ), value.a );\n}\nvec4
LinearToGamma( in vec4 value, in float gammaFactor ) {\n\treturn vec4( pow(
value.rgb, vec3( 1.0 / gammaFactor ) ), value.a );\n}\nvec4 sRGBToLinear( in vec4
value ) {\n\treturn vec4( mix( pow( value.rgb * 0.9478672986 + vec3( 0.0521327014
), vec3( 2.4 ) ), value.rgb * 0.0773993808, vec3( lessThanEqual( value.rgb, vec3(
0.04045 ) ) ) ), value.a );\n}\nvec4 LinearToSRGB( in vec4 value ) {\n\treturn
vec4( mix( pow( value.rgb, vec3( 0.41666 ) ) * 1.055 - vec3( 0.055 ), value.rgb *
12.92, vec3( lessThanEqual( value.rgb, vec3( 0.0031308 ) ) ) ), value.a );\n}\nvec4
RGBToLinear( in vec4 value ) {\n\treturn vec4( value.rgb * exp2( value.a * 255.0 -
128.0 ), 1.0 );\n}\nvec4 LinearToRGBE( in vec4 value ) {\n\tfloat maxComponent =
max( max( value.r, value.g ), value.b );\n\tfloat fExp = clamp( ceil( log2(
maxComponent ) ), -128.0, 127.0 );\n\treturn vec4( value.rgb / exp2( fExp ), ( fExp
+ 128.0 ) / 255.0 );\n}\nvec4 RGBMToLinear( in vec4 value, in float maxRange )
{\n\treturn vec4( value.rgb * value.a * maxRange, 1.0 );\n}\nvec4 LinearToRGBM( in
vec4 value, in float maxRange ) {\n\tfloat maxRGB = max( value.r, max( value.g,
value.b ) );\n\tfloat M = clamp( maxRGB / maxRange, 0.0, 1.0 );\n\tM = ceil( M *
255.0 ) / 255.0;\n\treturn vec4( value.rgb / ( M * maxRange ), M );\n}\nvec4
RGBDToLinear( in vec4 value, in float maxRange ) {\n\treturn vec4( value.rgb * ( (
maxRange / 255.0 ) / value.a ), 1.0 );\n}\nvec4 LinearToRGBD( in vec4 value, in
float maxRange ) {\n\tfloat maxRGB = max( value.r, max( value.g, value.b )
);\n\tfloat D = max( maxRange / maxRGB, 1.0 );\n\tD = clamp( floor( D ) / 255.0,
0.0, 1.0 );\n\treturn vec4( value.rgb * ( D * ( 255.0 / maxRange ) ), D
);\n}\nconst mat3 cLogLuvM = mat3( 0.2209, 0.3390, 0.4184, 0.1138, 0.6780, 0.7319,
0.0102, 0.1130, 0.2969 );\nvec4 LinearToLogLuv( in vec4 value ) {\n\tvec3 Xp_Y_XYZp
= cLogLuvM * value.rgb;\n\tXp_Y_XYZp = max( Xp_Y_XYZp, vec3( 1e-6, 1e-6, 1e-6 )
);\n\tvec4 vResult;\n\tvResult.xy = Xp_Y_XYZp.xy / Xp_Y_XYZp.z;\n\tfloat Le = 2.0 *
log2(Xp_Y_XYZp.y) + 127.0;\n\tvResult.w = fract( Le );\n\tvResult.z = ( Le - (
floor( vResult.w * 255.0 ) ) / 255.0 ) / 255.0;\n\treturn vResult;\n}\nconst mat3
cLogLuvInverseM = mat3( 6.0014, -2.7008, -1.7996, -1.3320, 3.1029, -5.7721, 0.3008,
-1.0882, 5.6268 );\nvec4 LogLuvToLinear( in vec4 value ) {\n\tfloat Le = value.z *
255.0 + value.w;\n\tvec3 Xp_Y_XYZp;\n\tXp_Y_XYZp.y = exp2( ( Le - 127.0 ) / 2.0
);\n\tXp_Y_XYZp.z = Xp_Y_XYZp.y / value.y;\n\tXp_Y_XYZp.x = value.x *
Xp_Y_XYZp.z;\n\tvec3 vRGB = cLogLuvInverseM * Xp_Y_XYZp.rgb;\n\treturn vec4( max(
vRGB, 0.0 ), 1.0 );\n}",envmap_fragment:"#ifdef USE_ENVMAP\n\t#ifdef
ENV_WORLDPOS\n\t\tvec3 cameraToFrag;\n\t\tif ( isOrthographic )
{\n\t\t\tcameraToFrag = normalize( vec3( - viewMatrix[ 0 ][ 2 ], - viewMatrix[ 1 ][

```

```
[ 2 ] ) );\n\t\t} else {\n\t\tcameraToFrag = normalize(\nvWorldPosition - cameraPosition);\n\t\t}\n\t\tvec3 worldNormal = inverseTransformDirection( normal, viewMatrix );\n\t\t#ifdef ENVMAP_MODE_REFLECTION\n\t\t\tvec3 reflectVec = reflect( cameraToFrag, worldNormal );\n\t\t#else\n\t\t\tvec3 reflectVec = refract( cameraToFrag, worldNormal, refractionRatio );\n\t\t#endif\n\t\t}else{\n\t\t\tvec3 reflectVec = vReflect;\n\t\t#endif\n\t\t#ifdef ENVMAP_TYPE_CUBE\n\t\t\tvec4 envColor = textureCube( envMap, vec3( flipEnvMap * reflectVec.x, reflectVec.yz ));\n\t\t#elif defined( ENVMAP_TYPE_CUBE_UV )\n\t\t\tvec4 envColor = textureCubeUV( envMap, reflectVec, 0.0 );\n\t\t#else\n\t\t\tvec4 envColor = vec4( 0.0 );\n\t\t#endif\n\t\t#ifndef ENVMAP_TYPE_CUBE_UV\n\t\t\tenvColor = envMapTexelToLinear( envColor );\n\t\t#endif\n\t\t#ifdef ENVMAP_BLENDING_MULTIPLY\n\t\t\toutgoingLight = mix( outgoingLight, outgoingLight * envColor.xyz, specularStrength * reflectivity );\n\t\t#elif defined( ENVMAP_BLENDING_MIX )\n\t\t\toutgoingLight = mix( outgoingLight, envColor.xyz, specularStrength * reflectivity );\n\t\t#elif defined( ENVMAP_BLENDING_ADD )\n\t\t\toutgoingLight += envColor.xyz * specularStrength * reflectivity;\n\t\t#endif\n\t\t#endif", envmap_common_pars_fragment:"#ifdef USE_ENVMAP\n\t\tuniform float envMapIntensity;\n\t\tuniform float flipEnvMap;\n\t\tuniform int maxMipLevel;\n\t\t#ifdef ENVMAP_TYPE_CUBE\n\t\t\tuniform samplerCube envMap;\n\t\t#else\n\t\t\tuniform sampler2D envMap;\n\t\t#endif\n\t\t\n\t\tendif", envmap_pars_fragment:"#ifdef USE_ENVMAP\n\t\tuniform float reflectivity;\n\t\t#if defined( USE_BUMPMAP ) || defined( USE_NORMALMAP ) || defined( PHONG )\n\t\t\t#define ENV_WORLDPOS\n\t\t\tendif\n\t\t\t#ifdef ENV_WORLDPOS\n\t\t\t\tvarying vec3 vWorldPosition;\n\t\t\tuniform float refractionRatio;\n\t\t\t#else\n\t\t\t\tvarying vec3 vReflect;\n\t\t\tendif\n\t\t\t\n\t\t\tendif", envmap_pars_vertex:"#ifdef USE_ENVMAP\n\t\t\tif defined( USE_BUMPMAP ) || defined( USE_NORMALMAP ) || defined( PHONG )\n\t\t\t\tdefine ENV_WORLDPOS\n\t\t\t\tendif\n\t\t\t\t#ifdef ENV_WORLDPOS\n\t\t\t\t\tvariating vec3 vWorldPosition;\n\t\t\t\t#else\n\t\t\t\t\tvariating vec3 vReflect;\n\t\t\t\tuniform float refractionRatio;\n\t\t\t\tendif\n\t\t\t\t\n\t\t\t\tendif", envmap_physical_pars_fragment:"#if defined( USE_ENVMAP )\n\t\t\t\t#ifdef ENVMAP_MODE_REFRACTION\n\t\t\t\t\tuniform float refractionRatio;\n\t\t\t\t\tendif\n\t\t\t\t\tvec3 getLightProbeIndirectIrradiance( const in GeometricContext geometry, const in int maxMIPLevel ) {\n\t\t\t\t\t\tvec3 worldNormal = inverseTransformDirection( geometry.normal, viewMatrix );\n\t\t\t\t\t\t#ifdef ENVMAP_TYPE_CUBE\n\t\t\t\t\t\t\tqueryVec = vec3( flipEnvMap * worldNormal.x, worldNormal.yz );\n\t\t\t\t\t\t\t#endif\n\t\t\t\t\t\t\t#ifdef TEXTURE_LOD_EXT\n\t\t\t\t\t\t\t\tvec4 envMapColor = textureCubeLodEXT( envMap, queryVec, float( maxMIPLevel ) );\n\t\t\t\t\t\t\t\telse\n\t\t\t\t\t\t\t\t\tvec4 envMapColor = textureCube( envMap, queryVec, float( maxMIPLevel ) );\n\t\t\t\t\t\t\t\tendif\n\t\t\t\t\t\t\t\tenvMapColor.rgb = envMapTexelToLinear( envMapColor ).rgb;\n\t\t\t\t\t\t\t\telif defined( ENVMAP_TYPE_CUBE_UV )\n\t\t\t\t\t\t\t\t\tvec4 envMapColor = textureCubeUV( envMap, worldNormal, 1.0 );\n\t\t\t\t\t\t\t\t\telse\n\t\t\t\t\t\t\t\t\t\tvec4 envMapColor = vec4( 0.0 );\n\t\t\t\t\t\t\t\t\tendif\n\t\t\t\t\t\t\t\treturn PI * envMapColor.rgb * envMapIntensity;\n\t\t\t\t\t}\n\t\t\t\tfloat getSpecularMIPLevel( const in float roughness, const in int maxMIPLevel ) {\n\t\t\t\t\t\tfloat maxMIPLevelScalar = float( maxMIPLevel );\n\t\t\t\t\t\tfloat sigma = PI * roughness * roughness / ( 1.0 + roughness );\n\t\t\t\t\t\tfloat desiredMIPLevel = maxMIPLevelScalar + log2( sigma );\n\t\t\t\t\t\treturn clamp( desiredMIPLevel, 0.0, maxMIPLevelScalar );\n\t\t\t\t\t}\n\t\t\t\t\n\t\t\t\tvec3 getLightProbeIndirectRadiance( const in vec3 viewDir, const in vec3 normal, const in float roughness, const in int maxMIPLevel ) {\n\t\t\t\t\t\t#ifdef ENVMAP_MODE_REFLECTION\n\t\t\t\t\t\t\tvec3 reflectVec = reflect( -viewDir, normal );\n\t\t\t\t\t\t\treflectVec = normalize( mix( reflectVec, normal, roughness * roughness ) );\n\t\t\t\t\t\t\t#else\n\t\t\t\t\t\t\tvec3 reflectVec = refract( -viewDir, normal, refractionRatio );\n\t\t\t\t\t\t\tendif\n\t\t\t\t\t\t\treflectVec = inverseTransformDirection( reflectVec, viewMatrix );\n\t\t\t\t\t\t\tfloat specularMIPLevel = getSpecularMIPLevel( roughness, maxMIPLevel );\n\t\t\t\t\t\t\t#ifdef ENVMAP_TYPE_CUBE\n\t\t\t\t\t\t\t\tqueryReflectVec = vec3( flipEnvMap * reflectVec.x, reflectVec.yz );\n\t\t\t\t\t\t\t\t#ifdef TEXTURE_LOD_EXT\n\t\t\t\t\t\t\t\t\tvec4 envMapColor = textureCubeLodEXT( envMap, queryReflectVec, specularMIPLevel );\n\t\t\t\t\t\t\t\t\telse\n\t\t\t\t\t\t\t\t\t\tvec4 envMapColor = textureCube( envMap, queryReflectVec,
```

[illegible]

[illegible]

[illegible]

```

);\nmaterial.specularRoughness = max( roughnessFactor, 0.0525
);\nmaterial.specularRoughness += geometryRoughness;\nmaterial.specularRoughness =
min( material.specularRoughness, 1.0 );\n#endif
REFLECTIVITY\n\tmaterial.specularColor = mix( vec3( MAXIMUM_SPECULAR_COEFFICIENT *
pow2( reflectivity ) ), diffuseColor.rgb, metalnessFactor
);\n#else\n\tmaterial.specularColor = mix( vec3( DEFAULT_SPECULAR_COEFFICIENT ),
diffuseColor.rgb, metalnessFactor );\n#endif\n#endif
CLEARCOAT\n\tmaterial.clearcoat = clearcoat;\n\tmaterial.clearcoatRoughness =
clearcoatRoughness;\n\t#ifdef USE_CLEARCOATMAP\n\t\tmaterial.clearcoat *=
texture2D( clearcoatMap, vUv ).x;\n\t#endif\n\t#ifdef
USE_CLEARCOAT_ROUGHNESSMAP\n\t\tmaterial.clearcoatRoughness *= texture2D(
clearcoatRoughnessMap, vUv ).y;\n\t#endif\n\tmaterial.clearcoat = saturate(
material.clearcoat );\n\tmaterial.clearcoatRoughness = max(
material.clearcoatRoughness, 0.0525 );\n\tmaterial.clearcoatRoughness +=
geometryRoughness;\n\tmaterial.clearcoatRoughness = min(
material.clearcoatRoughness, 1.0 );\n#endif\n#endif
USE_SHEEN\n\tmaterial.sheenColor =
sheen;\n#endif",lights_physical_pars_fragment:"struct PhysicalMaterial {\n\tvec3
diffuseColor;\n\tfloat specularRoughness;\n\tvec3 specularColor;\n#ifdef
CLEARCOAT\n\tfloat clearcoat;\n\tfloat clearcoatRoughness;\n#endif\n#ifdef
USE_SHEEN\n\tvec3 sheenColor;\n#endif\n};\n#define MAXIMUM_SPECULAR_COEFFICIENT
0.16\n#define DEFAULT_SPECULAR_COEFFICIENT 0.04\nfloat clearcoatDHRApprox( const in
float roughness, const in float dotNL ) {\n\treturn DEFAULT_SPECULAR_COEFFICIENT +
( 1.0 - DEFAULT_SPECULAR_COEFFICIENT ) * ( pow( 1.0 - dotNL, 5.0 ) * pow( 1.0 -
roughness, 2.0 ) );\n}\n#if NUM_RECT_AREA_LIGHTS > 0\n\tvoid
RE_Direct_RectArea_Physical( const in RectAreaLight rectAreaLight, const in
GeometricContext geometry, const in PhysicalMaterial material, inout ReflectedLight
reflectedLight ) {\n\t\tvec3 normal = geometry.normal;\n\t\tvec3 viewDir =
geometry.viewDir;\n\t\tvec3 position = geometry.position;\n\t\tvec3 lightPos =
rectAreaLight.position;\n\t\tvec3 halfWidth = rectAreaLight.halfWidth;\n\t\tvec3
halfHeight = rectAreaLight.halfHeight;\n\t\tvec3 lightColor =
rectAreaLight.color;\n\t\tfloat roughness = material.specularRoughness;\n\t\tvec3
rectCoords[ 4 ];\n\t\trectCoords[ 0 ] = lightPos + halfWidth -
halfHeight;\n\t\trectCoords[ 1 ] = lightPos - halfWidth -
halfHeight;\n\t\trectCoords[ 2 ] = lightPos - halfWidth +
halfHeight;\n\t\trectCoords[ 3 ] = lightPos + halfWidth + halfHeight;\n\t\tvec2 uv
= LTC_Uv( normal, viewDir, roughness );\n\t\tvec4 t1 = texture2D( ltc_1, uv
);\n\t\tvec4 t2 = texture2D( ltc_2, uv );\n\t\tmat3 mInv = mat3( \n\t\t\tvec3( t1.x,
0, t1.y ),\n\t\t\tvec3( \t\t0, 1, \t\t0 ),\n\t\t\tvec3( t1.z, 0, t1.w
)\n\t\t\t);\n\t\tvec3 fresnel = ( material.specularColor * t2.x + ( vec3( 1.0 ) -
material.specularColor ) * t2.y );\n\t\treflectedLight.directSpecular += lightColor
* fresnel * LTC_Evaluate( normal, viewDir, position, mInv, rectCoords
);\n\t\treflectedLight.directDiffuse += lightColor * material.diffuseColor *
LTC_Evaluate( normal, viewDir, position, mat3( 1.0 ), rectCoords
);\n\t}\n#endif\n\tvoid RE_Direct_Physical( const in IncidentLight directLight, const
in GeometricContext geometry, const in PhysicalMaterial material, inout
ReflectedLight reflectedLight ) {\n\t\tfloat dotNL = saturate( dot( geometry.normal,
directLight.direction ) );\n\t\tvec3 irradiance = dotNL *
directLight.color;\n\t\t#ifdef PHYSICALLY_CORRECT_LIGHTS\n\t\t\tirradiance *=
PI;\n\t\t#endif\n\t\t#ifdef CLEARCOAT\n\t\t\tfloat ccDotNL = saturate( dot(
geometry.clearcoatNormal, directLight.direction ) );\n\t\t\tvec3 ccIrradiance =
ccDotNL * directLight.color;\n\t\t\t#ifdef
PHYSICALLY_CORRECT_LIGHTS\n\t\t\t\tccIrradiance *= PI;\n\t\t\t#endif\n\t\t\tfloat
clearcoatDHR = material.clearcoat * clearcoatDHRApprox(
material.clearcoatRoughness, ccDotNL );\n\t\t\treflectedLight.directSpecular +=
ccIrradiance * material.clearcoat * BRDF_Specular_GGX( directLight,
geometry.viewDir, geometry.clearcoatNormal, vec3( DEFAULT_SPECULAR_COEFFICIENT ),
material.clearcoatRoughness );\n\t\t#else\n\t\t\tfloat clearcoatDHR =

```

```
0.0;\n\t#endif\n\tif(USE_SHEEN)\n\t\treflectedLight.directSpecular += ( 1.0 - clearcoatDHR ) * irradiance * BRDF_Specular_Sheen(\n\t\t\tmaterial.specularRoughness,\n\t\t\t\tdirectLight.direction,\n\t\t\t\tgeometry,\n\t\t\t\tmaterial.sheenColor);\n\t\telse\n\t\t\treflectedLight.directSpecular += ( 1.0 - clearcoatDHR ) * irradiance * BRDF_Specular_GGX(directLight, geometry.viewDir, geometry.normal, material.specularColor, material.specularRoughness);\n\t#endif\n\treflectedLight.directDiffuse += ( 1.0 - clearcoatDHR ) * irradiance * BRDF_Diffuse_Lambert( material.diffuseColor );\n}\n\nvoid RE_IndirectDiffuse_Physical( const in vec3 irradiance, const in GeometricContext geometry, const in PhysicalMaterial material, inout ReflectedLight reflectedLight ) {\n\treflectedLight.indirectDiffuse += irradiance * BRDF_Diffuse_Lambert( material.diffuseColor );\n}\n\nvoid RE_IndirectSpecular_Physical( const in vec3 radiance, const in vec3 irradiance, const in vec3 clearcoatRadiance, const in GeometricContext geometry, const in PhysicalMaterial material, inout ReflectedLight reflectedLight) {\n\t#ifdef CLEARCOAT\n\t\tfloat ccDotNV = saturate( dot( geometry.clearcoatNormal, geometry.viewDir ) );\n\t\treflectedLight.indirectSpecular += clearcoatRadiance * material.clearcoat * BRDF_Specular_GGX_Environment( geometry.viewDir, geometry.clearcoatNormal, vec3( DEFAULT_SPECULAR_COEFFICIENT ), material.clearcoatRoughness );\n\t\tfloat ccDotNL = ccDotNV;\n\t\tfloat clearcoatDHR = material.clearcoat * clearcoatDHRApprox( material.clearcoatRoughness, ccDotNL );\n\t\telse\n\t\t\tfloat clearcoatDHR = 0.0;\n\t\tendif\n\t\tfloat clearcoatInv = 1.0 - clearcoatDHR;\n\t\tvec3 singleScattering = vec3( 0.0 );\n\t\tvec3 multiScattering = vec3( 0.0 );\n\t\tvec3 cosineWeightedIrradiance = irradiance * RECIPROCAL_PI;\n\t\tBRDF_Specular_Multiscattering_Environment( geometry, material.specularColor, material.specularRoughness, singleScattering, multiScattering );\n\t\tvec3 diffuse = material.diffuseColor * ( 1.0 - ( singleScattering + multiScattering ) );\n\t\treflectedLight.indirectSpecular += clearcoatInv * radiance * singleScattering;\n\t\treflectedLight.indirectSpecular += multiScattering * cosineWeightedIrradiance;\n\t\treflectedLight.indirectDiffuse += diffuse * cosineWeightedIrradiance;\n}\n\n#define RE_Direct\n\t\tRE_Direct_Physical\n#define RE_Direct_RectArea\n\t\tRE_Direct_RectArea_Physical\n#define RE_IndirectDiffuse\n\t\tRE_IndirectDiffuse_Physical\n#define RE_IndirectSpecular\n\t\tRE_IndirectSpecular_Physical\n\nfloat computeSpecularOcclusion( const in float dotNV, const in float ambientOcclusion, const in float roughness ) {\n\treturn saturate( pow( dotNV + ambientOcclusion, exp2( - 16.0 * roughness - 1.0 ) ) - 1.0 + ambientOcclusion );\n}\n\nlights_fragment_begin:\n\tGeometricContext geometry;\n\tgeometry.position = vViewPosition;\n\tgeometry.normal = normal;\n\tgeometry.viewDir = ( isOrthographic ) ? vec3( 0, 0, 1 ) : normalize( vViewPosition );\n\n#ifdef CLEARCOAT\n\tgeometry.clearcoatNormal = clearcoatNormal;\n#endif\n\nIncidentLight directLight;\n\nif ( NUM_POINT_LIGHTS > 0 ) && defined( RE_Direct )\n\tPointLight pointLight;\n\tif defined( USE_SHADOWMAP ) && NUM_POINT_LIGHT_SHADOWS > 0\n\t\tPointLightShadow pointLightShadow;\n\tendif\n\t#pragma unroll_loop_start\n\tfor ( int i = 0; i < NUM_POINT_LIGHTS; i ++ )\n\t{\n\t\tpointLight = pointLights[ i ];\n\t\tgetPointDirectLightIrradiance( pointLight, geometry, directLight );\n\t\tif defined( USE_SHADOWMAP ) && ( UNROLLED_LOOP_INDEX < NUM_POINT_LIGHT_SHADOWS )\n\t\t\tpointLightShadow = pointLightShadows[ i ];\n\t\t\tif ( directLight.color == all( bvec2( directLight.visible, receiveShadow ) ) )\n\t\t\t\tgetPointShadow( pointShadowMap[ i ], pointLightShadow.shadowMapSize, pointLightShadow.shadowBias, pointLightShadow.shadowRadius, vPointShadowCoord[ i ], pointLightShadow.shadowCameraNear, pointLightShadow.shadowCameraFar ) : 1.0;\n\t\tendif\n\t\tRE_Direct( directLight, geometry, material, reflectedLight );\n\t}\n\t#pragma unroll_loop_end\n\nif ( NUM_SPOT_LIGHTS > 0 ) && defined( RE_Direct )\n\tSpotLight spotLight;\n\tif defined( USE_SHADOWMAP ) &&
```

<https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js>

[illegible]

```

normalize( vTBN * mapN );\n\t#else\n\t\ttnormal = perturbNormal2Arb( -vViewPosition,
normal, mapN, faceDirection );\n\t#endif\n#elif defined( USE_BUMPMAP )\n\ttnormal =
perturbNormalArb( -vViewPosition, normal, dHdxy_fwd(), faceDirection
);\n#endif",normalmap_pars_fragment:"#ifdef USE_NORMALMAP\n\t\tuniform sampler2D
normalMap;\n\t\tuniform vec2 normalScale;\n#endif\n#if defined
OBJECTSPACE_NORMALMAP\n\t\tuniform mat3 normalMatrix;\n#endif\n#if ! defined (
USE_TANGENT ) && ( defined ( TANGENTSPACE_NORMALMAP ) || defined (
USE_CLEARCOAT_NORMALMAP ) )\n\t\tvec3 perturbNormal2Arb( vec3 eye_pos, vec3
surf_norm, vec3 mapN, float faceDirection ) {\n\t\t\tvec3 q0 = vec3( dFdx( eye_pos.x
), dFdx( eye_pos.y ), dFdx( eye_pos.z ) );\n\t\t\tvec3 q1 = vec3( dFdy( eye_pos.x ),
dFdy( eye_pos.y ), dFdy( eye_pos.z ) );\n\t\t\tvec2 st0 = dFdx( vUv.st );\n\t\t\tvec2
st1 = dFdy( vUv.st );\n\t\t\tvec3 N = surf_norm;\n\t\t\tvec3 q1perp = cross( q1, N
);\n\t\t\tvec3 q0perp = cross( N, q0 );\n\t\t\tvec3 T = q1perp * st0.x + q0perp *
st1.x;\n\t\t\tvec3 B = q1perp * st0.y + q0perp * st1.y;\n\t\t\tfloat det = max( dot( T,
T ), dot( B, B ) );\n\t\t\tfloat scale = ( det == 0.0 ) ? 0.0 : faceDirection *
inversesqrt( det );\n\t\t\treturn normalize( T * ( mapN.x * scale ) + B * ( mapN.y *
scale ) + N * mapN.z );\n\t\t}\n#endif",clearcoat_normal_fragment_begin:"#ifdef
CLEARCOAT\n\t\tvec3 clearcoatNormal =
geometryNormal;\n#endif",clearcoat_normal_fragment_maps:"#ifdef
USE_CLEARCOAT_NORMALMAP\n\t\tvec3 clearcoatMapN = texture2D( clearcoatNormalMap, vUv
).xyz * 2.0 - 1.0;\n\t\tclearcoatMapN.xy *= clearcoatNormalScale;\n\t\t#endif
USE_TANGENT\n\t\t\tclearcoatNormal = normalize( vTBN * clearcoatMapN
);\n\t\t#else\n\t\t\tclearcoatNormal = perturbNormal2Arb( - vViewPosition,
clearcoatNormal, clearcoatMapN, faceDirection
);\n\t\t#endif\n#endif",clearcoat_pars_fragment:"#ifdef USE_CLEARCOATMAP\n\t\tuniform
sampler2D clearcoatMap;\n#endif\n#if defined USE_CLEARCOAT_ROUGHNESSMAP\n\t\tuniform
sampler2D clearcoatRoughnessMap;\n#endif\n#if defined USE_CLEARCOAT_NORMALMAP\n\t\tuniform
sampler2D clearcoatNormalMap;\n\t\tuniform vec2
clearcoatNormalScale;\n#endif",packing:"vec3 packNormalToRGB( const in vec3 normal
) {\n\t\treturn normalize( normal ) * 0.5 + 0.5;\n}\nvec3 unpackRGBToNormal( const in
vec3 rgb ) {\n\t\treturn 2.0 * rgb.xyz - 1.0;\n}\nconst float PackUpscale = 256. /
255.;const float UnpackDownscale = 255. / 256.;\nconst vec3 PackFactors = vec3(
256. * 256. * 256., 256. * 256., 256. );\nconst vec4 UnpackFactors =
UnpackDownscale / vec4( PackFactors, 1. );\nconst float ShiftRight8 = 1. /
256.;\nvec4 packDepthToRGBA( const in float v ) {\n\t\tvec4 r = vec4( fract( v *
PackFactors ), v );\n\t\ttr.yzw -= r.xyz * ShiftRight8;\n\t\treturn r *
PackUpscale;\n}\nfloat unpackRGBAToDepth( const in vec4 v ) {\n\t\treturn dot( v,
UnpackFactors );\n}\nvec4 pack2HalfToRGBA( vec2 v ) {\n\t\tvec4 r = vec4( v.x, fract(
v.x * 255.0 ), v.y, fract( v.y * 255.0 ));\n\t\treturn vec4( r.x - r.y / 255.0, r.y,
r.z - r.w / 255.0, r.w);\n}\nvec2 unpackRGBATo2Half( vec4 v ) {\n\t\treturn vec2( v.x
+ ( v.y / 255.0 ), v.z + ( v.w / 255.0 ) );\n}\nfloat viewZToOrthographicDepth(
const in float viewZ, const in float near, const in float far ) {\n\t\treturn ( viewZ
+ near ) / ( near - far );\n}\nfloat orthographicDepthToViewZ( const in float
linearClipZ, const in float near, const in float far ) {\n\t\treturn linearClipZ * (
near - far ) - near;\n}\nfloat viewZToPerspectiveDepth( const in float viewZ, const
in float near, const in float far ) {\n\t\treturn ( ( near + viewZ ) * far ) / ( ( far
- near ) * viewZ );\n}\nfloat perspectiveDepthToViewZ( const in float invClipZ,
const in float near, const in float far ) {\n\t\treturn ( near * far ) / ( ( far -
near ) * invClipZ - far );\n}",premultiplied_alpha_fragment:"#ifdef
PREMULTIPLIED_ALPHA\n\t\tgl_FragColor.rgb *=
gl_FragColor.a;\n#endif",project_vertex:"vec4 mvPosition = vec4( transformed, 1.0
);\n#if defined USE_INSTANCING\n\t\tmvPosition = instanceMatrix *
mvPosition;\n#endif\n\t\tmvPosition = modelViewMatrix * mvPosition;\n\t\tngl_Position =
projectionMatrix * mvPosition;",dithering_fragment:"#ifdef
DITHERING\n\t\tgl_FragColor.rgb = dithering( gl_FragColor.rgb
);\n#endif",dithering_pars_fragment:"#ifdef DITHERING\n\t\tvec3 dithering( vec3 color
) {\n\t\t\tfloat grid_position = rand( gl_FragCoord.xy );\n\t\t\tvec3 dither_shift_RGB
= vec3( 0.25 / 255.0, -0.25 / 255.0, 0.25 / 255.0 );\n\t\t\ttdither_shift_RGB = mix(

```

```

2.0 * dither_shift_RGB, -2.0 * dither_shift_RGB, grid_position );\n\t\treturn color
+ dither_shift_RGB;\n\t}\n#endif",roughnessmap_fragment:"float roughnessFactor =
roughness;\n#ifdef USE_ROUGHNESSMAP\n\tvec4 texelRoughness = texture2D(
roughnessMap, vUv );\n\t\troughnessFactor *=
texelRoughness.g;\n#endif",roughnessmap_pars_fragment:"#ifdef
USE_ROUGHNESSMAP\n\t\tuniform sampler2D
roughnessMap;\n#endif",shadowmap_pars_fragment:"#ifdef USE_SHADOWMAP\n\t\tif
NUM_DIR_LIGHT_SHADOWS > 0\n\t\t\tuniform sampler2D directionalShadowMap[
NUM_DIR_LIGHT_SHADOWS ];\n\t\t\tvarying vec4 vDirectionalShadowCoord[
NUM_DIR_LIGHT_SHADOWS ];\n\t\t\tstruct DirectionalLightShadow {\n\t\t\t\tfloat
shadowBias;\n\t\t\t\tfloat shadowNormalBias;\n\t\t\t\tfloat shadowRadius;\n\t\t\t\tvec2
shadowMapSize;\n\t\t\t};\n\t\t\tuniform DirectionalLightShadow directionalLightShadows[
NUM_DIR_LIGHT_SHADOWS ];\n\t\t\t#ifdef NUM_SPOT_LIGHT_SHADOWS > 0\n\t\t\t\tuniform
sampler2D spotShadowMap[ NUM_SPOT_LIGHT_SHADOWS ];\n\t\t\t\tvarying vec4
vSpotShadowCoord[ NUM_SPOT_LIGHT_SHADOWS ];\n\t\t\t\tstruct SpotLightShadow
{\n\t\t\t\t\tfloat shadowBias;\n\t\t\t\t\tfloat shadowNormalBias;\n\t\t\t\t\tfloat
shadowRadius;\n\t\t\t\t\tvec2 shadowMapSize;\n\t\t\t\t};\n\t\t\t\tuniform SpotLightShadow
spotLightShadows[ NUM_SPOT_LIGHT_SHADOWS ];\n\t\t\t\t#ifdef NUM_POINT_LIGHT_SHADOWS > 0\n\t\t\t\t\tuniform sampler2D pointShadowMap[
NUM_POINT_LIGHT_SHADOWS ];\n\t\t\t\t\tvarying vec4 vPointShadowCoord[
NUM_POINT_LIGHT_SHADOWS ];\n\t\t\t\t\tstruct PointLightShadow {\n\t\t\t\t\t\tfloat
shadowBias;\n\t\t\t\t\t\tfloat shadowNormalBias;\n\t\t\t\t\t\tfloat shadowRadius;\n\t\t\t\t\t\tvec2
shadowMapSize;\n\t\t\t\t\t\tfloat shadowCameraNear;\n\t\t\t\t\t\tfloat
shadowCameraFar;\n\t\t\t\t\t};\n\t\t\t\t\tuniform PointLightShadow pointLightShadows[
NUM_POINT_LIGHT_SHADOWS ];\n\t\t\t\t\t#ifdef\n\t\t\t\t\t\tfloat texture2DCompare( sampler2D depths,
vec2 uv, float compare ) {\n\t\t\t\t\t\t\treturn step( compare, unpackRGBAToDepth( texture2D(
depths, uv ) ) );\n\t\t\t\t\t}\n\t\t\t\t\tvec2 texture2DDistribution( sampler2D shadow, vec2 uv )
{\n\t\t\t\t\t\t\treturn unpackRGBATo2Half( texture2D( shadow, uv ) );\n\t\t\t\t\t}\n\t\t\t\t\tfloat VSMShadow
(sampler2D shadow, vec2 uv, float compare ){\n\t\t\t\t\t\t\tfloat occlusion = 1.0;\n\t\t\t\t\t\t\tvec2
distribution = texture2DDistribution( shadow, uv );\n\t\t\t\t\t\t\tfloat hard_shadow = step(
compare , distribution.x );\n\t\t\t\t\t\t\tif (hard_shadow != 1.0 ) {\n\t\t\t\t\t\t\t\tfloat distance =
compare - distribution.x ;\n\t\t\t\t\t\t\t\tfloat variance = max( 0.00000, distribution.y *
distribution.y );\n\t\t\t\t\t\t\t\tfloat softness_probability = variance / (variance +
distance * distance );\n\t\t\t\t\t\t\t\tfloat softness_probability = clamp( ( softness_probability -
0.3 ) / ( 0.95 - 0.3 ), 0.0, 1.0 );\n\t\t\t\t\t\t\t\tfloat occlusion = clamp( max( hard_shadow,
softness_probability ), 0.0, 1.0 );\n\t\t\t\t\t\t\t}\n\t\t\t\t\t\t\treturn occlusion;\n\t\t\t\t\t}\n\t\t\t\t\tfloat
getShadow( sampler2D shadowMap, vec2 shadowMapSize, float shadowBias, float
shadowRadius, vec4 shadowCoord ) {\n\t\t\t\t\t\tfloat shadow = 1.0;\n\t\t\t\t\t\tfloat shadowCoord.xyz /=
shadowCoord.w;\n\t\t\t\t\t\tfloat shadowCoord.z += shadowBias;\n\t\t\t\t\t\tbvec4 inFrustumVec = bvec4 (
shadowCoord.x >= 0.0, shadowCoord.x <= 1.0, shadowCoord.y >= 0.0, shadowCoord.y <=
1.0 );\n\t\t\t\t\t\tbool inFrustum = all( inFrustumVec );\n\t\t\t\t\t\tbvec2 frustumTestVec =
bvec2( inFrustum, shadowCoord.z <= 1.0 );\n\t\t\t\t\t\tbool frustumTest = all(
frustumTestVec );\n\t\t\t\t\t\tif ( frustumTest ) {\n\t\t\t\t\t\t\t#ifdef SHADOWMAP_TYPE_PCF
)\n\t\t\t\t\t\t\t\tvec2 texelSize = vec2( 1.0 ) / shadowMapSize;\n\t\t\t\t\t\t\t\tfloat dx0 = -
texelSize.x * shadowRadius;\n\t\t\t\t\t\t\t\tfloat dy0 = - texelSize.y *
shadowRadius;\n\t\t\t\t\t\t\t\tfloat dx1 = + texelSize.x * shadowRadius;\n\t\t\t\t\t\t\t\tfloat dy1 = +
texelSize.y * shadowRadius;\n\t\t\t\t\t\t\t\tfloat dx2 = dx0 / 2.0;\n\t\t\t\t\t\t\t\tfloat dy2 = dy0 /
2.0;\n\t\t\t\t\t\t\t\tfloat dx3 = dx1 / 2.0;\n\t\t\t\t\t\t\t\tfloat dy3 = dy1 / 2.0;\n\t\t\t\t\t\t\t\tfloat shadow =
(\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy + vec2( dx0, dy0 ),
shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy + vec2( 0.0,
dy0 ), shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy +
vec2( dx1, dy0 ), shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap,
shadowCoord.xy + vec2( dx2, dy2 ), shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare(
shadowMap, shadowCoord.xy + vec2( 0.0, dy2 ), shadowCoord.z )
+ \n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy + vec2( dx3, dy2 ),
shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy + vec2( dx0,
0.0 ), shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap, shadowCoord.xy +
vec2( dx2, 0.0 ), shadowCoord.z ) +\n\t\t\t\t\t\t\t\t\ttexture2DCompare( shadowMap,

```

[illegible]


```

NUM_SPOT_LIGHT_SHADOWS; i ++ ) {\n\t\tspotLight = spotLightShadows[ i
];\n\t\tshadow *= receiveShadow ? getShadow( spotShadowMap[ i ],
spotLight.shadowMapSize, spotLight.shadowBias, spotLight.shadowRadius,
vSpotShadowCoord[ i ] ) : 1.0;\n\t}\n\t#pragma unroll_loop_end\n\t#endif\n\tif
NUM_POINT_LIGHT_SHADOWS > 0\n\tPointLightShadow pointLight;\n\t#pragma
unroll_loop_start\n\tfor ( int i = 0; i < NUM_POINT_LIGHT_SHADOWS; i ++ )
{\n\t\tpointLight = pointLightShadows[ i ];\n\t\tshadow *= receiveShadow ?
getPointShadow( pointShadowMap[ i ], pointLight.shadowMapSize,
pointLight.shadowBias, pointLight.shadowRadius, vPointShadowCoord[ i ],
pointLight.shadowCameraNear, pointLight.shadowCameraFar ) : 1.0;\n\t}\n\t#pragma
unroll_loop_end\n\t#endif\n\t#endif\n\treturn shadow;\n}",skinbase_vertex:"#ifdef
USE_SKINNING\n\tmat4 boneMatX = getBoneMatrix( skinIndex.x );\n\tmat4 boneMatY =
getBoneMatrix( skinIndex.y );\n\tmat4 boneMatZ = getBoneMatrix( skinIndex.z
);\n\tmat4 boneMatW = getBoneMatrix( skinIndex.w
);\n#endif",skinning_pars_vertex:"#ifdef USE_SKINNING\n\tuniform mat4
bindMatrix;\n\tuniform mat4 bindMatrixInverse;\n\t#ifdef BONE_TEXTURE\n\t\tuniform
highp sampler2D boneTexture;\n\t\tuniform int boneTextureSize;\n\t\tmat4
getBoneMatrix( const in float i ) {\n\t\t\tfloat j = i * 4.0;\n\t\t\tfloat x = mod(
j, float( boneTextureSize ) );\n\t\t\tfloat y = floor( j / float( boneTextureSize )
);\n\t\t\tfloat dx = 1.0 / float( boneTextureSize );\n\t\t\tfloat dy = 1.0 / float(
boneTextureSize );\n\t\t\tty = dy * ( y + 0.5 );\n\t\t\tvec4 v1 = texture2D(
boneTexture, vec2( dx * ( x + 0.5 ), y ) );\n\t\t\tvec4 v2 = texture2D(
boneTexture, vec2( dx * ( x + 1.5 ), y ) );\n\t\t\tvec4 v3 = texture2D(
boneTexture, vec2( dx * ( x + 2.5 ), y ) );\n\t\t\tvec4 v4 = texture2D(
boneTexture, vec2( dx * ( x + 3.5 ), y ) );\n\t\t\tmat4 bone = mat4( v1, v2, v3, v4
);\n\t\t\treturn bone;\n\t\t}\n\t}else\n\t\tuniform mat4 boneMatrices[ MAX_BONES
];\n\t\tmat4 getBoneMatrix( const in float i ) {\n\t\t\tmat4 bone = boneMatrices[
int(i) ];\n\t\t\treturn bone;\n\t\t}\n\t#endif\n#endif",skinning_vertex:"#ifdef
USE_SKINNING\n\tvec4 skinVertex = bindMatrix * vec4( transformed, 1.0 );\n\tvec4
skinned = vec4( 0.0 );\n\t\tskinned += boneMatX * skinVertex *
skinWeight.x;\n\t\tskinned += boneMatY * skinVertex * skinWeight.y;\n\t\tskinned +=
boneMatZ * skinVertex * skinWeight.z;\n\t\tskinned += boneMatW * skinVertex *
skinWeight.w;\n\t\ttransformed = ( bindMatrixInverse * skinned
).xyz;\n#endif",skinnormal_vertex:"#ifdef USE_SKINNING\n\tmat4 skinMatrix = mat4(
0.0 );\n\t\tskinMatrix += skinWeight.x * boneMatX;\n\t\tskinMatrix += skinWeight.y *
boneMatY;\n\t\tskinMatrix += skinWeight.z * boneMatZ;\n\t\tskinMatrix += skinWeight.w *
boneMatW;\n\t\tskinMatrix = bindMatrixInverse * skinMatrix *
bindMatrix;\n\t\tobjectNormal = vec4( skinMatrix * vec4( objectNormal, 0.0 )
).xyz;\n\t\t#ifdef USE_TANGENT\n\t\t\tobjectTangent = vec4( skinMatrix * vec4(
objectTangent, 0.0 ) ).xyz;\n\t\t#endif\n#endif",specularmap_fragment:"float
specularStrength;\n\t#ifdef USE_SPECULARMAP\n\t\tvec4 texelSpecular = texture2D(
specularMap, vUv );\n\t\tspecularStrength =
texelSpecular.r;\n\t}else\n\t\tspecularStrength =
1.0;\n#endif",specularmap_pars_fragment:"#ifdef USE_SPECULARMAP\n\tuniform
sampler2D specularMap;\n\t#endif",tonemapping_fragment:"#if defined( TONE_MAPPING
)\n\tgl_FragColor.rgb = toneMapping( gl_FragColor.rgb
);\n#endif",tonemapping_pars_fragment:"#ifndef saturate\n#define saturate(a) clamp(
a, 0.0, 1.0 )\n#endif\n\tuniform float toneMappingExposure;\n\tvec3 LinearToneMapping(
vec3 color ) {\n\t\treturn toneMappingExposure * color;\n\t}\n\tvec3 ReinhardToneMapping(
vec3 color ) {\n\t\tcolor *= toneMappingExposure;\n\t\treturn saturate( color / ( vec3(
1.0 ) + color ) );\n\t}\n\tvec3 OptimizedCineonToneMapping( vec3 color ) {\n\t\tcolor *=
toneMappingExposure;\n\t\tcolor = max( vec3( 0.0 ), color - 0.004 );\n\t\treturn pow( (
color * ( 6.2 * color + 0.5 ) ) / ( color * ( 6.2 * color + 1.7 ) + 0.06 ), vec3(
2.2 ) );\n\t}\n\tvec3 RRTAndODTfit( vec3 v ) {\n\t\tvec3 a = v * ( v + 0.0245786 ) -
0.000090537;\n\t\tvec3 b = v * ( 0.983729 * v + 0.4329510 ) + 0.238081;\n\t\treturn a /
b;\n\t}\n\tvec3 ACESFilmicToneMapping( vec3 color ) {\n\t\tconst mat3 ACESInputMat =
mat3( \n\t\t\tvec3( 0.59719, 0.07600, 0.02840 ),\n\t\t\tvec3( 0.35458, 0.90834, 0.13383
),\n\t\t\tvec3( 0.04823, 0.01566, 0.83777 )\n\t\t);\n\t\tconst mat3 ACESOutputMat =

```

```

mat3(\n\t\tvec3(\t1.60475, -0.10208, -0.00327 ),\t\tvec3( -0.53108,\t1.10813,
-0.07276 ),\n\t\tvec3( -0.07367, -0.00605,\t1.07602 )\n\t);\n\tcolor *=
toneMappingExposure / 0.6;\n\tcolor = ACESInputMat * color;\n\tcolor =
RRTAndODTFit( color );\n\tcolor = ACESOutputMat * color;\n\treturn saturate( color
);\n}\nvec3 CustomToneMapping( vec3 color ) { return color;
}",transmissionmap_fragment:"#ifdef USE_TRANSMISSIONMAP\n\ttotalTransmission *=
texture2D( transmissionMap, vUv ).r;\n#endif",transmissionmap_pars_fragment:"#ifdef
USE_TRANSMISSIONMAP\n\tuniform sampler2D
transmissionMap;\n#endif",uv_pars_fragment:"#if ( defined( USE_UV ) && ! defined(
UVS_VERTEX_ONLY ) )\n\tvarying vec2 vUv;\n#endif",uv_pars_vertex:"#ifdef
USE_UV\n\t#ifdef UVS_VERTEX_ONLY\n\t\tvec2 vUv;\n\t#else\n\t\tvarying vec2
vUv;\n\t#endif\n\tuniform mat3 uvTransform;\n#endif",uv_vertex:"#ifdef
USE_UV\n\tvUv = ( uvTransform * vec3( uv, 1 ) ).xy;\n#endif",uv2_pars_fragment:"#if
defined( USE_LIGHTMAP ) || defined( USE_AOMAP )\n\tvarying vec2
vUv2;\n#endif",uv2_pars_vertex:"#if defined( USE_LIGHTMAP ) || defined( USE_AOMAP
)\n\tattribute vec2 uv2;\n\tvarying vec2 vUv2;\n\tuniform mat3
uv2Transform;\n#endif",uv2_vertex:"#if defined( USE_LIGHTMAP ) || defined(
USE_AOMAP )\n\tvUv2 = ( uv2Transform * vec3( uv2, 1 )
).xy;\n#endif",worldpos_vertex:"#if defined( USE_ENVMAP ) || defined( DISTANCE ) ||
defined( USE_SHADOWMAP )\n\tvec4 worldPosition = vec4( transformed, 1.0
);\n\t#ifdef USE_INSTANCING\n\t\tworldPosition = instanceMatrix *
worldPosition;\n\t#endif\n\tworldPosition = modelMatrix *
worldPosition;\n#endif",background_frag:"uniform sampler2D t2D;\nvarying vec2
vUv;\nvoid main() {\n\tvec4 texColor = texture2D( t2D, vUv );\n\tgl_FragColor =
mapTexelToLinear( texColor );\n\t#include <tonemapping_fragment>\n\t#include
<encodings_fragment>\n}",background_vert:"varying vec2 vUv;\nuniform mat3
uvTransform;\nvoid main() {\n\tvUv = ( uvTransform * vec3( uv, 1 )
).xy;\n\tgl_Position = vec4( position.xy, 1.0, 1.0 );\n}",cube_frag:"#include
<envmap_common_pars_fragment>\nuniform float opacity;\nvarying vec3
vWorldDirection;\n#include <cube_uv_reflection_fragment>\nvoid main() {\n\tvec3
vReflect = vWorldDirection;\n\t#include <envmap_fragment>\n\tgl_FragColor =
envColor;\n\tgl_FragColor.a *= opacity;\n\t#include
<tonemapping_fragment>\n\t#include <encodings_fragment>\n}",cube_vert:"varying vec3
vWorldDirection;\n#include <common>\nvoid main() {\n\tvWorldDirection =
transformDirection( position, modelMatrix );\n\t#include <begin_vertex>\n\t#include
<project_vertex>\n\tgl_Position.z = gl_Position.w;\n}",depth_frag:"#if
DEPTH_PACKING == 3200\n\tuniform float opacity;\n#endif\n#include
<common>\n#include <packing>\n#include <uv_pars_fragment>\n#include
<map_pars_fragment>\n#include <alphamap_pars_fragment>\n#include
<logdepthbuf_pars_fragment>\n#include <clipping_planes_pars_fragment>\nvarying vec2
vHighPrecisionZW;\nvoid main() {\n\t#include <clipping_planes_fragment>\n\tvec4
diffuseColor = vec4( 1.0 );\n\t#if DEPTH_PACKING == 3200\n\t\ttdiffuseColor.a =
opacity;\n\t#endif\n\t#include <map_fragment>\n\t#include
<alphamap_fragment>\n\t#include <alphatest_fragment>\n\t#include
<logdepthbuf_fragment>\n\tfloat fragCoordZ = 0.5 * vHighPrecisionZW[0] /
vHighPrecisionZW[1] + 0.5;\n\t#if DEPTH_PACKING == 3200\n\t\tgl_FragColor = vec4(
vec3( 1.0 - fragCoordZ ), opacity );\n\t#elif DEPTH_PACKING ==
3201\n\t\tgl_FragColor = packDepthToRGBA( fragCoordZ
);\n\t#endif\n}",depth_vert:"#include <common>\n#include <uv_pars_vertex>\n#include
<displacementmap_pars_vertex>\n#include <morphtarget_pars_vertex>\n#include
<skinning_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvarying vec2 vHighPrecisionZW;\nvoid main()
{\n\t#include <uv_vertex>\n\t#include <skinbase_vertex>\n\t#ifdef
USE_DISPLACEMENTMAP\n\t\t#include <beginnormal_vertex>\n\t\t#include
<morphnormal_vertex>\n\t\t#include <skinnormal_vertex>\n\t#endif\n\t#include
<begin_vertex>\n\t#include <morphtarget_vertex>\n\t#include
<skinning_vertex>\n\t#include <displacementmap_vertex>\n\t#include
<project_vertex>\n\t#include <logdepthbuf_vertex>\n\t#include

```



```

<logdepthbuf_pars_fragment>\n#include <clipping_planes_pars_fragment>\nvoid main()
{\n\t#include <clipping_planes_fragment>\n\tvec4 diffuseColor = vec4( diffuse,
opacity );\n\t#include <logdepthbuf_fragment>\n\t#include
<map_fragment>\n\t#include <color_fragment>\n\t#include
<alphamap_fragment>\n\t#include <alphatest_fragment>\n\t#include
<specularmap_fragment>\n\tReflectedLight reflectedLight = ReflectedLight( vec3( 0.0
), vec3( 0.0 ), vec3( 0.0 ), vec3( 0.0 ) );\n\t#ifdef USE_LIGHTMAP\n\t\tvec4
lightMapTexel= texture2D( lightMap, vUv2 );\n\t\treflectedLight.indirectDiffuse +=
lightMapTexelToLinear( lightMapTexel ).rgb *
lightMapIntensity;\n\t#else\n\t\treflectedLight.indirectDiffuse += vec3( 1.0
);\n\t#endif\n\t#include <aomap_fragment>\n\treflectedLight.indirectDiffuse *=
diffuseColor.rgb;\n\tvec3 outgoingLight =
reflectedLight.indirectDiffuse;\n\t#include <envmap_fragment>\n\tgl_FragColor =
vec4( outgoingLight, diffuseColor.a );\n\t#include
<tonemapping_fragment>\n\t#include <encodings_fragment>\n\t#include
<fog_fragment>\n\t#include <premultiplied_alpha_fragment>\n\t#include
<dithering_fragment>\n}\n",meshbasic_vert:"\n#include <common>\n#include
<uv_pars_vertex>\n#include <uv2_pars_vertex>\n#include
<envmap_pars_vertex>\n#include <color_pars_vertex>\n#include
<fog_pars_vertex>\n#include <morphtarget_pars_vertex>\n#include
<skinning_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvoid main() {\n\t#include <uv_vertex>\n\t#include
<uv2_vertex>\n\t#include <color_vertex>\n\t#include <skinbase_vertex>\n\t#ifdef
USE_ENVMAP\n\t\t#include <beginnormal_vertex>\n\t\t#include
<morphnormal_vertex>\n\t\t#include <skinnormal_vertex>\n\t\t#include
<defaultnormal_vertex>\n\t\t#endif\n\t\t#include <begin_vertex>\n\t\t#include
<morphtarget_vertex>\n\t\t#include <skinning_vertex>\n\t\t#include
<project_vertex>\n\t\t#include <logdepthbuf_vertex>\n\t\t#include
<worldpos_vertex>\n\t\t#include <clipping_planes_vertex>\n\t\t#include
<envmap_vertex>\n\t\t#include <fog_vertex>\n}\n",meshlambert_frag:"uniform vec3
diffuse;\nnuniform vec3 emissive;\nnuniform float opacity;\nvarying vec3
vLightFront;\nvarying vec3 vIndirectFront;\n#ifdef DOUBLE_SIDED\n\tvarying vec3
vLightBack;\n\tvarying vec3 vIndirectBack;\n#endif\n#include <common>\n#include
<packing>\n#include <dithering_pars_fragment>\n#include
<color_pars_fragment>\n#include <uv_pars_fragment>\n#include
<uv2_pars_fragment>\n#include <map_pars_fragment>\n#include
<alphamap_pars_fragment>\n#include <aomap_pars_fragment>\n#include
<lightmap_pars_fragment>\n#include <emissivemap_pars_fragment>\n#include
<envmap_common_pars_fragment>\n#include <envmap_pars_fragment>\n#include
<cube_uv_reflection_fragment>\n#include <bsdfs>\n#include
<lights_pars_begin>\n#include <fog_pars_fragment>\n#include
<shadowmap_pars_fragment>\n#include <shadowmask_pars_fragment>\n#include
<specularmap_pars_fragment>\n#include <logdepthbuf_pars_fragment>\n#include
<clipping_planes_pars_fragment>\nvoid main() {\n\t#include
<clipping_planes_fragment>\n\tvec4 diffuseColor = vec4( diffuse, opacity
);\n\tReflectedLight reflectedLight = ReflectedLight( vec3( 0.0 ), vec3( 0.0 ),
vec3( 0.0 ), vec3( 0.0 ) );\n\tvec3 totalEmissiveRadiance = emissive;\n\t#include
<logdepthbuf_fragment>\n\t#include <map_fragment>\n\t#include
<color_fragment>\n\t#include <alphamap_fragment>\n\t#include
<alphatest_fragment>\n\t#include <specularmap_fragment>\n\t#include
<emissivemap_fragment>\n\t#ifdef DOUBLE_SIDED\n\t\treflectedLight.indirectDiffuse
+= ( gl_FrontFacing ) ? vIndirectFront :
vIndirectBack;\n\t#else\n\t\treflectedLight.indirectDiffuse +=
vIndirectFront;\n\t#endif\n\t#include
<lightmap_fragment>\n\treflectedLight.indirectDiffuse *= BRDF_Diffuse_Lambert(
diffuseColor.rgb );\n\t#ifdef DOUBLE_SIDED\n\t\treflectedLight.directDiffuse = (
gl_FrontFacing ) ? vLightFront :
vLightBack;\n\t#else\n\t\treflectedLight.directDiffuse =

```

```

vLightFront;\n\t#endifif\n\t\treflectedLight.directDiffuse *= BRDF_Diffuse_Lambert(
diffuseColor.rgb ) * getShadowMask();\n\t\t#include <aomap_fragment>\n\t\tvec3
outgoingLight = reflectedLight.directDiffuse + reflectedLight.indirectDiffuse +
totalEmissiveRadiance;\n\t\t#include <envmap_fragment>\n\t\tgl_FragColor = vec4(
outgoingLight, diffuseColor.a );\n\t\t#include <tonemapping_fragment>\n\t\t#include
<encodings_fragment>\n\t\t#include <fog_fragment>\n\t\t#include
<premultiplied_alpha_fragment>\n\t\t#include
<dithering_fragment>\n\t\t",meshlambert_vert:"#define LAMBERT\nvarying vec3
vLightFront;\nvarying vec3 vIndirectFront;\n#ifdef DOUBLE_SIDED\n\t\tvarying vec3
vLightBack;\n\t\tvarying vec3 vIndirectBack;\n#endif\n#include <common>\n#include
<uv_pars_vertex>\n#include <uv2_pars_vertex>\n#include
<envmap_pars_vertex>\n#include <bsdfs>\n#include <lights_pars_begin>\n#include
<color_pars_vertex>\n#include <fog_pars_vertex>\n#include
<morphtarget_pars_vertex>\n#include <skinning_pars_vertex>\n#include
<shadowmap_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvoid main() {\n\t\t#include <uv_vertex>\n\t\t#include
<uv2_vertex>\n\t\t#include <color_vertex>\n\t\t#include
<beginnormal_vertex>\n\t\t#include <morphnormal_vertex>\n\t\t#include
<skinbase_vertex>\n\t\t#include <skinnormal_vertex>\n\t\t#include
<defaultnormal_vertex>\n\t\t#include <begin_vertex>\n\t\t#include
<morphtarget_vertex>\n\t\t#include <skinning_vertex>\n\t\t#include
<project_vertex>\n\t\t#include <logdepthbuf_vertex>\n\t\t#include
<clipping_planes_vertex>\n\t\t#include <worldpos_vertex>\n\t\t#include
<envmap_vertex>\n\t\t#include <lights_lambert_vertex>\n\t\t#include
<shadowmap_vertex>\n\t\t#include <fog_vertex>\n\t\t",meshmatcap_frag:"#define
MATCAP\nuniform vec3 diffuse;\nuniform float opacity;\nuniform sampler2D
matcap;\nvarying vec3 vViewPosition;\n#ifdef FLAT_SHADED\n\t\tvarying vec3
vNormal;\n#endif\n#include <common>\n#include <dithering_pars_fragment>\n#include
<color_pars_fragment>\n#include <uv_pars_fragment>\n#include
<map_pars_fragment>\n#include <alphamap_pars_fragment>\n#include
<fog_pars_fragment>\n#include <bumpmap_pars_fragment>\n#include
<normalmap_pars_fragment>\n#include <logdepthbuf_pars_fragment>\n#include
<clipping_planes_pars_fragment>\nvoid main() {\n\t\t#include
<clipping_planes_fragment>\n\t\tvec4 diffuseColor = vec4( diffuse, opacity
);\n\t\t#include <logdepthbuf_fragment>\n\t\t#include <map_fragment>\n\t\t#include
<color_fragment>\n\t\t#include <alphamap_fragment>\n\t\t#include
<alphatest_fragment>\n\t\t#include <normal_fragment_begin>\n\t\t#include
<normal_fragment_maps>\n\t\tvec3 viewDir = normalize( vViewPosition );\n\t\tvec3 x =
normalize( vec3( viewDir.z, 0.0, - viewDir.x ) );\n\t\tvec3 y = cross( viewDir, x
);\n\t\tvec2 uv = vec2( dot( x, normal ), dot( y, normal ) ) * 0.495 + 0.5;\n\t\t#ifdef
USE_MATCAP\n\t\t\tvec4 matcapColor = texture2D( matcap, uv );\n\t\t\tmatcapColor =
matcapTexelToLinear( matcapColor );\n\t\t#else\n\t\t\tvec4 matcapColor = vec4( 1.0
);\n\t\t#endif\n\t\tvec3 outgoingLight = diffuseColor.rgb *
matcapColor.rgb;\n\t\tgl_FragColor = vec4( outgoingLight, diffuseColor.a
);\n\t\t#include <tonemapping_fragment>\n\t\t#include <encodings_fragment>\n\t\t#include
<fog_fragment>\n\t\t#include <premultiplied_alpha_fragment>\n\t\t#include
<dithering_fragment>\n\t\t",meshmatcap_vert:"#define MATCAP\nvarying vec3
vViewPosition;\n#ifdef FLAT_SHADED\n\t\tvarying vec3 vNormal;\n#endif\n#include
<common>\n#include <uv_pars_vertex>\n#include <color_pars_vertex>\n#include
<displacementmap_pars_vertex>\n#include <fog_pars_vertex>\n#include
<morphtarget_pars_vertex>\n#include <skinning_pars_vertex>\n#include
<logdepthbuf_pars_vertex>\n#include <clipping_planes_pars_vertex>\nvoid main()
{\n\t\t#include <uv_vertex>\n\t\t#include <color_vertex>\n\t\t#include
<beginnormal_vertex>\n\t\t#include <morphnormal_vertex>\n\t\t#include
<skinbase_vertex>\n\t\t#include <skinnormal_vertex>\n\t\t#include
<defaultnormal_vertex>\n\t\t#ifdef FLAT_SHADED\n\t\t\tvec3 vNormal = normalize(
transformedNormal );\n\t\t#endif\n\t\t#include <begin_vertex>\n\t\t#include
<morphtarget_vertex>\n\t\t#include <skinning_vertex>\n\t\t#include

```

```

<displacementmap_vertex>\n\t#include <project_vertex>\n\t#include
<logdepthbuf_vertex>\n\t#include <clipping_planes_vertex>\n\t#include
<fog_vertex>\n\tvViewPosition = - mvPosition.xyz;\n}",meshtoon_frag:"#define
TOON\nuniform vec3 diffuse;\nuniform vec3 emissive;\nuniform float
opacity;\n#include <common>\n#include <packing>\n#include
<dithering_pars_fragment>\n#include <color_pars_fragment>\n#include
<uv_pars_fragment>\n#include <uv2_pars_fragment>\n#include
<map_pars_fragment>\n#include <alphamap_pars_fragment>\n#include
<aomap_pars_fragment>\n#include <lightmap_pars_fragment>\n#include
<emissivemap_pars_fragment>\n#include <gradientmap_pars_fragment>\n#include
<fog_pars_fragment>\n#include <bsdfs>\n#include <lights_pars_begin>\n#include
<lights_toon_pars_fragment>\n#include <shadowmap_pars_fragment>\n#include
<bumpmap_pars_fragment>\n#include <normalmap_pars_fragment>\n#include
<logdepthbuf_pars_fragment>\n#include <clipping_planes_pars_fragment>\nvoid main()
{\n\t#include <clipping_planes_fragment>\n\tvec4 diffuseColor = vec4( diffuse,
opacity );\n\tReflectedLight reflectedLight = ReflectedLight( vec3( 0.0 ), vec3(
0.0 ), vec3( 0.0 ), vec3( 0.0 ) );\n\tvec3 totalEmissiveRadiance =
emissive;\n\t#include <logdepthbuf_fragment>\n\t#include <map_fragment>\n\t#include
<color_fragment>\n\t#include <alphamap_fragment>\n\t#include
<alphatest_fragment>\n\t#include <normal_fragment_begin>\n\t#include
<normal_fragment_maps>\n\t#include <emissivemap_fragment>\n\t#include
<lights_toon_fragment>\n\t#include <lights_fragment_begin>\n\t#include
<lights_fragment_maps>\n\t#include <lights_fragment_end>\n\t#include
<aomap_fragment>\n\tvec3 outgoingLight = reflectedLight.directDiffuse +
reflectedLight.indirectDiffuse + totalEmissiveRadiance;\n\tgl_FragColor = vec4(
outgoingLight, diffuseColor.a );\n\t#include <tonemapping_fragment>\n\t#include
<encodings_fragment>\n\t#include <fog_fragment>\n\t#include
<premultiplied_alpha_fragment>\n\t#include
<dithering_fragment>\n}",meshtoon_vert:"#define TOON\nvarying vec3
vViewPosition;\n#ifdef FLAT_SHADED\n\tvarying vec3 vNormal;\n#endif\n#include
<common>\n#include <uv_pars_vertex>\n#include <uv2_pars_vertex>\n#include
<displacementmap_pars_vertex>\n#include <color_pars_vertex>\n#include
<fog_pars_vertex>\n#include <morphtarget_pars_vertex>\n#include
<skinning_pars_vertex>\n#include <shadowmap_pars_vertex>\n#include
<logdepthbuf_pars_vertex>\n#include <clipping_planes_pars_vertex>\nvoid main()
{\n\t#include <uv_vertex>\n\t#include <uv2_vertex>\n\t#include
<color_vertex>\n\t#include <beginnormal_vertex>\n\t#include
<morphnormal_vertex>\n\t#include <skinbase_vertex>\n\t#include
<skinnormal_vertex>\n\t#include <defaultnormal_vertex>\n#ifdef
FLAT_SHADED\n\tvNormal = normalize( transformedNormal );\n#endif\n\t#include
<begin_vertex>\n\t#include <morphtarget_vertex>\n\t#include
<skinning_vertex>\n\t#include <displacementmap_vertex>\n\t#include
<project_vertex>\n\t#include <logdepthbuf_vertex>\n\t#include
<clipping_planes_vertex>\n\tvViewPosition = - mvPosition.xyz;\n\t#include
<worldpos_vertex>\n\t#include <shadowmap_vertex>\n\t#include
<fog_vertex>\n}",meshphong_frag:"#define PHONG\nuniform vec3 diffuse;\nuniform vec3
emissive;\nuniform vec3 specular;\nuniform float shininess;\nuniform float
opacity;\n#include <common>\n#include <packing>\n#include
<dithering_pars_fragment>\n#include <color_pars_fragment>\n#include
<uv_pars_fragment>\n#include <uv2_pars_fragment>\n#include
<map_pars_fragment>\n#include <alphamap_pars_fragment>\n#include
<aomap_pars_fragment>\n#include <lightmap_pars_fragment>\n#include
<emissivemap_pars_fragment>\n#include <envmap_common_pars_fragment>\n#include
<envmap_pars_fragment>\n#include <cube_uv_reflection_fragment>\n#include
<fog_pars_fragment>\n#include <bsdfs>\n#include <lights_pars_begin>\n#include
<lights_phong_pars_fragment>\n#include <shadowmap_pars_fragment>\n#include
<bumpmap_pars_fragment>\n#include <normalmap_pars_fragment>\n#include
<specularmap_pars_fragment>\n#include <logdepthbuf_pars_fragment>\n#include

```

```

<clipping_planes_pars_fragment>\nvoid main() {\n\t#include
<clipping_planes_fragment>\n\tvec4 diffuseColor = vec4( diffuse, opacity
);\n\tReflectedLight reflectedLight = ReflectedLight( vec3( 0.0 ), vec3( 0.0 ),
vec3( 0.0 ), vec3( 0.0 ) );\n\tvec3 totalEmissiveRadiance = emissive;\n\t#include
<logdepthbuf_fragment>\n\t#include <map_fragment>\n\t#include
<color_fragment>\n\t#include <alphamap_fragment>\n\t#include
<alphatest_fragment>\n\t#include <specularmap_fragment>\n\t#include
<normal_fragment_begin>\n\t#include <normal_fragment_maps>\n\t#include
<emissivemap_fragment>\n\t#include <lights_phong_fragment>\n\t#include
<lights_fragment_begin>\n\t#include <lights_fragment_maps>\n\t#include
<lights_fragment_end>\n\t#include <aomap_fragment>\n\tvec3 outgoingLight =
reflectedLight.directDiffuse + reflectedLight.indirectDiffuse +
reflectedLight.directSpecular + reflectedLight.indirectSpecular +
totalEmissiveRadiance;\n\t#include <envmap_fragment>\n\tgl_FragColor = vec4(
outgoingLight, diffuseColor.a );\n\t#include <tonemapping_fragment>\n\t#include
<encodings_fragment>\n\t#include <fog_fragment>\n\t#include
<premultiplied_alpha_fragment>\n\t#include
<dithering_fragment>\n}",meshphong_vert:"#define PHONG\nvarying vec3
vViewPosition;\n#ifdef FLAT_SHADED\n\tvarying vec3 vNormal;\n#endif\n#include
<common>\n#include <uv_pars_vertex>\n#include <uv2_pars_vertex>\n#include
<displacementmap_pars_vertex>\n#include <envmap_pars_vertex>\n#include
<color_pars_vertex>\n#include <fog_pars_vertex>\n#include
<morphtarget_pars_vertex>\n#include <skinning_pars_vertex>\n#include
<shadowmap_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvoid main() {\n\t#include <uv_vertex>\n\t#include
<uv2_vertex>\n\t#include <color_vertex>\n\t#include
<beginnormal_vertex>\n\t#include <morphnormal_vertex>\n\t#include
<skinbase_vertex>\n\t#include <skinnormal_vertex>\n\t#include
<defaultnormal_vertex>\n#ifdef FLAT_SHADED\n\tvNormal = normalize(
transformedNormal );\n#endif\n\t#include <begin_vertex>\n\t#include
<morphtarget_vertex>\n\t#include <skinning_vertex>\n\t#include
<displacementmap_vertex>\n\t#include <project_vertex>\n\t#include
<logdepthbuf_vertex>\n\t#include <clipping_planes_vertex>\n\tvViewPosition = -
mvPosition.xyz;\n\t#include <worldpos_vertex>\n\t#include
<envmap_vertex>\n\t#include <shadowmap_vertex>\n\t#include
<fog_vertex>\n}",meshphysical_frag:"#define STANDARD\n#ifdef PHYSICAL\n\tdefine
REFLECTIVITY\n\tdefine CLEARCOAT\n\tdefine TRANSMISSION\n#endif\nuniform vec3
diffuse;\nuniform vec3 emissive;\nuniform float roughness;\nuniform float
metalness;\nuniform float opacity;\n#ifdef TRANSMISSION\n\tuniform float
transmission;\n#endif\n#ifdef REFLECTIVITY\n\tuniform float
reflectivity;\n#endif\n#ifdef CLEARCOAT\n\tuniform float clearcoat;\n\tuniform
float clearcoatRoughness;\n#endif\n#ifdef USE_SHEEN\n\tuniform vec3
sheen;\n#endif\nvarying vec3 vViewPosition;\n#ifdef FLAT_SHADED\n\tvarying vec3
vNormal;\n\t#ifdef USE_TANGENT\n\t\tvarying vec3 vTangent;\n\t\tvarying vec3
vBitangent;\n\t#endif\n#endif\n#include <common>\n#include <packing>\n#include
<dithering_pars_fragment>\n#include <color_fragment>\n#include
<uv_pars_fragment>\n#include <uv2_pars_fragment>\n#include
<map_pars_fragment>\n#include <alphamap_pars_fragment>\n#include
<aomap_pars_fragment>\n#include <lightmap_pars_fragment>\n#include
<emissivemap_pars_fragment>\n#include <transmissionmap_pars_fragment>\n#include
<bsdfs>\n#include <cube_uv_reflection_fragment>\n#include
<envmap_common_pars_fragment>\n#include <envmap_physical_pars_fragment>\n#include
<fog_pars_fragment>\n#include <lights_pars_begin>\n#include
<lights_physical_pars_fragment>\n#include <shadowmap_pars_fragment>\n#include
<bumpmap_pars_fragment>\n#include <normalmap_pars_fragment>\n#include
<clearcoat_pars_fragment>\n#include <roughnessmap_pars_fragment>\n#include
<metalnessmap_pars_fragment>\n#include <logdepthbuf_pars_fragment>\n#include
<clipping_planes_pars_fragment>\nvoid main() {\n\t#include

```


<https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js>

```

<morphtarget_pars_vertex>\n#include <skinning_pars_vertex>\n#include
<logdepthbuf_pars_vertex>\n#include <clipping_planes_pars_vertex>\nvoid main()
{\n\t#include <uv_vertex>\n\t#include <beginnormal_vertex>\n\t#include
<morphnormal_vertex>\n\t#include <skinbase_vertex>\n\t#include
<skinnormal_vertex>\n\t#include <defaultnormal_vertex>\n#ifdef
FLAT_SHADED\n\t\tvNormal = normalize( transformedNormal );\n\t\tif
USE_TANGENT\n\t\t\tvTangent = normalize( transformedTangent );\n\t\t\tvBitangent =
normalize( cross( vNormal, vTangent ) * tangent.w );\n\t\tendif\n\tendif\n\t#include
<begin_vertex>\n\t#include <morphtarget_vertex>\n\t#include
<skinning_vertex>\n\t#include <displacementmap_vertex>\n\t#include
<project_vertex>\n\t#include <logdepthbuf_vertex>\n\t#include
<clipping_planes_vertex>\n\tif defined( FLAT_SHADED ) || defined( USE_BUMPMAP ) ||
defined( TANGENTSPACE_NORMALMAP )\n\t\tvViewPosition = -
mvPosition.xyz;\n\t\tendif\n\t},points_frag:"uniform vec3 diffuse;\nunifrom float
opacity;\n#include <common>\n#include <color_pars_fragment>\n#include
<map_particle_pars_fragment>\n#include <fog_pars_fragment>\n#include
<logdepthbuf_pars_fragment>\n#include <clipping_planes_pars_fragment>\nvoid main()
{\n\t#include <clipping_planes_fragment>\n\t\tvec3 outgoingLight = vec3( 0.0
);\n\t\tvec4 diffuseColor = vec4( diffuse, opacity );\n\t\t#include
<logdepthbuf_fragment>\n\t\t#include <map_particle_fragment>\n\t\t#include
<color_fragment>\n\t\t#include <alphatest_fragment>\n\t\toutgoingLight =
diffuseColor.rgb;\n\t\tgl_FragColor = vec4( outgoingLight, diffuseColor.a
);\n\t\t#include <tonemapping_fragment>\n\t\t#include <encodings_fragment>\n\t\t#include
<fog_fragment>\n\t\t#include <premultiplied_alpha_fragment>\n\t},points_vert:"uniform
float size;\nunifrom float scale;\n#include <common>\n#include
<color_pars_vertex>\n#include <fog_pars_vertex>\n#include
<morphtarget_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvoid main() {\n\t#include <color_vertex>\n\t#include
<begin_vertex>\n\t\t#include <morphtarget_vertex>\n\t\t#include
<project_vertex>\n\t\t\tgl_PointSize = size;\n\t\t\tif USE_SIZEATTENUATION\n\t\t\t\tbool
isPerspective = isPerspectiveMatrix( projectionMatrix );\n\t\t\t\tif ( isPerspective )
gl_PointSize *= ( scale / - mvPosition.z );\n\t\t\t\tendif\n\t\t\t#include
<logdepthbuf_vertex>\n\t\t\t#include <clipping_planes_vertex>\n\t\t\t#include
<worldpos_vertex>\n\t\t\t#include <fog_vertex>\n\t\t},shadow_frag:"uniform vec3
color;\nunifrom float opacity;\n#include <common>\n#include <packing>\n#include
<fog_pars_fragment>\n#include <bsdfs>\n#include <lights_pars_begin>\n#include
<shadowmap_pars_fragment>\n#include <shadowmask_pars_fragment>\nvoid main()
{\n\t\tgl_FragColor = vec4( color, opacity * ( 1.0 - getShadowMask() ) );\n\t\t#include
<tonemapping_fragment>\n\t\t#include <encodings_fragment>\n\t\t#include
<fog_fragment>\n\t\t},shadow_vert:"\n#include <common>\n#include
<fog_pars_vertex>\n#include <shadowmap_pars_vertex>\nvoid main() {\n\t#include
<begin_vertex>\n\t\t#include <project_vertex>\n\t\t#include
<worldpos_vertex>\n\t\t#include <beginnormal_vertex>\n\t\t#include
<morphnormal_vertex>\n\t\t#include <skinbase_vertex>\n\t\t#include
<skinnormal_vertex>\n\t\t#include <defaultnormal_vertex>\n\t\t#include
<shadowmap_vertex>\n\t\t#include <fog_vertex>\n\t\t},sprite_frag:"uniform vec3
diffuse;\nunifrom float opacity;\n#include <common>\n#include
<uv_pars_fragment>\n#include <map_pars_fragment>\n#include
<alphamap_pars_fragment>\n#include <fog_pars_fragment>\n#include
<logdepthbuf_pars_fragment>\n#include <clipping_planes_pars_fragment>\nvoid main()
{\n\t#include <clipping_planes_fragment>\n\t\tvec3 outgoingLight = vec3( 0.0
);\n\t\tvec4 diffuseColor = vec4( diffuse, opacity );\n\t\t#include
<logdepthbuf_fragment>\n\t\t#include <map_fragment>\n\t\t#include
<alphamap_fragment>\n\t\t#include <alphatest_fragment>\n\t\toutgoingLight =
diffuseColor.rgb;\n\t\tgl_FragColor = vec4( outgoingLight, diffuseColor.a
);\n\t\t#include <tonemapping_fragment>\n\t\t#include <encodings_fragment>\n\t\t#include
<fog_fragment>\n\t\t},sprite_vert:"uniform float rotation;\nunifrom vec2
center;\n#include <common>\n#include <uv_pars_vertex>\n#include

```

```

<fog_pars_vertex>\n#include <logdepthbuf_pars_vertex>\n#include
<clipping_planes_pars_vertex>\nvoid main() {\n\t#include <uv_vertex>\n\tvec4
mvPosition = modelViewMatrix * vec4( 0.0, 0.0, 0.0, 1.0 );\n\tvec2
scale;\n\ttscale.x = length( vec3( modelMatrix[ 0 ].x, modelMatrix[ 0 ].y,
modelMatrix[ 0 ].z ) );\n\ttscale.y = length( vec3( modelMatrix[ 1 ].x, modelMatrix[
1 ].y, modelMatrix[ 1 ].z ) );\n\t#ifdef USE_SIZEATTENUATION\n\t\tbool
isPerspective = isPerspectiveMatrix( projectionMatrix );\n\t\tif ( isPerspective )
scale *= - mvPosition.z;\n\t#endif\n\tvec2 alignedPosition = ( position.xy - (
center - vec2( 0.5 ) ) ) * scale;\n\tvec2 rotatedPosition;\n\trotatedPosition.x =
cos( rotation ) * alignedPosition.x - sin( rotation ) *
alignedPosition.y;\n\trotatedPosition.y = sin( rotation ) * alignedPosition.x +
cos( rotation ) * alignedPosition.y;\n\tmvPosition.xy +=
rotatedPosition;\n\tgl_Position = projectionMatrix * mvPosition;\n\t#include
<logdepthbuf_vertex>\n\t#include <clipping_planes_vertex>\n\t#include
<fog_vertex>\n}"} , ui={common:{diffuse:{value:new tn(15658734)},opacity:
{value:1},map:{value:null},uvTransform:{value:new yt},uv2Transform:{value:new
yt},alphaMap:{value:null}},specularmap:{specularMap:{value:null}},envmap:{envMap:
{value:null},flipEnvMap:{value:-1},reflectivity:{value:1},refractionRatio:
{value:.98},maxMipLevel:{value:0}},aomap:{aoMap:{value:null},aoMapIntensity:
{value:1}},lightmap:{lightMap:{value:null},lightMapIntensity:
{value:1}},emissivemap:{emissiveMap:{value:null}},bumpmap:{bumpMap:
{value:null},bumpScale:{value:1}},normalmap:{normalMap:{value:null},normalScale:
{value:new vt(1,1)}},displacementmap:{displacementMap:
{value:null},displacementScale:{value:1},displacementBias:{value:0}},roughnessmap:
{roughnessMap:{value:null}},metalnessmap:{metalnessMap:{value:null}},gradientmap:
{gradientMap:{value:null}},fog:{fogDensity:{value:25e-5},fogNear:{value:1},fogFar:
{value:2e3},fogColor:{value:new tn(16777215)}},lights:{ambientLightColor:{value:
[]},lightProbe:{value:[]},directionalLights:{value:[],properties:{direction:
{}},color:{}},directionalLightShadows:{value:[],properties:{shadowBias:
{}},shadowNormalBias:{},shadowRadius:{},shadowMapSize:{}},directionalShadowMap:
{value:[],directionalShadowMatrix:{value:[],spotLights:{value:[],properties:
{color:{},position:{},direction:{},distance:{},coneCos:{},penumbraCos:{},decay:
{}}},spotLightShadows:{value:[],properties:{shadowBias:{},shadowNormalBias:
{}},shadowRadius:{},shadowMapSize:{}},spotShadowMap:{value:[],spotShadowMatrix:
{value:[],pointLights:{value:[],properties:{color:{},position:{},decay:
{}},distance:{}}},pointLightShadows:{value:[],properties:{shadowBias:
{}},shadowNormalBias:{},shadowRadius:{},shadowMapSize:{},shadowCameraNear:
{}},shadowCameraFar:{}}},pointShadowMap:{value:[],pointShadowMatrix:{value:
[]},hemisphereLights:{value:[],properties:{direction:{},skyColor:{},groundColor:
{}}},rectAreaLights:{value:[],properties:{color:{},position:{},width:{},height:
{}}},lrc_1:{value:null},lrc_2:{value:null}},points:{diffuse:{value:new
tn(15658734)},opacity:{value:1},size:{value:1},scale:{value:1},map:
{value:null},alphaMap:{value:null},uvTransform:{value:new yt}},sprite:{diffuse:
{value:new tn(15658734)},opacity:{value:1},center:{value:new vt(.5,.5)},rotation:
{value:0},map:{value:null},alphaMap:{value:null},uvTransform:{value:new yt}}},di=
{basic:
{uniforms:Yn([ui.common,ui.specularmap,ui.envmap,ui.aomap,ui.lightmap,ui.fog]),vert
exShader:hi.meshbasic_vert,fragmentShader:hi.meshbasic_frag},lambert:
{uniforms:Yn([ui.common,ui.specularmap,ui.envmap,ui.aomap,ui.lightmap,ui.emissivema
p,ui.fog,ui.lights,{emissive:{value:new
tn(0)}}]),vertexShader:hi.meshlambert_vert,fragmentShader:hi.meshlambert_frag},phon
g:
{uniforms:Yn([ui.common,ui.specularmap,ui.envmap,ui.aomap,ui.lightmap,ui.emissivema
p,ui.bumpmap,ui.normalmap,ui.displacementmap,ui.fog,ui.lights,{emissive:{value:new
tn(0)},specular:{value:new tn(1118481)},shininess:
{value:30}}]),vertexShader:hi.meshphong_vert,fragmentShader:hi.meshphong_frag},stan
dard:
{uniforms:Yn([ui.common,ui.envmap,ui.aomap,ui.lightmap,ui.emissivemap,ui.bumpmap,ui

```

```

.normalmap,ui.displacementmap,ui.roughnessmap,ui.metalnessmap,ui.fog,ui.lights,
{emissive:{value:new tn(0)},roughness:{value:1},metalness:
{value:0},envMapIntensity:
{value:1}}]),vertexShader:hi.meshphysical_vert,fragmentShader:hi.meshphysical_frag}
,toon:
{uniforms:Yn([ui.common,ui.aomap,ui.lightmap,ui.emissivemap,ui.bumpmap,ui.normalmap
,ui.displacementmap,ui.gradientmap,ui.fog,ui.lights,{emissive:{value:new
tn(0)}}]),vertexShader:hi.meshtoon_vert,fragmentShader:hi.meshtoon_frag},matcap:
{uniforms:Yn([ui.common,ui.bumpmap,ui.normalmap,ui.displacementmap,ui.fog,{matcap:
{value:null}}]),vertexShader:hi.meshmatcap_vert,fragmentShader:hi.meshmatcap_frag},
points:
{uniforms:Yn([ui.points,ui.fog]),vertexShader:hi.points_vert,fragmentShader:hi.poin
ts_frag},dashed:{uniforms:Yn([ui.common,ui.fog,{scale:{value:1},dashSize:
{value:1}},totalSize:
{value:2}}]),vertexShader:hi.linedashed_vert,fragmentShader:hi.linedashed_frag},dep
th:
{uniforms:Yn([ui.common,ui.displacementmap]),vertexShader:hi.depth_vert,fragmentSha
der:hi.depth_frag},normal:
{uniforms:Yn([ui.common,ui.bumpmap,ui.normalmap,ui.displacementmap,{opacity:
{value:1}}]),vertexShader:hi.normal_vert,fragmentShader:hi.normal_frag},sprite:
{uniforms:Yn([ui.sprite,ui.fog]),vertexShader:hi.sprite_vert,fragmentShader:hi.spri
te_frag},background:{uniforms:{uvTransform:{value:new yt},t2D:
{value:null}},vertexShader:hi.background_vert,fragmentShader:hi.background_frag},cu
be:{uniforms:Yn([ui.envmap,{opacity:
{value:1}}]),vertexShader:hi.cube_vert,fragmentShader:hi.cube_frag},equirect:
{uniforms:{tEquirect:
{value:null}},vertexShader:hi.equirect_vert,fragmentShader:hi.equirect_frag},distan
ceRGBA:{uniforms:Yn([ui.common,ui.displacementmap,{referencePosition:{value:new
Lt},nearDistance:{value:1},farDistance:
{value:1e3}}]),vertexShader:hi.distanceRGBA_vert,fragmentShader:hi.distanceRGBA_fra
g},shadow:{uniforms:Yn([ui.lights,ui.fog,{color:{value:new tn(0)},opacity:
{value:1}}]),vertexShader:hi.shadow_vert,fragmentShader:hi.shadow_frag}};function
pi(t,e,n,i,r){const s=new tn(0);let a,o,c=0,h=null,u=0,d=null;function p(t,e)
{n.buffers.color.setClear(t.r,t.g,t.b,e,r)}return{getClearColor:function(){return
s},setClearColor:function(t,e=1){s.set(t),c=e,p(s,c)},getClearAlpha:function()
{return c},setClearAlpha:function(t){c=t,p(s,c)},render:function(n,r,m,f){let
g=!0===r.isScene?r.background:null;g&&g.isTexture&&(g=e.get(g));const
v=t.xr,y=v.getSession&&v.getSession();y&&"additive"===y.environmentBlendMode&&
(g=null),null===g?p(s,c):g&&g.isColor&&(p(g,1),f=!0),
(t.autoClear||f)&&t.clear(t.autoClearColor,t.autoClearDepth,t.autoClearStencil),g&&
(g.isCubeTexture||g.mapping===1)?(void 0===o&&(o=new Wn(new qn(1,1,1),new
Jn({name:"BackgroundCubeMaterial",uniforms:Xn(di.cube.uniforms),vertexShader:di.cub
e.vertexShader,fragmentShader:di.cube.fragmentShader,side:1,depthTest:!1,depthWrite
:!1,fog:!1})),o.geometry.deleteAttribute("normal"),o.geometry.deleteAttribute("uv")
,o.onBeforeRender=function(t,e,n)
{this.matrixWorld.copyPosition(n.matrixWorld)},Object.defineProperty(o.material,"en
vMap",{get:function(){return
this.uniforms.envMap.value}}),i.update(o)),o.material.uniforms.envMap.value=g,o.mat
erial.uniforms.flipEnvMap.value=g.isCubeTexture&&g._needsFlipEnvMap?
-1:1,h===g&&u===g.version&&d===t.toneMapping||
(o.material.needsUpdate=!0,h=g,u=g.version,d=t.toneMapping),n.unshift(o,o.geometry,
o.material,0,0,null)):g&&g.isTexture&&(void 0===a&&(a=new Wn(new ci(2,2),new
Jn({name:"BackgroundMaterial",uniforms:Xn(di.background.uniforms),vertexShader:di.b
ackground.vertexShader,fragmentShader:di.background.fragmentShader,side:0,depthTest
:!1,depthWrite:!1,fog:!1})),a.geometry.deleteAttribute("normal"),Object.definePrope
rty(a.material,"map",{get:function(){return
this.uniforms.t2D.value}}),i.update(a)),a.material.uniforms.t2D.value=g,!0===g.matr
ixAutoUpdate&&g.updateMatrix(),a.material.uniforms.uvTransform.value.copy(g.matrix)

```

```

,h===g&&u===g.version&&d===t.toneMapping||
(a.material.needsUpdate=!0,h=g,u=g.version,d=t.toneMapping),n.unshift(a,a.geometry,
a.material,0,0,null))}}function mi(t,e,n,i){const
r=t.getParameter(34921),s=i.isWebGL2?
null:e.get("OES_vertex_array_object"),a=i.isWebGL2||null!==s,o={},l=d(null);let
c=1;function h(e){return i.isWebGL2?
t.bindVertexArray(e):s.bindVertexArrayOES(e)}function u(e){return i.isWebGL2?
t.deleteVertexArray(e):s.deleteVertexArrayOES(e)}function d(t){const e=[],n=[],i=
[];for(let
t=0;t<r;t++)e[t]=0,n[t]=0,i[t]=0;return{geometry:null,program:null,wireframe:!1,new
Attributes:e,enabledAttributes:n,attributeDivisors:i,object:t,attributes:
{},index:null}}function p(){const t=c.newAttributes;for(let
e=0,n=t.length;e<n;e++)t[e]=0}function m(t){f(t,0)}function f(n,r){const
s=c.newAttributes,a=c.enabledAttributes,o=c.attributeDivisors;if(s[n]=1,0===a[n]&&
(t.enableVertexAttribArray(n),a[n]=1),o[n]!==r){(i.isWebGL2?
t.e.get("ANGLE_instanced_arrays"))
[i.isWebGL2?"vertexAttribDivisor":"vertexAttribDivisorANGLE"](n,r),o[n]=r}}function
g(){const e=c.newAttributes,n=c.enabledAttributes;for(let
i=0,r=n.length;i<r;i++)n[i]!==e[i]&&(t.disableVertexAttribArray(i),n[i]=0)}function
v(e,n,r,s,a,o){!0!==i.isWebGL2||5124!==r&&5125!==r?
t.vertexAttribPointer(e,n,r,s,a,o):t.vertexAttribIPointer(e,n,r,a,o)}function y()
{x(),c!==l&&(c=1,h(c.object))}function x()
{!l.geometry=null,!l.program=null,!l.wireframe=!1}return{setup:function(r,l,u,y,x){let
_=!1;if(a){const e=function(e,n,r){const a=!0===r.wireframe;let l=o[e.id];void
0===l&&(l={},o[e.id]=l);let c=l[n.id];void 0===c&&(c={},l[n.id]=c);let h=c[a];void
0===h&&(h=d(i.isWebGL2?
t.createVertexArray():s.createVertexArrayOES()),c[a]=h);return h}(y,u,l);c!==e&&
(c=e,h(c.object)),_=function(t,e){const n=c.attributes,i=t.attributes;let
r=0;for(const t in i){const e=n[t],s=i[t];if(void
0===e)return!0;if(e.attribute!==s)return!0;if(e.data!==s.data)return!0;r++}return
c.attributesNum!==r||c.index!==e}(y,x),_&&function(t,e){const n=
{},i=t.attributes;let r=0;for(const t in i){const e=i[t],s=
{};s.attribute=e,e.data&&
(s.data=e.data),n[t]=s,r++}c.attributes=n,c.attributesNum=r,c.index=e}
(y,x)}else{const
t=!0===l.wireframe;c.geometry===y.id&&c.program===u.id&&c.wireframe===t||
(c.geometry=y.id,c.program=u.id,c.wireframe=t,_=!0)}!0===r.isInstancedMesh&&
(_=!0),null!==x&&n.update(x,34963),_&&(!function(r,s,a,o){if(!1===i.isWebGL2&&
(r.isInstancedMesh||o.isInstancedBufferGeometry)&&null===e.get("ANGLE_instanced_arr
ays"))return;p();const
l=o.attributes,c=a.getAttributes(),h=s.defaultAttributeValues;for(const e in c)
{const i=c[e];if(i>=0){const s=l[e];if(void 0!==s){const
e=s.normalized,r=s.itemSize,a=n.get(s);if(void 0===a)continue;const
l=a.buffer,c=a.type,h=a.bytesPerElement;if(s.isInterleavedBufferAttribute){const
n=s.data,a=n.Stride,u=s.offset;n&&n.isInstancedInterleavedBuffer?
(f(i,n.meshPerAttribute),void 0===o._maxInstanceCount&&
(o._maxInstanceCount=n.meshPerAttribute*n.count)):m(i),t.bindBuffer(34962,l),v(i,r,
c,e,a*h,u*h)}else s.isInstancedBufferAttribute?(f(i,s.meshPerAttribute),void
0===o._maxInstanceCount&&
(o._maxInstanceCount=s.meshPerAttribute*s.count)):m(i),t.bindBuffer(34962,l),v(i,r,
c,e,0,0)}else if("instanceMatrix"===e){const e=n.get(r.instanceMatrix);if(void
0===e)continue;const
s=e.buffer,a=e.type;f(i+0,1),f(i+1,1),f(i+2,1),f(i+3,1),t.bindBuffer(34962,s),t.ver
texAttribPointer(i+0,4,a,!1,64,0),t.vertexAttribPointer(i+1,4,a,!1,64,16),t.vertexA
ttribPointer(i+2,4,a,!1,64,32),t.vertexAttribPointer(i+3,4,a,!1,64,48)}else
if("instanceColor"===e){const e=n.get(r.instanceColor);if(void 0===e)continue;const
s=e.buffer,a=e.type;f(i,1),t.bindBuffer(34962,s),t.vertexAttribPointer(i,3,a,!1,12,
0)}else if(void 0!==h){const n=h[e];if(void 0!==n)switch(n.length){case

```

```

2:t.vertexAttrib2fv(i,n);break;case 3:t.vertexAttrib3fv(i,n);break;case
4:t.vertexAttrib4fv(i,n);break;default:t.vertexAttrib1fv(i,n)}}}g()
(r,l,u,y),null!=x&&t.bindBuffer(34963,n.get(x).buffer)),reset:y,resetDefaultState
:x,dispose:function(){y();for(const t in o){const e=o[t];for(const t in e){const
n=e[t];for(const t in n)u(n[t].object),delete n[t];delete e[t]}delete
o[t]}},releaseStatesOfGeometry:function(t){if(void 0===o[t.id])return;const
e=o[t.id];for(const t in e){const n=e[t];for(const t in n)u(n[t].object),delete
n[t];delete e[t]}delete o[t.id]},releaseStatesOfProgram:function(t){for(const e in
o){const n=o[e];if(void 0===n[t.id])continue;const i=n[t.id];for(const t in
i)u(i[t].object),delete i[t];delete
n[t.id]}},initAttributes:p,enableAttribute:m,disableUnusedAttributes:g}}function
fi(t,e,n,i){const r=i.isWebGL2;let s;this.setMode=function(t)
{s=t,this.render=function(e,i)
{t.drawArrays(s,e,i),n.update(i,s,1)},this.renderInstances=function(i,a,o)
{if(0===o)return;let l,c;if(r)l=t,c="drawArraysInstanced";else
if(l=e.get("ANGLE_instanced_arrays"),c="drawArraysInstancedANGLE",null===l)return
void console.error("THREE.WebGLBufferRenderer: using THREE.InstancedBufferGeometry
but hardware does not support extension ANGLE_instanced_arrays.");l[c]
(s,i,a,o),n.update(a,s,o)}}function gi(t,e,n){let i;function r(e){if("highp"===e)
{if(t.getShaderPrecisionFormat(35633,36338).precision>0&&t.getShaderPrecisionFormat
(35632,36338).precision>0)return"highp";e="mediump"}return"mediump"===e&&t.getShade
rPrecisionFormat(35633,36337).precision>0&&t.getShaderPrecisionFormat(35632,36337).
precision>0?"mediump":"lowp"}const s="undefined"!==typeof WebGL2RenderingContext&&t
instanceof WebGL2RenderingContext||"undefined"!==typeof
WebGL2ComputeRenderingContext&&t instanceof WebGL2ComputeRenderingContext;let
a=void 0!==(n.precision?n.precision:"highp");const o=r(a);o!==a&&
(console.warn("THREE.WebGLRenderer:",a,"not supported,
using",o,"instead."),a=o);const
l=!0===n.logarithmicDepthBuffer,c=t.getParameter(34930),h=t.getParameter(35660),u=t
.getParameter(3379),d=t.getParameter(34076),p=t.getParameter(34921),m=t.getParame
r(36347),f=t.getParameter(36348),g=t.getParameter(36349),v=h>0,y=s||e.has("OES_text
ure_float");return{isWebGL2:s,getMaxAnisotropy:function(){if(void 0!==(i)return
i;if(!0===e.has("EXT_texture_filter_anisotropic")){const
n=e.get("EXT_texture_filter_anisotropic");i=t.getParameter(n.MAX_TEXTURE_MAX_ANISOT
ROPY_EXT)}else i=0;return
i},getMaxPrecision:r,precision:a,logarithmicDepthBuffer:l,maxTextures:c,maxVertexTe
xtures:h,maxTextureSize:u,maxCubemapSize:d,maxAttributes:p,maxVertexUniforms:m,maxV
aryings:f,maxFragmentUniforms:g,vertexTextures:v,floatFragmentTextures:y,floatVerte
xTextures:v&&y,maxSamples:s?t.getParameter(36183):0}}function vi(t){const
e=this;let n=null,i=0,r=!1,s=!1;const a=new Ne,o=new yt,l=
{value:null,needsUpdate:!1};function c(){l.value!==n&&
(l.value=n,l.needsUpdate=i>0),e.numPlanes=i,e.numIntersection=0}function h(t,n,i,r)
{const s=null!==(t?t.length:0);let c=null;if(0!==(s){if(c=l.value,!0!==(r||null===c)
{const e=i+4*s,r=n.matrixWorldInverse;o.getNormalMatrix(r),(null===c||c.length<e)&&
(c=new Float32Array(e));for(let
e=0,n=i;e!==(s;+e,n+=4)a.copy(t[e]).applyMatrix4(r,o),a.normal.toArray(c,n),c[n+3]=
a.constant}l.value=c,l.needsUpdate=!0}return
e.numPlanes=s,e.numIntersection=0,c}this.uniform=l,this.numPlanes=0,this.numInterse
ction=0,this.init=function(t,e,s){const a=0!==(t.length||e||0!==(i||r);return
r=e,n=h(t,s,0),i=t.length,a},this.beginShadows=function()
{s=!0,h(null)},this.endShadows=function(){s=!1,c()},this.setState=function(e,a,o)
{const
u=e.clippingPlanes,d=e.clipIntersection,p=e.clipShadows,m=t.get(e);if(!r||null===u|
|0===u.length||s&&!p)s?h(null):c();else{const t=s?0:i,e=4*t;let
r=m.clippingState||null;l.value=r,r=h(u,a,e,o);for(let
t=0;t!==(e;+t)r[t]=n[t];m.clippingState=r,this.numIntersection=d?
this.numPlanes:0,this.numPlanes+=t}}function yi(t){let e=new WeakMap;function
n(t,e){return e===a?t.mapping=r:e===o&&(t.mapping=s),t}function i(t){const

```

```

n=t.target;n.removeEventListener("dispose",i);const r=e.get(n);void 0!==r&&
(e.delete(n),r.dispose())}return{get:function(r){if(r&&r.isTexture){const
s=r.mapping;if(s===a||s===o){if(e.has(r)){return n(e.get(r).texture,r.mapping)}
{const s=r.image;if(s&&s.height>0){const a=t.getRenderTarget(),o=new
ni(s.height/2);return
o.fromEquiRectangularTexture(t,r),e.set(r,o),t.setRenderTarget(a),r.addEventListene
r("dispose",i),n(o.texture,r.mapping)}return null}}return r},dispose:function()
{e=new WeakMap}}function xi(t){const e={};function n(n){if(void 0!==e[n])return
e[n];let i;switch(n)
{case"WEBGL_depth_texture":i=t.getExtension("WEBGL_depth_texture")||t.getExtension(
"MOZ_WEBGL_depth_texture")||t.getExtension("WEBKIT_WEBGL_depth_texture");break;case
"EXT_texture_filter_anisotropic":i=t.getExtension("EXT_texture_filter_anisotropic")
||t.getExtension("MOZ_EXT_texture_filter_anisotropic")||t.getExtension("WEBKIT_EXT_
texture_filter_anisotropic");break;case"WEBGL_compressed_texture_s3tc":i=t.getExten
sion("WEBGL_compressed_texture_s3tc")||t.getExtension("MOZ_WEBGL_compressed_texture
_s3tc")||t.getExtension("WEBKIT_WEBGL_compressed_texture_s3tc");break;case"WEBGL_co
mpressed_texture_pvrtc":i=t.getExtension("WEBGL_compressed_texture_pvrtc")||t.getEx
tension("WEBKIT_WEBGL_compressed_texture_pvrtc");break;default:i=t.getExtension(n)}
return e[n]=i,i}return{has:function(t){return null!==n(t)},init:function(t)
{t.isWebGL2?n("EXT_color_buffer_float"):
(n("WEBGL_depth_texture"),n("OES_texture_float"),n("OES_texture_half_float"),n("OES
_texture_half_float_linear"),n("OES_standard_derivatives"),n("OES_element_index_uin
t"),n("OES_vertex_array_object"),n("ANGLE_instanced_arrays")),n("OES_texture_float_
linear"),n("EXT_color_buffer_half_float")},get:function(t){const e=n(t);return
null===e&&console.warn("THREE.WebGLRenderer: "+t+" extension not
supported."),e}}function _i(t,e,n,i){const r={},s=new WeakMap;function a(t){const
o=t.target;null!==o.index&&e.remove(o.index);for(const t in
o.attributes)e.remove(o.attributes[t]);o.removeEventListener("dispose",a),delete
r[o.id];const l=s.get(o);l&&
(e.remove(l),s.delete(o)),i.releaseStatesOfGeometry(o),!0===o.isInstancedBufferGeom
etry&&delete o._maxInstanceCount,n.memory.geometries--}function o(t){const n=
[],i=t.index,r=t.attributes.position;let a=0;if(null!==i){const
t=i.array;a=i.version;for(let e=0,i=t.length;e<i;e+=3){const
i=t[e+0],r=t[e+1],s=t[e+2];n.push(i,r,r,s,s,i)}}else{const
t=r.array;a=r.version;for(let e=0,i=t.length/3-1;e<i;e+=3){const
t=e+0,i=e+1,r=e+2;n.push(t,i,i,r,r,t)}}const o=new(gn(n)>65535?dn:hn)
(n,l);o.version=a;const
l=s.get(t);l&&e.remove(l),s.set(t,o)}return{get:function(t,e){return!0===r[e.id]||
(e.addEventListener("dispose",a),r[e.id]=!0,n.memory.geometries++),e},update:functi
on(t){const n=t.attributes;for(const t in n)e.update(n[t],34962);const
i=t.morphAttributes;for(const t in i){const n=i[t];for(let
t=0,i=n.length;t<i;t++)e.update(n[t],34962)}},getWireframeAttribute:function(t)
{const e=s.get(t);if(e){const n=t.index;null!==n&&e.version<n.version&&o(t)}else
o(t);return s.get(t)}}function wi(t,e,n,i){const r=i.isWebGL2;let
s,a,o;this.setMode=function(t){s=t},this.setIndex=function(t)
{a=t.type,o=t.bytesPerElement},this.render=function(e,i)
{t.drawElements(s,i,a,e*o),n.update(i,s,l)},this.renderInstances=function(i,l,c)
{if(0===c)return;let h,u;if(r)h=t,u="drawElementsInstanced";else
if(h=e.get("ANGLE_instanced_arrays"),u="drawElementsInstancedANGLE",null===h)return
void console.error("THREE.WebGLIndexedBufferRenderer: using
THREE.InstancedBufferGeometry but hardware does not support extension
ANGLE_instanced_arrays.");h[u](s,l,a,i*o,c),n.update(l,s,c)}}function bi(t){const
e={frame:0,calls:0,triangles:0,points:0,lines:0};return{memory:
{geometries:0,textures:0},render:e,programs:null,autoReset:!0,reset:function()
{e.frame++,e.calls=0,e.triangles=0,e.points=0,e.lines=0},update:function(t,n,i)
{switch(e.calls++,n){case 4:e.triangles+=i*(t/3);break;case 1:e.lines+=i*
(t/2);break;case 3:e.lines+=i*(t-1);break;case 2:e.lines+=i*t;break;case
0:e.points+=i*t;break;default:console.error("THREE.WebGLInfo: Unknown draw

```

```

mode:",n)}}}}function Mi(t,e){return t[0]-e[0]}function Si(t,e){return
Math.abs(e[1])-Math.abs(t[1])}function Ti(t){const e={},n=new Float32Array(8),i=
[];for(let t=0;t<8;t++)i[t]=[t,0];return{update:function(r,s,a,o){const
l=r.morphTargetInfluences,c=void 0===l?0:l.length;let h=e[s.id];if(void 0===h){h=
[];for(let t=0;t<c;t++)h[t]=[t,0];e[s.id]=h}for(let t=0;t<c;t++){const
e=h[t];e[0]=t,e[1]=l[t]}h.sort(Si);for(let t=0;t<8;t++)t<c&&h[t][1]?i[t][0]=h[t]
[0],i[t][1]=h[t][1):(i[t][0]=Number.MAX_SAFE_INTEGER,i[t][1]=0);i.sort(Mi);const
u=a.morphTargets&&s.morphAttributes.position,d=a.morphNormals&&s.morphAttributes.no
rmal;let p=0;for(let t=0;t<8;t++){const
e=i[t],r=e[0],a=e[1];r!==Number.MAX_SAFE_INTEGER&&a?
(u&&s.getAttribute("morphTarget"+t)!=u[r]&&s.setAttribute("morphTarget"+t,u[r]),d&
&s.getAttribute("morphNormal"+t)!=d[r]&&s.setAttribute("morphNormal"+t,d[r]),n[t]=
a,p+=a):
(u&&!0===s.hasAttribute("morphTarget"+t)&&s.deleteAttribute("morphTarget"+t),d&&!0=
==s.hasAttribute("morphNormal"+t)&&s.deleteAttribute("morphNormal"+t),n[t]=0)}const
m=s.morphTargetsRelative?1:1-
p;o.getUniforms().setValue(t,"morphTargetBaseInfluence",m),o.getUniforms().setValue
(t,"morphTargetInfluences",n)}}}function Ei(t,e,n,i){let r=new WeakMap;function
s(t){const
e=t.target;e.removeEventListener("dispose",s),n.remove(e.instanceMatrix),null!==e.i
nstanceColor&&n.remove(e.instanceColor)}return{update:function(t){const
a=i.render.frame,o=t.geometry,l=e.get(t,o);return r.get(l)!=a&&
(e.update(l),r.set(l,a)),t.isInstancedMesh&&
(!l===t.hasEventListener("dispose",s)&&t.addEventListener("dispose",s),n.update(t.i
nstanceMatrix,34962),null!==t.instanceColor&&n.update(t.instanceColor,34962)),l},di
spose:function(){r=new WeakMap}}di.physical={uniforms:Yn([di.standard.uniforms,
{clearcoat:{value:0},clearcoatMap:{value:null},clearcoatRoughness:
{value:0},clearcoatRoughnessMap:{value:null},clearcoatNormalScale:{value:new
vt(1,1)},clearcoatNormalMap:{value:null},sheen:{value:new tn(0)},transmission:
{value:0},transmissionMap:
{value:null}}]),vertexShader:hi.meshphysical_vert,fragmentShader:hi.meshphysical_fr
ag};class Ai extends bt{constructor(t=null,e=1,n=1,i=1){super(null),this.image=
{data:t,width:e,height:n,depth:i},this.magFilter=p,this.minFilter=p,this.wrapR=u,th
is.generateMipmaps=!1,this.flipY=!1,this.unpackAlignment=1,this.needsUpdate=!0}}Ai.
prototype.isDataTexture2DArray=!0;class Li extends
bt{constructor(t=null,e=1,n=1,i=1){super(null),this.image=
{data:t,width:e,height:n,depth:i},this.magFilter=p,this.minFilter=p,this.wrapR=u,th
is.generateMipmaps=!1,this.flipY=!1,this.unpackAlignment=1,this.needsUpdate=!0}}Li.
prototype.isDataTexture3D=!0;const Ri=new bt,Ci=new Ai,Pi=new Li,Di=new ei,Ii=
[],Ni=[],Bi=new Float32Array(16),zi=new Float32Array(9),Fi=new
Float32Array(4);function Oi(t,e,n){const i=t[0];if(i<=0||i>0)return t;const
r=e*n;let s=Ii[r];if(void 0===s&&(s=new Float32Array(r),Ii[r]=s),0!==e)
{i.toArray(s,0);for(let i=1,r=0;i!==e;++i)r+=n,t[i].toArray(s,r)}return s}function
Hi(t,e){if(t.length!==e.length)return!1;for(let
n=0,i=t.length;n<i;n++)if(t[n]!==e[n])return!1;return!0}function Gi(t,e){for(let
n=0,i=e.length;n<i;n++)t[n]=e[n]}function Ui(t,e){let n=Ni[e];void 0===n&&(n=new
Int32Array(e),Ni[e]=n);for(let i=0;i!==e;++i)n[i]=t.allocateTextureUnit();return
n}function ki(t,e){const n=this.cache;n[0]!==e&&
(t.uniform1f(this.addr,e),n[0]=e)}function Vi(t,e){const n=this.cache;if(void
0!==e.x)n[0]===e.x&&n[1]===e.y||
(t.uniform2f(this.addr,e.x,e.y),n[0]=e.x,n[1]=e.y);else{if(Hi(n,e))return;t.uniform
2fv(this.addr,e),Gi(n,e)}}function Wi(t,e){const n=this.cache;if(void
0!==e.x)n[0]===e.x&&n[1]===e.y&&n[2]===e.z||
(t.uniform3f(this.addr,e.x,e.y,e.z),n[0]=e.x,n[1]=e.y,n[2]=e.z);else if(void
0!==e.r)n[0]===e.r&&n[1]===e.g&&n[2]===e.b||
(t.uniform3f(this.addr,e.r,e.g,e.b),n[0]=e.r,n[1]=e.g,n[2]=e.b);else{if(Hi(n,e))ret
urn;t.uniform3fv(this.addr,e),Gi(n,e)}}function ji(t,e){const n=this.cache;if(void
0!==e.x)n[0]===e.x&&n[1]===e.y&&n[2]===e.z&&n[3]===e.w||

```



```

(t.uniform4f(this.addr,e.x,e.y,e.z,e.w),n[0]=e.x,n[1]=e.y,n[2]=e.z,n[3]=e.w);else{if
f(Hi(n,e))return;t.uniform4fv(this.addr,e),Gi(n,e)}}function qi(t,e){const
n=this.cache,i=e.elements;if(void 0===i)
{if(Hi(n,e))return;t.uniformMatrix2fv(this.addr,!1,e),Gi(n,e)}else{if(Hi(n,i))retur
n;Fi.set(i),t.uniformMatrix2fv(this.addr,!1,Fi),Gi(n,i)}}function Xi(t,e){const
n=this.cache,i=e.elements;if(void 0===i)
{if(Hi(n,e))return;t.uniformMatrix3fv(this.addr,!1,e),Gi(n,e)}else{if(Hi(n,i))retur
n;zi.set(i),t.uniformMatrix3fv(this.addr,!1,zi),Gi(n,i)}}function Yi(t,e){const
n=this.cache,i=e.elements;if(void 0===i)
{if(Hi(n,e))return;t.uniformMatrix4fv(this.addr,!1,e),Gi(n,e)}else{if(Hi(n,i))retur
n;Bi.set(i),t.uniformMatrix4fv(this.addr,!1,Bi),Gi(n,i)}}function Zi(t,e){const
n=this.cache;n[0]!==e&&(t.uniform1i(this.addr,e),n[0]=e)}function Ji(t,e){const
n=this.cache;Hi(n,e)|(t.uniform2iv(this.addr,e),Gi(n,e))}function Qi(t,e){const
n=this.cache;Hi(n,e)|(t.uniform3iv(this.addr,e),Gi(n,e))}function Ki(t,e){const
n=this.cache;Hi(n,e)|(t.uniform4iv(this.addr,e),Gi(n,e))}function $i(t,e){const
n=this.cache;n[0]!==e&&(t.uniform1ui(this.addr,e),n[0]=e)}function tr(t,e){const
n=this.cache;Hi(n,e)|(t.uniform2uiv(this.addr,e),Gi(n,e))}function er(t,e){const
n=this.cache;Hi(n,e)|(t.uniform3uiv(this.addr,e),Gi(n,e))}function nr(t,e){const
n=this.cache;Hi(n,e)|(t.uniform4uiv(this.addr,e),Gi(n,e))}function ir(t,e,n){const
i=this.cache,r=n.allocateTextureUnit();i[0]!==r&&
(t.uniform1i(this.addr,r),i[0]=r),n.safeSetTexture2D(e|Ri,r)}function rr(t,e,n)
{const i=this.cache,r=n.allocateTextureUnit();i[0]!==r&&
(t.uniform1i(this.addr,r),i[0]=r),n.setTexture3D(e|Pi,r)}function sr(t,e,n){const
i=this.cache,r=n.allocateTextureUnit();i[0]!==r&&
(t.uniform1i(this.addr,r),i[0]=r),n.safeSetTextureCube(e|Di,r)}function ar(t,e,n)
{const i=this.cache,r=n.allocateTextureUnit();i[0]!==r&&
(t.uniform1i(this.addr,r),i[0]=r),n.setTexture2DArray(e|Ci,r)}function or(t,e)
{t.uniform1fv(this.addr,e)}function lr(t,e){const
n=Oi(e,this.size,2);t.uniform2fv(this.addr,n)}function cr(t,e){const
n=Oi(e,this.size,3);t.uniform3fv(this.addr,n)}function hr(t,e){const
n=Oi(e,this.size,4);t.uniform4fv(this.addr,n)}function ur(t,e){const
n=Oi(e,this.size,4);t.uniformMatrix2fv(this.addr,!1,n)}function dr(t,e){const
n=Oi(e,this.size,9);t.uniformMatrix3fv(this.addr,!1,n)}function pr(t,e){const
n=Oi(e,this.size,16);t.uniformMatrix4fv(this.addr,!1,n)}function mr(t,e)
{t.uniform1liv(this.addr,e)}function fr(t,e){t.uniform2iv(this.addr,e)}function
gr(t,e){t.uniform3iv(this.addr,e)}function vr(t,e)
{t.uniform4iv(this.addr,e)}function yr(t,e){t.uniform1uiv(this.addr,e)}function
xr(t,e){t.uniform2uiv(this.addr,e)}function _r(t,e)
{t.uniform3uiv(this.addr,e)}function wr(t,e){t.uniform4uiv(this.addr,e)}function
br(t,e,n){const i=e.length,r=Ui(n,i);t.uniform1liv(this.addr,r);for(let
t=0;t!==i;++t)n.safeSetTexture2D(e[t]|Ri,r[t])}function Mr(t,e,n){const
i=e.length,r=Ui(n,i);t.uniform1liv(this.addr,r);for(let
t=0;t!==i;++t)n.safeSetTextureCube(e[t]|Di,r[t])}function Sr(t,e,n)
{this.id=t,this.addr=n,this.cache=[],this.setValue=function(t){switch(t){case
5126:return ki;case 35664:return Vi;case 35665:return Wi;case 35666:return ji;case
35674:return qi;case 35675:return Xi;case 35676:return Yi;case 5124:case
35670:return Zi;case 35667:case 35671:return Ji;case 35668:case 35672:return
Qi;case 35669:case 35673:return Ki;case 5125:return $i;case 36294:return tr;case
36295:return er;case 36296:return nr;case 35678:case 36198:case 36298:case
36306:case 35682:return ir;case 35679:case 36299:case 36307:return rr;case
35680:case 36300:case 36308:case 36293:return sr;case 36289:case 36303:case
36311:case 36292:return ar}}(e.type)}function Tr(t,e,n)
{this.id=t,this.addr=n,this.cache=[],this.size=e.size,this.setValue=function(t)
{switch(t){case 5126:return or;case 35664:return lr;case 35665:return cr;case
35666:return hr;case 35674:return ur;case 35675:return dr;case 35676:return pr;case
5124:case 35670:return mr;case 35667:case 35671:return fr;case 35668:case
35672:return gr;case 35669:case 35673:return vr;case 5125:return yr;case
36294:return xr;case 36295:return _r;case 36296:return wr;case 35678:case

```

```

36198:case 36298:case 36306:case 35682:return br;case 35680:case 36300:case
36308:case 36293:return Mr}}(e.type)}function Er(t){this.id=t,this.seq=[],this.map=
{}}Tr.prototype.updateCache=function(t){const e=this.cache;t instanceof
Float32Array&&e.length!==t.length&&(this.cache=new
Float32Array(t.length)),Gi(e,t)},Er.prototype.setValue=function(t,e,n){const
i=this.seq;for(let r=0,s=i.length;r!==s;++r){const
s=i[r];s.setValue(t,e[s.id],n)};const Ar=/(\w+)(\s)?(\[|\.)?/g;function Lr(t,e)
{t.seq.push(e),t.map[e.id]=e}function Rr(t,e,n){const
i=t.name,r=i.length;for(Ar.lastIndex=0;;){const s=Ar.exec(i),a=Ar.lastIndex;let
o=s[1];const l=""===s[2],c=s[3];if(l&&(o|0),void 0===c||"[]"===c&&a+2===r)
{Lr(n,void 0===c?new Sr(o,t,e):new Tr(o,t,e));break}{let t=n.map[o];void 0===t&&
(t=new Er(o),Lr(n,t)),n=t}}function Cr(t,e){this.seq=[],this.map={};const
n=t.getProgramParameter(e,35718);for(let i=0;i<n;++i){const
n=t.getActiveUniform(e,i);Rr(n,t.getUniformLocation(e,n.name),this)}}function
Pr(t,e,n){const i=t.createShader(e);return
t.shaderSource(i,n),t.compileShader(i),i}Cr.prototype.setValue=function(t,e,n,i)
{const r=this.map[e];void
0!==r&&r.setValue(t,n,i)},Cr.prototype.setOptional=function(t,e,n){const
i=e[n];void 0!==i&&this.setValue(t,n,i)},Cr.upload=function(t,e,n,i){for(let
r=0,s=e.length;r!==s;++r){const
s=e[r],a=n[s.id];!1!==a.needsUpdate&&s.setValue(t,a.value,i)},Cr.seqWithValue=func
tion(t,e){const n=[];for(let i=0,r=t.length;i!==r;++i){const r=t[i];r.id in
e&&n.push(r)}return n};let Dr=0;function Ir(t){switch(t){case X:return["Linear","(
value)"];case Y:return["sRGB","( value)"];case J:return["RGBE","( value)"];case
Q:return["RGBM","( value, 7.0)"];case K:return["RGBM","( value, 16.0)"];case
$:return["RGBD","( value, 256.0)"];case Z:return["Gamma","( value, float(
GAMMA_FACTOR))"];case 3003:return["LogLuv","( value)"];default:return
console.warn("THREE.WebGLProgram: Unsupported encoding:",t),["Linear","( value
)"]}function Nr(t,e,n){const
i=t.getShaderParameter(e,35713),r=t.getShaderInfoLog(e).trim();if(i&&"===r)return"
";return"THREE.WebGLShader: gl.getShaderInfoLog() "+n+"\n"+r+function(t){const
e=t.split("\n");for(let t=0;t<e.length;t++)e[t]=t+1+": "+e[t];return e.join("\n")}
(t.getShaderSource(e))}function Br(t,e){const n=Ir(e);return"vec4 "+t+"( vec4 value
) { return "+n[0]+"ToLinear"+n[1]+"; }"}function zr(t,e){const n=Ir(e);return"vec4
"+t+"( vec4 value ) { return LinearTo"+n[0]+n[1]+"; }"}function Fr(t,e){let
n;switch(e){case 1:n="Linear";break;case 2:n="Reinhard";break;case
3:n="OptimizedCineon";break;case 4:n="ACESFilmic";break;case
5:n="Custom";break;default:console.warn("THREE.WebGLProgram: Unsupported
toneMapping:",e),n="Linear"}return"vec3 "+t+"( vec3 color ) { return
"+n+"ToneMapping( color ); }"}function Or(t){return"!==t"}function Hr(t,e){return
t.replace(/NUM_DIR_LIGHTS/g,e.numDirLights).replace(/NUM_SPOT_LIGHTS/g,e.numSpotLig
hts).replace(/NUM_RECT_AREA_LIGHTS/g,e.numRectAreaLights).replace(/NUM_POINT_LIGHTS
/g,e.numPointLights).replace(/NUM_HEMI_LIGHTS/g,e.numHemiLights).replace(/NUM_DIR_L
IGHT_SHADOWS/g,e.numDirLightShadows).replace(/NUM_SPOT_LIGHT_SHADOWS/g,e.numSpotLig
htShadows).replace(/NUM_POINT_LIGHT_SHADOWS/g,e.numPointLightShadows)}function
Gr(t,e){return
t.replace(/NUM_CLIPPING_PLANES/g,e.numClippingPlanes).replace(/UNION_CLIPPING_PLANE
S/g,e.numClippingPlanes-e.numClipIntersection)}const Ur=/^[ \t]*#include +
<([\w\d.\/]+)>/gm;function kr(t){return t.replace(Ur,Vr)}function Vr(t,e){const
n=hi[e];if(void 0===n)throw new Error("Can not resolve #include <"+e+">");return
kr(n)}const Wr=/#pragma unroll_loop\[s\]+?for \( int i \= (\d+)\; i < (\d+)\; i \+ \+
\) \{([\s\S]+?)\}(?=\})\}/g,jr=/#pragma unroll_loop_start\s+for\s*\((\s*int\s+i\s*=\s*
(\d+)\s*;\s*i\s* < \s*(\d+)\s*;\s*i\s* \+ \+ \s*)\)\s*\{([\s\S]+?)\}\s+/#pragma
unroll_loop_end/g;function qr(t){return t.replace(jr,Yr).replace(Wr,Xr)}function
Xr(t,e,n,i){return console.warn("WebGLProgram: #pragma unroll_loop shader syntax is
deprecated. Please use #pragma unroll_loop_start syntax
instead."),Yr(t,e,n,i)}function Yr(t,e,n,i){let r="";for(let
t=parseInt(e);t<parseInt(n);t++)r+=i.replace(/\s*i\s*/g,"[ "+t+"

```

```

}).replace(/UNROLLED_LOOP_INDEX/g,t);return r}function Zr(t){let e="precision
+t.precision+" float;\nprecision "+t.precision+"
int;";return"highp"===t.precision?e+="\n#define
HIGH_PRECISION":"mediump"===t.precision?e+="\n#define
MEDIUM_PRECISION":"lowp"===t.precision&&(e+="\n#define LOW_PRECISION"),e}function
Jr(t,e,n,i){const a=t.getContext(),o=n.defines;let
h=n.vertexShader,u=n.fragmentShader;const d=function(t){let
e="SHADOWMAP_TYPE_BASIC";return 1===t.shadowMapType?
e="SHADOWMAP_TYPE_PCF":2===t.shadowMapType?
e="SHADOWMAP_TYPE_PCF_SOFT":3===t.shadowMapType&&(e="SHADOWMAP_TYPE_VSM"),e}
(n),p=function(t){let e="ENVMAP_TYPE_CUBE";if(t.envMap)switch(t.envMapMode){case
r:case s:e="ENVMAP_TYPE_CUBE";break;case l:case c:e="ENVMAP_TYPE_CUBE_UV"}return e}
(n),m=function(t){let e="ENVMAP_MODE_REFLECTION";if(t.envMap)switch(t.envMapMode)
{case s:case c:e="ENVMAP_MODE_REFRACTION"}return e}(n),f=function(t){let
e="ENVMAP_BLENDING_NONE";if(t.envMap)switch(t.combine){case
0:e="ENVMAP_BLENDING_MULTIPLY";break;case 1:e="ENVMAP_BLENDING_MIX";break;case
2:e="ENVMAP_BLENDING_ADD"}return e}(n),g=t.gammaFactor>0?
t.gammaFactor:1,v=n.isWebGL2?"":function(t)
{return[t.extensionDerivatives|t.envMapCubeUV|t.bumpMap|t.tangentSpaceNormalMap|
t.clearcoatNormalMap|t.flatShading|"physical"===t.shaderID?"#extension
GL_OES_standard_derivatives : enable":"",
(t.extensionFragDepth|t.logarithmicDepthBuffer)&&t.rendererExtensionFragDepth?"#ex
tension GL_EXT_frag_depth :
enable":"",t.extensionDrawBuffers&&t.rendererExtensionDrawBuffers?"#extension
GL_EXT_draw_buffers : require":"",
(t.extensionShaderTextureLOD|t.envMap)&&t.rendererExtensionShaderTextureLod?"#exte
nsion GL_EXT_shader_texture_lod : enable":""].filter(Or).join("\n")}
(n),y=function(t){const e=[];for(const n in t){const i=t[n];!!i===i&&e.push("#define
"+n+" "+i)}return e.join("\n")}(o),x=a.createProgram();let
_,w,b=n.glslVersion?"#version "+n.glslVersion+"\n":"";n.isRawShaderMaterial?(_=
[y].filter(Or).join("\n"),_.length>0&&(_+="\n"),w=
[v,y].filter(Or).join("\n"),w.length>0&&(w+="\n")):(_=[Zr(n),"#define SHADER_NAME
"+n.shaderName,y,n.instancing?"#define
USE_INSTANCING":"",n.instancingColor?"#define
USE_INSTANCING_COLOR":"",n.supportsVertexTextures?"#define
VERTEX_TEXTURES":"",n.useFog&&n.fog?"#define USE_FOG":"",n.useFog&&n.fogExp2?"#define
FOG_EXP2":"",n.map?"#define USE_MAP":"",n.envMap?"#define
USE_ENVMAP":"",n.envMap?"#define "+m:"n.lightMap?"#define
USE_LIGHTMAP":"",n.aoMap?"#define USE_AOMAP":"",n.emissiveMap?"#define
USE_EMISSIVEMAP":"",n.bumpMap?"#define USE_BUMPMAP":"",n.normalMap?"#define
USE_NORMALMAP":"",n.normalMap&&n.objectSpaceNormalMap?"#define
OBJECTSPACE_NORMALMAP":"",n.normalMap&&n.tangentSpaceNormalMap?"#define
TANGENTSPACE_NORMALMAP":"",n.clearcoatMap?"#define
USE_CLEARCOATMAP":"",n.clearcoatRoughnessMap?"#define
USE_CLEARCOAT_ROUGHNESSMAP":"",n.clearcoatNormalMap?"#define
USE_CLEARCOAT_NORMALMAP":"",n.displacementMap&&n.supportsVertexTextures?"#define
USE_DISPLACEMENTMAP":"",n.specularMap?"#define
USE_SPECULARMAP":"",n.roughnessMap?"#define
USE_ROUGHNESSMAP":"",n.metalnessMap?"#define
USE_METALNESSMAP":"",n.alphaMap?"#define
USE_ALPHAMAP":"",n.transmissionMap?"#define
USE_TRANSMISSIONMAP":"",n.vertexTangents?"#define
USE_TANGENT":"",n.vertexColors?"#define USE_COLOR":"",n.vertexAlphas?"#define
USE_COLOR_ALPHA":"",n.vertexUvs?"#define USE_UV":"",n.uvsVertexOnly?"#define
UVS_VERTEX_ONLY":"",n.flatShading?"#define FLAT_SHADED":"",n.skinning?"#define
USE_SKINNING":"",n.useVertexTexture?"#define
BONE_TEXTURE":"",n.morphTargets?"#define

```

[illegible]

```

Fr("toneMapping",n.toneMapping):"" ,n.dithering?"#define
DITHERING:"" ,hi.encodings_pars_fragment,n.map?
Br("mapTexelToLinear",n.mapEncoding):"" ,n.matcap?
Br("matcapTexelToLinear",n.matcapEncoding):"" ,n.envMap?
Br("envMapTexelToLinear",n.envMapEncoding):"" ,n.emissiveMap?
Br("emissiveMapTexelToLinear",n.emissiveMapEncoding):"" ,n.lightMap?
Br("lightMapTexelToLinear",n.lightMapEncoding):"" ,zr("linearToOutputTexel",n.output
Encoding),n.depthPacking?"#define DEPTH_PACKING
"+n.depthPacking:"" ,"\"n\"].filter(Or).join(\"n\")) ,h=kr(h),h=Hr(h,n),h=Gr(h,n),u=kr(u
),u=Hr(u,n),u=Gr(u,n),h=qr(h),u=qr(u),n.isWebGL2&&!0!==(n.isRawShaderMaterial&&
(b="#version 300 es\"n\",_=[\"#define attribute in\", \"#define varying out\", \"#define
texture2D texture\"].join(\"n\")+\"n\"+_ ,w=[\"#define varying
in\",n.glslVersion===it?"\" :\"out highp vec4
pc_fragColor;\" ,n.glslVersion===it?"\" :\"#define gl_FragColor pc_fragColor\", \"#define
gl_FragDepthEXT gl_FragDepth\", \"#define texture2D texture\", \"#define textureCube
texture\", \"#define texture2DProj textureProj\", \"#define texture2DLodEXT
textureLod\", \"#define texture2DProjLodEXT textureProjLod\", \"#define textureCubeLodEXT
textureLod\", \"#define texture2DGradEXT textureGrad\", \"#define texture2DProjGradEXT
textureProjGrad\", \"#define textureCubeGradEXT textureGrad\"].join(\"n\")+\"n\"+w);const
M=b+w+u,S=Pr(a,35633,b+_h),T=Pr(a,35632,M);if(a.attachShader(x,S),a.attachShader(x
,T),void 0!==(n.index0AttributeName)?
a.bindAttribLocation(x,0,n.index0AttributeName):!0===n.morphTargets&&a.bindAttribLo
cation(x,0,\"position\"),a.linkProgram(x),t.debug.checkShaderErrors){const
t=a.getProgramInfoLog(x).trim(),e=a.getShaderInfoLog(S).trim(),n=a.getShaderInfoLog
(T).trim();let i=!0,r=!0;if(!1===a.getProgramParameter(x,35714)){i=!1;const
e=Nr(a,S,\"vertex\"),n=Nr(a,T,\"fragment\");console.error(\"THREE.WebGLProgram: shader
error:
\",a.getError(),\"35715\",a.getProgramParameter(x,35715),\"gl.getProgramInfoLog\",t,e,n)
}else\"!==(t?console.warn(\"THREE.WebGLProgram:
gl.getProgramInfoLog()\",t):\"!==(e&&\"!==(n|| (r=!1);r&&(this.diagnostics=
{runnable:i,programLog:t,vertexShader:{log:e,prefix:_},fragmentShader:
{log:n,prefix:w}}))}let E,A;return
a.deleteShader(S),a.deleteShader(T),this.getUniforms=function(){return void 0===E&&
(E=new Cr(a,x)),E},this.getAttributes=function(){return void 0===A&&
(A=function(t,e){const n={},i=t.getProgramParameter(e,35721);for(let r=0;r<i;r++)
{const i=t.getActiveAttrib(e,r).name;n[i]=t.getAttribLocation(e,i)}return n}
(a,x)),A},this.destroy=function()
{i.releaseStatesOfProgram(this),a.deleteProgram(x),this.program=void
0},this.name=n.shaderName,this.id=Dr++,this.cacheKey=e,this.usedTimes=1,this.progra
m=x,this.vertexShader=S,this.fragmentShader=T,this}function Qr(t,e,n,i,r,s){const
a=
[],o=i.isWebGL2,h=i.logarithmicDepthBuffer,u=i.floatVertexTextures,d=i.maxVertexUni
forms,p=i.vertexTextures;let m=i.precision;const f=
{MeshDepthMaterial:\"depth\",MeshDistanceMaterial:\"distanceRGBA\",MeshNormalMaterial:
\"normal\",MeshBasicMaterial:\"basic\",MeshLambertMaterial:\"lambert\",MeshPhongMaterial:
\"phong\",MeshToonMaterial:\"toon\",MeshStandardMaterial:\"physical\",MeshPhysicalMaterial:
\"physical\",MeshMatcapMaterial:\"matcap\",LineBasicMaterial:\"basic\",LineDashedMateria
l:\"dashed\",PointsMaterial:\"points\",ShadowMaterial:\"shadow\",SpriteMaterial:\"sprite\"}
,g=
[\"precision\", \"isWebGL2\", \"supportsVertexTextures\", \"outputEncoding\", \"instancing\", \"ins
tancingColor\", \"map\", \"mapEncoding\", \"matcap\", \"matcapEncoding\", \"envMap\", \"envMapMode\",
\"envMapEncoding\", \"envMapCubeUV\", \"lightMap\", \"lightMapEncoding\", \"aoMap\", \"emissiveMap\",
\"emissiveMapEncoding\", \"bumpMap\", \"normalMap\", \"objectSpaceNormalMap\", \"tangentSpaceNor
malMap\", \"clearcoatMap\", \"clearcoatRoughnessMap\", \"clearcoatNormalMap\", \"displacementMa
p\", \"specularMap\", \"roughnessMap\", \"metalnessMap\", \"gradientMap\", \"alphaMap\", \"combine\",
\"vertexColors\", \"vertexAlphas\", \"vertexTangents\", \"vertexUvs\", \"uvsVertexOnly\", \"fog\", \"us
eFog\", \"fogExp2\", \"flatShading\", \"sizeAttenuation\", \"logarithmicDepthBuffer\", \"skinning\",
\"maxBones\", \"useVertexTexture\", \"morphTargets\", \"morphNormals\", \"premultipliedAlpha\",

```

```

numDirLights","numPointLights","numSpotLights","numHemiLights","numRectAreaLights",
"numDirLightShadows","numPointLightShadows","numSpotLightShadows","shadowMapEnabled",
"shadowMapType","toneMapping","physicallyCorrectLights","alphaTest","doubleSided",
"flipSided","numClippingPlanes","numClipIntersection","depthPacking","dithering",
"sheen","transmissionMap"];function v(t){let e;return t&&t.isTexture?
e=t.encoding:t&&t.isWebGLRenderTarget?
(console.warn("THREE.WebGLPrograms.getTextureEncodingFromMap: don't use render
targets as textures. Use their .texture property
instead."),e=t.texture.encoding):e=X,e)return{getParameters:function(r,a,g,y,x)
{const _=y.fog,w=r.isMeshStandardMaterial?
y.environment:null,b=e.get(r.envMap|w),M=f[r.type],S=x.isSkinnedMesh?function(t)
{const e=t.skeleton.bones;if(u)return 1024;{const t=d,n=Math.floor((t-
20)/4),i=Math.min(n,e.length);return i<e.length?(console.warn("THREE.WebGLRenderer:
Skeleton has "+e.length+" bones. This GPU supports "+i+"."),0):i}}(x):0;let
T,E;if(null!==(r.precision&&
(m=i.getMaxPrecision(r.precision),m!==(r.precision&&console.warn("THREE.WebGLProgram
.getParameters:",r.precision,"not supported, using",m,"instead.")),M){const
t=di[M];T=t.vertexShader,E=t.fragmentShader}else
T=r.vertexShader,E=r.fragmentShader;const
A=t.getRenderTarget();return{isWebGL2:o,shaderID:M,shaderName:r.type,vertexShader:T
,fragmentShader:E,defines:r.defines,isRawShaderMaterial:!0===r.isRawShaderMaterial,
glslVersion:r.glslVersion,precision:m,instancing:!0===x.isInstancedMesh,instancingC
olor:!0===x.isInstancedMesh&&null!==(x.instanceColor,supportsVertexTextures:p,output
Encoding:null!==(A?
v(A.texture):t.outputEncoding,map:!!r.map,mapEncoding:v(r.map),matcap:!!r.matcap,ma
tcapEncoding:v(r.matcap),envMap:!!b,envMapMode:b&&b.mapping,envMapEncoding:v(b),env
MapCubeUV:!!b&&
(b.mapping===1||b.mapping===c),lightMap:!!r.lightMap,lightMapEncoding:v(r.lightMap)
,aoMap:!!r.aoMap,emissiveMap:!!r.emissiveMap,emissiveMapEncoding:v(r.emissiveMap),b
umpMap:!!r.bumpMap,normalMap:!!r.normalMap,objectSpaceNormalMap:1===r.normalMapType
,tangentSpaceNormalMap:0===r.normalMapType,clearcoatMap:!!r.clearcoatMap,clearcoatR
oughnessMap:!!r.clearcoatRoughnessMap,clearcoatNormalMap:!!r.clearcoatNormalMap,dis
placementMap:!!r.displacementMap,roughnessMap:!!r.roughnessMap,metalnessMap:!!r.met
alnessMap,specularMap:!!r.specularMap,alphaMap:!!r.alphaMap,gradientMap:!!r.gradien
tMap,sheen:!!r.sheen,transmissionMap:!!r.transmissionMap,combine:r.combine,vertexTa
ngents:r.normalMap&&r.vertexTangents,vertexColors:r.vertexColors,vertexAlphas:!0===
r.vertexColors&&x.geometry&&x.geometry.attributes.color&&4===x.geometry.attributes.
color.itemSize,vertexUvs:!!
(r.map| r.bumpMap| r.normalMap| r.specularMap| r.alphaMap| r.emissiveMap| r.roughne
ssMap| r.metalnessMap| r.clearcoatMap| r.clearcoatRoughnessMap| r.clearcoatNormalMa
p| r.displacementMap| r.transmissionMap),uvsVertexOnly:
(r.map| r.bumpMap| r.normalMap| r.specularMap| r.alphaMap| r.emissiveMap| r.roughne
ssMap| r.metalnessMap| r.clearcoatNormalMap| r.transmissionMap| !r.displacementMap)
,fog:!!_,useFog:r.fog,fogExp2:_&&_.isFogExp2,flatShading:!!r.flatShading,sizeAttenu
ation:r.sizeAttenuation,logarithmicDepthBuffer:h,skinning:r.skinning&&S>0,maxBones:
S,useVertexTexture:u,morphTargets:r.morphTargets,morphNormals:r.morphNormals,numDir
Lights:a.directional.length,numPointLights:a.point.length,numSpotLights:a.spot.leng
th,numRectAreaLights:a.rectArea.length,numHemiLights:a.hemi.length,numDirLightShado
ws:a.directionalShadowMap.length,numPointLightShadows:a.pointShadowMap.length,numSp
otLightShadows:a.spotShadowMap.length,numClippingPlanes:s.numPlanes,numClipIntersec
tion:s.numIntersection,dithering:r.dithering,shadowMapEnabled:t.shadowMap.enabled&&
g.length>0,shadowMapType:t.shadowMap.type,toneMapping:r.toneMapped?
t.toneMapping:0,physicallyCorrectLights:t.physicallyCorrectLights,premultipliedAlph
a:r.premultipliedAlpha,alphaTest:r.alphaTest,doubleSided:2===r.side,flipSided:1===r
.side,depthPacking:void
0!==(r.depthPacking&&r.depthPacking,index0AttributeName:r.index0AttributeName,extens
ionDerivatives:r.extensions&&r.extensions.derivatives,extensionFragDepth:r.extensio
ns&&r.extensions.fragDepth,extensionDrawBuffers:r.extensions&&r.extensions.drawBuff

```

```

ers,extensionShaderTextureLOD:r.extensions&&r.extensions.shaderTextureLOD,renderere
xtensionFragDepth:o||n.has("EXT_frag_depth"),renderereExtensionDrawBuffers:o||n.has(
"WEBGL_draw_buffers"),renderereExtensionShaderTextureLod:o||n.has("EXT_shader_textur
e_lod"),customProgramCacheKey:r.customProgramCacheKey({}),getProgramCacheKey:functi
on(e){const n=[];if(e.shaderID?n.push(e.shaderID):
(n.push(e.fragmentShader),n.push(e.vertexShader)),void 0!==e.defines)for(const t in
e.defines)n.push(t),n.push(e.defines[t]);if(!1===e.isRawShaderMaterial){for(let
t=0;t<g.length;t++)n.push(e[g[t]]);n.push(t.outputEncoding),n.push(t.gammaFactor)}r
eturn n.push(e.customProgramCacheKey),n.join(),getUniforms:function(t){const
e=f[t.type];let n;if(e){const t=di[e];n=Zn.clone(t.uniforms)}else
n=t.uniforms;return n},acquireProgram:function(e,n){let i;for(let
t=0,e=a.length;t<e;t++){const e=a[t];if(e.cacheKey===n)
{i=e,++i.usedTimes;break}}return void 0===i&&(i=new
Jr(t,n,e,r),a.push(i)),i},releaseProgram:function(t){if(0===--t.usedTimes){const
e=a.indexOf(t);a[e]=a[a.length-1],a.pop(),t.destroy(),programs:a}}function Kr()
{let t=new WeakMap;return{get:function(e){let n=t.get(e);return void 0===n&&(n=
{}),t.set(e,n)},n},remove:function(e){t.delete(e)},update:function(e,n,i){t.get(e)
[n]=i},dispose:function(){t=new WeakMap}}function $r(t,e){return
t.groupOrder!==e.groupOrder?t.groupOrder-
e.groupOrder:t.renderOrder!==e.renderOrder?t.renderOrder-
e.renderOrder:t.program!==e.program?t.program.id-
e.program.id:t.material.id!==e.material.id?t.material.id-e.material.id:t.z!==e.z?
t.z-e.z:t.id-e.id}function ts(t,e){return t.groupOrder!==e.groupOrder?t.groupOrder-
e.groupOrder:t.renderOrder!==e.renderOrder?t.renderOrder-e.renderOrder:t.z!==e.z?
e.z-t.z:t.id-e.id}function es(t){const e=[];let n=0;const i=[],r=[],s=
{id:-1};function a(i,r,a,o,l,c){let h=e[n];const u=t.get(a);return void 0===h?(h=
{id:i.id,object:i,geometry:r,material:a,program:u.program|s,groupOrder:o,renderOrd
er:i.renderOrder,z:l,group:c},e[n]=h):
(h.id=i.id,h.object=i,h.geometry=r,h.material=a,h.program=u.program|s,h.groupOrder
=o,h.renderOrder=i.renderOrder,h.z=l,h.group=c),n++,h}return{opaque:i,transparent:r
,init:function(){n=0,i.length=0,r.length=0},push:function(t,e,n,s,o,l){const
c=a(t,e,n,s,o,l);(!0===n.transparent?r:i).push(c)},unshift:function(t,e,n,s,o,l)
{const c=a(t,e,n,s,o,l);(!0===n.transparent?r:i).unshift(c)},finish:function()
{for(let t=n,i=e.length;t<i;t++){const
n=e[t];if(null===n.id)break;n.id=null,n.object=null,n.geometry=null,n.material=null
,n.program=null,n.group=null}},sort:function(t,e)
{i.length>1&&i.sort(t||$r),r.length>1&&r.sort(e||ts)}}function ns(t){let e=new
WeakMap;return{get:function(n,i){let r;return!1===e.has(n)?(r=new es(t),e.set(n,
[r])):i>=e.get(n).length?(r=new es(t),e.get(n).push(r)):r=e.get(n)
[i],r},dispose:function(){e=new WeakMap}}function is(){const t=
{};return{get:function(e){if(void 0!==t[e.id])return t[e.id];let n;switch(e.type)
{case"DirectionalLight":n={direction:new Lt,color:new tn};break;case"SpotLight":n=
{position:new Lt,direction:new Lt,color:new
tn,distance:0,coneCos:0,penumbraCos:0,decay:0};break;case"PointLight":n=
{position:new Lt,color:new tn,distance:0,decay:0};break;case"HemisphereLight":n=
{direction:new Lt,skyColor:new tn,groundColor:new tn};break;case"RectAreaLight":n=
{color:new tn,position:new Lt,halfWidth:new Lt,halfHeight:new Lt}}return
t[e.id]=n,n}}let rs=0;function ss(t,e){return(e.castShadow?1:0)-(t.castShadow?
1:0)}function as(t,e){const n=new is,i=function(){const t={};return{get:function(e)
{if(void 0!==t[e.id])return t[e.id];let n;switch(e.type)
{case"DirectionalLight":case"SpotLight":n=
{shadowBias:0,shadowNormalBias:0,shadowRadius:1,shadowMapSize:new
vt};break;case"PointLight":n=
{shadowBias:0,shadowNormalBias:0,shadowRadius:1,shadowMapSize:new
vt,shadowCameraNear:1,shadowCameraFar:1e3}}return t[e.id]=n,n}}(),r=
{version:0,hash:
{directionalLength:-1,pointLength:-1,spotLength:-1,rectAreaLength:-1,hemiLength:-1,
numDirectionalShadows:-1,numPointShadows:-1,numSpotShadows:-1},ambient:

```

```

[0,0,0],probe:[],directional:[],directionalShadow:[],directionalShadowMap:
[],directionalShadowMatrix:[],spot:[],spotShadow:[],spotShadowMap:
[],spotShadowMatrix:[],rectArea:[],rectAreaLTC1:null,rectAreaLTC2:null,point:
[],pointShadow:[],pointShadowMap:[],pointShadowMatrix:[],hemi:[]};for(let
t=0;t<9;t++)r.probe.push(new Lt);const s=new Lt,a=new se,o=new
se;return{setup:function(s){let a=0,o=0,l=0;for(let
t=0;t<9;t++)r.probe[t].set(0,0,0);let
c=0,h=0,u=0,d=0,p=0,m=0,f=0,g=0;s.sort(ss);for(let t=0,e=s.length;t<e;t++){const
e=s[t],v=e.color,y=e.intensity,x=e.distance,_=e.shadow&&e.shadow.map?
e.shadow.map.texture:null;if(e.isAmbientLight)a+=v.r*y,o+=v.g*y,l+=v.b*y;else
if(e.isLightProbe)for(let
t=0;t<9;t++)r.probe[t].addScaledVector(e.sh.coefficients[t],y);else
if(e.isDirectionalLight){const
t=n.get(e);if(t.color.copy(e.color).multiplyScalar(e.intensity),e.castShadow){const
t=e.shadow,n=i.get(e);n.shadowBias=t.bias,n.shadowNormalBias=t.normalBias,n.shadowR
adius=t.radius,n.shadowMapSize=t.mapSize,r.directionalShadow[c]=n,r.directionalShad
owMap[c]=_,r.directionalShadowMatrix[c]=e.shadow.matrix,m++}r.directional[c]=t,c++}
else if(e.isSpotLight){const
t=n.get(e);if(t.position.setFromMatrixPosition(e.matrixWorld),t.color.copy(v).multi
plyScalar(y),t.distance=x,t.coneCos=Math.cos(e.angle),t.penumbraCos=Math.cos(e angl
e*(1-e.penumbra)),t.decay=e.decay,e.castShadow){const
t=e.shadow,n=i.get(e);n.shadowBias=t.bias,n.shadowNormalBias=t.normalBias,n.shadowR
adius=t.radius,n.shadowMapSize=t.mapSize,r.spotShadow[u]=n,r.spotShadowMap[u]=_,r.s
potShadowMatrix[u]=e.shadow.matrix,g++}r.spot[u]=t,u++}else if(e.isRectAreaLight)
{const
t=n.get(e);t.color.copy(v).multiplyScalar(y),t.halfWidth.set(.5*e.width,0,0),t.half
Height.set(0,.5*e.height,0),r.rectArea[d]=t,d++}else if(e.isPointLight){const
t=n.get(e);if(t.color.copy(e.color).multiplyScalar(e.intensity),t.distance=e.distan
ce,t.decay=e.decay,e.castShadow){const
t=e.shadow,n=i.get(e);n.shadowBias=t.bias,n.shadowNormalBias=t.normalBias,n.shadowR
adius=t.radius,n.shadowMapSize=t.mapSize,n.shadowCameraNear=t.camera.near,n.shadowC
ameraFar=t.camera.far,r.pointShadow[h]=n,r.pointShadowMap[h]=_,r.pointShadowMatrix[
h]=e.shadow.matrix,f++}r.point[h]=t,h++}else if(e.isHemisphereLight){const
t=n.get(e);t.skyColor.copy(e.color).multiplyScalar(y),t.groundColor.copy(e.groundCo
lor).multiplyScalar(y),r.hemi[p]=t,p++}}d>0&&
(e.isWebGL2||!0===t.has("OES_texture_float_linear"))?
(r.rectAreaLTC1=ui.LTC_FLOAT_1,r.rectAreaLTC2=ui.LTC_FLOAT_2):!0===t.has("OES_textu
re_half_float_linear"?
(r.rectAreaLTC1=ui.LTC_HALF_1,r.rectAreaLTC2=ui.LTC_HALF_2):console.error("THREE.We
bGLRenderer: Unable to use RectAreaLight. Missing WebGL
extensions.")),r.ambient[0]=a,r.ambient[1]=o,r.ambient[2]=l;const
v=r.hash;v.directionalLength===c&&v.pointLength===h&&v.spotLength===u&&v.rectAreaLe
ngth===d&&v.hemiLength===p&&v.numDirectionalShadows===m&&v.numPointShadows===f&&v.n
umSpotShadows===g||
(r.directional.length=c,r.spot.length=u,r.rectArea.length=d,r.point.length=h,r.hemi
.length=p,r.directionalShadow.length=m,r.directionalShadowMap.length=m,r.pointShado
w.length=f,r.pointShadowMap.length=f,r.spotShadow.length=g,r.spotShadowMap.length=g
,r.directionalShadowMatrix.length=m,r.pointShadowMatrix.length=f,r.spotShadowMatrix
.length=g,v.directionalLength=c,v.pointLength=h,v.spotLength=u,v.rectAreaLength=d,v
.hemiLength=p,v.numDirectionalShadows=m,v.numPointShadows=f,v.numSpotShadows=g,r.ve
rsion=rs++}},setupView:function(t,e){let n=0,i=0,l=0,c=0,h=0;const
u=e.matrixWorldInverse;for(let e=0,d=t.length;e<d;e++){const
d=t[e];if(d.isDirectionalLight){const
t=r.directional[n];t.direction.setFromMatrixPosition(d.matrixWorld),s.setFromMatrix
Position(d.target.matrixWorld),t.direction.sub(s),t.direction.transformDirection(u)
,n++}else if(d.isSpotLight){const
t=r.spot[l];t.position.setFromMatrixPosition(d.matrixWorld),t.position.applyMatrix4
(u),t.direction.setFromMatrixPosition(d.matrixWorld),s.setFromMatrixPosition(d.targ

```



```
et.matrixWorld),t.direction.sub(s),t.direction.transformDirection(u),l++)}else if(d.isRectAreaLight){const t=r.rectArea[c];t.position.setFromMatrixPosition(d.matrixWorld),t.position.applyMatrix4(u),o.identity(),a.copy(d.matrixWorld),a.premultiply(u),o.extractRotation(a),t.halfWidth.set(.5*d.width,0,0),t.halfHeight.set(0,.5*d.height,0),t.halfWidth.applyMatrix4(o),t.halfHeight.applyMatrix4(o),c++)}else if(d.isPointLight){const t=r.point[i];t.position.setFromMatrixPosition(d.matrixWorld),t.position.applyMatrix4(u),i++}else if(d.isHemisphereLight){const t=r.hemi[h];t.direction.setFromMatrixPosition(d.matrixWorld),t.direction.transformDirection(u),t.direction.normalize(),h++}}},state:r}}function os(t,e){const n=new as(t,e),i=[],r=[];return{init:function(){i.length=0,r.length=0},state:{lightsArray:i,shadowsArray:r,lights:n},setupLights:function() {n.setup(i),setupLightsView:function(t){n.setupView(i,t)},pushLight:function(t){i.push(t)},pushShadow:function(t){r.push(t)}}function ls(t,e){let n=new WeakMap;return{get:function(i,r=0){let s;return!l===n.has(i)?(s=new os(t,e),n.set(i,[s])):r>n.get(i).length?(s=new os(t,e),n.get(i).push(s)):s=n.get(i)[r],s},dispose:function(){n=new WeakMap}}}}class cs extends Xe{constructor(t){super(),this.type="MeshDepthMaterial",this.depthPacking=3200,this.skinning=!1,this.morphTargets=!1,this.map=null,this.alphaMap=null,this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,this.wireframe=!1,this.wireframeLinewidth=1,this.fog=!1,this.setValues(t)}copy(t){return super.copy(t),this.depthPacking=t.depthPacking,this.skinning=t.skinning,this.morphTargets=t.morphTargets,this.map=t.map,this.alphaMap=t.alphaMap,this.displacementMap=t.displacementMap,this.displacementScale=t.displacementScale,this.displacementBias=t.displacementBias,this.wireframe=t.wireframe,this.wireframeLinewidth=t.wireframeLinewidth,this}}cs.prototype.isMeshDepthMaterial=!0;class hs extends Xe{constructor(t){super(),this.type="MeshDistanceMaterial",this.referencePosition=new Lt,this.nearDistance=1,this.farDistance=1e3,this.skinning=!1,this.morphTargets=!1,this.map=null,this.alphaMap=null,this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,this.fog=!1,this.setValues(t)}copy(t){return super.copy(t),this.referencePosition.copy(t.referencePosition),this.nearDistance=t.nearDistance,this.farDistance=t.farDistance,this.skinning=t.skinning,this.morphTargets=t.morphTargets,this.map=t.map,this.alphaMap=t.alphaMap,this.displacementMap=t.displacementMap,this.displacementScale=t.displacementScale,this.displacementBias=t.displacementBias,this}}hs.prototype.isMeshDistanceMaterial=!0;function us(t,e,n){let i=new ai;const r=new vt,s=new vt,a=new St,o=[],l=[],c={},h=n.maxTextureSize,u={0:1,1:0,2:2},d=new Jn({defines:{SAMPLE_RATE:2/8,HALF_SAMPLE_RATE:1/8},uniforms:{shadow_pass:{value:null},resolution:{value:new vt},radius:{value:4}},vertexShader:"void main() {\n\tgl_Position = vec4( position, 1.0 );\n}",fragmentShader:"uniform sampler2D shadow_pass;\n\nuniform vec2 resolution;\n\nuniform float radius;\n\n#include <packing>\n\nvoid main() {\n\tfloat mean = 0.0;\n\tfloat squared_mean = 0.0;\n\tfloat depth = unpackRGBAToDepth( texture2D( shadow_pass, ( gl_FragCoord.xy ) / resolution ) );\n\tfor ( float i = -1.0; i < 1.0 ; i += SAMPLE_RATE) {\n\t\t#ifdef HORIZONTAL_PASS\n\t\t\tvec2 distribution = unpackRGBAToHalf( texture2D( shadow_pass, ( gl_FragCoord.xy + vec2( i, 0.0 ) * radius ) / resolution ) );\n\t\tmean += distribution.x;\n\t\squared_mean += distribution.y * distribution.y + distribution.x * distribution.x;\n\t\t#else\n\t\t\tfloat depth = unpackRGBAToDepth( texture2D( shadow_pass, ( gl_FragCoord.xy + vec2( 0.0, i ) * radius ) / resolution ) );\n\t\tmean += depth;\n\t\t\squared_mean += depth * depth;\n\t\t#endif\n\t}\n\tmean = mean * HALF_SAMPLE_RATE;\n\t\squared_mean = squared_mean * HALF_SAMPLE_RATE;\n\tfloat std_dev = sqrt( squared_mean - mean * mean );\n\tgl_FragColor = pack2HalfToRGBA( vec2( mean, std_dev ) );\n}"},m=d.clone();m.defines.HORIZONTAL_PASS=1;const f=new En,f.setAttribute("position",new sn(new Float32Array([-1,-1,.5,3,-1,.5,-1,3,.5]),3));const v=new Wn(f,d),y=this,function
```

```

x(n,i){const
r=e.update(v);d.uniforms.shadow_pass.value=n.map.texture,d.uniforms.resolution.valu
e=n.mapSize,d.uniforms.radius.value=n.radius,t.setRenderTarget(n.mapPass),t.clear()
,t.renderBufferDirect(i,null,r,d,v,null),m.uniforms.shadow_pass.value=n.mapPass.tex
ture,m.uniforms.resolution.value=n.mapSize,m.uniforms.radius.value=n.radius,t.setRe
nderTarget(n.map),t.clear(),t.renderBufferDirect(i,null,r,m,v,null)}function
_(t,e,n){const i=t<<0|e<<1|n<<2;let r=o[i];return void 0===r&&(r=new
cs({depthPacking:3201,morphTargets:t,skinning:e}),o[i]=r),r}function w(t,e,n){const
i=t<<0|e<<1|n<<2;let r=l[i];return void 0===r&&(r=new
hs({morphTargets:t,skinning:e}),l[i]=r),r}function b(e,n,i,r,s,a,o){let
l=null,h=_,d=e.customDepthMaterial;if(!0===r.isPointLight&&
(h=w,d=e.customDistanceMaterial),void 0===d){let t=!1;!0===i.morphTargets&&
(t=n.morphAttributes&&n.morphAttributes.position&&n.morphAttributes.position.length
>0);let r=!1;!0===e.isSkinnedMesh&&(!0===i.skinning?
r=!0:console.warn("THREE.WebGLShadowMap: THREE.SkinnedMesh with material.skinning
set to false:",e));l=h(t,r,!0===e.isInstancedMesh)}else
l=d;if(t.localClippingEnabled&&!0===i.clipShadows&&0!==i.clippingPlanes.length)
{const t=l.uuid,e=i.uuid;let n=c[t];void 0===n&&(n={},c[t]=n);let r=n[e];void
0===r&&(r=l.clone(),n[e]=r),l=r}return
l.visible=i.visible,l.wireframe=i.wireframe,l.side=3===o?null!==i.shadowSide?
i.shadowSide:i.side:null!==i.shadowSide?
i.shadowSide:u[i.side],l.clipShadows=i.clipShadows,l.clippingPlanes=i.clippingPlane
s,l.clipIntersection=i.clipIntersection,l.wireframeLinewidth=i.wireframeLinewidth,l
.linewidth=i.linewidth,!0===r.isPointLight&&!0===l.isMeshDistanceMaterial&&
(l.referencePosition.setFromMatrixPosition(r.matrixWorld),l.nearDistance=s,l.farDis
tance=a),l}function M(n,r,s,a,o)
{if(!1===n.visible)return;if(n.layers.test(r.layers)&&
(n.isMesh||n.isLine||n.isPoints)&&(n.castShadow||n.receiveShadow&&3===o)&&
(!n.frustumCulled||i.intersectsObject(n)))
{n.modelViewMatrix.multiplyMatrices(s.matrixWorldInverse,n.matrixWorld);const
i=e.update(n),r=n.material;if(Array.isArray(r)){const e=i.groups;for(let
l=0,c=e.length;l<c;l++){const c=e[l],h=r[c.materialIndex];if(h&&h.visible){const
e=b(n,i,h,a,s.near,s.far,o);t.renderBufferDirect(s,null,i,e,n,c)}}}else
if(r.visible){const
e=b(n,i,r,a,s.near,s.far,o);t.renderBufferDirect(s,null,i,e,n,null)}const
l=n.children;for(let
t=0,e=l.length;t<e;t++)M(l[t],r,s,a,o)}this.enabled=!1,this.autoUpdate=!0,this.need
sUpdate=!1,this.type=1,this.render=function(e,n,o)
{if(!1===y.enabled)return;if(!1===y.autoUpdate&&!1===y.needsUpdate)return;if(0===e.
length)return;const
l=t.getRenderTarget(),c=t.getActiveCubeFace(),u=t.getActiveMipmapLevel(),d=t.state;
d.setBlending(0),d.buffers.color.setClear(1,1,1,1),d.buffers.depth.setTest(!0),d.se
tScissorTest(!1);for(let l=0,c=e.length;l<c;l++){const c=e[l],u=c.shadow;if(void
0===u){console.warn("THREE.WebGLShadowMap:",c,"has no
shadow.");continue}if(!1===u.autoUpdate&&!1===u.needsUpdate)continue;r.copy(u.mapSi
ze);const m=u.getFrameExtents();if(r.multiply(m),s.copy(u.mapSize),(r.x>h||r.y>h)&&
(r.x>h&&(s.x=Math.floor(h/m.x),r.x=s.x*m.x,u.mapSize.x=s.x),r.y>h&&
(s.y=Math.floor(h/m.y),r.y=s.y*m.y,u.mapSize.y=s.y)),null===u.map&&!u.isPointLightS
hadow&&3===this.type){const t={minFilter:g,magFilter:g,format:E};u.map=new
Tt(r.x,r.y,t),u.map.texture.name=c.name+".shadowMap",u.mapPass=new
Tt(r.x,r.y,t),u.camera.updateProjectionMatrix()}if(null===u.map){const t=
{minFilter:p,magFilter:p,format:E};u.map=new
Tt(r.x,r.y,t),u.map.texture.name=c.name+".shadowMap",u.camera.updateProjectionMatri
x()}t.setRenderTarget(u.map),t.clear();const f=u.getViewportCount();for(let
t=0;t<f;t++){const
e=u.getViewport(t);a.set(s.x*e.x,s.y*e.y,s.x*e.z,s.y*e.w),d.viewport(a),u.updateMat
rices(c,t),i=u.getFrustum(),M(n,o,u.camera,c,this.type)}u.isPointLightShadow||3!===t
his.type||x(u,o),u.needsUpdate=!1}y.needsUpdate=!1,t.setRenderTarget(l,c,u)}}functi

```

```

on ds(t,e,i){const r=i.isWebGL2;const s=new function(){let e=!1;const n=new St;let
i=null;const r=new St(0,0,0,0);return{setMask:function(n){i===n||e||
(t.colorMask(n,n,n,n),i=n)},setLocked:function(t){e=t},setClear:function(e,i,s,a,o)
{!0===o&&(e*=a,i*=a,s*=a),n.set(e,i,s,a),!1===r.equals(n)&&
(t.clearColor(e,i,s,a),r.copy(n))},reset:function()
{e=!1,i=null,r.set(-1,0,0,0)}}},a=new function(){let
e=!1,n=null,i=null,r=null;return{setTest:function(t){t?
z(2929):F(2929)},setMask:function(i){n===i||e||
(t.depthMask(i),n=i)},setFunc:function(e){if(i!==e){if(e)switch(e){case
0:t.depthFunc(512);break;case 1:t.depthFunc(519);break;case
2:t.depthFunc(513);break;case 3:t.depthFunc(515);break;case
4:t.depthFunc(514);break;case 5:t.depthFunc(518);break;case
6:t.depthFunc(516);break;case
7:t.depthFunc(517);break;default:t.depthFunc(515)}}else
t.depthFunc(515);i=e}},setLocked:function(t){e=t},setClear:function(e){r!==e&&
(t.clearDepth(e),r=e)},reset:function(){e=!1,n=null,i=null,r=null}}},o=new
function(){let
e=!1,n=null,i=null,r=null,s=null,a=null,o=null,l=null,c=null;return{setTest:functio
n(t){e||t?z(2960):F(2960)},setMask:function(i){n===i||e||
(t.stencilMask(i),n=i)},setFunc:function(e,n,a){i===e&&r===n&&s===a||
(t.stencilFunc(e,n,a),i=e,r=n,s=a)},setOp:function(e,n,i){a===e&&o===n&&l===i||
(t.stencilOp(e,n,i),a=e,o=n,l=i)},setLocked:function(t){e=t},setClear:function(e)
{c!==e&&(t.clearStencil(e),c=e)},reset:function()
{e=!1,n=null,i=null,r=null,s=null,a=null,o=null,l=null,c=null}}};let l={},c=null,h=
{},u=null,d=!1,p=null,m=null,f=null,g=null,v=null,y=null,x=null,_=!1,w=null,b=null,
M=null,S=null,T=null;const E=t.getParameter(35661);let A=!1,L=0;const
R=t.getParameter(7938);-1!==R.indexOf("WebGL")?(L=parseFloat(/^WebGL (\d)/.exec(R)
[1]),A=L>=1):-1!==R.indexOf("OpenGL ES")&&(L=parseFloat(/^OpenGL ES (\d)/.exec(R)
[1]),A=L>=2);let C=null,P={};const D=new
St(0,0,t.canvas.width,t.canvas.height),I=new
St(0,0,t.canvas.width,t.canvas.height);function N(e,n,i){const r=new
Uint8Array(4),s=t.createTexture();t.bindTexture(e,s),t.texParameteri(e,10241,9728),
t.texParameteri(e,10240,9728);for(let
e=0;e<i;e++)t.texImage2D(n+e,0,6408,1,1,0,6408,5121,r);return s}const B={};function
z(e){!0!==l[e]&&(t.enable(e),l[e]=!0)}function F(e){!1!==l[e]&&
(t.disable(e),l[e]=!1)}B[3553]=N(3553,3553,1),B[34067]=N(34067,34069,6),s.setClear(
0,0,0,1),a.setClear(1),o.setClear(0),z(2929),a.setFunc(3),U(!1),k(1),z(2884),G(0);c
onst O=[n:32774,101:32778,102:32779];if(r)O[103]=32775,O[104]=32776;else{const
t=e.get("EXT_blend_minmax");null!==t&&(O[103]=t.MIN_EXT,O[104]=t.MAX_EXT)}const H=
{200:0,201:1,202:768,204:770,210:776,208:774,206:772,203:769,205:771,209:775,207:77
3};function G(e,i,r,s,a,o,l,c){if(0!==e){if(!l===d&&
(z(3042),d=!0),5===e)a=a||i,o=o||r,l=l||s,i===m&&a===v||
(t.blendEquationSeparate(O[i],O[a]),m=i,v=a),r===f&&s===g&&o===y&&l===x||
(t.blendFuncSeparate(H[r],H[s],H[o],H[l]),f=r,g=s,y=o,x=l),p=e,_=null;else
if(e!==p||c!==_){if(m===n&&v===n||t.blendEquation(32774),m=n,v=n),c)switch(e){case
1:t.blendFuncSeparate(1,771,1,771);break;case 2:t.blendFunc(1,1);break;case
3:t.blendFuncSeparate(0,0,769,771);break;case
4:t.blendFuncSeparate(0,768,0,770);break;default:console.error("THREE.WebGLState:
Invalid blending: ",e)}else switch(e){case
1:t.blendFuncSeparate(770,771,1,771);break;case 2:t.blendFunc(770,1);break;case
3:t.blendFunc(0,769);break;case
4:t.blendFunc(0,768);break;default:console.error("THREE.WebGLState: Invalid
blending: ",e)}f=null,g=null,y=null,x=null,p=e,_=c}}else!0===d&&
(F(3042),d=!1)}function U(e){w!==e&&(e?
t.frontFace(2304):t.frontFace(2305),w=e)}function k(e){0!==e?(z(2884),e!==b&&
(1===e?t.cullFace(1029):2===e?
t.cullFace(1028):t.cullFace(1032))):F(2884),b=e}function V(e,n,i){e?
(z(32823),S===n&&T===i||t.polygonOffset(n,i),S=n,T=i):F(32823)}function W(e){void

```

```

0===e&&(e=33984+E-1),C!==e&&(t.activeTexture(e),C=e)}return{buffers:
{color:s,depth:a,stencil:o},enable:z,disable:F,bindFramebuffer:function(e,n)
{null===n&&null!==c&&(n=c),h[e]!==n&&(t.bindFramebuffer(e,n),h[e]=n,r&&(36009===e&&
(h[36160]=n),36160===e&&(h[36009]=n)))},bindXRFramebuffer:function(e){e!==c&&
(t.bindFramebuffer(36160,e),c=e)},useProgram:function(e){return u!==e&&
(t.useProgram(e),u=e,!0)},setBlending:G,setMaterial:function(t,e){2===t.side?
F(2884):z(2884);let n=1===t.side;e&&(n=!n),U(n),1===t.blending&&!1===t.transparent?
G(0):G(t.blending,t.blendEquation,t.blendSrc,t.blendDst,t.blendEquationAlpha,t.blendSrcAlpha,t.blendDstAlpha,t.premultipliedAlpha),a.setFunc(t.depthFunc),a.setTest(t.depthTest),a.setMask(t.depthWrite),s.setMask(t.colorWrite);const
i=t.stencilWrite;o.setTest(i),i&&
(o.setMask(t.stencilWriteMask),o.setFunc(t.stencilFunc,t.stencilRef,t.stencilFuncMask),o.setOp(t.stencilFail,t.stencilZFail,t.stencilZPass)),V(t.polygonOffset,t.polygonOffsetFactor,t.polygonOffsetUnits),!0===t.alphaToCoverage?
z(32926):F(32926)},setFlipSided:U,setCullFace:k,setLineWidth:function(e){e!==M&&
(A&&t.lineWidth(e),M=e)},setPolygonOffset:V,setScissorTest:function(t){t?
z(3089):F(3089)},activeTexture:W,bindTexture:function(e,n){null===C&&W();let
i=P[C];void 0===i&&(i={type:void 0,texture:void 0},P[C]=i),i.type===e&&i.texture===n||
(t.bindTexture(e,n||B[e]),i.type=e,i.texture=n)},unbindTexture:function(){const
e=P[C];void 0!==e&&void 0!==e.type&&(t.bindTexture(e.type,null),e.type=void 0,e.texture=void 0)},compressedTexImage2D:function()
{try{t.compressedTexImage2D.apply(t,arguments)}catch(t)
{console.error("THREE.WebGLState:",t)}},texImage2D:function()
{try{t.texImage2D.apply(t,arguments)}catch(t)
{console.error("THREE.WebGLState:",t)}},texImage3D:function()
{try{t.texImage3D.apply(t,arguments)}catch(t)
{console.error("THREE.WebGLState:",t)}},scissor:function(e){!1===D.equals(e)&&
(t.scissor(e.x,e.y,e.z,e.w),D.copy(e)),viewport:function(e){!1===I.equals(e)&&
(t.viewport(e.x,e.y,e.z,e.w),I.copy(e))},reset:function()
{t.disable(3042),t.disable(2884),t.disable(2929),t.disable(32823),t.disable(3089),t.disable(2960),t.disable(32926),t.blendEquation(32774),t.blendFunc(1,0),t.blendFuncSeparate(1,0,1,0),t.colorMask(!0,!0,!0,!0),t.clearColor(0,0,0,0),t.depthMask(!0),t.depthFunc(513),t.clearDepth(1),t.stencilMask(4294967295),t.stencilFunc(519,0,4294967295),t.stencilOp(7680,7680,7680),t.clearStencil(0),t.cullFace(1029),t.frontFace(2305),t.polygonOffset(0,0),t.activeTexture(33984),t.bindFramebuffer(36160,null),!0===r&&
(t.bindFramebuffer(36009,null),t.bindFramebuffer(36008,null)),t.useProgram(null),t.lineWidth(1),t.scissor(0,0,t.canvas.width,t.canvas.height),t.viewport(0,0,t.canvas.width,t.canvas.height),l={},C=null,P={},c=null,h={},u=null,d=!1,p=null,m=null,f=null,g=null,v=null,y=null,x=null,_=!1,w=null,b=null,M=null,S=null,T=null,D.set(0,0,t.canvas.width,t.canvas.height),I.set(0,0,t.canvas.width,t.canvas.height),s.reset(),a.reset(),o.reset())}}function ps(t,e,n,i,r,s,a)
{const
o=r.isWebGL2,l=r.maxTextures,c=r.maxCubemapSize,x=r.maxTextureSize,R=r.maxSamples,C=new WeakMap;let P,D=!1;try{D="undefined"!==typeof OffscreenCanvas&&null!==new OffscreenCanvas(1,1).getContext("2d")}catch(t){}function I(t,e){return D?new OffscreenCanvas(t,e):document.createElementNS("http://www.w3.org/1999/xhtml","canvas")}function N(t,e,n,i){let r=1;if((t.width>i||t.height>i)&&
(r=i/Math.max(t.width,t.height)),r<1||!0===e){if("undefined"!==typeof HTMLImageElement&&t instanceof HTMLImageElement||"undefined"!==typeof HTMLCanvasElement&&t instanceof HTMLCanvasElement||"undefined"!==typeof ImageBitmap&&t instanceof ImageBitmap){const i=e?
ft:Math.floor,s=i(r*t.width),a=i(r*t.height);void 0===P&&(P=I(s,a));const o=n?
I(s,a):P;o.width=s,o.height=a;return
o.getContext("2d").drawImage(t,0,0,s,a),console.warn("THREE.WebGLRenderer: Texture has been resized from ("+t.width+"x"+t.height+") to ("+s+"x"+a+")."),o}return"data" in t&&console.warn("THREE.WebGLRenderer: Image in

```

```

DataTexture is too big ("++t.width+"x"+t.height+")."),t}return t}function B(t)
{return pt(t.width)&&pt(t.height)}function z(t,e){return
t.generateMipmaps&&t.minFilter!==p&&t.minFilter!==g}function F(e,n,r,s)
{t.generateMipmap(e);i.get(n).__maxMipLevel=Math.log2(Math.max(r,s))}function
O(n,i,r){if(!l===o)return i;if(null!==n){if(void 0!==t[n])return
t[n];console.warn("THREE.WebGLRenderer: Attempt to use non-existing WebGL internal
format '"+n+"'")}}let s=i;return 6403===i&&(5126===r&&(s=33326),5131===r&&
(s=33325),5121===r&&(s=33321)),6407===i&&(5126===r&&(s=34837),5131===r&&
(s=34843),5121===r&&(s=32849)),6408===i&&(5126===r&&(s=34836),5131===r&&
(s=34842),5121===r&&
(s=32856)),33325!==s&&33326!==s&&34842!==s&&34836!==s||e.get("EXT_color_buffer_floa
t"),s}function H(t){return t===p||t===m||t===f?9728:9729}function G(e){const
n=e.target;n.removeEventListener("dispose",G),function(e){const n=i.get(e);if(void
0===n.__webglInit)return;t.deleteTexture(n.__webglTexture),i.remove(e)}
(n),n.isVideoTexture&&C.delete(n),a.memory.textures--}function U(e){const
n=e.target;n.removeEventListener("dispose",U),function(e){const
n=e.texture,r=i.get(e),s=i.get(n);if(!e)return;void
0!==s.__webglTexture&&t.deleteTexture(s.__webglTexture);e.depthTexture&&e.depthText
ure.dispose();if(e.isWebGLCubeRenderTarget)for(let
e=0;e<6;e++)t.deleteFramebuffer(r.__webglFramebuffer[e]),r.__webglDepthbuffer&&t.de
leteRenderbuffer(r.__webglDepthbuffer[e]);else
t.deleteFramebuffer(r.__webglFramebuffer),r.__webglDepthbuffer&&t.deleteRenderbuffe
r(r.__webglDepthbuffer),r.__webglMultisampledFramebuffer&&t.deleteFramebuffer(r.__w
ebglMultisampledFramebuffer),r.__webglColorRenderbuffer&&t.deleteRenderbuffer(r.__w
ebglColorRenderbuffer),r.__webglDepthRenderbuffer&&t.deleteRenderbuffer(r.__webglDe
pthRenderbuffer);i.remove(n),i.remove(e)}(n),a.memory.textures--}let k=0;function
V(t,e){const r=i.get(t);if(t.isVideoTexture&&function(t){const
e=a.render.frame;C.get(t)!==e&&(C.set(t,e),t.update())}
(t),t.version>0&&r.__version!==t.version){const n=t.image;if(void
0===n)console.warn("THREE.WebGLRenderer: Texture marked for update but image is
undefined");else if(!l===n.complete)return void
Z(r,t,e);console.warn("THREE.WebGLRenderer: Texture marked for update but image is
incomplete")}}n.activeTexture(33984+e),n.bindTexture(3553,r.__webglTexture)}functio
n W(e,r){const a=i.get(e);e.version>0&&a.__version!==e.version?function(e,i,r)
{if(6!==i.image.length)return;Y(e,i),n.activeTexture(33984+r),n.bindTexture(34067,e
.__webglTexture),t.pixelStorei(37440,i.flipY),t.pixelStorei(37441,i.premultiplyAlph
a),t.pixelStorei(3317,i.unpackAlignment),t.pixelStorei(37443,0);const a=i&&
(i.isCompressedTexture||i.image[0].isCompressedTexture),l=i.image[0]&&i.image[0].is
DataTexture,h=[];for(let t=0;t<6;t++)h[t]=a||l?l?
i.image[t].image:i.image[t]:N(i.image[t],!l,!0,c);const
u=h[0],d=B(u)||o,p=s.convert(i.format),m=s.convert(i.type),f=0(i.internalFormat,p,m
);let g;if(X(34067,i,d),a){for(let t=0;t<6;t++){g=h[t].mipmaps;for(let
e=0;e<g.length;e++){const r=g[e];i.format!==E&&i.format!==T?null!==p?
n.compressedTexImage2D(34069+t,e,f,r.width,r.height,0,r.data):console.warn("THREE.W
ebGLRenderer: Attempt to load unsupported compressed texture format in
.setTextureCube()"):n.texImage2D(34069+t,e,f,r.width,r.height,0,p,m,r.data)}}e.__ma
xMipLevel=g.length-1}else{g=i.mipmaps;for(let t=0;t<6;t++)if(l)
{n.texImage2D(34069+t,0,f,h[t].width,h[t].height,0,p,m,h[t].data);for(let
e=0;e<g.length;e++){const
i=g[e].image[t].image;n.texImage2D(34069+t,e+1,f,i.width,i.height,0,p,m,i.data)}}el
se{n.texImage2D(34069+t,0,f,p,m,h[t]);for(let e=0;e<g.length;e++){const
i=g[e];n.texImage2D(34069+t,e+1,f,p,m,i.image[t])}}e.__maxMipLevel=g.length}z(i,d)&
&F(34067,i,u.width,u.height);e.__version=i.version,i.onUpdate&&i.onUpdate(i)}
(a,e,r):(n.activeTexture(33984+r),n.bindTexture(34067,a.__webglTexture))}const j=
{[h]:10497,[u]:33071,[d]:33648},q={[p]:9728,[m]:9984,[f]:9986,[g]:9729,[v]:9985,
[y]:9987};function X(n,s,a){if(a?
(t.texParameteri(n,10242,j[s.wrapS]),t.texParameteri(n,10243,j[s.wrapT]),32879!==n&
&35866!==n||t.texParameteri(n,32882,j[s.wrapR]),t.texParameteri(n,10240,q[s.magFilt

```

```

er]),t.texParameteri(n,10241,q[s.minFilter])):
(t.texParameteri(n,10242,33071),t.texParameteri(n,10243,33071),32879!==(n&&35866!==(n
||t.texParameteri(n,32882,33071),s.wrapS===u&&s.wrapT===u||console.warn("THREE.WebG
LRenderer: Texture is not power of two. Texture.wrapS and Texture.wrapT should be
set to
THREE.ClampToEdgeWrapping."),t.texParameteri(n,10240,H(s.magFilter)),t.texParameter
i(n,10241,H(s.minFilter)),s.minFilter!==p&&s.minFilter!==g&&console.warn("THREE.Web
GLRenderer: Texture is not power of two. Texture.minFilter should be set to
THREE.NearestFilter or
THREE.LinearFilter.")),!0===e.has("EXT_texture_filter_anisotropic")){const
a=e.get("EXT_texture_filter_anisotropic");if(s.type===b&&!l===e.has("OES_texture_fl
oat_linear"))return;if(!l===o&&s.type===M&&!l===e.has("OES_texture_half_float_linea
r"))return;(s.anisotropy>1||i.get(s).__currentAnisotropy)&&
(t.texParameterf(n,a.TEXTURE_MAX_ANISOTROPY_EXT,Math.min(s.anisotropy,r.getMaxAniso
tropy())),i.get(s).__currentAnisotropy=s.anisotropy)}}function Y(e,n){void
0===e.__webglInit&&
(e.__webglInit=!0,n.addEventListener("dispose",G),e.__webglTexture=t.createTexture(
),a.memory.textures++)}function Z(e,i,r){let a=3553;i.isDataTexture2DArray&&
(a=35866),i.isDataTexture3D&&
(a=32879),Y(e,i),n.activeTexture(33984+r),n.bindTexture(a,e.__webglTexture),t.pixel
Storei(37440,i.flipY),t.pixelStorei(37441,i.premultiplyAlpha),t.pixelStorei(3317,i.
unpackAlignment),t.pixelStorei(37443,0);const l=function(t){return!o&&
(t.wrapS!==u||t.wrapT!==u||t.minFilter!==p&&t.minFilter!==g)}
(i)&&!l===B(i.image),c=N(i.image,l,!1,x),h=B(c)||o,d=s.convert(i.format);let
m,f=s.convert(i.type),v=O(i.internalFormat,d,f);X(a,i,h);const
y=i.mipmaps;if(i.isDepthTexture)v=6402,o?v=i.type===b?36012:i.type===w?
33190:i.type===S?35056:33189:i.type===b&&console.error("WebGLRenderer: Floating
point depth texture requires
WebGL2."),i.format===A&&6402===v&&i.type!==_&&i.type!==w&&
(console.warn("THREE.WebGLRenderer: Use UnsignedShortType or UnsignedIntType for
DepthFormat DepthTexture."),i.type=_,f=s.convert(i.type)),i.format===L&&6402===v&&
(v=34041,i.type!==S&&(console.warn("THREE.WebGLRenderer: Use UnsignedInt248Type for
DepthStencilFormat
DepthTexture."),i.type=S,f=s.convert(i.type))),n.texImage2D(3553,0,v,c.width,c.heig
ht,0,d,f,null);else if(i.isDataTexture)if(y.length>0&&h){for(let
t=0,e=y.length;t<e;t++)m=y[t],n.texImage2D(3553,t,v,m.width,m.height,0,d,f,m.data);
i.generateMipmaps=!l,e.__maxMipLevel=y.length-1}else
n.texImage2D(3553,0,v,c.width,c.height,0,d,f,c.data),e.__maxMipLevel=0;else
if(i.isCompressedTexture){for(let
t=0,e=y.length;t<e;t++)m=y[t],i.format!==E&&i.format!==T?null!==d?
n.compressedTexImage2D(3553,t,v,m.width,m.height,0,m.data):console.warn("THREE.WebG
LRenderer: Attempt to load unsupported compressed texture format in
.uploadTexture()"):n.texImage2D(3553,t,v,m.width,m.height,0,d,f,m.data);e.__maxMipL
evel=y.length-1}else
if(i.isDataTexture2DArray)n.texImage3D(35866,0,v,c.width,c.height,c.depth,0,d,f,c.d
ata),e.__maxMipLevel=0;else
if(i.isDataTexture3D)n.texImage3D(32879,0,v,c.width,c.height,c.depth,0,d,f,c.data),
e.__maxMipLevel=0;else if(y.length>0&&h){for(let
t=0,e=y.length;t<e;t++)m=y[t],n.texImage2D(3553,t,v,d,f,m);i.generateMipmaps=!l,e.
__maxMipLevel=y.length-1}else
n.texImage2D(3553,0,v,d,f,c),e.__maxMipLevel=0;z(i,h)&&F(a,i,c.width,c.height),e.__
version=i.version,i.onUpdate&&i.onUpdate(i)}function J(e,r,a,o){const
l=r.texture,c=s.convert(l.format),h=s.convert(l.type),u=O(l.internalFormat,c,h);328
79===o||35866===o?
n.texImage3D(o,0,u,r.width,r.height,r.depth,0,c,h,null):n.texImage2D(o,0,u,r.width,
r.height,0,c,h,null),n.bindFramebuffer(36160,e),t.framebufferTexture2D(36160,a,o,i.
get(l).__webglTexture,0),n.bindFramebuffer(36160,null)}function Q(e,n,i)
{if(t.bindRenderbuffer(36161,e),n.depthBuffer&&!n.stencilBuffer){let r=33189;if(i)

```

```

{const e=n.depthTexture;e&&e.isDepthTexture&&(e.type===b?r=36012:e.type===w&&
(r=33190));const
i=$(n);t.renderbufferStorageMultisample(36161,i,r,n.width,n.height)}else
t.renderbufferStorage(36161,r,n.width,n.height);t.framebufferRenderbuffer(36160,360
96,36161,e)}else if(n.depthBuffer&&n.stencilBuffer){if(i){const
e=$(n);t.renderbufferStorageMultisample(36161,e,35056,n.width,n.height)}else
t.renderbufferStorage(36161,34041,n.width,n.height);t.framebufferRenderbuffer(36160
,33306,36161,e)}else{const
e=n.texture,r=s.convert(e.format),a=s.convert(e.type),o=0(e.internalFormat,r,a);if(
i){const e=$(n);t.renderbufferStorageMultisample(36161,e,o,n.width,n.height)}else
t.renderbufferStorage(36161,o,n.width,n.height)}t.bindRenderbuffer(36161,null)}func
tion K(e){const r=i.get(e),s=!0===e.isWebGLCubeRenderTarget;if(e.depthTexture)
{if(s)throw new Error("target.depthTexture not supported in Cube render
targets");!function(e,r){if(r&&r.isWebGLCubeRenderTarget)throw new Error("Depth
Texture with cube render targets is not
supported");if(n.bindFramebuffer(36160,e),!r.depthTexture||!r.depthTexture.isDepthT
exture)throw new Error("renderTarget.depthTexture must be an instance of
THREE.DepthTexture");i.get(r.depthTexture).__webglTexture&&r.depthTexture.image.wid
th===r.width&&r.depthTexture.image.height===r.height||
(r.depthTexture.image.width=r.width,r.depthTexture.image.height=r.height,r.depthTex
ture.needsUpdate=!0),V(r.depthTexture,0);const
s=i.get(r.depthTexture).__webglTexture;if(r.depthTexture.format===A)t.framebufferTe
xture2D(36160,36096,3553,s,0);else{if(r.depthTexture.format!==L)throw new
Error("Unknown depthTexture format");t.framebufferTexture2D(36160,33306,3553,s,0)}}
(r.__webglFramebuffer,e)}else if(s){r.__webglDepthbuffer=[];for(let
i=0;i<6;i++)n.bindFramebuffer(36160,r.__webglFramebuffer[i]),r.__webglDepthbuffer[i
]=t.createRenderbuffer(),Q(r.__webglDepthbuffer[i],e,!1)}else
n.bindFramebuffer(36160,r.__webglFramebuffer),r.__webglDepthbuffer=t.createRenderbu
ffer(),Q(r.__webglDepthbuffer,e,!1);n.bindFramebuffer(36160,null)}function $(t)
{return o&&t.isWebGLMultisampleRenderTarget?Math.min(R,t.samples):0}let
tt=!1,et=!1;this.allocateTextureUnit=function(){const t=k;return
t>=l&&console.warn("THREE.WebGLTextures: Trying to use "+t+" texture units while
this GPU supports only "+l,k+=1,t),this.resetTextureUnits=function()
{k=0},this.setTexture2D=V,this.setTexture2DArray=function(t,e){const
r=i.get(t);t.version>0&&r.__version!==t.version?Z(r,t,e):
(n.activeTexture(33984+e),n.bindTexture(35866,r.__webglTexture))},this.setTexture3D
=function(t,e){const r=i.get(t);t.version>0&&r.__version!==t.version?Z(r,t,e):
(n.activeTexture(33984+e),n.bindTexture(32879,r.__webglTexture))},this.setTextureCu
be=W,this.setupRenderTarget=function(e){const
r=e.texture,l=i.get(e),c=i.get(r);e.addEventListener("dispose",U),c.__webglTexture=
t.createTexture(),c.__version=r.version,a.memory.textures++;const
h=!0===e.isWebGLCubeRenderTarget,u=!0===e.isWebGLMultisampleRenderTarget,d=r.isData
Texture3D||r.isDataTexture2DArray,p=B(e)||o;if(!o||r.format!==T||r.type!==b&&r.type
!==M||r.format=E,console.warn("THREE.WebGLRenderer: Rendering to textures with RGB
format is not supported. Using RGBA format instead.")),h){l.__webglFramebuffer=
[];for(let e=0;e<6;e++)l.__webglFramebuffer[e]=t.createFramebuffer()}else
if(l.__webglFramebuffer=t.createFramebuffer(),u)if(o)
{l.__webglMultisampledFramebuffer=t.createFramebuffer(),l.__webglColorRenderbuffer=
t.createRenderbuffer(),t.bindRenderbuffer(36161,l.__webglColorRenderbuffer);const
i=s.convert(r.format),a=s.convert(r.type),o=0(r.internalFormat,i,a),c=$(e);t.render
bufferStorageMultisample(36161,c,o,e.width,e.height),n.bindFramebuffer(36160,l.__we
bglMultisampledFramebuffer),t.framebufferRenderbuffer(36160,36064,36161,l.__webglCo
lorRenderbuffer),t.bindRenderbuffer(36161,null),e.depthBuffer&&
(l.__webglDepthRenderbuffer=t.createRenderbuffer(),Q(l.__webglDepthRenderbuffer,e,!
0)),n.bindFramebuffer(36160,null)}else console.warn("THREE.WebGLRenderer:
WebGLMultisampleRenderTarget can only be used with WebGL2.");if(h)
{n.bindTexture(34067,c.__webglTexture),X(34067,r,p);for(let
t=0;t<6;t++)J(l.__webglFramebuffer[t],e,36064,34069+t);z(r,p)&&F(34067,r,e.width,e.

```

```

height),n.bindTexture(34067,null)}else{let t=3553;if(d)if(o){t=r.isDataTexture3D?
32879:35866}else console.warn("THREE.DataTexture3D and THREE.DataTexture2DArray
only supported with
WebGL2.");n.bindTexture(t,c.__webglTexture),X(t,r,p),J(1.__webglFramebuffer,e,36064
,t),z(r,p)&&F(3553,r,e.width,e.height),n.bindTexture(3553,null)}e.depthBuffer&&K(e)
},this.updateRenderTargetMipmap=function(t){const e=t.texture;if(z(e,B(t)||o))
{const r=t.isWebGLCubeRenderTarget?
34067:3553,s=i.get(e).__webglTexture;n.bindTexture(r,s),F(r,e,t.width,t.height),n.b
indTexture(r,null)}}},this.updateMultisampleRenderTarget=function(e)
{if(e.isWebGLMultisampleRenderTarget)if(o){const r=e.width,s=e.height;let
a=16384;e.depthBuffer&&(a|=256),e.stencilBuffer&&(a|=1024);const
o=i.get(e);n.bindFramebuffer(36008,o.__webglMultisampledFramebuffer),n.bindFramebuf
fer(36009,o.__webglFramebuffer),t.blitFramebuffer(0,0,r,s,0,0,r,s,a,9728),n.bindFra
mebuffer(36008,null),n.bindFramebuffer(36009,o.__webglMultisampledFramebuffer)}else
console.warn("THREE.WebGLRenderer: WebGLMultisampleRenderTarget can only be used
with WebGL2.")},this.safeSetTexture2D=function(t,e){t&&t.isWebGLRenderTarget&&
(!1===t&&(console.warn("THREE.WebGLTextures.safeSetTexture2D: don't use render
targets as textures. Use their .texture property
instead."),t=!0),t=t.texture),V(t,e)},this.safeSetTextureCube=function(t,e)
{t&&t.isWebGLCubeRenderTarget&&(!1===t&&
(console.warn("THREE.WebGLTextures.safeSetTextureCube: don't use cube render
targets as textures. Use their .texture property
instead."),t=!0),t=t.texture),W(t,e)}}function ms(t,e,n){const
i=n.isWebGL2;return{convert:function(t){let n;if(t===x)return
5121;if(1017===t)return 32819;if(1018===t)return 32820;if(1019===t)return
33635;if(1010===t)return 5120;if(1011===t)return 5122;if(t===_)return
5123;if(1013===t)return 5124;if(t===w)return 5125;if(t===b)return
5126;if(t===M)return i?5131:(n=e.get("OES_texture_half_float"),null!==(n?
n.HALF_FLOAT_OES:null));if(1021===t)return 6406;if(t===T)return 6407;if(t===E)return
6408;if(1024===t)return 6409;if(1025===t)return 6410;if(t===A)return
6402;if(t===L)return 34041;if(1028===t)return 6403;if(1029===t)return
36244;if(1030===t)return 33319;if(1031===t)return 33320;if(1032===t)return
36248;if(1033===t)return 36249;if(t===R||t===C||t===P||t===D)
{if(n=e.get("WEBGL_compressed_texture_s3tc"),null===n)return null;if(t===R)return
n.COMPRESSED_RGB_S3TC_DXT1_EXT;if(t===C)return
n.COMPRESSED_RGBA_S3TC_DXT1_EXT;if(t===P)return
n.COMPRESSED_RGBA_S3TC_DXT3_EXT;if(t===D)return
n.COMPRESSED_RGBA_S3TC_DXT5_EXT}if(t===I||t===N||t===B||t===z)
{if(n=e.get("WEBGL_compressed_texture_pvrtc"),null===n)return null;if(t===I)return
n.COMPRESSED_RGB_PVRTC_4BPPV1_IMG;if(t===N)return
n.COMPRESSED_RGB_PVRTC_2BPPV1_IMG;if(t===B)return
n.COMPRESSED_RGBA_PVRTC_4BPPV1_IMG;if(t===z)return
n.COMPRESSED_RGBA_PVRTC_2BPPV1_IMG}if(36196===t)return
n=e.get("WEBGL_compressed_texture_etc1"),null!==(n?
n.COMPRESSED_RGB_ETC1_WEBGL:null;if((t===F||t===O)&&
(n=e.get("WEBGL_compressed_texture_etc"),null!==(n)){if(t===F)return
n.COMPRESSED_RGB8_ETC2;if(t===O)return n.COMPRESSED_RGBA8_ETC2_EAC}return
37808===t||37809===t||37810===t||37811===t||37812===t||37813===t||37814===t||37815=
===t||37816===t||37817===t||37818===t||37819===t||37820===t||37821===t||37840===t||3
7841===t||37842===t||37843===t||37844===t||37845===t||37846===t||37847===t||37848==
=t||37849===t||37850===t||37851===t||37852===t||37853===t?
(n=e.get("WEBGL_compressed_texture_astc"),null!==(n?t:null):36492===t?
(n=e.get("EXT_texture_compression_bptc"),null!==(n?t:null):t===S?i?34042:
(n=e.get("WEBGL_depth_texture"),null!==(n?n.UNSIGNED_INT_24_8_WEBGL:null):void
0)}}}class fs extends Kn{constructor(t=[]){
super(),this.cameras=t}}fs.prototype.isArrayCamera=!0;class gs extends
Ce{constructor(){super(),this.type="Group"}}gs.prototype.isGroup=!0;const vs=
{type:"move"};class ys{constructor()

```



```

{this._targetRay=null,this._grip=null,this._hand=null}getHandSpace(){return
null===this._hand&&(this._hand=new
gs,this._hand.matrixAutoUpdate=!1,this._hand.visible=!1,this._hand.joints=
{}),this._hand.inputState={pinching:!1}),this._hand}getTargetRaySpace(){return
null===this._targetRay&&(this._targetRay=new
gs,this._targetRay.matrixAutoUpdate=!1,this._targetRay.visible=!1,this._targetRay.h
asLinearVelocity=!1,this._targetRay.linearVelocity=new
Lt,this._targetRay.hasAngularVelocity=!1,this._targetRay.angularVelocity=new
Lt),this._targetRay}getGripSpace(){return null===this._grip&&(this._grip=new
gs,this._grip.matrixAutoUpdate=!1,this._grip.visible=!1,this._grip.hasLinearVelocit
y=!1,this._grip.linearVelocity=new
Lt,this._grip.hasAngularVelocity=!1,this._grip.angularVelocity=new
Lt),this._grip}dispatchEvent(t){return
null!==this._targetRay&&this._targetRay.dispatchEvent(t),null!==this._grip&&this._g
rip.dispatchEvent(t),null!==this._hand&&this._hand.dispatchEvent(t),this}disconnect
(t){return
this.dispatchEvent({type:"disconnected",data:t}),null!==this._targetRay&&
(this._targetRay.visible=!1),null!==this._grip&&
(this._grip.visible=!1),null!==this._hand&&
(this._hand.visible=!1),this}update(t,e,n){let i=null,r=null,s=null;const
a=this._targetRay,o=this._grip,l=this._hand;if(t&&"visible-
blurred"!==e.session.visibilityState)if(null!==a&&
(i=e.getPose(t.targetRaySpace,n),null!==i&&
(a.matrix.fromArray(i.transform.matrix),a.matrix.decompose(a.position,a.rotation,a.
scale),i.linearVelocity?
(a.hasLinearVelocity=!0,a.linearVelocity.copy(i.linearVelocity)):a.hasLinearVelocit
y=!1,i.angularVelocity?
(a.hasAngularVelocity=!0,a.angularVelocity.copy(i.angularVelocity)):a.hasAngularVel
ocity=!1,this.dispatchEvent(vs))),l&&t.hand){s=!0;for(const i of t.hand.values())
{const t=e.getJointPose(i,n);if(void 0===l.joints[i.jointName]){const t=new
gs;t.matrixAutoUpdate=!1,t.visible=!1,l.joints[i.jointName]=t,l.add(t)}const
r=l.joints[i.jointName];null!==t&&
(r.matrix.fromArray(t.transform.matrix),r.matrix.decompose(r.position,r.rotation,r.
scale),r.jointRadius=t.radius),r.visible=null!==t}const i=l.joints["index-finger-
tip"],r=l.joints["thumb-
tip"],a=i.position.distanceTo(r.position),o=.02,c=.005;l.inputState.pinching&&a>o+c
?
(l.inputState.pinching=!1,this.dispatchEvent({type:"pinchend",handedness:t.handedne
ss,target:this})):!l.inputState.pinching&&a<=o-c&&
(l.inputState.pinching=!0,this.dispatchEvent({type:"pinchstart",handedness:t.handed
ness,target:this}))else null!==o&&t.gripSpace&&
(r=e.getPose(t.gripSpace,n),null!==r&&
(o.matrix.fromArray(r.transform.matrix),o.matrix.decompose(o.position,o.rotation,o.
scale),r.linearVelocity?
(o.hasLinearVelocity=!0,o.linearVelocity.copy(r.linearVelocity)):o.hasLinearVelocit
y=!1,r.angularVelocity?
(o.hasAngularVelocity=!0,o.angularVelocity.copy(r.angularVelocity)):o.hasAngularVel
ocity=!1));return null!==a&&(a.visible=null!==i),null!==o&&
(o.visible=null!==r),null!==l&&(l.visible=null!==s),this}}class xs extends
rt{constructor(t,e){super();const n=this,i=t.state;let r=null,s=1,a=null,o="local-
floor",l=null;const c=[],h=new Map,u=new Kn;u.layers.enable(1),u.viewport=new
St;const d=new Kn;d.layers.enable(2),d.viewport=new St;const p=[u,d],m=new
fs;m.layers.enable(1),m.layers.enable(2);let f=null,g=null;function v(t){const
e=h.get(t.inputSource);e&&e.dispatchEvent({type:t.type,data:t.inputSource})}functio
n y(){h.forEach((function(t,e)
{t.disconnect(e)})),h.clear(),f=null,g=null,i.bindXRFramebuffer(null),t.setRenderTa
rget(t.getRenderTarget()),S.stop(),n.isPresenting=!1,n.dispatchEvent({type:"session
end"})}}function x(t){const e=r.inputSources;for(let

```

```

t=0;t<c.length;t++)h.set(e[t],c[t]);for(let e=0;e<t.removed.length;e++){const
n=t.removed[e],i=h.get(n);i&&
(i.dispatchEvent({type:"disconnected",data:n}),h.delete(n))}for(let
e=0;e<t.added.length;e++){const
n=t.added[e],i=h.get(n);i&&i.dispatchEvent({type:"connected",data:n})}this.enabled
=!!1,this.isPresenting=!!1,this.getController=function(t){let e=c[t];return void
0===e&&(e=new ys,c[t]=e),e.getTargetRaySpace()},this.getControllerGrip=function(t)
{let e=c[t];return void 0===e&&(e=new
ys,c[t]=e),e.getGripSpace()},this.getHand=function(t){let e=c[t];return void
0===e&&(e=new
ys,c[t]=e),e.getHandSpace()},this.setFramebufferScaleFactor=function(t)
{s=t,!0===n.isPresenting&&console.warn("THREE.WebXRManager: Cannot change
framebuffer scale while presenting.")},this.setReferenceSpaceType=function(t)
{o=t,!0===n.isPresenting&&console.warn("THREE.WebXRManager: Cannot change reference
space type while presenting.")},this.getReferenceSpace=function(){return
a},this.getSession=function(){return r},this.setSession=async function(t)
{if(r=t,null!==r)
{r.addEventListener("select",v),r.addEventListener("selectstart",v),r.addEventListe
ner("selectend",v),r.addEventListener("squeeze",v),r.addEventListener("squeezestart
",v),r.addEventListener("squeezeend",v),r.addEventListener("end",y),r.addEventListe
ner("inputsourceschange",x);const
t=e.getContextAttributes();!0!==t.xrCompatible&&await e.makeXRCompatible();const i=
{antialias:t.antialias,alpha:t.alpha,depth:t.depth,stencil:t.stencil,framebufferSca
leFactor:s},l=new XRWebGLLayer(r,e,i);r.updateRenderState({baseLayer:l}),a=await
r.requestReferenceSpace(o),S.setContext(r),S.start(),n.isPresenting=!0,n.dispatchEv
ent({type:"sessionstart"})}};const _=new Lt,w=new Lt;function b(t,e){null===e?
t.matrixWorld.copy(t.matrix):t.matrixWorld.multiplyMatrices(e.matrixWorld,t.matrix)
,t.matrixWorldInverse.copy(t.matrixWorld).invert()}this.getCamera=function(t)
{m.near=d.near=u.near=t.near,m.far=d.far=u.far=t.far,f===m.near&&g===m.far||
(r.updateRenderState({depthNear:m.near,depthFar:m.far}),f=m.near,g=m.far);const
e=t.parent,n=m.cameras;b(m,e);for(let
t=0;t<n.length;t++)b(n[t],e);t.matrixWorld.copy(m.matrixWorld),t.matrix.copy(m.matr
ix),t.matrix.decompose(t.position,t.quaternion,t.scale);const i=t.children;for(let
t=0,e=i.length;t<e;t++)i[t].updateMatrixWorld(!0);return 2===n.length?
function(t,e,n)
{_.setFromMatrixPosition(e.matrixWorld),w.setFromMatrixPosition(n.matrixWorld);cons
t
i=_.distanceTo(w),r=e.projectionMatrix.elements,s=n.projectionMatrix.elements,a=r[1
4]/(r[10]-1),o=r[14]/(r[10]+1),l=(r[9]+1)/r[5],c=(r[9]-1)/r[5],h=(r[8]-1)/r[0],u=
(s[8]+1)/s[0],d=a*h,p=a*u,m=i/(-h+u),f=m*-
h;e.matrixWorld.decompose(t.position,t.quaternion,t.scale),t.translateX(f),t.transl
ateZ(m),t.matrixWorld.compose(t.position,t.quaternion,t.scale),t.matrixWorldInverse
.copy(t.matrixWorld).invert();const g=a+m,v=o+m,y=d-f,x=p+(i-
f),b=l*o/v*g,M=c*o/v*g;t.projectionMatrix.makePerspective(y,x,b,M,g,v)}
(m,u,d):m.projectionMatrix.copy(u.projectionMatrix),m};let M=null;const S=new
oi;S.setAnimationLoop((function(t,e){if(l=e.getViewerPose(a),null!==l){const
t=l.views,e=r.renderState.baseLayer;i.bindXRFramebuffer(e.framebuffer);let
n=!1;t.length!==m.cameras.length&&(m.cameras.length=0,n=!0);for(let
i=0;i<t.length;i++){const
r=t[i],s=e.setViewport(r),a=p[i];a.matrix.fromArray(r.transform.matrix),a.projectio
nMatrix.fromArray(r.projectionMatrix),a.viewport.set(s.x,s.y,s.width,s.height),0===
i&&m.matrix.copy(a.matrix),!0===n&&m.cameras.push(a)}}const
n=r.inputSources;for(let t=0;t<c.length;t++){const
i=c[t],r=n[t];i.update(r,e,a)}M&&M(t,e)}),this.setAnimationLoop=function(t)
{M=t},this.dispose=function(){}}function _s(t){function e(e,n)
{e.opacity.value=n.opacity,n.color&&e.diffuse.value.copy(n.color),n.emissive&&e.emi
ssive.value.copy(n.emissive).multiplyScalar(n.emissiveIntensity),n.map&&
(e.map.value=n.map),n.alphaMap&&(e.alphaMap.value=n.alphaMap),n.specularMap&&

```

```

(e.specularMap.value=n.specularMap);const i=t.get(n).envMap;if(i)
{e.envMap.value=i,e.flipEnvMap.value=i.isCubeTexture&&i._needsFlipEnvMap?
-1:1,e.reflectivity.value=n.reflectivity,e.refractionRatio.value=n.refractionRatio;
const r=t.get(i).__maxMipLevel;void 0!==r&&(e.maxMipLevel.value=r)}let
r,s;n.lightMap&&
(e.lightMap.value=n.lightMap,e.lightMapIntensity.value=n.lightMapIntensity),n.aoMap
&&(e.aoMap.value=n.aoMap,e.aoMapIntensity.value=n.aoMapIntensity),n.map?
r=n.map:n.specularMap?r=n.specularMap:n.displacementMap?
r=n.displacementMap:n.normalMap?r=n.normalMap:n.bumpMap?r=n.bumpMap:n.roughnessMap?
r=n.roughnessMap:n.metalnessMap?r=n.metalnessMap:n.alphaMap?
r=n.alphaMap:n.emissiveMap?r=n.emissiveMap:n.clearcoatMap?
r=n.clearcoatMap:n.clearcoatNormalMap?
r=n.clearcoatNormalMap:n.clearcoatRoughnessMap&&(r=n.clearcoatRoughnessMap),void
0!==r&&(r.isWebGLRenderTarget&&
(r=r.texture),!0===r.matrixAutoUpdate&&r.updateMatrix(),e.uvTransform.value.copy(r.
matrix)),n.aoMap?s=n.aoMap:n.lightMap&&(s=n.lightMap),void 0!==s&&
(s.isWebGLRenderTarget&&
(s=s.texture),!0===s.matrixAutoUpdate&&s.updateMatrix(),e.uv2Transform.value.copy(s
.matrix))}function n(e,n)
{e.roughness.value=n.roughness,e.metalness.value=n.metalness,n.roughnessMap&&
(e.roughnessMap.value=n.roughnessMap),n.metalnessMap&&
(e.metalnessMap.value=n.metalnessMap),n.emissiveMap&&
(e.emissiveMap.value=n.emissiveMap),n.bumpMap&&
(e.bumpMap.value=n.bumpMap,e.bumpScale.value=n.bumpScale,1===n.side&&
(e.bumpScale.value*=-1)),n.normalMap&&
(e.normalMap.value=n.normalMap,e.normalScale.value.copy(n.normalScale),1===n.side&&
e.normalScale.value.negate()),n.displacementMap&&
(e.displacementMap.value=n.displacementMap,e.displacementScale.value=n.displacement
Scale,e.displacementBias.value=n.displacementBias);t.get(n).envMap&&
(e.envMapIntensity.value=n.envMapIntensity)}return{refreshFogUniforms:function(t,e)
{t.fogColor.value.copy(e.color),e.isFog?
(t.fogNear.value=e.near,t.fogFar.value=e.far):e.isFogExp2&&
(t.fogDensity.value=e.density)},refreshMaterialUniforms:function(t,i,r,s)
{i.isMeshBasicMaterial?e(t,i):i.isMeshLambertMaterial?(e(t,i),function(t,e)
{e.emissiveMap&&(t.emissiveMap.value=e.emissiveMap)}(t,i)):i.isMeshToonMaterial?
(e(t,i),function(t,e){e.gradientMap&&
(t.gradientMap.value=e.gradientMap);e.emissiveMap&&
(t.emissiveMap.value=e.emissiveMap);e.bumpMap&&
(t.bumpMap.value=e.bumpMap,t.bumpScale.value=e.bumpScale,1===e.side&&
(t.bumpScale.value*=-1));e.normalMap&&
(t.normalMap.value=e.normalMap,t.normalScale.value.copy(e.normalScale),1===e.side&&
t.normalScale.value.negate());e.displacementMap&&
(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacement
Scale,t.displacementBias.value=e.displacementBias)}(t,i)):i.isMeshPhongMaterial?
(e(t,i),function(t,e)
{t.specular.value.copy(e.specular),t.shininess.value=Math.max(e.shininess,1e-
4),e.emissiveMap&&(t.emissiveMap.value=e.emissiveMap);e.bumpMap&&
(t.bumpMap.value=e.bumpMap,t.bumpScale.value=e.bumpScale,1===e.side&&
(t.bumpScale.value*=-1));e.normalMap&&
(t.normalMap.value=e.normalMap,t.normalScale.value.copy(e.normalScale),1===e.side&&
t.normalScale.value.negate());e.displacementMap&&
(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacement
Scale,t.displacementBias.value=e.displacementBias)}(t,i)):i.isMeshStandardMaterial?
(e(t,i),i.isMeshPhysicalMaterial?function(t,e)
{n(t,e),t.reflectivity.value=e.reflectivity,t.clearcoat.value=e.clearcoat,t.clearco
atRoughness.value=e.clearcoatRoughness,e.sheen&&t.sheen.value.copy(e.sheen);e.clear
coatMap&&(t.clearcoatMap.value=e.clearcoatMap);e.clearcoatRoughnessMap&&
(t.clearcoatRoughnessMap.value=e.clearcoatRoughnessMap);e.clearcoatNormalMap&&

```

```

(t.clearcoatNormalScale.value.copy(e.clearcoatNormalScale),t.clearcoatNormalMap.value=e.clearcoatNormalMap,1===e.side&&t.clearcoatNormalScale.value.negate());t.transmission.value=e.transmission,e.transmissionMap&&(t.transmissionMap.value=e.transmissionMap)}(t,i):n(t,i)):i.isMeshMatcapMaterial?(e(t,i),function(t,e){e.matcap&&(t.matcap.value=e.matcap);e.bumpMap&&(t.bumpMap.value=e.bumpMap,t.bumpScale.value=e.bumpScale,1===e.side&&(t.bumpScale.value*=-1));e.normalMap&&(t.normalMap.value=e.normalMap,t.normalScale.value.copy(e.normalScale),1===e.side&&t.normalScale.value.negate());e.displacementMap&&(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacementScale,t.displacementBias.value=e.displacementBias)}(t,i)):i.isMeshDepthMaterial?(e(t,i),function(t,e){e.displacementMap&&(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacementScale,t.displacementBias.value=e.displacementBias)}(t,i)):i.isMeshDistanceMaterial?(e(t,i),function(t,e){e.displacementMap&&(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacementScale,t.displacementBias.value=e.displacementBias);t.referencePosition.value.copy(e.referencePosition),t.nearDistance.value=e.nearDistance,t.farDistance.value=e.farDistance}(t,i)):i.isMeshNormalMaterial?(e(t,i),function(t,e){e.bumpMap&&(t.bumpMap.value=e.bumpMap,t.bumpScale.value=e.bumpScale,1===e.side&&(t.bumpScale.value*=-1));e.normalMap&&(t.normalMap.value=e.normalMap,t.normalScale.value.copy(e.normalScale),1===e.side&&t.normalScale.value.negate());e.displacementMap&&(t.displacementMap.value=e.displacementMap,t.displacementScale.value=e.displacementScale,t.displacementBias.value=e.displacementBias)}(t,i)):i.isLineBasicMaterial?(function(t,e){t.diffuse.value.copy(e.color),t.opacity.value=e.opacity}(t,i),i.isLineDashedMaterial&&function(t,e){t.dashSize.value=e.dashSize,t.totalSize.value=e.dashSize+e.gapSize,t.scale.value=e.scale}(t,i)):i.isPointsMaterial?function(t,e,n,i){t.diffuse.value.copy(e.color),t.opacity.value=e.opacity,t.size.value=e.size*n,t.scale.value=.5*i,e.map&&(t.map.value=e.map);e.alphaMap&&(t.alphaMap.value=e.alphaMap);let r;e.map?r=e.map:e.alphaMap&&(r=e.alphaMap);void 0!==r&&(!0===r.matrixAutoUpdate&&r.updateMatrix(),t.uvTransform.value.copy(r.matrix))}(t,i,r,s):i.isSpriteMaterial?function(t,e){t.diffuse.value.copy(e.color),t.opacity.value=e.opacity,t.rotation.value=e.rotation,e.map&&(t.map.value=e.map);e.alphaMap&&(t.alphaMap.value=e.alphaMap);let n;e.map?n=e.map:e.alphaMap&&(n=e.alphaMap);void 0!==n&&(!0===n.matrixAutoUpdate&&n.updateMatrix(),t.uvTransform.value.copy(n.matrix))}(t,i):i.isShadowMaterial?(t.color.value.copy(i.color),t.opacity.value=i.opacity):i.isShaderMaterial&&(i.uniformsNeedUpdate=!1)}}function ws(t){const e=void 0!==(t=t||{}).canvas?t.canvas:function(){const t=document.createElementNS("http://www.w3.org/1999/xhtml","canvas");return t.style.display="block",t}(),n=void 0!==(t.context?t.context:null,i=void 0!==(t.alpha&&t.alpha,r=void 0!==(t.depth||t.depth,s=void 0!==(t.stencil||t.stencil,a=void 0!==(t.antialias&&t.antialias,o=void 0!==(t.premultipliedAlpha||t.premultipliedAlpha,l=void 0!==(t.preserveDrawingBuffer&&t.preserveDrawingBuffer,c=void 0!==(t.powerPreference?t.powerPreference:"default",h=void 0!==(t.failIfMajorPerformanceCaveat&&t.failIfMajorPerformanceCaveat);let u=null,d=null;const p=[],m=[];this.domElement=e,this.debug={checkShaderErrors:!0},this.autoClear=!0,this.autoClearColor=!0,this.autoClearDepth=!0,this.autoClearStencil=!0,this.sortObjects=!0,this.clippingPlanes=[],this.localClippingEnabled=!1,this.gammaFactor=2,this.outputEncoding=X,this.physicallyCorrectLights=!1,this.toneMapping=0,this.toneMappingExposure=1;const f=this;let g=!1,v=0,y=0,_=null,w=-1,S=null;const T=new St,A=new St;let L=null,R=e.width,C=e.height,P=1,D=null,I=null;const N=new St(0,0,R,C),B=new

```

```

St(0,0,R,C);let z=!1;const F=new ai;let O=!1,H=!1;const G=new se,U=new Lt,k=
{background:null,fog:null,environment:null,overrideMaterial:null,isScene:!0};functi
on V(){return null===?P:1}let
W,j,q,Y,Z,J,Q,K,$,tt,et,nt,it,rt,st,at,ot,lt,ct,ht,ut,dt,pt=n;function ft(t,n)
{for(let i=0;i<t.length;i++){const r=t[i],s=e.getContext(r,n);if(null!==s)return
s}return null}try{const t=
{alpha:i,depth:r,stencil:s,antialias:a,premultipliedAlpha:o,preserveDrawingBuffer:1
,powerPreference:c,failIfMajorPerformanceCaveat:h};if(e.addEventListener("webglcont
extlost",xt,!1),e.addEventListener("webglcontextrestored",_t,!1),null===pt){const
e=["webgl2","webgl","experimental-
webgl"];if(!0===f.isWebGL1Renderer&&e.shift(),pt=ft(e,t),null===pt)throw ft(e)?new
Error("Error creating WebGL context with your selected attributes."):new
Error("Error creating WebGL context.")}void 0===pt.getShaderPrecisionFormat&&
(pt.getShaderPrecisionFormat=function()
{return{rangeMin:1,rangeMax:1,precision:1}})}catch(t){throw
console.error("THREE.WebGLRenderer: "+t.message),t}function gt(){W=new xi(pt),j=new
gi(pt,W,t),W.init(j),ut=new ms(pt,W,j),q=new ds(pt,W,j),Y=new bi(pt),Z=new Kr,J=new
ps(pt,W,q,Z,j,ut,Y),Q=new yi(f),K=new li(pt,j),dt=new mi(pt,W,K,j),$=new
_i(pt,K,Y,dt),tt=new Ei(pt,$,K,Y),lt=new Ti(pt),st=new vi(Z),et=new
Qr(f,Q,W,j,dt,st),nt=new _s(Z),it=new ns(Z),rt=new ls(W,j),ot=new
pi(f,Q,q,tt,o),at=new us(f,tt,j),ct=new fi(pt,W,Y,j),ht=new
wi(pt,W,Y,j),Y.programs=et.programs,f.capabilities=j,f.extensions=W,f.properties=Z,
f.renderLists=it,f.shadowMap=at,f.state=q,f.info=Y}gt();const yt=new
xs(f,pt);function xt(t){t.preventDefault(),console.log("THREE.WebGLRenderer:
Context Lost."),g=!0}function _t(){console.log("THREE.WebGLRenderer: Context
Restored."),g=!1;const
t=Y.autoReset,e=at.enabled,n=at.autoUpdate,i=at.needsUpdate,r=at.type;gt(),Y.autoRe
set=t,at.enabled=e,at.autoUpdate=n,at.needsUpdate=i,at.type=r}function wt(t){const
e=t.target;e.removeEventListener("dispose",wt),function(t){const
e=Z.get(t).programs,void 0!==e&&e.forEach(function(t){et.releaseProgram(t)}))}
(t),Z.remove(t)}(e)}this.xr=yt,this.getContext=function(){return
pt},this.getContextAttributes=function(){return
pt.getContextAttributes()},this.forceContextLoss=function(){const
t=W.get("WEBGL_lose_context");t&&t.loseContext()},this.forceContextRestore=function
(){const
t=W.get("WEBGL_lose_context");t&&t.restoreContext()},this.getPixelRatio=function()
{return P},this.setPixelRatio=function(t){void 0!==t&&
(P=t,this.setSize(R,C,!1))},this.getSize=function(t){return void 0===t&&
(console.warn("WebGLRenderer: .getSize() now requires a Vector2 as an
argument"),t=new vt),t.set(R,C)},this.setSize=function(t,n,i){yt.isPresenting?
console.warn("THREE.WebGLRenderer: Can't change size while VR device is
presenting.):(R=t,C=n,e.width=Math.floor(t*P),e.height=Math.floor(n*P),!1!==i&&
(e.style.width=t+"px",e.style.height=n+"px"),this.setViewport(0,0,t,n))},this.getDr
awingBufferSize=function(t){return void 0===t&&(console.warn("WebGLRenderer:
.getDrawingBufferSize() now requires a Vector2 as an argument"),t=new
vt),t.set(R*P,C*P).floor(),this.setDrawingBufferSize=function(t,n,i)
{R=t,C=n,P=i,e.width=Math.floor(t*i),e.height=Math.floor(n*i),this.setViewport(0,0,
t,n)},this.getCurrentViewport=function(t){return void 0===t&&
(console.warn("WebGLRenderer: .getCurrentViewport() now requires a Vector4 as an
argument"),t=new St),t.copy(T)},this.getViewport=function(t){return
t.copy(N)},this.setViewport=function(t,e,n,i){t.isVector4?
N.set(t.x,t.y,t.z,t.w):N.set(t,e,n,i),q.viewport(T.copy(N).multiplyScalar(P).floor(
))},this.getScissor=function(t){return t.copy(B)},this.setScissor=function(t,e,n,i)
{t.isVector4?
B.set(t.x,t.y,t.z,t.w):B.set(t,e,n,i),q.scissor(A.copy(B).multiplyScalar(P).floor(
))},this.getScissorTest=function(){return z},this.setScissorTest=function(t)
{q.setScissorTest(z=t)},this.setOpaqueSort=function(t)
{D=t},this.setTransparentSort=function(t){I=t},this.getClearColor=function(t)

```

```

{return void 0===t&&(console.warn("WebGLRenderer: .getClearColor() now requires a
Color as an argument"),t=new
tn),t.copy(ot.getClearColor()),this.setClearColor=function()
{ot.setClearColor.apply(ot,arguments)},this.getClearAlpha=function(){return
ot.getClearAlpha()},this.setClearAlpha=function()
{ot.setClearAlpha.apply(ot,arguments)},this.clear=function(t,e,n){let i=0;(void
0===t||t)&&(i|=16384),(void 0===e||e)&&(i|=256),(void 0===n||n)&&
(i|=1024),pt.clear(i)},this.clearColor=function()
{this.clear(!0,!1,!1)},this.clearDepth=function()
{this.clear(!1,!0,!1)},this.clearStencil=function()
{this.clear(!1,!1,!0)},this.dispose=function()
{e.removeEventListener("webglcontextlost",xt,!1),e.removeEventListener("webglcontext
restored",_t,!1),it.dispose(),rt.dispose(),Z.dispose(),Q.dispose(),tt.dispose(),dt
.dispose(),yt.dispose(),yt.removeEventListener("sessionstart",Mt),yt.removeEventLis
tener("sessionend",Tt),Et.stop(),this.renderBufferImmediate=function(t,e)
{dt.initAttributes();const n=Z.get(t);t.hasPositions&&!n.position&&
(n.position=pt.createBuffer()),t.hasNormals&&!n.normal&&
(n.normal=pt.createBuffer()),t.hasUvs&&!n.uv&&
(n.uv=pt.createBuffer()),t.hasColors&&!n.color&&(n.color=pt.createBuffer());const
i=e.getAttributes();t.hasPositions&&
(pt.bindBuffer(34962,n.position),pt.bufferData(34962,t.positionArray,35048),dt.enab
leAttribute(i.position),pt.vertexAttribPointer(i.position,3,5126,!1,0,0)),t.hasNorm
als&&
(pt.bindBuffer(34962,n.normal),pt.bufferData(34962,t.normalArray,35048),dt.enableAt
tribute(i.normal),pt.vertexAttribPointer(i.normal,3,5126,!1,0,0)),t.hasUvs&&
(pt.bindBuffer(34962,n.uv),pt.bufferData(34962,t.uvArray,35048),dt.enableAttribute(
i.uv),pt.vertexAttribPointer(i.uv,2,5126,!1,0,0)),t.hasColors&&
(pt.bindBuffer(34962,n.color),pt.bufferData(34962,t.colorArray,35048),dt.enableAttr
ibute(i.color),pt.vertexAttribPointer(i.color,3,5126,!1,0,0)),dt.disableUnusedAttri
butes(),pt.drawArrays(4,0,t.count),t.count=0},this.renderBufferDirect=function(t,e,
n,i,r,s){null===e&&(e=k);const a=r.isMesh&&r.matrixWorld.determinant()
<0,o=It(t,e,i,r);q.setMaterial(i,a);let l=n.index;const
c=n.attributes.position;if(null===l){if(void 0===c||0===c.count)return}else
if(0===l.count)return;let h,u=1;!0===i.wireframe&&
(l=$.getWireframeAttribute(n),u=2),
(i.morphTargets||i.morphNormals)&&lt.t.update(r,n,i,o),dt.setup(r,i,o,n,l);let
d=ct;null!==l&&(h=K.get(l),d=ht,d.setIndex(h));const p=null!==l?
l.count:c.count,m=n.drawRange.start*u,f=n.drawRange.count*u,g=null!==s?
s.start*u:0,v=null!==s?
s.count*u:1/0,y=Math.max(m,g),x=Math.min(p,m+f,g+v)-1,_=Math.max(0,x-y+1);if(0!==_)
{if(r.isMesh)!0===i.wireframe?
(q.setLineWidth(i.wireframeLineWidth*V()),d.setMode(1)):d.setMode(4);else
if(r.isLine){let t=i.linewidth;void 0===t&&
(t=1),q.setLineWidth(t*V()),r.isLineSegments?d.setMode(1):r.isLineLoop?
d.setMode(2):d.setMode(3)}else r.isPoints?
d.setMode(0):r.isSprite&&d.setMode(4);if(r.isInstancedMesh)d.renderInstances(y,_,r.
count);else if(n.isInstancedBufferGeometry){const
t=Math.min(n.instanceCount,n._maxInstanceCount);d.renderInstances(y,_,t)}else
d.render(y,_)},this.compile=function(t,e)
{d=rt.get(t),d.init(),t.traverseVisible((function(t)
{t.isLight&&t.layers.test(e.layers)&&
(d.pushLight(t),t.castShadow&&d.pushShadow(t))}),d.setupLights(),t.traverse((funct
ion(e){const n=e.material;if(n){if(Array.isArray(n))for(let i=0;i<n.length;i++)
{Pt(n[i],t,e)}else Pt(n,t,e)}}));let bt=null;function Mt(){Et.stop()}function Tt()
{Et.start()}const Et=new oi;function At(t,e,n,i)
{if(!1===t.visible)return;if(t.layers.test(e.layers))if(t.isGroup)n=t.renderOrder;e
lse if(t.isLOD)!0===t.autoUpdate&&t.update(e);else
if(t.isLight)d.pushLight(t),t.castShadow&&d.pushShadow(t);else if(t.isSprite)

```

```

{if(!t.frustumCulled||F.intersectsSprite(t))
{i&&U.setFromMatrixPosition(t.matrixWorld).applyMatrix4(G);const
e=tt.update(t),r=t.material;r.visible&&u.push(t,e,r,n,U.z,null)}}else
if(t.isImmediateRenderObject)i&&U.setFromMatrixPosition(t.matrixWorld).applyMatrix4
(G),u.push(t,null,t.material,n,U.z,null);else if((t.isMesh||t.isLine||t.isPoints)&&
(t.isSkinnedMesh&&t.skeleton.frame!==Y.render.frame&&
(t.skeleton.update(),t.skeleton.frame=Y.render.frame),!t.frustumCulled||F.intersect
sObject(t)))i&&U.setFromMatrixPosition(t.matrixWorld).applyMatrix4(G);const
e=tt.update(t),r=t.material;if(Array.isArray(r)){const i=e.groups;for(let
s=0,a=i.length;s<a;s++){const
a=i[s],o=r[a.materialIndex];o&&o.visible&&u.push(t,e,o,n,U.z,a)}}else
r.visible&&u.push(t,e,r,n,U.z,null)}const r=t.children;for(let
t=0,s=r.length;t<s;t++)At(r[t],e,n,i)}function Rt(t,e,n){const i=!0===e.isScene?
e.overrideMaterial:null;for(let r=0,s=t.length;r<s;r++){const
s=t[r],a=s.object,o=s.geometry,l=null===i?
s.material:i,c=s.group;if(n.isArrayCamera){const t=n.cameras;for(let
n=0,i=t.length;n<i;n++){const i=t[n];a.layers.test(i.layers)&&
(q.viewport(T.copy(i.viewport)),d.setupLightsView(i),Ct(a,e,i,o,l,c))}}else
Ct(a,e,n,o,l,c)}}function Ct(t,e,n,i,r,s)
{if(t.onBeforeRender(f,e,n,i,r,s),t.modelViewMatrix.multiplyMatrices(n.matrixWorldI
nverse,t.matrixWorld),t.normalMatrix.getNormalMatrix(t.modelViewMatrix),t.isImmedia
teRenderObject){const i=It(n,e,r,t);q.setMaterial(r),dt.reset(),function(t,e)
{t.render((function(t){f.renderBufferImmediate(t,e)}))}(t,i)}else
f.renderBufferDirect(n,e,i,r,t,s);t.onAfterRender(f,e,n,i,r,s)}function Pt(t,e,n)
{!0!==e.isScene&&(e=k);const
i=Z.get(t),r=d.state.lights,s=d.state.shadowsArray,a=r.state.version,o=et.getParame
ters(t,r.state,s,e,n),l=et.getProgramCacheKey(o);let
c=i.programs;i.environment=t.isMeshStandardMaterial?
e.environment:null,i.fog=e.fog,i.envMap=Q.get(t.envMap||i.environment),void 0===c&&
(t.addEventListener("dispose",wt),c=new Map,i.programs=c);let h=c.get(l);if(void
0!==h){if(i.currentProgram===h&&i.lightsStateVersion===a)return Dt(t,o),h}else
o.uniforms=et.getUniforms(t),t.onBuild(o,f),t.onBeforeCompile(o,f),h=et.acquireProg
ram(o,l),c.set(l,h),i.uniforms=o.uniforms;const u=i.uniforms;
(t.isShaderMaterial||t.isRawShaderMaterial)&&!0!==t.clipping||
(u.clippingPlanes=st.uniform),Dt(t,o),i.needsLights=function(t){return
t.isMeshLambertMaterial||t.isMeshToonMaterial||t.isMeshPhongMaterial||t.isMeshStand
ardMaterial||t.isShadowMaterial||t.isShaderMaterial&&!0===t.lights}
(t),i.lightsStateVersion=a,i.needsLights&&
(u.ambientLightColor.value=r.state.ambient,u.lightProbe.value=r.state.probe,u.direc
tionalLights.value=r.state.directional,u.directionalLightShadows.value=r.state.dire
ctionalShadow,u.spotLights.value=r.state.spot,u.spotLightShadows.value=r.state.spot
Shadow,u.rectAreaLights.value=r.state.rectArea,u.ltc_1.value=r.state.rectAreaLTC1,u
.ltc_2.value=r.state.rectAreaLTC2,u.pointLights.value=r.state.point,u.pointLightSha
dows.value=r.state.pointShadow,u.hemisphereLights.value=r.state.hemi,u.directionalS
hadowMap.value=r.state.directionalShadowMap,u.directionalShadowMatrix.value=r.state
.directionalShadowMatrix,u.spotShadowMap.value=r.state.spotShadowMap,u.spotShadowMa
trix.value=r.state.spotShadowMatrix,u.pointShadowMap.value=r.state.pointShadowMap,u
.pointShadowMatrix.value=r.state.pointShadowMatrix);const
p=h.getUniforms(),m=Cr.seqWithValue(p.seq,u);return
i.currentProgram=h,i.uniformsList=m,h}function Dt(t,e){const
n=Z.get(t);n.outputEncoding=e.outputEncoding,n.instancing=e.instancing,n.numClippin
gPlanes=e.numClippingPlanes,n.numIntersection=e.numClipIntersection,n.vertexAlphas=
e.vertexAlphas}function It(t,e,n,i){!0!==e.isScene&&
(e=k),J.resetTextureUnits();const r=e.fog,s=n.isMeshStandardMaterial?
e.environment:null,a=null===_?
f.outputEncoding:_.texture.encoding,o=Q.get(n.envMap||s),l=!0===n.vertexColors&&i.g
eometry&&i.geometry.attributes.color&&4===i.geometry.attributes.color.itemSize,c=Z.
get(n),h=d.state.lights;if(!0===O&&(!0===H||t!==S)){const

```

```

e=t===S&&n.id===w;st.setState(n,t,e)}let u=!1;n.version===c.__version?
c.needsLights&&c.lightsStateVersion!==h.state.version||c.outputEncoding!==a||i.isIn
stancedMesh&&!1===c.instancing?u=!0:i.isInstancedMesh||!0!==c.instancing?
c.envMap!==o||n.fog&&c.fog!==r?u=!0:void
0===c.numClippingPlanes||c.numClippingPlanes===st.numPlanes&&c.numIntersection===st
.numIntersection?c.vertexAlphas!==l&&(u=!0):u=!0:u=!0:
(u=!0,c.__version=n.version);let p=c.currentProgram;!0===u&&(p=Pt(n,e,i));let
m=!1,g=!1,v=!1;const y=p.getUniforms(),x=c.uniforms;if(q.useProgram(p.program)&&
(m=!0,g=!0,v=!0),n.id!==w&&(w=n.id,g=!0),m||S!==t)
{if(y.setValue(pt,"projectionMatrix",t.projectionMatrix),j.logarithmicDepthBuffer&&
y.setValue(pt,"logDepthBufFC",2/(Math.log(t.far+1)/Math.LN2)),S!==t&&
(S=t,g=!0,v=!0),n.isShaderMaterial||n.isMeshPhongMaterial||n.isMeshToonMaterial||n.
isMeshStandardMaterial||n.envMap){const e=y.map.cameraPosition;void
0!==e&&e.setValue(pt,U.setFromMatrixPosition(t.matrixWorld))}
(n.isMeshPhongMaterial||n.isMeshToonMaterial||n.isMeshLambertMaterial||n.isMeshBasi
cMaterial||n.isMeshStandardMaterial||n.isShaderMaterial)&&y.setValue(pt,"isOrthogra
phic",!0===t.isOrthographicCamera),
(n.isMeshPhongMaterial||n.isMeshToonMaterial||n.isMeshLambertMaterial||n.isMeshBasi
cMaterial||n.isMeshStandardMaterial||n.isShaderMaterial||n.isShadowMaterial||n.skin
ning)&&y.setValue(pt,"viewMatrix",t.matrixWorldInverse)}if(n.skinning)
{y.setOptional(pt,i,"bindMatrix"),y.setOptional(pt,i,"bindMatrixInverse");const
t=i.skeleton;if(t){const e=t.bones;if(j.floatVertexTextures)
{if(null===t.boneTexture){let n=Math.sqrt(4*e.length);n=mt(n),n=Math.max(n,4);const
i=new Float32Array(n*n*4);i.set(t.boneMatrices);const r=new
ii(i,n,n,E,b);t.boneMatrices=i,t.boneTexture=r,t.boneTextureSize=n}y.setValue(pt,"b
oneTexture",t.boneTexture,J),y.setValue(pt,"boneTextureSize",t.boneTextureSize)}els
e y.setOptional(pt,t,"boneMatrices")}}var
M,T;return(g||c.receiveShadow!==i.receiveShadow)&&
(c.receiveShadow=i.receiveShadow,y.setValue(pt,"receiveShadow",i.receiveShadow)),g&
&(y.setValue(pt,"toneMappingExposure",f.toneMappingExposure),c.needsLights&&(T=v,
(M=x).ambientLightColor.needsUpdate=T,M.lightProbe.needsUpdate=T,M.directionalLight
s.needsUpdate=T,M.directionalLightShadows.needsUpdate=T,M.pointLights.needsUpdate=T
,M.pointLightShadows.needsUpdate=T,M.spotLights.needsUpdate=T,M.spotLightShadows.ne
edsUpdate=T,M.rectAreaLights.needsUpdate=T,M.hemisphereLights.needsUpdate=T),r&&n.f
og&&nt.refreshFogUniforms(x,r),nt.refreshMaterialUniforms(x,n,P,C),Cr.upload(pt,c.u
niformsList,x,J),n.isShaderMaterial&&!0===n.uniformsNeedUpdate&&
(Cr.upload(pt,c.uniformsList,x,J),n.uniformsNeedUpdate=!1),n.isSpriteMaterial&&y.se
tValue(pt,"center",i.center),y.setValue(pt,"modelViewMatrix",i.modelViewMatrix),y.s
etValue(pt,"normalMatrix",i.normalMatrix),y.setValue(pt,"modelMatrix",i.matrixWorld
),p)}Et.setAnimationLoop((function(t){bt&&bt(t)}),"undefined"!==typeof
window&&Et.setContext(window),this.setAnimationLoop=function(t)
{bt=t,yt.setAnimationLoop(t),null===t?
Et.stop():Et.start()},yt.addEventListener("sessionstart",Mt),yt.addEventListener("s
essionend",Tt),this.render=function(t,e){let n,i;if(void 0!==arguments[2]&&
(console.warn("THREE.WebGLRenderer.render(): the renderTarget argument has been
removed. Use .setRenderTarget() instead."),n=arguments[2]),void 0!==arguments[3]&&
(console.warn("THREE.WebGLRenderer.render(): the forceClear argument has been
removed. Use .clear() instead."),i=arguments[3]),void 0!==e&&!0!==e.isCamera)return
void console.error("THREE.WebGLRenderer.render: camera is not an instance of
THREE.Camera.");if(!0===g)return;!0===t.autoUpdate&&t.updateMatrixWorld(),null===e.
parent&&e.updateMatrixWorld(),!0===yt.enabled&&!0===yt.isPresenting&&
(e=yt.getCamera(e)),!0===t.isScene&&t.onBeforeRender(f,t,e,n||_),d=rt.get(t,m.lengt
h),d.init(),m.push(d),G.multiplyMatrices(e.projectionMatrix,e.matrixWorldInverse),F
.setFromProjectionMatrix(G),H=this.localClippingEnabled,O=st.init(this.clippingPlan
es,H,e),u=it.get(t,p.length),u.init(),p.push(u),At(t,e,O,f.sortObjects),u.finish(),
!0===f.sortObjects&&u.sort(D,I),!0===O&&st.beginShadows();const
r=d.state.shadowsArray;at.render(r,t,e),d.setupLights(),d.setupLightsView(e),!0===O
&&st.endShadows(),!0===this.info.autoReset&&this.info.reset(),void

```



```

0!==n&&this.setRenderTarget(n),ot.render(u,t,e,i);const
s=u.opaque,a=u.transparent;s.length>0&&Rt(s,t,e),a.length>0&&Rt(a,t,e),null!==(J.updateRenderTargetMipmap(_),J.updateMultisampleRenderTarget(_)),!0===t.isScene&&
t.onAfterRender(f,t,e),q.buffers.depth.setTest(!0),q.buffers.depth.setMask(!0),q.bu
ffers.color.setMask(!0),q.setPolygonOffset(!1),dt.resetDefaultState(),w=-1,S=null,m
.pop(),d=m.length>0?m[m.length-1]:null,p.pop(),u=p.length>0?p[p.length-
1]:null},this.getActiveCubeFace=function(){return
v},this.getActiveMipmapLevel=function(){return y},this.getRenderTarget=function()
{return _},this.setRenderTarget=function(t,e=0,n=0){_=t,v=e,y=n,t&&void
0===Z.get(t).__webglFramebuffer&&J.setupRenderTarget(t);let i=null,r=!1,s=!1;if(t)
{const n=t.texture;(n.isDataTexture3D|n.isDataTexture2DArray)&&(s=!0);const
a=Z.get(t).__webglFramebuffer;t.isWebGLCubeRenderTarget?
(i=a[e],r=!0):i=t.isWebGLMultisampleRenderTarget?
Z.get(t).__webglMultisampledFramebuffer:a,T.copy(t.viewport),A.copy(t.scissor),L=t.
scissorTest}else
T.copy(N).multiplyScalar(P).floor(),A.copy(B).multiplyScalar(P).floor(),L=z;if(q.bi
ndFramebuffer(36160,i),q.viewport(T),q.scissor(A),q.setScissorTest(L),r){const
i=Z.get(t.texture);pt.framebufferTexture2D(36160,36064,34069+e,i.__webglTexture,n)}
else if(s){const
i=Z.get(t.texture),r=e||0;pt.framebufferTextureLayer(36160,36064,i.__webglTexture,n
||0,r)},this.readRenderTargetPixels=function(t,e,n,i,r,s,a)
{if(!t||!t.isWebGLRenderTarget)return void
console.error("THREE.WebGLRenderer.readRenderTargetPixels: renderTarget is not
THREE.WebGLRenderTarget.");let
o=Z.get(t).__webglFramebuffer;if(t.isWebGLCubeRenderTarget&&void 0!==a&&(o=o[a]),o)
{q.bindFramebuffer(36160,o);try{const
a=t.texture,o=a.format,l=a.type;if(o!==E&&ut.convert(o)!==pt.getParameter(35739))re
turn void console.error("THREE.WebGLRenderer.readRenderTargetPixels: renderTarget
is not in RGBA or implementation defined format.");const c=l===M&&
(W.has("EXT_color_buffer_half_float")||j.isWebGL2&&W.has("EXT_color_buffer_float"))
;if(!(l===x||ut.convert(l)===pt.getParameter(35738)||l===b&&
(j.isWebGL2||W.has("OES_texture_float")||W.has("WEBGL_color_buffer_float"))||c))ret
urn void console.error("THREE.WebGLRenderer.readRenderTargetPixels: renderTarget is
not in UnsignedByteType or implementation defined
type.");36053===pt.checkFramebufferStatus(36160)?e>=0&&e<=t.width-
i&&n>=0&&n<=t.height-
r&&pt.readPixels(e,n,i,r,ut.convert(o),ut.convert(l),s):console.error("THREE.WebGLR
enderer.readRenderTargetPixels: readPixels from renderTarget failed. Framebuffer
not complete.")}finally{const t=null!==(Z.get(_).__webglFramebuffer:null;q.bindFramebuffer(36160,t))},this.copyFramebuffer
ToTexture=function(t,e,n=0){const i=Math.pow(2,-
n),r=Math.floor(e.image.width*i),s=Math.floor(e.image.height*i),a=ut.convert(e.form
at);J.setTexture2D(e,0),pt.copyTexImage2D(3553,n,a,t.x,t.y,r,s,0),q.unbindTexture()
},this.copyTextureToTexture=function(t,e,n,i=0){const
r=e.image.width,s=e.image.height,a=ut.convert(n.format),o=ut.convert(n.type);J.setT
exture2D(n,0),pt.pixelStorei(37440,n.flipY),pt.pixelStorei(37441,n.premultiplyAlpha
),pt.pixelStorei(3317,n.unpackAlignment),e.isDataTexture?
pt.texSubImage2D(3553,i,t.x,t.y,r,s,a,o,e.image.data):e.isCompressedTexture?
pt.compressedTexSubImage2D(3553,i,t.x,t.y,e.mipmaps[0].width,e.mipmaps[0].height,a,
e.mipmaps[0].data):pt.texSubImage2D(3553,i,t.x,t.y,a,o,e.image),0===i&&n.generateMi
pmaps&&pt.generateMipmap(3553),q.unbindTexture(),this.copyTextureToTexture3D=funct
ion(t,e,n,i,r=0){if(f.isWebGL1Renderer)return void
console.warn("THREE.WebGLRenderer.copyTextureToTexture3D: can only be used with
WebGL2.");const{width:s,height:a,data:o}=n.image,l=ut.convert(i.format),c=ut.conver
t(i.type);let
h;if(i.isDataTexture3D)J.setTexture3D(i,0),h=32879;else{if(!i.isDataTexture2DArray)
return void console.warn("THREE.WebGLRenderer.copyTextureToTexture3D: only supports
THREE.DataTexture3D and

```

```

THREE.DataTexture2DArray.");J.setTexture2DArray(i,0),h=35866}pt.pixelStorei(37440,i
.flipY),pt.pixelStorei(37441,i.premultiplyAlpha),pt.pixelStorei(3317,i.unpackAlignm
ent);const
u=pt.getParameter(3314),d=pt.getParameter(32878),p=pt.getParameter(3316),m=pt.getPa
rameter(3315),g=pt.getParameter(32877);pt.pixelStorei(3314,s),pt.pixelStorei(32878,
a),pt.pixelStorei(3316,t.min.x),pt.pixelStorei(3315,t.min.y),pt.pixelStorei(32877,t
.min.z),pt.texSubImage3D(h,r,e.x,e.y,e.z,t.max.x-t.min.x+1,t.max.y-
t.min.y+1,t.max.z-
t.min.z+1,l,c,o),pt.pixelStorei(3314,u),pt.pixelStorei(32878,d),pt.pixelStorei(3316
,p),pt.pixelStorei(3315,m),pt.pixelStorei(32877,g),0==r&i.generateMipmaps&pt.gen
erateMipmap(h),q.unbindTexture(),this.initTexture=function(t)
{J.setTexture2D(t,0),q.unbindTexture(),this.resetState=function()
{v=0,y=0,_=null,q.reset(),dt.reset(),"undefined"!=typeof
__THREE_DEVTOOLS__&&__THREE_DEVTOOLS__.dispatchEvent(new CustomEvent("observe",
{detail:this}))}class bs extends ws{bs.prototype.isWebGL1Renderer=!0;class
Ms{constructor(t,e=25e-5){this.name="",this.color=new tn(t),this.density=e}clone()
{return new Ms(this.color,this.density)}toJSON()
{return{type:"FogExp2",color:this.color.getHex(),density:this.density}}Ms.prototype
.isFogExp2=!0;class Ss{constructor(t,e=1,n=1e3){this.name="",this.color=new
tn(t),this.near=e,this.far=n}clone(){return new
Ss(this.color,this.near,this.far)}toJSON()
{return{type:"Fog",color:this.color.getHex(),near:this.near,far:this.far}}Ss.proto
type.isFog=!0;class Ts extends Ce{constructor()
{super(),this.type="Scene",this.background=null,this.environment=null,this.fog=null
,this.overrideMaterial=null,this.autoUpdate=!0,"undefined"!=typeof
__THREE_DEVTOOLS__&&__THREE_DEVTOOLS__.dispatchEvent(new CustomEvent("observe",
{detail:this}))}copy(t,e){return super.copy(t,e),null!==t.background&&
(this.background=t.background.clone()),null!==t.environment&&
(this.environment=t.environment.clone()),null!==t.fog&&
(this.fog=t.fog.clone()),null!==t.overrideMaterial&&
(this.overrideMaterial=t.overrideMaterial.clone()),this.autoUpdate=t.autoUpdate,thi
s.matrixAutoUpdate=t.matrixAutoUpdate,this}toJSON(t){const e=super.toJSON(t);return
null!==this.background&&
(e.object.background=this.background.toJSON(t)),null!==this.environment&&
(e.object.environment=this.environment.toJSON(t)),null!==this.fog&&
(e.object.fog=this.fog.toJSON()),e}}Ts.prototype.isScene=!0;class
Es{constructor(t,e){this.array=t,this.stride=e,this.count=void 0!==(t?
t.length/e:0,this.usage=et,this.updateRange=
{offset:0,count:-1},this.version=0,this.uuid=ct(),this.onUploadCallback=function()
{}}set needsUpdate(t){!0===t&&this.version++}setUsage(t){return
this.usage=t,this}copy(t){return this.array=new
t.array.constructor(t.array),this.count=t.count,this.stride=t.stride,this.usage=t.u
sage,this}copyAt(t,e,n){t*=this.stride,n*=e.stride;for(let
i=0,r=this.stride;i<r;i++)this.array[t+i]=e.array[n+i];return this}set(t,e=0)
{return this.array.set(t,e),this}clone(t){void 0===t.arrayBuffers&&(t.arrayBuffers=
{}),void 0===this.array.buffer._uuid&&(this.array.buffer._uuid=ct()),void
0===t.arrayBuffers[this.array.buffer._uuid]&&
(t.arrayBuffers[this.array.buffer._uuid]=this.array.slice(0).buffer);const e=new
this.array.constructor(t.arrayBuffers[this.array.buffer._uuid]),n=new
Es(e,this.stride);return n.setUsage(this.usage),n}onUpload(t){return
this.onUploadCallback=t,this}toJSON(t){return void 0===t.arrayBuffers&&
(t.arrayBuffers={}),void 0===this.array.buffer._uuid&&
(this.array.buffer._uuid=ct()),void 0===t.arrayBuffers[this.array.buffer._uuid]&&
(t.arrayBuffers[this.array.buffer._uuid]=Array.prototype.slice.call(new
Uint32Array(this.array.buffer))),
{uuid:this.uuid,buffer:this.array.buffer._uuid,type:this.array.constructor.name,str
ide:this.stride}}Es.prototype.isInterleavedBuffer=!0;const As=new Lt;class
Ls{constructor(t,e,n,i)

```

```

{this.name="",this.data=t,this.itemSize=e,this.offset=n,this.normalized=!0===i}get
count(){return this.data.count}get array(){return this.data.array}set
needsUpdate(t){this.data.needsUpdate=t}applyMatrix4(t){for(let
e=0,n=this.data.count;e<n;e++)As.x=this.getX(e),As.y=this.getY(e),As.z=this.getZ(e)
,As.applyMatrix4(t),this.setXYZ(e,As.x,As.y,As.z);return this}applyNormalMatrix(t)
{for(let
e=0,n=this.count;e<n;e++)As.x=this.getX(e),As.y=this.getY(e),As.z=this.getZ(e),As.a
pplyNormalMatrix(t),this.setXYZ(e,As.x,As.y,As.z);return this}transformDirection(t)
{for(let
e=0,n=this.count;e<n;e++)As.x=this.getX(e),As.y=this.getY(e),As.z=this.getZ(e),As.t
ransformDirection(t),this.setXYZ(e,As.x,As.y,As.z);return this}setX(t,e){return
this.data.array[t*this.data.stride+this.offset]=e,this}setY(t,e){return
this.data.array[t*this.data.stride+this.offset+1]=e,this}setZ(t,e){return
this.data.array[t*this.data.stride+this.offset+2]=e,this}setW(t,e){return
this.data.array[t*this.data.stride+this.offset+3]=e,this}getX(t){return
this.data.array[t*this.data.stride+this.offset]}getY(t){return
this.data.array[t*this.data.stride+this.offset+1]}getZ(t){return
this.data.array[t*this.data.stride+this.offset+2]}getW(t){return
this.data.array[t*this.data.stride+this.offset+3]}setXY(t,e,n){return
t=t*this.data.stride+this.offset,this.data.array[t+0]=e,this.data.array[t+1]=n,this
}setXYZ(t,e,n,i){return
t=t*this.data.stride+this.offset,this.data.array[t+0]=e,this.data.array[t+1]=n,this
.data.array[t+2]=i,this}setXYZW(t,e,n,i,r){return
t=t*this.data.stride+this.offset,this.data.array[t+0]=e,this.data.array[t+1]=n,this
.data.array[t+2]=i,this.data.array[t+3]=r,this}clone(t){if(void 0===t)
{console.log("THREE.InterleavedBufferAttribute.clone(): Cloning an interleaved
buffer attribute will deinterleave buffer data.");const t=[];for(let
e=0;e<this.count;e++){const n=e*this.data.stride+this.offset;for(let
e=0;e<this.itemSize;e++)t.push(this.data.array[n+e])}return new sn(new
this.array.constructor(t),this.itemSize,this.normalized)}return void
0===t.interleavedBuffers&&(t.interleavedBuffers={}),void
0===t.interleavedBuffers[this.data.uuid]&&
(t.interleavedBuffers[this.data.uuid]=this.data.clone(t)),new
Ls(t.interleavedBuffers[this.data.uuid],this.itemSize,this.offset,this.normalized)}
toJSON(t){if(void 0===t){console.log("THREE.InterleavedBufferAttribute.toJSON():
Serializing an interleaved buffer attribute will deinterleave buffer data.");const
t=[];for(let e=0;e<this.count;e++){const n=e*this.data.stride+this.offset;for(let
e=0;e<this.itemSize;e++)t.push(this.data.array[n+e])}return{itemSize:this.itemSize,
type:this.array.constructor.name,array:t,normalized:this.normalized}}return void
0===t.interleavedBuffers&&(t.interleavedBuffers={}),void
0===t.interleavedBuffers[this.data.uuid]&&
(t.interleavedBuffers[this.data.uuid]=this.data.toJSON(t)),
{isInterleavedBufferAttribute:!0,itemSize:this.itemSize,data:this.data.uuid,offset:
this.offset,normalized:this.normalized}}}}Ls.prototype.isInterleavedBufferAttribute=
!0;class Rs extends Xe{constructor(t)
{super(),this.type="SpriteMaterial",this.color=new
tn(16777215),this.map=null,this.alphaMap=null,this.rotation=0,this.sizeAttenuation=
!0,this.transparent=!0,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.map=t.map,this.alphaMap=t.alphaMap,this
.rotation=t.rotation,this.sizeAttenuation=t.sizeAttenuation,this}}let
Cs;Rs.prototype.isSpriteMaterial=!0;const Ps=new Lt,Ds=new Lt,Is=new Lt,Ns=new
vt,Bs=new vt,zs=new se,Fs=new Lt,Os=new Lt,Hs=new Lt,Gs=new vt,Us=new vt,ks=new
vt;class Vs extends Ce{constructor(t){if(super(),this.type="Sprite",void 0===Cs)
{Cs=new En;const t=new
Float32Array([-0.5,-0.5,0,0,0,.5,-0.5,0,1,0,.5,.5,0,1,1,-0.5,.5,0,0,1]),e=new
Es(t,5);Cs.setIndex([0,1,2,0,2,3]),Cs.setAttribute("position",new
Ls(e,3,0,!1)),Cs.setAttribute("uv",new
Ls(e,2,3,!1))}this.geometry=Cs,this.material=void 0!==t?t:new Rs,this.center=new

```

```

vt(.5,.5)}raycast(t,e){null===t.camera&&console.error('THREE.Sprite:
"Raycaster.camera" needs to be set in order to raycast against
sprites. '),Ds.setFromMatrixScale(this.matrixWorld),zs.copy(t.camera.matrixWorld),th
is.modelViewMatrix.multiplyMatrices(t.camera.matrixWorldInverse,this.matrixWorld),I
s.setFromMatrixPosition(this.modelViewMatrix),t.camera.isPerspectiveCamera&&!l===th
is.material.sizeAttenuation&&Ds.multiplyScalar(-Is.z);const
n=this.material.rotation;let i,r;0!==n&&(r=Math.cos(n),i=Math.sin(n));const
s=this.center;Ws(Fs.set(-.5,-.5,0),Is,s,Ds,i,r),Ws(Os.set(.5,-.5,0),Is,s,Ds,i,r),Ws
(Hs.set(.5,.5,0),Is,s,Ds,i,r),Gs.set(0,0),Us.set(1,0),ks.set(1,1);let
a=t.ray.intersectTriangle(Fs,Os,Hs,!l,Ps);if(null===a&&
(Ws(Os.set(-.5,.5,0),Is,s,Ds,i,r),Us.set(0,1),a=t.ray.intersectTriangle(Fs,Hs,Os,!l
,Ps),null===a))return;const
o=t.ray.origin.distanceTo(Ps);o<t.near||o>t.far||e.push({distance:o,point:Ps.clone(
),uv:je.getUV(Ps,Fs,Os,Hs,Gs,Us,ks,new vt),face:null,object:this})}copy(t){return
super.copy(t),void
0!==t.center&&this.center.copy(t.center),this.material=t.material,this}}function
Ws(t,e,n,i,r,s){Ns.subVectors(t,n).addScalar(.5).multiply(i),void 0!==r?
(Bs.x=s*Ns.x-
r*Ns.y,Bs.y=r*Ns.x+s*Ns.y):Bs.copy(Ns),t.copy(e),t.x+=Bs.x,t.y+=Bs.y,t.applyMatrix4
(zs)}Vs.prototype.isSprite=!0;const js=new Lt,qs=new Lt;class Xs extends
Ce{constructor()
{super(),this._currentLevel=0,this.type="LOD",Object.defineProperties(this,{levels:
{enumerable:!0,value:[]},isLOD:{value:!0}}),this.autoUpdate=!0}copy(t)
{super.copy(t,!l);const e=t.levels;for(let t=0,n=e.length;t<n;t++){const
n=e[t];this.addLevel(n.object.clone(),n.distance)}return
this.autoUpdate=t.autoUpdate,this}addLevel(t,e=0){e=Math.abs(e);const
n=this.levels;let i;for(i=0;i<n.length&&!(e<n[i].distance);i++);return
n.splice(i,0,{distance:e,object:t}),this.add(t),this}getCurrentLevel(){return
this._currentLevel}getObjectForDistance(t){const e=this.levels;if(e.length>0){let
n,i;for(n=1,i=e.length;n<i&&!(t<e[n].distance);n++);return e[n-1].object}return
null}raycast(t,e){if(this.levels.length>0)
{js.setFromMatrixPosition(this.matrixWorld);const
n=t.ray.origin.distanceTo(js);this.getObjectForDistance(n).raycast(t,e)}}update(t)
{const e=this.levels;if(e.length>1)
{js.setFromMatrixPosition(t.matrixWorld),qs.setFromMatrixPosition(this.matrixWorld)
;const n=js.distanceTo(qs)/t.zoom;let
i,r;for(e[0].object.visible=!0,i=1,r=e.length;i<r&&n>=e[i].distance;i++)e[i-
1].object.visible=!1,e[i].object.visible=!0;for(this._currentLevel=i-
1;i<r;i++)e[i].object.visible=!1}}toJSON(t){const
e=super.toJSON(t);!l===this.autoUpdate&&(e.object.autoUpdate=!1),e.object.levels=
[];const n=this.levels;for(let t=0,i=n.length;t<i;t++){const
i=n[t];e.object.levels.push({object:i.object.uuid,distance:i.distance})}return
e}}const Ys=new Lt,Zs=new St,Js=new St,Qs=new Lt,Ks=new se;class $s extends
Wn{constructor(t,e)
{super(t,e),this.type="SkinnedMesh",this.bindMode="attached",this.bindMatrix=new
se,this.bindMatrixInverse=new se}copy(t){return
super.copy(t),this.bindMode=t.bindMode,this.bindMatrix.copy(t.bindMatrix),this.bind
MatrixInverse.copy(t.bindMatrixInverse),this.skeleton=t.skeleton,this}bind(t,e)
{this.skeleton=t,void 0===e&&
(this.updateMatrixWorld(!0),this.skeleton.calculateInverses(),e=this.matrixWorld),t
his.bindMatrix.copy(e),this.bindMatrixInverse.copy(e).invert()}pose()
{this.skeleton.pose()}normalizeSkinWeights(){const t=new
St,e=this.geometry.attributes.skinWeight;for(let n=0,i=e.count;n<i;n++)
{t.x=e.getX(n),t.y=e.getY(n),t.z=e.getZ(n),t.w=e.getW(n);const
i=1/t.manhattanLength();i!==1/0?
t.multiplyScalar(i):t.set(1,0,0,0),e.setXYZW(n,t.x,t.y,t.z,t.w)}}updateMatrixWorld(
t){super.updateMatrixWorld(t),"attached"===this.bindMode?
this.bindMatrixInverse.copy(this.matrixWorld).invert():"detached"===this.bindMode?

```

```

this.bindMatrixInverse.copy(this.bindMatrix).invert():console.warn("THREE.SkinnedMesh: Unrecognized bindMode: "+this.bindMode)}boneTransform(t,e){const n=this.skeleton,i=this.geometry;Zs.fromBufferAttribute(i.attributes.skinIndex,t),Js.fromBufferAttribute(i.attributes.skinWeight,t),Ys.fromBufferAttribute(i.attributes.position,t).applyMatrix4(this.bindMatrix),e.set(0,0,0);for(let t=0;t<4;t++){const i=Js.getComponent(t);if(0!==i){const r=Zs.getComponent(t);Ks.multiplyMatrices(n.bones[r].matrixWorld,n.boneInverses[r]),e.addScaledVector(Qs.copy(Ys).applyMatrix4(Ks),i)}}return e.applyMatrix4(this.bindMatrixInverse)}}$s.prototype.isSkinnedMesh=!0;class ta extends Ce{constructor(){super(),this.type="Bone"}}ta.prototype.isBone=!0;const ea=new se,na=new se;class ia{constructor(t=[],e=[]){this.uuid=ct(),this.bones=t.slice(0),this.boneInverses=e,this.boneMatrices=null,this.boneTexture=null,this.boneTextureSize=0,this.frame=-1,this.init()}init(){const t=this.bones,e=this.boneInverses;if(this.boneMatrices=new Float32Array(16*t.length),0===e.length)this.calculateInverses();else if(t.length!==e.length){console.warn("THREE.Skeleton: Number of inverse bone matrices does not match amount of bones."),this.boneInverses=[];for(let t=0,e=this.bones.length;t<e;t++)this.boneInverses.push(new se)}calculateInverses(){this.boneInverses.length=0;for(let t=0,e=this.bones.length;t<e;t++){const e=new se;this.bones[t]&&e.copy(this.bones[t].matrixWorld).invert(),this.boneInverses.push(e)}}pose(){for(let t=0,e=this.bones.length;t<e;t++){const e=this.bones[t];e&&e.matrixWorld.copy(this.boneInverses[t]).invert()}for(let t=0,e=this.bones.length;t<e;t++){const e=this.bones[t];e&&(e.parent&&e.parent.isBone?(e.matrix.copy(e.parent.matrixWorld).invert(),e.matrix.multiply(e.matrixWorld)):e.matrix.copy(e.matrixWorld),e.matrix.decompose(e.position,e.quaternion,e.scale))}}update(){const t=this.bones,e=this.boneInverses,n=this.boneMatrices,i=this.boneTexture;for(let i=0,r=t.length;i<r;i++){const r=t[i]?t[i].matrixWorld:na;ea.multiplyMatrices(r,e[i]),ea.toArray(n,16*i)}null!==i&&(i.needsUpdate=!0)}clone(){return new ia(this.bones,this.boneInverses)}getBoneByName(t){for(let e=0,n=this.bones.length;e<n;e++){const n=this.bones[e];if(n.name===t)return n}}dispose(){null!==this.boneTexture&&(this.boneTexture.dispose(),this.boneTexture=null)}fromJSON(t,e){this.uuid=t.uuid;for(let n=0,i=t.bones.length;n<i;n++){const i=t.bones[n];let r=e[i];void 0===r&&(console.warn("THREE.Skeleton: No bone found with UUID:",i),r=new ta),this.bones.push(r),this.boneInverses.push((new se).fromArray(t.boneInverses[n]))}return this.init(),this}toJSON(){const t={metadata:{version:4.5,type:"Skeleton",generator:"Skeleton.toJSON"},bones:[],boneInverses:[]};t.uuid=this.uuid;const e=this.bones,n=this.boneInverses;for(let i=0,r=e.length;i<r;i++){const r=e[i];t.bones.push(r.uuid);const s=n[i];t.boneInverses.push(s.toArray())}return t}}const ra=new se,sa=new se,aa=[],oa=new Wn;class la extends Wn{constructor(t,e,n){super(t,e),this.instanceMatrix=new sn(new Float32Array(16*n),16),this.instanceColor=null,this.count=n,this.frustumCulled=!1}copy(t){return super.copy(t),this.instanceMatrix.copy(t.instanceMatrix),null!==t.instanceColor&&(this.instanceColor=t.instanceColor.clone()),this.count=t.count,this}getColorAt(t,e){e.fromArray(this.instanceColor.array,3*t)}getMatrixAt(t,e){e.fromArray(this.instanceMatrix.array,16*t)}raycast(t,e){const n=this.matrixWorld,i=this.count;if(oa.geometry=this.geometry,oa.material=this.material,void 0!==oa.material)for(let r=0;r<i;r++){this.getMatrixAt(r,ra),sa.multiplyMatrices(n,ra),oa.matrixWorld=sa,oa.raycast(t,aa);for(let t=0,n=aa.length;t<n;t++){const n=aa[t];n.instanceId=r,n.object=this,e.push(n)}aa.length=0}}setColorAt(t,e){null===this.instanceColor&&(this.instanceColor=new sn(new Float32Array(3*this.count),3)),e.toArray(this.instanceColor.array,3*t)}setMatrixAt(

```

```

t,e){e.toArray(this.instanceMatrix.array,16*t)}updateMorphTargets(){dispose()
{this.dispatchEvent({type:"dispose"})}}la.prototype.isInstancedMesh=!0;class ca
extends Xe{constructor(t){super(),this.type="LineBasicMaterial",this.color=new
tn(16777215),this.linewidth=1,this.linecap="round",this.linejoin="round",this.morph
Targets=!1,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.linewidth=t.linewidth,this.linecap=t.li
necap,this.linejoin=t.linejoin,this.morphTargets=t.morphTargets,this}}ca.prototype.
isLineBasicMaterial=!0;const ha=new Lt,ua=new Lt,da=new se,pa=new re,ma=new
Jt;class fa extends Ce{constructor(t=new En,e=new ca)
{super(),this.type="Line",this.geometry=t,this.material=e,this.updateMorphTargets()
}copy(t){return
super.copy(t),this.material=t.material,this.geometry=t.geometry,this}computeLineDis
tances(){const t=this.geometry;if(t.isBufferGeometry)if(null===t.index){const
e=t.attributes.position,n=[0];for(let
t=1,i=e.count;t<i;t++)ha.fromBufferAttribute(e,t-1),ua.fromBufferAttribute(e,t),n[t]=n[t-
1],n[t]+=ha.distanceTo(ua);t.setAttribute("lineDistance",new mn(n,1))}else
console.warn("THREE.Line.computeLineDistances(): Computation only possible with
non-indexed BufferGeometry.");else
t.isGeometry&&console.error("THREE.Line.computeLineDistances() no longer supports
THREE.Geometry. Use THREE.BufferGeometry instead.");return this}raycast(t,e){const
n=this.geometry,i=this.matrixWorld,r=t.params.Line.threshold,s=n.drawRange;if(null=
==n.boundingSphere&&n.computeBoundingSphere(),ma.copy(n.boundingSphere),ma.applyMat
rix4(i),ma.radius+=r,!1===t.ray.intersectsSphere(ma))return;da.copy(i).invert(),pa.
copy(t.ray).applyMatrix4(da);const
a=r/((this.scale.x+this.scale.y+this.scale.z)/3),o=a*a,l=new Lt,c=new Lt,h=new
Lt,u=new Lt,d=this.isLineSegments?2:1;if(n.isBufferGeometry){const
i=n.index,r=n.attributes.position;if(null!==i){for(let
n=Math.max(0,s.start),a=Math.min(i.count,s.start+s.count)-1;n<a;n+=d){const
s=i.getX(n),a=i.getX(n+1);l.fromBufferAttribute(r,s),c.fromBufferAttribute(r,a);if(
pa.distanceSqToSegment(l,c,u,h)>o)continue;u.applyMatrix4(this.matrixWorld);const
d=t.ray.origin.distanceTo(u);d<t.near||d>t.far||e.push({distance:d,point:h.clone().
applyMatrix4(this.matrixWorld),index:n,face:null,faceIndex:null,object:this})}}else
{for(let n=Math.max(0,s.start),i=Math.min(r.count,s.start+s.count)-1;n<i;n+=d)
{l.fromBufferAttribute(r,n),c.fromBufferAttribute(r,n+1);if(pa.distanceSqToSegment(
l,c,u,h)>o)continue;u.applyMatrix4(this.matrixWorld);const
i=t.ray.origin.distanceTo(u);i<t.near||i>t.far||e.push({distance:i,point:h.clone().
applyMatrix4(this.matrixWorld),index:n,face:null,faceIndex:null,object:this})}}els
e n.isGeometry&&console.error("THREE.Line.raycast() no longer supports
THREE.Geometry. Use THREE.BufferGeometry instead.")}updateMorphTargets(){const
t=this.geometry;if(t.isBufferGeometry){const
e=t.morphAttributes,n=Object.keys(e);if(n.length>0){const t=e[n[0]];if(void 0!==t)
{this.morphTargetInfluences=[],this.morphTargetDictionary={};for(let
e=0,n=t.length;e<n;e++){const
n=t[e].name||String(e);this.morphTargetInfluences.push(0),this.morphTargetDictionar
y[n]=e}}}}else{const e=t.morphTargets;void
0!==e&&e.length>0&&console.error("THREE.Line.updateMorphTargets() does not support
THREE.Geometry. Use THREE.BufferGeometry instead.")}}fa.prototype.isLine=!0;const
ga=new Lt,va=new Lt;class ya extends fa{constructor(t,e)
{super(t,e),this.type="LineSegments"}computeLineDistances(){const
t=this.geometry;if(t.isBufferGeometry)if(null===t.index){const
e=t.attributes.position,n=[];for(let
t=0,i=e.count;t<i;t+=2)ga.fromBufferAttribute(e,t),va.fromBufferAttribute(e,t+1),n[
t]=0===t?0:n[t-1],n[t+1]=n[t]+ga.distanceTo(va);t.setAttribute("lineDistance",new
mn(n,1))}else console.warn("THREE.LineSegments.computeLineDistances(): Computation
only possible with non-indexed BufferGeometry.");else
t.isGeometry&&console.error("THREE.LineSegments.computeLineDistances() no longer
supports THREE.Geometry. Use THREE.BufferGeometry instead.");return

```

```

this}}ya.prototype.isLineSegments=!0;class xa extends fa{constructor(t,e)
{super(t,e),this.type="LineLoop"}}xa.prototype.isLineLoop=!0;class _a extends
Xe{constructor(t){super(),this.type="PointsMaterial",this.color=new
tn(16777215),this.map=null,this.alphaMap=null,this.size=1,this.sizeAttenuation=!0,t
his.morphTargets=!1,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.map=t.map,this.alphaMap=t.alphaMap,this
.size=t.size,this.sizeAttenuation=t.sizeAttenuation,this.morphTargets=t.morphTarget
s,this}}_a.prototype.isPointsMaterial=!0;const wa=new se,ba=new re,Ma=new Jt,Sa=new
Lt;class Ta extends Ce{constructor(t=new En,e=new _a)
{super(),this.type="Points",this.geometry=t,this.material=e,this.updateMorphTargets
()}copy(t){return
super.copy(t),this.material=t.material,this.geometry=t.geometry,this}raycast(t,e)
{const
n=this.geometry,i=this.matrixWorld,r=t.params.Points.threshold,s=n.drawRange;if(nul
l===n.boundingSphere&&n.computeBoundingSphere(),Ma.copy(n.boundingSphere),Ma.applyM
atrix4(i),Ma.radius+=r,!1===t.ray.intersectsSphere(Ma))return;wa.copy(i).invert(),b
a.copy(t.ray).applyMatrix4(wa);const
a=r/((this.scale.x+this.scale.y+this.scale.z)/3),o=a*a;if(n.isBufferGeometry){const
r=n.index,a=n.attributes.position;if(null!==r){for(let
n=Math.max(0,s.start),l=Math.min(r.count,s.start+s.count);n<l;n++){const
s=r.getX(n);Sa.fromBufferAttribute(a,s),Ea(Sa,s,o,i,t,e,this)}}else{for(let
n=Math.max(0,s.start),r=Math.min(a.count,s.start+s.count);n<r;n++)Sa.fromBufferAttr
ibute(a,n),Ea(Sa,n,o,i,t,e,this)}}else console.error("THREE.Points.raycast() no
longer supports THREE.Geometry. Use THREE.BufferGeometry
instead.")}updateMorphTargets(){const t=this.geometry;if(t.isBufferGeometry){const
e=t.morphAttributes,n=Object.keys(e);if(n.length>0){const t=e[n[0]];if(void 0!==(t)
{this.morphTargetInfluences=[],this.morphTargetDictionary={};for(let
e=0,n=t.length;e<n;e++){const
n=t[e].name||String(e);this.morphTargetInfluences.push(0),this.morphTargetDictionar
y[n]=e}}}}else{const e=t.morphTargets;void
0!==(e&&e.length>0&&console.error("THREE.Points.updateMorphTargets() does not
support THREE.Geometry. Use THREE.BufferGeometry instead."))}}function
Ea(t,e,n,i,r,s,a){const o=ba.distanceSqToPoint(t);if(o<n){const n=new
Lt;ba.closestPointToPoint(t,n),n.applyMatrix4(i);const
l=r.ray.origin.distanceTo(n);if(l<r.near||l>r.far)return;s.push({distance:l,distanc
eToRay:Math.sqrt(o),point:n,index:e,face:null,object:a})}}Ta.prototype.isPoints=!0;
class Aa extends bt{constructor(t,e,n,i,r,s,a,o,l)
{super(t,e,n,i,r,s,a,o,l),this.format=void 0!==(a?T:this.minFilter=void 0!==(s?
s:g,this.magFilter=void 0!==(r?r:g,this.generateMipmaps=!1;const
c=this;"requestVideoFrameCallback"in t&&t.requestVideoFrameCallback((function e()
{c.needsUpdate=!0,t.requestVideoFrameCallback(e)}))}clone(){return new
this.constructor(this.image).copy(this)}update(){const
t=this.image;!1===t.requestVideoFrameCallback"in
t&&t.readyState>=t.HAVE_CURRENT_DATA&&
(this.needsUpdate=!0)}}Aa.prototype.isVideoTexture=!0;class La extends
bt{constructor(t,e,n,i,r,s,a,o,l,c,h,u){super(null,s,a,o,l,c,i,r,h,u),this.image=
{width:e,height:n},this.mipmaps=t,this.flipY=!1,this.generateMipmaps=!1}}La.prototy
pe.isCompressedTexture=!0;class Ra extends bt{constructor(t,e,n,i,r,s,a,o,l)
{super(t,e,n,i,r,s,a,o,l),this.needsUpdate=!0}}Ra.prototype.isCanvasTexture=!0;clas
s Ca extends bt{constructor(t,e,n,i,r,s,a,o,l,c){if((c=void 0!==(c?
c:A)!==(A&&c!==(L)throw new Error("DepthTexture format must be either
THREE.DepthFormat or THREE.DepthStencilFormat");void 0===n&&c===A&&(n=_),void
0===n&&c===L&&(n=S),super(null,i,r,s,a,o,c,n,l),this.image=
{width:t,height:e},this.magFilter=void 0!==(a?a:p,this.minFilter=void 0!==(o?
o:p,this.flipY=!1,this.generateMipmaps=!1}}Ca.prototype.isDepthTexture=!0;class Pa
extends En{constructor(t=1,e=8,n=0,i=2*Math.PI)
{super(),this.type="CircleGeometry",this.parameters=
{radius:t,segments:e,thetaStart:n,thetaLength:i},e=Math.max(3,e);const r=[],s=[],a=

```

```

[],o=[],l=new Lt,c=new vt;s.push(0,0,0),a.push(0,0,1),o.push(.5,.5);for(let
r=0,h=3;r<=e;r++,h+=3){const
u=n+r/e*i;l.x=t*Math.cos(u),l.y=t*Math.sin(u),s.push(l.x,l.y,l.z),a.push(0,0,1),c.x
=(s[h]/t+1)/2,c.y=(s[h+1]/t+1)/2,o.push(c.x,c.y)}for(let
t=1;t<=e;t++)r.push(t,t+1,0);this.setIndex(r),this.setAttribute("position",new
mn(s,3)),this.setAttribute("normal",new mn(a,3)),this.setAttribute("uv",new
mn(o,2))}}class Da extends En{constructor(t=1,e=1,n=1,i=8,r=1,s=!1,a=0,o=2*Math.PI)
{super(),this.type="CylinderGeometry",this.parameters=
{radiusTop:t,radiusBottom:e,height:n,radialSegments:i,heightSegments:r,openEnded:s,
thetaStart:a,thetaLength:o};const l=this;i=Math.floor(i),r=Math.floor(r);const c=
[],h=[],u=[],d=[];let p=0;const m=[],f=n/2;let g=0;function v(n){const r=p,s=new
vt,m=new Lt;let v=0;const y=!0===n?t:e,x=!0===n?l:-1;for(let
t=1;t<=i;t++)h.push(0,f*x,0),u.push(0,x,0),d.push(.5,.5),p++;const _=p;for(let
t=0;t<=i;t++){const
e=t/i*o+a,n=Math.cos(e),r=Math.sin(e);m.x=y*r,m.y=f*x,m.z=y*n,h.push(m.x,m.y,m.z),u
.push(0,x,0),s.x=.5*n+.5,s.y=.5*r*x+.5,d.push(s.x,s.y),p++}for(let t=0;t<i;t++)
{const e=r+t,i=_+t;!0===n?
c.push(i,i+1,e):c.push(i+1,i,e),v+=3}l.addGroup(g,v,!0===n?1:2),g+=v)!function()
{const s=new Lt,v=new Lt;let y=0;const x=(e-t)/n;for(let l=0;l<=r;l++){const c=
[],g=l/r,y=g*(e-t)+t;for(let t=0;t<=i;t++){const
e=t/i,r=e*o+a,l=Math.sin(r),m=Math.cos(r);v.x=y*l,v.y=-
g*n+f,v.z=y*m,h.push(v.x,v.y,v.z),s.set(l,x,m).normalize(),u.push(s.x,s.y,s.z),d.pu
sh(e,1-g),c.push(p++)}m.push(c)}for(let t=0;t<i;t++)for(let e=0;e<r;e++){const
n=m[e][t],i=m[e+1][t],r=m[e+1][t+1],s=m[e]
[t+1];c.push(n,i,s),c.push(i,r,s),y+=6}l.addGroup(g,y,0),g+=y}(),!l===s&&
(t>0&&v(!0),e>0&&v(!1)),this.setIndex(c),this.setAttribute("position",new
mn(h,3)),this.setAttribute("normal",new mn(u,3)),this.setAttribute("uv",new
mn(d,2))}}class Ia extends Da{constructor(t=1,e=1,n=8,i=1,r=!1,s=0,a=2*Math.PI)
{super(0,t,e,n,i,r,s,a),this.type="ConeGeometry",this.parameters=
{radius:t,height:e,radialSegments:n,heightSegments:i,openEnded:r,thetaStart:s,theta
Length:a}}class Na extends En{constructor(t,e,n=1,i=0)
{super(),this.type="PolyhedronGeometry",this.parameters=
{vertices:t,indices:e,radius:n,detail:i};const r=[],s=[];function a(t,e,n,i){const
r=i+1,s=[];for(let i=0;i<=r;i++){s[i]=[];const
a=t.clone().lerp(n,i/r),o=e.clone().lerp(n,i/r),l=r-i;for(let t=0;t<=l;t++)s[i]
[t]=0===t&&i===r?a:a.clone().lerp(o,t/l)}for(let t=0;t<r;t++)for(let e=0;e<2*(r-
t)-1;e++){const n=Math.floor(e/2);e%2===0?(o(s[t][n+1]),o(s[t+1][n]),o(s[t][n])):
(o(s[t][n+1]),o(s[t+1][n+1]),o(s[t+1][n]))}}function o(t)
{r.push(t.x,t.y,t.z)}function l(e,n){const
i=3*e;n.x=t[i+0],n.y=t[i+1],n.z=t[i+2]}function c(t,e,n,i){i<0&&l===t.x&&(s[e]=t.x-
1),0===n.x&&0===n.z&&(s[e]=i/2/Math.PI+.5)}function h(t){return Math.atan2(t.z,-
t.x)}!function(t){const n=new Lt,i=new Lt,r=new Lt;for(let
s=0;s<e.length;s+=3)l(e[s+0],n),l(e[s+1],i),l(e[s+2],r),a(n,i,r,t)}(i),function(t)
{const e=new Lt;for(let
n=0;n<r.length;n+=3){e.x=r[n+0],e.y=r[n+1],e.z=r[n+2],e.normalize().multiplyScalar(t
),r[n+0]=e.x,r[n+1]=e.y,r[n+2]=e.z}(n),function(){const t=new Lt;for(let
n=0;n<r.length;n+=3){t.x=r[n+0],t.y=r[n+1],t.z=r[n+2];const i=h(t)/2/Math.PI+.5,a=
(e=t,Math.atan2(-e.y,Math.sqrt(e.x*e.x+e.z*e.z))/Math.PI+.5);s.push(i,1-a)}var e;
(function(){const t=new Lt,e=new Lt,n=new Lt,i=new Lt,a=new vt,o=new vt,l=new
vt;for(let u=0,d=0;u<r.length;u+=9,d+=6)
{t.set(r[u+0],r[u+1],r[u+2]),e.set(r[u+3],r[u+4],r[u+5]),n.set(r[u+6],r[u+7],r[u+8]
),a.set(s[d+0],s[d+1]),o.set(s[d+2],s[d+3]),l.set(s[d+4],s[d+5]),i.copy(t).add(e).a
dd(n).divideScalar(3);const p=h(i);c(a,d+0,t,p),c(o,d+2,e,p),c(l,d+4,n,p)}})
(),function(){for(let t=0;t<s.length;t+=6){const
e=s[t+0],n=s[t+2],i=s[t+4],r=Math.max(e,n,i),a=Math.min(e,n,i);r>.9&&a<.1&&(e<.2&&
(s[t+0]+=1),n<.2&&(s[t+2]+=1),i<.2&&(s[t+4]+=1))}}()
(),this.setAttribute("position",new mn(r,3)),this.setAttribute("normal",new
mn(r.slice(),3)),this.setAttribute("uv",new mn(s,2)),0===i?

```



```

this.computeVertexNormals():this.normalizeNormals()}}class Ba extends
Na{constructor(t=1,e=0){const n=
(1+Math.sqrt(5))/2,i=1/n;super([-1,-1,-1,-1,-1,1,-1,1,-1,-1,1,1,1,-1,-1,1,-1,1,1,1,
-1,1,1,1,0,-i,-n,0,-i,n,0,i,-n,0,i,n,-i,-n,0,-i,n,0,i,-n,0,i,n,0,-n,0,-i,n,0,-i,-
n,0,i,n,0,i],
[3,11,7,3,7,15,3,15,13,7,19,17,7,17,6,7,6,15,17,4,8,17,8,10,17,10,6,8,0,16,8,16,2,8
,2,10,0,12,1,0,1,18,0,18,16,6,10,2,6,2,13,6,13,15,2,16,18,2,18,3,2,3,13,18,1,9,18,9
,11,18,11,3,4,14,12,4,12,0,4,0,8,11,9,5,11,5,19,11,19,7,19,5,14,19,14,4,19,4,17,1,1
2,14,1,14,5,1,5,9],t,e),this.type="DodecahedronGeometry",this.parameters=
{radius:t,detail:e}}const za=new Lt,Fa=new Lt,Oa=new Lt,Ha=new je;class Ga extends
En{constructor(t,e){if(super(),this.type="EdgesGeometry",this.parameters=
{thresholdAngle:e},e=void 0!==e?e:1,!0===t.isGeometry)return void
console.error("THREE.EdgesGeometry no longer supports THREE.Geometry. Use
THREE.BufferGeometry instead.");const
n=Math.pow(10,4),i=Math.cos(ot*e),r=t.getIndex(),s=t.getAttribute("position"),a=r?
r.count:s.count,o=[0,0,0],l=["a","b","c"],c=new Array(3),h={},u=[];for(let
t=0;t<a;t+=3){r?(o[0]=r.getX(t),o[1]=r.getX(t+1),o[2]=r.getX(t+2)):
(o[0]=t,o[1]=t+1,o[2]=t+2);const{a:e,b:a,c:d}=Ha;if(e.fromBufferAttribute(s,o[0]),a
.fromBufferAttribute(s,o[1]),d.fromBufferAttribute(s,o[2]),Ha.getNormal(Oa),c[0]=`$
{Math.round(e.x*n)},${Math.round(e.y*n)},${Math.round(e.z*n)}`,`c[1]=`${Math.round(a
.x*n)},${Math.round(a.y*n)},${Math.round(a.z*n)}`,`c[2]=`${Math.round(d.x*n)},${Math
.round(d.y*n)},${Math.round(d.z*n)}`,`c[0]!==c[1]&&c[1]!==c[2]&&c[2]!==c[0])for(let
t=0;t<3;t++){const e=
(t+1)%3,n=c[t],r=c[e],s=Ha[l[t]],a=Ha[l[e]],d=`${n}_${r}`,p=`${r}_${n}`;p in
h&&h[p]?((Oa.dot(h[p].normal)<=i&&
(u.push(s.x,s.y,s.z),u.push(a.x,a.y,a.z)),h[p]=null):d in h||(h[d]=
{index0:o[t],index1:o[e],normal:Oa.clone()}))}for(const t in h){if(h[t])
{const{index0:e,index1:n}=h[t];za.fromBufferAttribute(s,e),Fa.fromBufferAttribute(s
,n),u.push(za.x,za.y,za.z),u.push(Fa.x,Fa.y,Fa.z)}this.setAttribute("position",new
mn(u,3))}}const Ua=function(t,e,n){n=n|2;const i=e&&e.length,r=i?
e[0]*n:t.length;let s=ka(t,0,r,n,!0);const a=[];if(!s||s.next===s.prev)return a;let
o,l,c,h,u,d,p;if(i&&(s=function(t,e,n,i){const r=[];let
s,a,o,l,c;for(s=0,a=e.length;s<a;s++)o=e[s]*i,l=s<a-1?
e[s+1]*i:t.length,c=ka(t,o,l,i,!1),c===c.next&&
(c.steiner=!0),r.push($a(c));for(r.sort(Za),s=0;s<r.length;s++)Ja(r[s],n),n=Va(n,n.
next);return n})(t,e,s,n)),t.length>80*n){o=c=t[0],l=h=t[1];for(let
e=n;e<r;e+=n)u=t[e],d=t[e+1],u<o&&(o=u),d<l&&(l=d),u>c&&(c=u),d>h&&
(h=d);p=Math.max(c-o,h-l),p=0!==p?1/p:0}return Wa(s,a,n,o,l,p),a};function
ka(t,e,n,i,r){let s,a;if(r===function(t,e,n,i){let r=0;for(let s=e,a=n-
i;s<n;s+=i)r+=(t[a]-t[s])*(t[s+1]+t[a+1]),a=s;return r}
(t,e,n,i)>0)for(s=e;s<n;s+=i)a=co(s,t[s],t[s+1],a);else for(s=n-i;s>=e;s-
=i)a=co(s,t[s],t[s+1],a);return a&&io(a,a.next)&&(ho(a),a=a.next),a}function
Va(t,e){if(!t)return t;e||(e=t);let
n,i=t;do{if(n=!1,i.steiner||!io(i,i.next)&&0!==no(i.prev,i,i.next))i=i.next;else{if
(ho(i),i=e=i.prev,i===i.next)break;n=!0}}while(n||i!==e);return e}function
Wa(t,e,n,i,r,s,a){if(!t)return;!a&&s&&function(t,e,n,i){let r=t;do{null===r.z&&
(r.z=Ka(r.x,r.y,e,n,i)),r.prevZ=r.prev,r.nextZ=r.next,r=r.next}while(r!==t);r.prevZ
.nextZ=null,r.prevZ=null,function(t){let
e,n,i,r,s,a,o,l,c=1;do{for(n=t,t=null,s=null,a=0;n){for(a++,i=n,o=0,e=0;e<c&&
(o++,i=i.nextZ,i);e++);for(l=c;o>0||l>0&&i;)0!==o&&(0===l||!i||n.z<=i.z)?
(r=n,n=n.nextZ,o--):(r=i,i=i.nextZ,l--),s?
s.nextZ=r:t=r,r.prevZ=s,s=r;n=i}s.nextZ=null,c*=2}while(a>1)}(r)}(t,i,r,s);let
o,l,c=t;for(;t.prev!==t.next;)if(o=t.prev,l=t.next,s?
qa(t,i,r,s):ja(t))e.push(o.i/n),e.push(t.i/n),e.push(l.i/n),ho(t),t=l.next,c=l.next
;else if((t=l)===c){a?l===a?
Wa(t=Xa(Va(t),e,n),e,n,i,r,s,2):2===a&&Ya(t,e,n,i,r,s):Wa(Va(t),e,n,i,r,s,1);break}
}function ja(t){const e=t.prev,n=t,i=t.next;if(no(e,n,i)>=0)return!1;let
r=t.next.next;for(;r!==t.prev;)}

```

```

{if(to(e.x,e.y,n.x,n.y,i.x,i.y,r.x,r.y)&&no(r.prev,r,r.next)>=0)return!1;r=r.next}r
return!0}function qa(t,e,n,i){const
r=t.prev,s=t,a=t.next;if(no(r,s,a)>=0)return!1;const o=r.x<s.x?r.x<a.x?
r.x:a.x:s.x<a.x?s.x:a.x,l=r.y<s.y?r.y<a.y?r.y:a.y:s.y<a.y?s.y:a.y,c=r.x>s.x?
r.x>a.x?r.x:a.x:s.x>a.x?s.x:a.x,h=r.y>s.y?r.y>a.y?r.y:a.y:s.y>a.y?
s.y:a.y,u=Ka(o,l,e,n,i),d=Ka(c,h,e,n,i);let
p=t.prevZ,m=t.nextZ;for(;p&&p.z>=u&&m&&m.z<=d;)
{if(p!==(t.prev&&p!==(t.next&&to(r.x,r.y,s.x,s.y,a.x,a.y,p.x,p.y)&&no(p.prev,p,p.next
)>=0)return!1;if(p=p.prevZ,m!==(t.prev&&m!==(t.next&&to(r.x,r.y,s.x,s.y,a.x,a.y,m.x,m
.y)&&no(m.prev,m,m.next)>=0)return!1;m=m.nextZ}for(;p&&p.z>=u;)
{if(p!==(t.prev&&p!==(t.next&&to(r.x,r.y,s.x,s.y,a.x,a.y,p.x,p.y)&&no(p.prev,p,p.next
)>=0)return!1;p=p.prevZ}for(;m&&m.z<=d;)
{if(m!==(t.prev&&m!==(t.next&&to(r.x,r.y,s.x,s.y,a.x,a.y,m.x,m.y)&&no(m.prev,m,m.next
)>=0)return!1;m=m.nextZ}return!0}function Xa(t,e,n){let i=t;do{const
r=i.prev,s=i.next.next;!io(r,s)&&ro(r,i,i.next,s)&&oo(r,s)&&oo(s,r)&&
(e.push(r.i/n),e.push(i.i/n),e.push(s.i/n),ho(i),ho(i.next),i=t=s),i=i.next}while(i
!=t);return Va(i)}function Ya(t,e,n,i,r,s){let a=t;do{let
t=a.next.next;for(;t!=a.prev;){if(a.i!=t.i&&eo(a,t)){let o=lo(a,t);return
a=Va(a,a.next),o=Va(o,o.next),Wa(a,e,n,i,r,s),void
Wa(o,e,n,i,r,s)}t=t.next}a=a.next}while(a!=t)}function Za(t,e){return t.x-
e.x}function Ja(t,e){if(e=function(t,e){let n=e;const i=t.x,r=t.y;let
s,a=-1/0;do{if(r<=n.y&&r>=n.next.y&&n.next.y!=n.y){const t=n.x+(r-n.y)*(n.next.x-
n.x)/(n.next.y-n.y);if(t<=i&&t>a){if(a=t,t===i){if(r===n.y)return
n;if(r===n.next.y)return n.next}s=n.x<n.next.x?
n:n.next}}n=n.next}while(n!=e);if(!s)return null;if(i===a)return s;const
o=s,l=s.x,c=s.y;let h,u=1/0;n=s;do{>=n.x&&n.x>=l&&i!=n.x&&to(r<c?i:a,r,l,c,r<c?
a:i,r,n.x,n.y)&&(h=Math.abs(r-n.y)/(i-n.x),oo(n,t)&&(h<u||h===u&&
(n.x>s.x||n.x===s.x&&Qa(s,n))&&(s=n,u=h)),n=n.next}while(n!=o);return s}(t,e))
{const n=lo(e,t);Va(e,e.next),Va(n,n.next)}function Qa(t,e){return
no(t.prev,t,e.prev)<0&&no(e.next,t,t.next)<0}function Ka(t,e,n,i,r)
{return(t=1431655765&((t=858993459&((t=252645135&((t=16711935&((t=32767*(t-
n)*r)|t<<8))|t<<4))|t<<2))|t<<1))|(e=1431655765&((e=858993459&((e=252645135&
((e=16711935&((e=32767*(e-i)*r)|e<<8))|e<<4))|e<<2))|e<<1))<<1}function $a(t){let
e=t,n=t;do{(e.x<n.x||e.x===n.x&&e.y<n.y)&&(n=e),e=e.next}while(e!=t);return
n}function to(t,e,n,i,r,s,a,o){return(r-a)*(e-o)-(t-a)*(s-o)>=0&&(t-a)*(i-o)-(n-a)*
(e-o)>=0&&(n-a)*(s-o)-(r-a)*(i-o)>=0}function eo(t,e){return
t.next.i!=e.i&&t.prev.i!=e.i&&!function(t,e){let
n=t;do{if(n.i!=t.i&&n.next.i!=t.i&&n.i!=e.i&&n.next.i!=e.i&&ro(n,n.next,t,e))re
turn!0;n=n.next}while(n!=t);return!1}(t,e)&&(oo(t,e)&&oo(e,t)&&function(t,e){let
n=t,i=!1;const r=(t.x+e.x)/2,s=
(t.y+e.y)/2;do{n.y>s!>n.next.y>s&&n.next.y!=n.y&&r<(n.next.x-n.x)*(s-
n.y)/(n.next.y-n.y)+n.x&&(i=!i),n=n.next}while(n!=t);return i}(t,e)&&
(no(t.prev,t,e.prev)||no(t,e.prev,e)||io(t,e)&&no(t.prev,t,t.next)>0&&no(e.prev,e,
e.next)>0)}function no(t,e,n){return(e.y-t.y)*(n.x-e.x)-(e.x-t.x)*(n.y-
e.y)}function io(t,e){return t.x===e.x&&t.y===e.y}function ro(t,e,n,i){const
r=ao(no(t,e,n)),s=ao(no(t,e,i)),a=ao(no(n,i,t)),o=ao(no(n,i,e));return
r!=s&&a!=o||(!(!0!=r||!so(t,n,e))||(!(!0!=s||!so(t,i,e))||(!
(0!=a||!so(n,t,i))||(!(!0!=o||!so(n,e,i))))))}function so(t,e,n){return
e.x<=Math.max(t.x,n.x)&&e.x>=Math.min(t.x,n.x)&&e.y<=Math.max(t.y,n.y)&&e.y>=Math.m
in(t.y,n.y)}function ao(t){return t>0?1:t<0?-1:0}function oo(t,e){return
no(t.prev,t,t.next)<0?no(t,e,t.next)>=0&&no(t,t.prev,e)>=0:no(t,e,t.prev)
<0||no(t,t.next,e)<0}function lo(t,e){const n=new uo(t,i,t.x,t.y),i=new
uo(e,i,e.x,e.y),r=t.next,s=e.prev;return
t.next=e,e.prev=t,n.next=r,r.prev=n,i.next=n,n.prev=i,s.next=i,i.prev=s,i}function
co(t,e,n,i){const r=new uo(t,e,n);return i?
(r.next=i.next,r.prev=i,i.next.prev=r,i.next=r):(r.prev=r,r.next=r),r}function
ho(t){t.next.prev=t.prev,t.prev.next=t.next,t.prevZ&&
(t.prevZ.nextZ=t.nextZ),t.nextZ&&(t.nextZ.prevZ=t.prevZ)}function uo(t,e,n)

```

```

{this.i=t,this.x=e,this.y=n,this.prev=null,this.next=null,this.z=null,this.prevZ=null,
this.nextZ=null,this.steiner=!1}class po{static area(t){const e=t.length;let
n=0;for(let i=e-1,r=0;r<e;i=r++)n+=t[i].x*t[r].y-t[r].x*t[i].y;return.5*n}static
isClockWise(t){return po.area(t)<0}static triangulateShape(t,e){const n=[],i=[],r=
[];mo(t),fo(n,t);let s=t.length;e.forEach(mo);for(let
t=0;t<e.length;t++)i.push(s),s+=e[t].length,fo(n,e[t]);const a=Ua(n,i);for(let
t=0;t<a.length;t+=3)r.push(a.slice(t,t+3));return r}}function mo(t){const
e=t.length;e>2&&t[e-1].equals(t[0])&&t.pop()}function fo(t,e){for(let
n=0;n<e.length;n++)t.push(e[n].x),t.push(e[n].y)}class go extends
En{constructor(t,e){super(),this.type="ExtrudeGeometry",this.parameters=
{shapes:t,options:e},t=Array.isArray(t)?t:[t];const n=this,i=[],r=[];for(let
e=0,n=t.length;e<n;e++){s(t[e])}function s(t){const s=[],a=void
0!==(e.curveSegments?e.curveSegments:12,o=void 0!==(e.steps?e.steps:1);let l=void
0!==(e.depth?e.depth:100,c=void 0==(e.bevelEnabled||e.bevelEnabled,h=void
0!==(e.bevelThickness?e.bevelThickness:6,u=void 0!==(e.bevelSize?e.bevelSize:h-
2,d=void 0!==(e.bevelOffset?e.bevelOffset:0,p=void 0!==(e.bevelSegments?
e.bevelSegments:3);const m=e.extrudePath,f=void 0!==(e.UVGenerator?
e.UVGenerator:vo);void 0!==(e.amount&&(console.warn("THREE.ExtrudeBufferGeometry:
amount has been renamed to depth."),l=e.amount));let g,v,y,x,_,w=!1;m&&
(g=m.getSpacedPoints(o),w=!0,c=!1,v=m.computeFrenetFrames(o,!1),y=new Lt,x=new
Lt,_=new Lt),c||(p=0,h=0,u=0,d=0);const b=t.extractPoints(a);let M=b.shape;const
S=b.holes;if(!po.isClockWise(M)){M=M.reverse();for(let t=0,e=S.length;t<e;t++)
{const e=S[t];po.isClockWise(e)&&(S[t]=e.reverse())}}const
T=po.triangulateShape(M,S),E=M;for(let t=0,e=S.length;t<e;t++){const
e=S[t];M=M.concat(e)}function A(t,e,n){return
e||console.error("THREE.ExtrudeGeometry: vec does not
exist"),e.clone().multiplyScalar(n).add(t)}const L=M.length,R=T.length;function
C(t,e,n){let i,r,s;const a=t.x-e.x,o=t.y-e.y,l=n.x-t.x,c=n.y-t.y,h=a*a+o*o,u=a*c-
o*l;if(Math.abs(u)>Number.EPSILON){const u=Math.sqrt(h),d=Math.sqrt(l*l+c*c),p=e.x-
o*u,m=e.y+a/u,f=((n.x-c/d-p)*c-(n.y+l/d-m)*l)/(a*c-o*l);i=p+a*f-t.x,r=m+o*f-
t.y;const g=i*i+r*r;if(g<=2)return new vt(i,r);s=Math.sqrt(g/2)}else{let
t=!1;a>Number.EPSILON?l>Number.EPSILON&&(t=!0):a<-Number.EPSILON?l<-
Number.EPSILON&&(t=!0):Math.sign(o)===Math.sign(c)&&(t=!0),t?(i=-
o,r=a,s=Math.sqrt(h)):(i=a,r=o,s=Math.sqrt(h/2))}return new vt(i/s,r/s)}const P=
[];for(let t=0,e=E.length,n=e-1,i=t+1;t<e;t++,n++,i++)n===e&&(n=0),i===e&&
(i=0),P[t]=C(E[t],E[n],E[i]);const D=[];let I,N=P.concat();for(let
t=0,e=S.length;t<e;t++){const e=S[t];I=[];for(let t=0,n=e.length,i=n-
1,r=t+1;t<n;t++,i++,r++)i===n&&(i=0),r===n&&
(r=0),I[t]=C(e[t],e[i],e[r]);D.push(I),N=N.concat(I)}for(let t=0;t<p;t++){const
e=t/p,n=h*Math.cos(e*Math.PI/2),i=u*Math.sin(e*Math.PI/2)+d;for(let
t=0,e=E.length;t<e;t++){const e=A(E[t],P[t],i);F(e.x,e.y,-n)}for(let
t=0,e=S.length;t<e;t++){const e=S[t];I=D[t];for(let t=0,r=e.length;t<r;t++){const
r=A(e[t],I[t],i);F(r.x,r.y,-n)}}const B=u+d;for(let t=0;t<L;t++){const e=c?
A(M[t],N[t],B):M[t];w?
(x.copy(v.normals[0]).multiplyScalar(e.x),y.copy(v.binormals[0]).multiplyScalar(e.y
),_.copy(g[0]).add(x).add(y),F(_x,_y,_z)):F(e.x,e.y,0)}for(let
t=1;t<=o;t++)for(let e=0;e<L;e++){const n=c?A(M[e],N[e],B):M[e];w?
(x.copy(v.normals[t]).multiplyScalar(n.x),y.copy(v.binormals[t]).multiplyScalar(n.y
),_.copy(g[t]).add(x).add(y),F(_x,_y,_z)):F(n.x,n.y,l/o*t)}for(let t=p-1;t>=0;t-
-){const e=t/p,n=h*Math.cos(e*Math.PI/2),i=u*Math.sin(e*Math.PI/2)+d;for(let
t=0,e=E.length;t<e;t++){const e=A(E[t],P[t],i);F(e.x,e.y,l+n)}for(let
t=0,e=S.length;t<e;t++){const e=S[t];I=D[t];for(let t=0,r=e.length;t<r;t++){const
r=A(e[t],I[t],i);w?F(r.x,r.y+g[o-1].y,g[o-1].x+n):F(r.x,r.y,l+n)}}}function z(t,e)
{let n=t.length;for(--n>=0;){const i=n;let r=n-1;r<0&&(r=t.length-1);for(let
t=0,n=o+2*p;t<n;t++){const n=L*t,s=L*(t+1);H(e+i+n,e+r+n,e+r+s,e+i+s)}}function
F(t,e,n){s.push(t),s.push(e),s.push(n)}function O(t,e,r){G(t),G(e),G(r);const
s=i.length/3,a=f.generateTopUV(n,i,s-3,s-2,s-1);U(a[0]),U(a[1]),U(a[2])}function
H(t,e,r,s){G(t),G(e),G(s),G(e),G(r),G(s);const

```

```

a=i.length/3,o=f.generateSideWallUV(n,i,a-6,a-3,a-2,a-
1);U(o[0]),U(o[1]),U(o[3]),U(o[1]),U(o[2]),U(o[3])}function G(t)
{i.push(s[3*t+0]),i.push(s[3*t+1]),i.push(s[3*t+2])}function U(t)
{r.push(t.x),r.push(t.y)}!function(){const t=i.length/3;if(c){let t=0,e=L*t;for(let
t=0;t<R;t++){const n=T[t];O(n[2]+e,n[1]+e,n[0]+e)}t=o+2*p,e=L*t;for(let
t=0;t<R;t++){const n=T[t];O(n[0]+e,n[1]+e,n[2]+e)}}else{for(let t=0;t<R;t++){const
e=T[t];O(e[2],e[1],e[0])}for(let t=0;t<R;t++){const
e=T[t];O(e[0]+L*o,e[1]+L*o,e[2]+L*o)}}n.addGroup(t,i.length/3-t,0)}(),function()
{const t=i.length/3;let e=0;z(E,e),e+=E.length;for(let t=0,n=S.length;t<n;t++)
{const n=S[t];z(n,e),e+=n.length}n.addGroup(t,i.length/3-t,1)}
())}this.setAttribute("position",new mn(i,3)),this.setAttribute("uv",new
mn(r,2)),this.computeVertexNormals()}toJSON(){const
t=En.prototype.toJSON.call(this);return function(t,e,n){if(n.shapes=
[],Array.isArray(t))for(let e=0,i=t.length;e<i;e++){const
i=t[e];n.shapes.push(i.uuid)}else n.shapes.push(t.uuid);void 0!==e.extrudePath&&
(n.options.extrudePath=e.extrudePath.toJSON());return n}
(this.parameters.shapes,this.parameters.options,t))}const vo=
{generateTopUV:function(t,e,n,i,r){const
s=e[3*n],a=e[3*n+1],o=e[3*i],l=e[3*i+1],c=e[3*r],h=e[3*r+1];return[new vt(s,a),new
vt(o,l),new vt(c,h)]},generateSideWallUV:function(t,e,n,i,r,s){const
a=e[3*n],o=e[3*n+1],l=e[3*n+2],c=e[3*i],h=e[3*i+1],u=e[3*i+2],d=e[3*r],p=e[3*r+1],m
=e[3*r+2],f=e[3*s],g=e[3*s+1],v=e[3*s+2];return Math.abs(o-h)<.01?[new vt(a,l-
1),new vt(c,l-u),new vt(d,l-m),new vt(f,l-v)]:[new vt(o,l-1),new vt(h,l-u),new
vt(p,l-m),new vt(g,l-v)]};class yo extends Na{constructor(t=1,e=0){const n=
(1+Math.sqrt(5))/2;super([-1,n,0,1,n,0,-1,-n,0,1,-n,0,0,-1,n,0,1,n,0,-1,-n,0,1,-
n,n,0,-1,n,0,1,-n,0,-1,-n,0,1],
[0,11,5,0,5,1,0,1,7,0,7,10,0,10,11,1,5,9,5,11,4,11,10,2,10,7,6,7,1,8,3,9,4,3,4,2,3,
2,6,3,6,8,3,8,9,4,9,5,2,4,11,6,2,10,8,6,7,9,8,1],t,e),this.type="IcosahedronGeometr
y",this.parameters={radius:t,detail:e}}class xo extends
En{constructor(t,e=12,n=0,i=2*Math.PI)
{super(),this.type="LatheGeometry",this.parameters=
{points:t,segments:e,phiStart:n,phiLength:i},e=Math.floor(e),i=ht(i,0,2*Math.PI);co
nst r=[],s=[],a=[],o=1/e,l=new Lt,c=new vt;for(let r=0;r<=e;r++){const
h=n+r*o*i,u=Math.sin(h),d=Math.cos(h);for(let n=0;n<=t.length-
1;n++)l.x=t[n].x*u,l.y=t[n].y,l.z=t[n].x*d,s.push(l.x,l.y,l.z),c.x=r/e,c.y=n/(t.len
gth-1),a.push(c.x,c.y)}for(let n=0;n<e;n++)for(let e=0;e<t.length-1;e++){const
i=e+n*t.length,s=i,a=i+t.length,o=i+t.length+1,l=i+1;r.push(s,a,l),r.push(a,o,l)}if
(this.setIndex(r),this.setAttribute("position",new
mn(s,3)),this.setAttribute("uv",new
mn(a,2)),this.computeVertexNormals(),i===2*Math.PI){const
n=this.attributes.normal.array,i=new Lt,r=new Lt,s=new Lt,a=e*t.length*3;for(let
e=0,o=0;e<t.length;e++,o+=3)i.x=n[o+0],i.y=n[o+1],i.z=n[o+2],r.x=n[a+o+0],r.y=n[a+o
+1],r.z=n[a+o+2],s.addVectors(i,r).normalize(),n[o+0]=n[a+o+0]=s.x,n[o+1]=n[a+o+1]=
s.y,n[o+2]=n[a+o+2]=s.z}}class _o extends Na{constructor(t=1,e=0)
{super([1,0,0,-1,0,0,0,1,0,0,-1,0,0,0,1,0,0,-1],
[0,2,4,0,4,3,0,3,5,0,5,2,1,2,5,1,5,3,1,3,4,1,4,2],t,e),this.type="OctahedronGeometr
y",this.parameters={radius:t,detail:e}}class wo extends En{constructor(t,e,n)
{super(),this.type="ParametricGeometry",this.parameters=
{func:t,slices:e,stacks:n};const i=[],r=[],s=[],a=[],o=1e-5,l=new Lt,c=new Lt,h=new
Lt,u=new Lt,d=new Lt;t.length<3&&console.error("THREE.ParametricGeometry: Function
must now modify a Vector3 as third parameter.");const p=e+1;for(let i=0;i<=n;i++)
{const p=i/n;for(let n=0;n<=e;n++){const i=n/e;t(i,p,c),r.push(c.x,c.y,c.z),i-o>=0?
(t(i-o,p,h),u.subVectors(c,h)):(t(i+o,p,h),u.subVectors(h,c)),p-o>=0?(t(i,p-
o,h),d.subVectors(c,h)):
(t(i,p+o,h),d.subVectors(h,c)),l.crossVectors(u,d).normalize(),s.push(l.x,l.y,l.z),
a.push(i,p)}}for(let t=0;t<n;t++)for(let n=0;n<e;n++){const e=t*p+n,r=t*p+n+1,s=
(t+1)*p+n+1,a=
(t+1)*p+n;i.push(e,r,a),i.push(r,s,a)}this.setIndex(i),this.setAttribute("position"

```

```

, new mn(r, 3)), this.setAttribute("normal", new mn(s, 3)), this.setAttribute("uv", new
mn(a, 2))}}class bo extends En{constructor(t=.5, e=1, n=8, i=1, r=0, s=2*Math.PI)
{super(), this.type="RingGeometry", this.parameters=
{innerRadius:t, outerRadius:e, thetaSegments:n, phiSegments:i, thetaStart:r, thetaLength
:s}, n=Math.max(3, n); const a=[], o=[], l=[], c=[]; let h=t; const u=(e-
t)/(i=Math.max(1, i)), d=new Lt, p=new vt; for(let t=0; t<=i; t++){for(let t=0; t<=n; t++)
{const
i=r+t/n*s; d.x=h*Math.cos(i), d.y=h*Math.sin(i), o.push(d.x, d.y, d.z), l.push(0, 0, 1), p.x
=(d.x/e+1)/2, p.y=(d.y/e+1)/2, c.push(p.x, p.y)}h+=u}for(let t=0; t<i; t++){const e=t*
(n+1); for(let t=0; t<n; t++){const
i=t+e, r=i, s=i+n+1, o=i+n+2, l=i+1; a.push(r, s, l), a.push(s, o, l)}}this.setIndex(a), this.
setAttribute("position", new mn(o, 3)), this.setAttribute("normal", new
mn(l, 3)), this.setAttribute("uv", new mn(c, 2))}}class Mo extends
En{constructor(t, e=12){super(), this.type="ShapeGeometry", this.parameters=
{shapes:t, curveSegments:e}; const n=[], i=[], r=[], s=[]; let
a=0, o=0; if(!l===Array.isArray(t))l(t); else for(let
e=0; e<t.length; e++)l(t[e]), this.addGroup(a, o, e), a+=o, o=0; function l(t){const
a=i.length/3, l=t.extractPoints(e); let c=l.shape; const
h=l.holes; !l===po.isClockWise(c)&&(c=c.reverse()); for(let t=0, e=h.length; t<e; t++)
{const e=h[t]; !0===po.isClockWise(e)&&(h[t]=e.reverse())}const
u=po.triangulateShape(c, h); for(let t=0, e=h.length; t<e; t++){const
e=h[t]; c=c.concat(e)}for(let t=0, e=c.length; t<e; t++){const
e=c[t]; i.push(e.x, e.y, 0), r.push(0, 0, 1), s.push(e.x, e.y)}for(let
t=0, e=u.length; t<e; t++){const
e=u[t], i=e[0]+a, r=e[1]+a, s=e[2]+a; n.push(i, r, s), o+=3}}this.setIndex(n), this.setAttr
ibute("position", new mn(i, 3)), this.setAttribute("normal", new
mn(r, 3)), this.setAttribute("uv", new mn(s, 2))}toJSON(){const
t=En.prototype.toJSON.call(this); return function(t, e){if(e.shapes=
[], Array.isArray(t))for(let n=0, i=t.length; n<i; n++){const
i=t[n]; e.shapes.push(i.uuid)}else e.shapes.push(t.uuid); return e}
(this.parameters.shapes, t)}}class So extends
En{constructor(t=1, e=8, n=6, i=0, r=2*Math.PI, s=0, a=Math.PI)
{super(), this.type="SphereGeometry", this.parameters=
{radius:t, widthSegments:e, heightSegments:n, phiStart:i, phiLength:r, thetaStart:s, thet
aLength:a}, e=Math.max(3, Math.floor(e)), n=Math.max(2, Math.floor(n)); const
o=Math.min(s+a, Math.PI); let l=0; const c=[], h=new Lt, u=new Lt, d=[], p=[], m=[], f=
[]; for(let d=0; d<=n; d++){const g=[], v=d/n; let y=0; 0===d&&0===s?
y=.5/e:d===n&&0===Math.PI&&(y=-.5/e); for(let n=0; n<=e; n++){const o=n/e; h.x=-
t*Math.cos(i+o*r)*Math.sin(s+v*a), h.y=t*Math.cos(s+v*a), h.z=t*Math.sin(i+o*r)*Math.
sin(s+v*a), p.push(h.x, h.y, h.z), u.copy(h).normalize(), m.push(u.x, u.y, u.z), f.push(o+y
, 1-v), g.push(l++)}c.push(g)}for(let t=0; t<n; t++)for(let i=0; i<e; i++){const e=c[t]
[i+1], r=c[t][i], a=c[t+1][i], l=c[t+1][i+1]; (0!==t||s>0)&&d.push(e, r, l), (t!==n-
1||o<Math.PI)&&d.push(r, a, l)}this.setIndex(d), this.setAttribute("position", new
mn(p, 3)), this.setAttribute("normal", new mn(m, 3)), this.setAttribute("uv", new
mn(f, 2))}}class To extends Na{constructor(t=1, e=0)
{super([1, 1, 1, -1, -1, 1, -1, 1, -1, 1, -1, -1],
[2, 1, 0, 0, 3, 2, 1, 3, 0, 2, 3, 1], t, e), this.type="TetrahedronGeometry", this.parameters=
{radius:t, detail:e}}class Eo extends go{constructor(t, e={}){const
n=e.font; if(!n||!n.isFont)return console.error("THREE.TextGeometry: font parameter
is not an instance of THREE.Font."), new En; const
i=n.generateShapes(t, e.size); e.depth=void 0!===e.height?e.height:50, void
0===e.bevelThickness&&(e.bevelThickness=10), void 0===e.bevelSize&&
(e.bevelSize=8), void 0===e.bevelEnabled&&
(e.bevelEnabled=!1), super(i, e), this.type="TextGeometry"}}class Ao extends
En{constructor(t=1, e=.4, n=8, i=6, r=2*Math.PI)
{super(), this.type="TorusGeometry", this.parameters=
{radius:t, tube:e, radialSegments:n, tubularSegments:i, arc:r}, n=Math.floor(n), i=Math.f
loor(i); const s=[], a=[], o=[], l=[], c=new Lt, h=new Lt, u=new Lt; for(let

```

```

s=0;s<=n;s++)for(let d=0;d<=i;d++){const p=d/i*r,m=s/n*Math.PI*2;h.x=
(t+e*Math.cos(m))*Math.cos(p),h.y=
(t+e*Math.cos(m))*Math.sin(p),h.z=e*Math.sin(m),a.push(h.x,h.y,h.z),c.x=t*Math.cos(
p),c.y=t*Math.sin(p),u.subVectors(h,c).normalize(),o.push(u.x,u.y,u.z),l.push(d/i),
l.push(s/n)}for(let t=1;t<=n;t++)for(let e=1;e<=i;e++){const n=(i+1)*t+e-1,r=(i+1)*
(t-1)+e-1,a=(i+1)*(t-1)+e,o=
(i+1)*t+e;s.push(n,r,o),s.push(r,a,o)}this.setIndex(s),this.setAttribute("position"
,new mn(a,3)),this.setAttribute("normal",new mn(o,3)),this.setAttribute("uv",new
mn(l,2))}class Lo extends En{constructor(t=1,e=.4,n=64,i=8,r=2,s=3)
{super(),this.type="TorusKnotGeometry",this.parameters=
{radius:t,tube:e,tubularSegments:n,radialSegments:i,p:r,q:s},n=Math.floor(n),i=Math
.floor(i);const a=[],o=[],l=[],c=[],h=new Lt,u=new Lt,d=new Lt,p=new Lt,m=new
Lt,f=new Lt,g=new Lt;for(let a=0;a<=n;++a){const
y=a/n*r*Math.PI*2;v(y,r,s,t,d),v(y+.01,r,s,t,p),f.subVectors(p,d),g.addVectors(p,d)
,m.crossVectors(f,g),g.crossVectors(m,f),m.normalize(),g.normalize();for(let
t=0;t<=i;++t){const r=t/i*Math.PI*2,s=-e*Math.cos(r),p=e*Math.sin(r);h.x=d.x+
(s*g.x+p*m.x),h.y=d.y+(s*g.y+p*m.y),h.z=d.z+
(s*g.z+p*m.z),o.push(h.x,h.y,h.z),u.subVectors(h,d).normalize(),l.push(u.x,u.y,u.z)
,c.push(a/n),c.push(t/i)}}for(let t=1;t<=n;t++)for(let e=1;e<=i;e++){const n=(i+1)*
(t-1)+(e-1),r=(i+1)*t+(e-1),s=(i+1)*t+e,o=(i+1)*(t-
1)+e;a.push(n,r,o),a.push(r,s,o)}function v(t,e,n,i,r){const
s=Math.cos(t),a=Math.sin(t),o=n/e*t,l=Math.cos(o);r.x=i*(2+l)*.5*s,r.y=i*
(2+l)*a*.5,r.z=i*Math.sin(o)*.5}this.setIndex(a),this.setAttribute("position",new
mn(o,3)),this.setAttribute("normal",new mn(l,3)),this.setAttribute("uv",new
mn(c,2))}class Ro extends En{constructor(t,e=64,n=1,i=8,r=!1)
{super(),this.type="TubeGeometry",this.parameters=
{path:t,tubularSegments:e,radius:n,radialSegments:i,closed:r};const
s=t.computeFrenetFrames(e,r);this.tangents=s.tangents,this.normals=s.normals,this.b
inormals=s.binormals;const a=new Lt,o=new Lt,l=new Lt;let c=new Lt;const h=[],u=
[],d=[],p=[];function m(r){c=t.getPointAt(r/e,c);const
l=s.normals[r],d=s.binormals[r];for(let t=0;t<=i;t++){const
e=t/i*Math.PI*2,r=Math.sin(e),s=-
Math.cos(e);o.x=s*l.x+r*d.x,o.y=s*l.y+r*d.y,o.z=s*l.z+r*d.z,o.normalize(),u.push(o.
x,o.y,o.z),a.x=c.x+n*o.x,a.y=c.y+n*o.y,a.z=c.z+n*o.z,h.push(a.x,a.y,a.z)}!function
(){for(let t=0;t<e;t++)m(t);m(!1===r?e:0),function(){for(let t=0;t<=e;t++)for(let
n=0;n<=i;n++)l.x=t/e,l.y=n/i,d.push(l.x,l.y)}(),function(){for(let
t=1;t<=e;t++)for(let e=1;e<=i;e++){const n=(i+1)*(t-1)+(e-1),r=(i+1)*t+(e-1),s=
(i+1)*t+e,a=(i+1)*(t-1)+e;p.push(n,r,a),p.push(r,s,a)}}()}
(),this.setIndex(p),this.setAttribute("position",new
mn(h,3)),this.setAttribute("normal",new mn(u,3)),this.setAttribute("uv",new
mn(d,2))}toJSON(){const t=En.prototype.toJSON.call(this);return
t.path=this.parameters.path.toJSON(),t}}class Co extends En{constructor(t)
{if(super(),this.type="WireframeGeometry",!0===t.isGeometry)return void
console.error("THREE.WireframeGeometry no longer supports THREE.Geometry. Use
THREE.BufferGeometry instead.");const e=[],n=[0,0],i={},r=new Lt;if(null!==t.index)
{const s=t.attributes.position,a=t.index;let o=t.groups;0===o.length&&(o=
[{start:0,count:a.count,materialIndex:0}]);for(let t=0,e=o.length;t<e;t++){const
e=o[t],r=e.start;for(let t=r,s=r+e.count;t<s;t+=3)for(let e=0;e<3;e++){const
r=a.getX(t+e),s=a.getX(t+(e+1)%3);n[0]=Math.min(r,s),n[1]=Math.max(r,s);const
o=n[0]+", "+n[1];void 0===i[o]&&(i[o]={index1:n[0],index2:n[1]})}}for(const t in i)
{const
n=i[t];r.fromBufferAttribute(s,n.index1),e.push(r.x,r.y,r.z),r.fromBufferAttribute(
s,n.index2),e.push(r.x,r.y,r.z)}else{const n=t.attributes.position;for(let
t=0,i=n.count/3;t<i;t++)for(let i=0;i<3;i++){const
s=3*t+i;r.fromBufferAttribute(n,s),e.push(r.x,r.y,r.z);const a=3*t+
(i+1)%3;r.fromBufferAttribute(n,a),e.push(r.x,r.y,r.z)}}this.setAttribute("position"
,new mn(e,3))}var
Po=Object.freeze({__proto__:null,BoxGeometry:qn,BoxBufferGeometry:qn,CircleGeometry

```

```

:Pa,CircleBufferGeometry:Pa,ConeGeometry:Ia,ConeBufferGeometry:Ia,CylinderGeometry:
Da,CylinderBufferGeometry:Da,DodecahedronGeometry:Ba,DodecahedronBufferGeometry:Ba,
EdgesGeometry:Ga,ExtrudeGeometry:go,ExtrudeBufferGeometry:go,IcosahedronGeometry:yo
,IcosahedronBufferGeometry:yo,LatheGeometry:xo,LatheBufferGeometry:xo,OctahedronGeo
metry:_o,OctahedronBufferGeometry:_o,ParametricGeometry:wo,ParametricBufferGeometry
:wo,PlaneGeometry:ci,PlaneBufferGeometry:ci,PolyhedronGeometry:Na,PolyhedronBufferG
eometry:Na,RingGeometry:bo,RingBufferGeometry:bo,ShapeGeometry:Mo,ShapeBufferGeomet
ry:Mo,SphereGeometry:So,SphereBufferGeometry:So,TetrahedronGeometry:To,TetrahedronB
ufferGeometry:To,TextGeometry:Eo,TextBufferGeometry:Eo,TorusGeometry:Ao,TorusBuffer
Geometry:Ao,TorusKnotGeometry:Lo,TorusKnotBufferGeometry:Lo,TubeGeometry:Ro,TubeBuf
ferGeometry:Ro,WireframeGeometry:Co));class Do extends Xe{constructor(t)
{super(),this.type="ShadowMaterial",this.color=new
tn(0),this.transparent=!0,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this}}Do.prototype.isShadowMaterial=!0;class
Io extends Jn{constructor(t)
{super(t),this.type="RawShaderMaterial"}}Io.prototype.isRawShaderMaterial=!0;class
No extends Xe{constructor(t){super(),this.defines=
{STANDARD:""},this.type="MeshStandardMaterial",this.color=new
tn(16777215),this.roughness=1,this.metalness=0,this.map=null,this.lightMap=null,thi
s.lightMapIntensity=1,this.aoMap=null,this.aoMapIntensity=1,this.emissive=new
tn(0),this.emissiveIntensity=1,this.emissiveMap=null,this.bumpMap=null,this.bumpSca
le=1,this.normalMap=null,this.normalMapType=0,this.normalScale=new
vt(1,1),this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,
this.roughnessMap=null,this.metalnessMap=null,this.alphaMap=null,this.envMap=null,t
his.envMapIntensity=1,this.refractionRatio=.98,this.wireframe=!1,this.wireframeLine
width=1,this.wireframeLinecap="round",this.wireframeLinejoin="round",this.skinning=
!1,this.morphTargets=!1,this.morphNormals=!1,this.flatShading=!1,this.vertexTangent
s=!1,this.setValues(t)}copy(t){return super.copy(t),this.defines=
{STANDARD:"",this.color.copy(t.color),this.roughness=t.roughness,this.metalness=t.
metalness,this.map=t.map,this.lightMap=t.lightMap,this.lightMapIntensity=t.lightMap
Intensity,this.aoMap=t.aoMap,this.aoMapIntensity=t.aoMapIntensity,this.emissive.cop
y(t.emissive),this.emissiveMap=t.emissiveMap,this.emissiveIntensity=t.emissiveInten
sity,this.bumpMap=t.bumpMap,this.bumpScale=t.bumpScale,this.normalMap=t.normalMap,t
his.normalMapType=t.normalMapType,this.normalScale.copy(t.normalScale),this.displac
ementMap=t.displacementMap,this.displacementScale=t.displacementScale,this.displace
mentBias=t.displacementBias,this.roughnessMap=t.roughnessMap,this.metalnessMap=t.me
talnessMap,this.alphaMap=t.alphaMap,this.envMap=t.envMap,this.envMapIntensity=t.env
MapIntensity,this.refractionRatio=t.refractionRatio,this.wireframe=t.wireframe,this
.wireframeLinewidth=t.wireframeLinewidth,this.wireframeLinecap=t.wireframeLinecap,t
his.wireframeLinejoin=t.wireframeLinejoin,this.skinning=t.skinning,this.morphTarget
s=t.morphTargets,this.morphNormals=t.morphNormals,this.flatShading=t.flatShading,th
is.vertexTangents=t.vertexTangents,this}}No.prototype.isMeshStandardMaterial=!0;cla
ss Bo extends No{constructor(t){super(),this.defines=
{STANDARD:"",PHYSICAL:""},this.type="MeshPhysicalMaterial",this.clearcoat=0,this.cl
earcoatMap=null,this.clearcoatRoughness=0,this.clearcoatRoughnessMap=null,this.clea
rcoatNormalScale=new
vt(1,1),this.clearcoatNormalMap=null,this.reflectivity=.5,Object.defineProperty(thi
s,"ior",{get:function()
{return(1+.4*this.reflectivity)/(1-.4*this.reflectivity)},set:function(t)
{this.reflectivity=ht(2.5*(t-
1)/(t+1),0,1)}}),this.sheen=null,this.transmission=0,this.transmissionMap=null,this
.setValues(t)}copy(t){return super.copy(t),this.defines=
{STANDARD:"",PHYSICAL:""},this.clearcoat=t.clearcoat,this.clearcoatMap=t.clearcoatM
ap,this.clearcoatRoughness=t.clearcoatRoughness,this.clearcoatRoughnessMap=t.clearc
oatRoughnessMap,this.clearcoatNormalMap=t.clearcoatNormalMap,this.clearcoatNormalSc
ale.copy(t.clearcoatNormalScale),this.reflectivity=t.reflectivity,t.sheen?
this.sheen=(this.sheen|new
tn).copy(t.sheen):this.sheen=null,this.transmission=t.transmission,this.transmissio

```

```

nMap=t.transmissionMap,this}}Bo.prototype.isMeshPhysicalMaterial=!0;class zo
extends Xe{constructor(t){super(),this.type="MeshPhongMaterial",this.color=new
tn(16777215),this.specular=new
tn(1118481),this.shininess=30,this.map=null,this.lightMap=null,this.lightMapIntensi
ty=1,this.aoMap=null,this.aoMapIntensity=1,this.emissive=new
tn(0),this.emissiveIntensity=1,this.emissiveMap=null,this.bumpMap=null,this.bumpSca
le=1,this.normalMap=null,this.normalMapType=0,this.normalScale=new
vt(1,1),this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,
this.specularMap=null,this.alphaMap=null,this.envMap=null,this.combine=0,this.refle
ctivity=1,this.refractionRatio=.98,this.wireframe=!1,this.wireframeLinewidth=1,this
.wireframeLinecap="round",this.wireframeLinejoin="round",this.skinning=!1,this.morp
hTargets=!1,this.morphNormals=!1,this.flatShading=!1,this.setValues(t)}copy(t)
{return
super.copy(t),this.color.copy(t.color),this.specular.copy(t.specular),this.shinines
s=t.shininess,this.map=t.map,this.lightMap=t.lightMap,this.lightMapIntensity=t.ligh
tMapIntensity,this.aoMap=t.aoMap,this.aoMapIntensity=t.aoMapIntensity,this.emissive
.copy(t.emissive),this.emissiveMap=t.emissiveMap,this.emissiveIntensity=t.emissiveI
ntensity,this.bumpMap=t.bumpMap,this.bumpScale=t.bumpScale,this.normalMap=t.normalM
ap,this.normalMapType=t.normalMapType,this.normalScale.copy(t.normalScale),this.dis
placementMap=t.displacementMap,this.displacementScale=t.displacementScale,this.disp
lacementBias=t.displacementBias,this.specularMap=t.specularMap,this.alphaMap=t.alph
aMap,this.envMap=t.envMap,this.combine=t.combine,this.reflectivity=t.reflectivity,t
his.refractionRatio=t.refractionRatio,this.wireframe=t.wireframe,this.wireframeLine
width=t.wireframeLinewidth,this.wireframeLinecap=t.wireframeLinecap,this.wireframeL
inejoin=t.wireframeLinejoin,this.skinning=t.skinning,this.morphTargets=t.morphTarge
ts,this.morphNormals=t.morphNormals,this.flatShading=t.flatShading,this}}zo.prototy
pe.isMeshPhongMaterial=!0;class Fo extends Xe{constructor(t){super(),this.defines=
{TOON:""},this.type="MeshToonMaterial",this.color=new
tn(16777215),this.map=null,this.gradientMap=null,this.lightMap=null,this.lightMapIn
tensity=1,this.aoMap=null,this.aoMapIntensity=1,this.emissive=new
tn(0),this.emissiveIntensity=1,this.emissiveMap=null,this.bumpMap=null,this.bumpSca
le=1,this.normalMap=null,this.normalMapType=0,this.normalScale=new
vt(1,1),this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,
this.alphaMap=null,this.wireframe=!1,this.wireframeLinewidth=1,this.wireframeLineca
p="round",this.wireframeLinejoin="round",this.skinning=!1,this.morphTargets=!1,this
.morphNormals=!1,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.map=t.map,this.gradientMap=t.gradientMa
p,this.lightMap=t.lightMap,this.lightMapIntensity=t.lightMapIntensity,this.aoMap=t.
aoMap,this.aoMapIntensity=t.aoMapIntensity,this.emissive.copy(t.emissive),this.emis
siveMap=t.emissiveMap,this.emissiveIntensity=t.emissiveIntensity,this.bumpMap=t.bum
pMap,this.bumpScale=t.bumpScale,this.normalMap=t.normalMap,this.normalMapType=t.nor
malMapType,this.normalScale.copy(t.normalScale),this.displacementMap=t.displacement
Map,this.displacementScale=t.displacementScale,this.displacementBias=t.displacement
Bias,this.alphaMap=t.alphaMap,this.wireframe=t.wireframe,this.wireframeLinewidth=t.
wireframeLinewidth,this.wireframeLinecap=t.wireframeLinecap,this.wireframeLinejoin=
t.wireframeLinejoin,this.skinning=t.skinning,this.morphTargets=t.morphTargets,this
.morphNormals=t.morphNormals,this}}Fo.prototype.isMeshToonMaterial=!0;class Oo
extends Xe{constructor(t)
{super(),this.type="MeshNormalMaterial",this.bumpMap=null,this.bumpScale=1,this.nor
malMap=null,this.normalMapType=0,this.normalScale=new
vt(1,1),this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,
this.wireframe=!1,this.wireframeLinewidth=1,this.fog=!1,this.skinning=!1,this.morph
Targets=!1,this.morphNormals=!1,this.flatShading=!1,this.setValues(t)}copy(t)
{return
super.copy(t),this.bumpMap=t.bumpMap,this.bumpScale=t.bumpScale,this.normalMap=t.no
rmalMap,this.normalMapType=t.normalMapType,this.normalScale.copy(t.normalScale),thi
s.displacementMap=t.displacementMap,this.displacementScale=t.displacementScale,this
.displacementBias=t.displacementBias,this.wireframe=t.wireframe,this.wireframeLinew

```



```

idth=t.wireframeLinewidth,this.skinning=t.skinning,this.morphTargets=t.morphTargets
,this.morphNormals=t.morphNormals,this.flatShading=t.flatShading,this}}Oo.prototype
.isMeshNormalMaterial=!0;class Ho extends Xe{constructor(t)
{super(),this.type="MeshLambertMaterial",this.color=new
tn(16777215),this.map=null,this.lightMap=null,this.lightMapIntensity=1,this.aoMap=n
ull,this.aoMapIntensity=1,this.emissive=new
tn(0),this.emissiveIntensity=1,this.emissiveMap=null,this.specularMap=null,this.alp
haMap=null,this.envMap=null,this.combine=0,this.reflectivity=1,this.refractionRatio
=.98,this.wireframe=!1,this.wireframeLinewidth=1,this.wireframeLinecap="round",this
.wireframeLinejoin="round",this.skinning=!1,this.morphTargets=!1,this.morphNormals=
!1,this.setValues(t)}copy(t){return
super.copy(t),this.color.copy(t.color),this.map=t.map,this.lightMap=t.lightMap,this
.lightMapIntensity=t.lightMapIntensity,this.aoMap=t.aoMap,this.aoMapIntensity=t.aoM
apIntensity,this.emissive.copy(t.emissive),this.emissiveMap=t.emissiveMap,this.emis
siveIntensity=t.emissiveIntensity,this.specularMap=t.specularMap,this.alphaMap=t.al
phaMap,this.envMap=t.envMap,this.combine=t.combine,this.reflectivity=t.reflectivity
,this.refractionRatio=t.refractionRatio,this.wireframe=t.wireframe,this.wireframeLi
newidth=t.wireframeLinewidth,this.wireframeLinecap=t.wireframeLinecap,this.wirefram
eLinejoin=t.wireframeLinejoin,this.skinning=t.skinning,this.morphTargets=t.morphTar
gets,this.morphNormals=t.morphNormals,this}}Ho.prototype.isMeshLambertMaterial=!0;c
lass Go extends Xe{constructor(t){super(),this.defines=
{MATCAP:""},this.type="MeshMatcapMaterial",this.color=new
tn(16777215),this.matcap=null,this.map=null,this.bumpMap=null,this.bumpScale=1,this
.normalMap=null,this.normalMapType=0,this.normalScale=new
vt(1,1),this.displacementMap=null,this.displacementScale=1,this.displacementBias=0,
this.alphaMap=null,this.skinning=!1,this.morphTargets=!1,this.morphNormals=!1,this
.flatShading=!1,this.setValues(t)}copy(t){return super.copy(t),this.defines=
{MATCAP:""},this.color.copy(t.color),this.matcap=t.matcap,this.map=t.map,this.bumpM
ap=t.bumpMap,this.bumpScale=t.bumpScale,this.normalMap=t.normalMap,this.normalMapTy
pe=t.normalMapType,this.normalScale.copy(t.normalScale),this.displacementMap=t.disp
lacementMap,this.displacementScale=t.displacementScale,this.displacementBias=t.disp
lacementBias,this.alphaMap=t.alphaMap,this.skinning=t.skinning,this.morphTargets=t.
morphTargets,this.morphNormals=t.morphNormals,this.flatShading=t.flatShading,this}}
Go.prototype.isMeshMatcapMaterial=!0;class Uo extends ca{constructor(t)
{super(),this.type="LineDashedMaterial",this.scale=1,this.dashSize=3,this.gapSize=1
,this.setValues(t)}copy(t){return
super.copy(t),this.scale=t.scale,this.dashSize=t.dashSize,this.gapSize=t.gapSize,th
is}}Uo.prototype.isLineDashedMaterial=!0;var
ko=Object.freeze({__proto__:null,ShadowMaterial:Do,SpriteMaterial:Rs,RawShaderMater
ial:Io,ShaderMaterial:Jn,PointsMaterial:_a,MeshPhysicalMaterial:Bo,MeshStandardMate
rial:No,MeshPhongMaterial:zo,MeshToonMaterial:Fo,MeshNormalMaterial:Oo,MeshLambertM
aterial:Ho,MeshDepthMaterial:cs,MeshDistanceMaterial:hs,MeshBasicMaterial:en,MeshMa
tcapMaterial:Go,LineDashedMaterial:Uo,LineBasicMaterial:ca,Material:Xe});const Vo=
{arraySlice:function(t,e,n){return Vo.isTypedArray(t)?new
t.constructor(t.subarray(e,void 0!==(n?
n:t.length)):t.slice(e,n)},convertArray:function(t,e,n)
{return!t||!n&&t.constructor===e?t:"number"===typeof e.BYTES_PER_ELEMENT?new
e(t):Array.prototype.slice.call(t)},isTypedArray:function(t){return
ArrayBuffer.isView(t)&&!(t instanceof DataView)},getKeyframeOrder:function(t){const
e=t.length,n=new Array(e);for(let t=0;t!==e;++t)n[t]=t;return n.sort((function(e,n)
{return t[e]-t[n]})),n},sortedArray:function(t,e,n){const i=t.length,r=new
t.constructor(i);for(let s=0,a=0;a!==i;++s){const i=n[s]*e;for(let
n=0;n!==e;++n)r[a++]=t[i+n]}return r},flattenJSON:function(t,e,n,i){let
r=1,s=t[0];for(;;void 0!==(s&&void 0===s[i]);s=t[r++];if(void 0===s)return;let
a=s[i];if(void 0!==(a&&Array.isArray(a)))do{a=s[i],void 0!==(a&&
(e.push(s.time),n.push.apply(n,a)),s=t[r++]}}while(void 0!==(s));else if(void
0!==(a.toArray))do{a=s[i],void 0!==(a&&
(e.push(s.time),a.toArray(n,n.length)),s=t[r++]}}while(void 0!==(s));else

```

```

do{a=s[i],void 0!==a&&(e.push(s.time),n.push(a)),s=t[r++]}}while(void
0!==s)},subclip:function(t,e,n,i,r=30){const s=t.clone();s.name=e;const a=
[];for(let t=0;t<s.tracks.length;++t){const e=s.tracks[t],o=e.getValueSize(),l=
[],c=[];for(let t=0;t<e.times.length;++t){const s=e.times[t]*r;if(!(s<n||s>=i))
{l.push(e.times[t]);for(let n=0;n<o;++n)c.push(e.values[t*o+n])}}0!==l.length&&
(e.times=Vo.convertArray(l,e.times.constructor),e.values=Vo.convertArray(c,e.values
.constructor),a.push(e))}s.tracks=a;let o=1/0;for(let
t=0;t<s.tracks.length;++t)o>s.tracks[t].times[0]&&(o=s.tracks[t].times[0]);for(let
t=0;t<s.tracks.length;++t)s.tracks[t].shift(-1*o);return
s.resetDuration(),s},makeClipAdditive:function(t,e=0,n=t,i=30){i<=0&&(i=30);const
r=n.tracks.length,s=e/i;for(let e=0;e<r;++e){const
i=n.tracks[e],r=i.ValueTypeName;if("bool"===r||"string"===r)continue;const
a=t.tracks.find((function(t){return t.name===i.name&&t.ValueTypeName===r}));if(void
0===a)continue;let o=0;const
l=i.getValueSize();i.createInterpolant.isInterpolantFactoryMethodGLTFCubicSpline&&
(o=l/3);let c=0;const
h=a.getValueSize();a.createInterpolant.isInterpolantFactoryMethodGLTFCubicSpline&&
(c=h/3);const u=i.times.length-1;let d;if(s<=i.times[0]){const t=o,e=l-
o;d=Vo.arraySlice(i.values,t,e)}else if(s>=i.times[u]){const t=u*l+o,e=t+l-
o;d=Vo.arraySlice(i.values,t,e)}else{const t=i.createInterpolant(),e=o,n=l-
o;t.evaluate(s),d=Vo.arraySlice(t.resultBuffer,e,n)}if("quaternion"===r){(new
At).fromArray(d).normalize().conjugate().toArray(d)}const p=a.times.length;for(let
t=0;t<p;++t){const
e=t*h+c;if("quaternion"===r)At.multiplyQuaternionsFlat(a.values,e,d,0,a.values,e);e
lse{const t=h-2*c;for(let n=0;n<t;++n)a.values[e+n]-=d[n]}}return
t.blendMode=q,t}};class Wo{constructor(t,e,n,i)
{this.parameterPositions=t,this._cachedIndex=0,this.resultBuffer=void 0!==i?i:new
e.constructor(n),this.sampleValues=e,this.valueSize=n,this.settings=null,this.Default
ltSettings_={}}evaluate(t){const e=this.parameterPositions;let
n=this._cachedIndex,i=e[n],r=e[n-1];t:{e:{let s;n:{i:iif(!(t<i))}{for(let s=n+2;;)
{if(void 0===i){if(t<r)break i;return
n=e.length,this._cachedIndex=n,this.afterEnd_(n-
1,t,r)}if(n===s)break;if(r=i,i=e[++n],t<i)break e}s=e.length;break n}if(t>=r)break
t;{const a=e[1];t<a&&(n=2,r=a);for(let s=n-2;;){if(void 0===r)return
this._cachedIndex=0,this.beforeStart_(0,t,i);if(n===s)break;if(i=r,r=e[--n-
1],t>=r)break e}s=n,n=0}}for(;n<s;){const i=n+s>>>1;t<e[i]?
s=i:n=i+1;if(i=e[n],r=e[n-1],void 0===r)return
this._cachedIndex=0,this.beforeStart_(0,t,i);if(void 0===i)return
n=e.length,this._cachedIndex=n,this.afterEnd_(n-
1,r,t)}this._cachedIndex=n,this.intervalChanged_(n,r,i)}return
this.interpolate_(n,r,t,i)}getSettings_(){return
this.settings||this.DefaultSettings_}copySampleValue_(t){const
e=this.resultBuffer,n=this.sampleValues,i=this.valueSize,r=t*i;for(let
t=0;t!==i;++t)e[t]=n[r+t];return e}interpolate_(){throw new Error("call to abstract
method")}}intervalChanged_()
{}}Wo.prototype.beforeStart_=Wo.prototype.copySampleValue_,Wo.prototype.afterEnd_=W
o.prototype.copySampleValue_;class jo extends Wo{constructor(t,e,n,i)
{super(t,e,n,i),this._weightPrev=-0,this._offsetPrev=-0,this._weightNext=-0,this._o
ffsetNext=-0,this.DefaultSettings_
{endingStart:k,endingEnd:k}}intervalChanged_(t,e,n){const
i=this.parameterPositions;let r=t-2,s=t+1,a=i[r],o=i[s];if(void
0===a)switch(this.getSettings_().endingStart){case V:r=t,a=2*e-n;break;case
W:r=i.length-2,a=e+i[r]-i[r+1];break;default:r=t,a=n}if(void
0===o)switch(this.getSettings_().endingEnd){case V:s=t,o=2*n-e;break;case
W:s=1,o=n+i[1]-i[0];break;default:s=t-1,o=e}const l=.5*(n-
e),c=this.valueSize;this._weightPrev=1/(e-a),this._weightNext=1/(o-
n),this._offsetPrev=r*c,this._offsetNext=s*c}interpolate_(t,e,n,i){const
r=this.resultBuffer,s=this.sampleValues,a=this.valueSize,o=t*a,l=o-

```

```

a,c=this._offsetPrev,h=this._offsetNext,u=this._weightPrev,d=this._weightNext,p=(n-
e)/(i-e),m=p*p,f=m*p,g=-u*f+2*u*m-u*p,v=(1+u)*f+(-1.5*2*u)*m+(-.5+u)*p+1,y=(-1-
d)*f+(1.5+d)*m+.5*p,x=d*f-d*m;for(let
t=0;t!==(a;+t)r[t]=g*s[c+t]+v*s[l+t]+y*s[o+t]+x*s[h+t];return r}}class qo extends
Wo{constructor(t,e,n,i){super(t,e,n,i)}interpolate_(t,e,n,i){const
r=this.resultBuffer,s=this.sampleValues,a=this.valueSize,o=t*a,l=o-a,c=(n-e)/(i-
e),h=1-c;for(let t=0;t!==(a;+t)r[t]=s[l+t]*h+s[o+t]*c;return r}}class Xo extends
Wo{constructor(t,e,n,i){super(t,e,n,i)}interpolate_(t){return
this.copySampleValue_(t-1)}}class Yo{constructor(t,e,n,i){if(void 0===t)throw new
Error("THREE.KeyframeTrack: track name is undefined");if(void
0===e||0===e.length)throw new Error("THREE.KeyframeTrack: no keyframes in track
named
"+t);this.name=t,this.times=Vo.convertArray(e,this.TimeBufferType),this.values=Vo.c
onvertArray(n,this.ValueBufferType),this.setInterpolation(i||this.DefaultInterpolat
ion)}static toJSON(t){const e=t.constructor;let
n;if(e.toJSON!==this.toJSON)n=e.toJSON(t);else{n=
{name:t.name,times:Vo.convertArray(t.times,Array),values:Vo.convertArray(t.values,A
rray)};const e=t.getInterpolation();e!==t.DefaultInterpolation&&
(n.interpolation=e)}return
n.type=t.ValueTypeName,n}InterpolantFactoryMethodDiscrete(t){return new
Xo(this.times,this.values,this.getValueSize(),t)}InterpolantFactoryMethodLinear(t)
{return new
qo(this.times,this.values,this.getValueSize(),t)}InterpolantFactoryMethodSmooth(t)
{return new jo(this.times,this.values,this.getValueSize(),t)}setInterpolation(t)
{let e;switch(t){case H:e=this.InterpolantFactoryMethodDiscrete;break;case
G:e=this.InterpolantFactoryMethodLinear;break;case
U:e=this.InterpolantFactoryMethodSmooth;if(void 0===e){const e="unsupported
interpolation for "+this.ValueTypeName+" keyframe track named "+this.name;if(void
0===this.createInterpolant){if(t===this.DefaultInterpolation)throw new
Error(e);this.setInterpolation(this.DefaultInterpolation)}return
console.warn("THREE.KeyframeTrack:",e),this}return
this.createInterpolant=e,this}getInterpolation(){switch(this.createInterpolant)
{case this.InterpolantFactoryMethodDiscrete:return H;case
this.InterpolantFactoryMethodLinear:return G;case
this.InterpolantFactoryMethodSmooth:return U}}getValueSize(){return
this.values.length/this.times.length}shift(t){if(0!==t){const e=this.times;for(let
n=0,i=e.length;n!==(i;+n)e[n]+=t}return this}scale(t){if(1!==t){const
e=this.times;for(let n=0,i=e.length;n!==(i;+n)e[n]*=t}return this}trim(t,e){const
n=this.times,i=n.length;let r=0,s=i-1;for(;r!==(i&&n[r]
<t;)+r;for(;-1!==s&&n[s]>e;)--s;if(+s,0!==(r||s!==(i){r=s&&(s=Math.max(s,1),r=s-
1);const
t=this.getValueSize();this.times=Vo.arraySlice(n,r,s),this.values=Vo.arraySlice(thi
s.values,r*t,s*t)}return this}validate(){let t=!0;const e=this.getValueSize();e-
Math.floor(e)!==0&&(console.error("THREE.KeyframeTrack: Invalid value size in
track.",this),t=!1);const n=this.times,i=this.values,r=n.length;0===r&&
(console.error("THREE.KeyframeTrack: Track is empty.",this),t=!1);let
s=null;for(let e=0;e!==(r;+e){const i=n[e];if("number"===typeof i&&isNaN(i))
{console.error("THREE.KeyframeTrack: Time is not a valid
number.",this,e,i),t=!1;break}if(null!==s&&s>i){console.error("THREE.KeyframeTrack:
Out of order keys.",this,e,i,s),t=!1;break}s=i}if(void
0!==(i&&Vo.isArray(i))for(let e=0,n=i.length;e!==(n;+e){const
n=i[e];if(isNaN(n)){console.error("THREE.KeyframeTrack: Value is not a valid
number.",this,e,n),t=!1;break}}return t}optimize(){const
t=Vo.arraySlice(this.times),e=Vo.arraySlice(this.values),n=this.getValueSize(),i=th
is.getInterpolation()===U,r=t.length-1;let s=1;for(let a=1;a<r;+a){let r=!1;const
o=t[a];if(o!==(t[a+1]&&(1!==(a||o!==(t[0]))if(i)r=!0;else{const t=a*n,i=t-
n,s=t+n;for(let a=0;a!==(n;+a){const n=e[t+a];if(n!==(e[i+a]||n!==(e[s+a])
{r=!0;break}}if(r){if(a!==(s){t[s]=t[a];const i=a*n,r=s*n;for(let

```

```

t=0;t!==n;++t)e[r+t]=e[i+t]]++s}}if(r>0){t[s]=t[r];for(let
t=r*n,i=s*n,a=0;a!==n;++a)e[i+a]=e[t+a];++s}return s!==t.length?
(this.times=Vo.arraySlice(t,0,s),this.values=Vo.arraySlice(e,0,s*n)):
(this.times=t,this.values=e),this.clone(){const
t=Vo.arraySlice(this.times,0),e=Vo.arraySlice(this.values,0),n=new(0,this.construct
or)(this.name,t,e);return
n.createInterpolant=this.createInterpolant,n}}Yo.prototype.TimeBufferType=Float32Ar
ray,Yo.prototype.ValueBufferType=Float32Array,Yo.prototype.DefaultInterpolation=G;c
lass Zo extends
Yo{}Zo.prototype.ValueTypeName="bool",Zo.prototype.ValueBufferType=Array,Zo.prototy
pe.DefaultInterpolation=H,Zo.prototype.InterpolantFactoryMethodLinear=void
0,Zo.prototype.InterpolantFactoryMethodSmooth=void 0;class Jo extends
Yo{}Jo.prototype.ValueTypeName="color";class Qo extends
Yo{}Qo.prototype.ValueTypeName="number";class Ko extends Wo{constructor(t,e,n,i)
{super(t,e,n,i)}interpolate_(t,e,n,i){const
r=this.resultBuffer,s=this.sampleValues,a=this.valueSize,o=(n-e)/(i-e);let
l=t*a;for(let t=l+a;l!==t;l+=4)At.slerpFlat(r,0,s,l-a,s,l,o);return r}}class $o
extends Yo{InterpolantFactoryMethodLinear(t){return new
Ko(this.times,this.values,this.getValueSize(),t)}}$o.prototype.ValueTypeName="quate
rnion",$o.prototype.DefaultInterpolation=G,$o.prototype.InterpolantFactoryMethodSmo
oth=void 0;class tl extends
Yo{}tl.prototype.ValueTypeName="string",tl.prototype.ValueBufferType=Array,tl.proto
type.DefaultInterpolation=H,tl.prototype.InterpolantFactoryMethodLinear=void
0,tl.prototype.InterpolantFactoryMethodSmooth=void 0;class el extends
Yo{}el.prototype.ValueTypeName="vector";class nl{constructor(t,e=-1,n,i=2500)
{this.name=t,this.tracks=n,this.duration=e,this.blendMode=i,this.uuid=ct(),this.dur
ation<0&&this.resetDuration()}static parse(t){const e=
[],n=t.tracks,i=1/(t.fps||1);for(let
t=0,r=n.length;t!==r;++t)e.push(il(n[t]).scale(i));const r=new
this(t.name,t.duration,e,t.blendMode);return r.uuid=t.uuid,r}static toJSON(t){const
e=[],n=t.tracks,i=
{name:t.name,duration:t.duration,tracks:e,uuid:t.uuid,blendMode:t.blendMode};for(le
t t=0,i=n.length;t!==i;++t)e.push(Yo.toJSON(n[t]));return i}static
CreateFromMorphTargetSequence(t,e,n,i){const r=e.length,s=[];for(let t=0;t<r;t++)
{let a=[],o=[];a.push((t+r-1)%r,t,(t+1)%r),o.push(0,1,0);const
l=Vo.getKeyframeOrder(a);a=Vo.sortedArray(a,l,l),o=Vo.sortedArray(o,l,l),i||0!==a[0
]||a.push(r),o.push(o[0]),s.push(new
Qo(".morphTargetInfluences["+e[t].name+"]",a,o).scale(1/n))}return new
this(t,-1,s)}static findByName(t,e){let n=t;if(!Array.isArray(t)){const
e=t;n=e.geometry&&e.geometry.animations||e.animations}for(let
t=0;t<n.length;t++)if(n[t].name===e)return n[t];return null}static
CreateClipsFromMorphTargetSequences(t,e,n){const i={},r=/^([\w-]*?)
([\d]+)$/;for(let e=0,n=t.length;e<n;e++){const
n=t[e],s=n.name.match(r);if(s&&s.length>1){const t=s[1];let e=i[t];e||(i[t]=e=
[]),e.push(n)}const s=[];for(const t in
i)s.push(this.CreateFromMorphTargetSequence(t,i[t],e,n));return s}static
parseAnimation(t,e){if(!t)return console.error("THREE.AnimationClip: No animation
in JSONLoader data."),null;const n=function(t,e,n,i,r){if(0!==n.length){const s=
[],a=[];Vo.flattenJSON(n,s,a,i),0!==s.length&&r.push(new t(e,s,a))},i=
[],r=t.name||"default",s=t.fps||30,a=t.blendMode;let o=t.length||-1;const
l=t.hierarchy||[];for(let t=0;t<l.length;t++){const
r=l[t].keys;if(r&&0!==r.length)if(r[0].morphTargets){const t={};let
e;for(e=0;e<r.length;e++)if(r[e].morphTargets)for(let
n=0;n<r[e].morphTargets.length;n++)t[r[e].morphTargets[n]]=-1;for(const n in t)
{const t=[],s=[];for(let i=0;i!==r[e].morphTargets.length;++i){const
i=r[e];t.push(i.time),s.push(i.morphTarget===n?1:0)}i.push(new
Qo(".morphTargetInfluence["+n+"]",t,s))}o=t.length*(s||1)}else{const
s=".bones["+e[t].name+"]";n(el,s+".position",r,"pos",i),n($o,s+".quaternion",r,"rot

```

```

",i),n(el,s+".scale",r,"scl",i)}}if(0===i.length)return null;return new
this(r,o,i,a)}resetDuration(){let t=0;for(let e=0,n=this.tracks.length;e!==n;++e)
{const n=this.tracks[e];t=Math.max(t,n.times[n.times.length-1])}return
this.duration=t,this}trim(){for(let
t=0;t<this.tracks.length;t++)this.tracks[t].trim(0,this.duration);return
this}validate(){let t=!0;for(let
e=0;e<this.tracks.length;e++)t=t&&this.tracks[e].validate();return t}optimize()
{for(let t=0;t<this.tracks.length;t++)this.tracks[t].optimize();return this}clone()
{const t=[];for(let
e=0;e<this.tracks.length;e++)t.push(this.tracks[e].clone());return new
this.constructor(this.name,this.duration,t,this.blendMode)}toJSON(){return
this.constructor.toJSON(this)}}function il(t){if(void 0===t.type)throw new
Error("THREE.KeyframeTrack: track type undefined, can not parse");const
e=function(t){switch(t.toLowerCase()){
case"scalar":case"double":case"float":case"number":case"integer":return
Qo;case"vector":case"vector2":case"vector3":case"vector4":return
el;case"color":return Jo;case"quaternion":return $o;case"bool":case"boolean":return
Zo;case"string":return tl}throw new Error("THREE.KeyframeTrack: Unsupported
typeName: "+t)}(t.type);if(void 0===t.times){const e=[],n=
[];Vo.flattenJSON(t.keys,e,n,"value"),t.times=e,t.values=n}return void 0!==e.parse?
e.parse(t):new e(t.name,t.times,t.values,t.interpolation)}const rl=
{enabled:!1,files:{},add:function(t,e){!1!==this.enabled&&
(this.files[t]=e)},get:function(t){if(!1!==this.enabled)return
this.files[t]},remove:function(t){delete this.files[t]},clear:function()
{this.files={}}};class sl{constructor(t,e,n){const i=this;let r,s=!1,a=0,o=0;const
l=[];this.onStart=void
0,this.onLoad=t,this.onProgress=e,this.onError=n,this.itemStart=function(t)
{o++,!1===s&&void 0!==i.onStart&&i.onStart(t,a,o),s=!0},this.itemEnd=function(t)
{a++,void 0!==i.onProgress&&i.onProgress(t,a,o),a===o&&(s=!1,void
0!==i.onLoad&&i.onLoad()),this.itemError=function(t){void
0!==i.onError&&i.onError(t)},this.resolveURL=function(t){return r?
r(t):t},this.setURLModifier=function(t){return
r=t,this},this.addHandler=function(t,e){return
l.push(t,e),this},this.removeHandler=function(t){const e=l.indexOf(t);return-
1!==e&&l.splice(e,2),this},this.getHandler=function(t){for(let
e=0,n=l.length;e<n;e+=2){const n=l[e],i=l[e+1];if(n.global&&
(n.lastIndex=0),n.test(t))return i}return null}}const al=new sl;class
ol{constructor(t){this.manager=void 0!==t?
t:al,this.crossOrigin="anonymous",this.withCredentials=!1,this.path="",this.resourc
ePath="",this.requestHeader={}}load(){loadAsync(t,e){const n=this;return new
Promise((function(i,r){n.load(t,i,e,r)}))}parse(){setCrossOrigin(t){return
this.crossOrigin=t,this}setWithCredentials(t){return
this.withCredentials=t,this}setPath(t){return this.path=t,this}setResourcePath(t)
{return this.resourcePath=t,this}setRequestHeader(t){return
this.requestHeader=t,this}}const ll=[];class cl extends ol{constructor(t)
{super(t)}load(t,e,n,i){void 0===t&&(t=""),void 0!==this.path&&
(t=this.path+t),t=this.manager.resolveURL(t);const r=this,s=rl.get(t);if(void
0!==s)return r.manager.itemStart(t),setTimeout((function()
{e&&e(s),r.manager.itemEnd(t)}),0),s;if(void 0!==ll[t])return void
ll[t].push({onLoad:e,onProgress:n,onError:i});const a=t.match(/^data:(.*?)
(;base64)?,(.*)$/);let o;if(a){const n=a[1],s=!!a[2];let
o=a[3];o=decodeURIComponent(o),s&&(o=atob(o));try{let i;const s=
(this.responseType||"").toLowerCase();switch(s){case"arraybuffer":case"blob":const
t=new Uint8Array(o.length);for(let
e=0;e<o.length;e++)t[e]=o.charCodeAt(e);i="blob"===s?new Blob([t.buffer],
{type:n}):t.buffer;break;case"document":const e=new
DOMParser;i=e.parseFromString(o,n);break;case"json":i=JSON.parse(o);break;default:i
=o}setTimeout((function(){e&&e(i),r.manager.itemEnd(t)}),0)}catch(e)

```

```

{setTimeout((function()
{i&&i(e),r.manager.itemError(t),r.manager.itemEnd(t)}),0)}else{ll[t]=
[],ll[t].push({onLoad:e,onProgress:n,onError:i}),o=new
XMLHttpRequest,o.open("GET",t,!0),o.addEventListener("load",(function(e){const
n=this.response,i=ll[t];if(delete ll[t],200===this.status||0===this.status)
{0===this.status&&console.warn("THREE.FileLoader: HTTP Status 0
received."),rl.add(t,n);for(let t=0,e=i.length;t<e;t++){const
e=i[t];e.onLoad&&e.onLoad(n)}r.manager.itemEnd(t)}else{for(let
t=0,n=i.length;t<n;t++){const
n=i[t];n.onError&&n.onError(e)}r.manager.itemError(t),r.manager.itemEnd(t)}},!1),o
.addEventListener("progress",(function(e){const n=ll[t];for(let
t=0,i=n.length;t<i;t++){const
i=n[t];i.onProgress&&i.onProgress(e)}},!1),o.addEventListener("error",(function(e)
{const n=ll[t];delete ll[t];for(let t=0,i=n.length;t<i;t++){const
i=n[t];i.onError&&i.onError(e)}r.manager.itemError(t),r.manager.itemEnd(t)}},!1),o.
addEventListener("abort",(function(e){const n=ll[t];delete ll[t];for(let
t=0,i=n.length;t<i;t++){const
i=n[t];i.onError&&i.onError(e)}r.manager.itemError(t),r.manager.itemEnd(t)}},!1),vo
id 0!==this.responseType&&(o.responseType=this.responseType),void
0!==this.withCredentials&&
(o.withCredentials=this.withCredentials),o.overrideMimeType&&o.overrideMimeType(voi
d 0!==this.mimeType?this.mimeType:"text/plain");for(const t in
this.requestHeader)o.setRequestHeader(t,this.requestHeader[t]);o.send(null)}return
r.manager.itemStart(t,o)setResponseType(t){return
this.responseType=t,this}setMimeType(t){return this.mimeType=t,this}}class hl
extends ol{constructor(t){super(t)}load(t,e,n,i){void 0!==this.path&&
(t=this.path+t),t=this.manager.resolveURL(t);const r=this,s=rl.get(t);if(void
0!==s)return r.manager.itemStart(t),setTimeout((function()
{e&&e(s),r.manager.itemEnd(t)}),0),s;const
a=document.createElementNS("http://www.w3.org/1999/xhtml","img");function o()
{a.removeEventListener("load",o,!1),a.removeEventListener("error",l,!1),rl.add(t,th
is),e&&e(this),r.manager.itemEnd(t)}function l(e)
{a.removeEventListener("load",o,!1),a.removeEventListener("error",l,!1),i&&i(e),r.m
anager.itemError(t),r.manager.itemEnd(t)}return
a.addEventListener("load",o,!1),a.addEventListener("error",l,!1),"data:"!==t.substr
(0,5)&&void 0!==this.crossOrigin&&
(a.crossOrigin=this.crossOrigin),r.manager.itemStart(t,a.src=t,a)}class ul extends
ol{constructor(t){super(t)}load(t,e,n,i){const r=new ei,s=new
hl(this.manager);s.setCrossOrigin(this.crossOrigin),s.setPath(this.path);let
a=0;function o(n){s.load(t[n],(function(t){r.images[n]=t,a++,6===a&&
(r.needsUpdate=!0,e&&e(r)}),void 0,i)}for(let e=0;e<t.length;++e)o(e);return
r}}class dl extends ol{constructor(t){super(t)}load(t,e,n,i){const r=this,s=new
ii,a=new cl(this.manager);return
a.setResponseType("arraybuffer"),a.setRequestHeader(this.requestHeader),a.setPath(t
his.path),a.setWithCredentials(r.withCredentials),a.load(t,(function(t){const
n=r.parse(t);n&&(void 0!==n.image?s.image=n.image:void 0!==n.data&&
(s.image.width=n.width,s.image.height=n.height,s.image.data=n.data),s.wrapS=void
0!==n.wrapS?n.wrapS:u,s.wrapT=void 0!==n.wrapT?n.wrapT:u,s.magFilter=void
0!==n.magFilter?n.magFilter:g,s.minFilter=void 0!==n.minFilter?
n.minFilter:g,s.anisotropy=void 0!==n.anisotropy?n.anisotropy:1,void
0!==n.encoding&&(s.encoding=n.encoding),void 0!==n.flipY&&(s.flipY=n.flipY),void
0!==n.format&&(s.format=n.format),void 0!==n.type&&(s.type=n.type),void
0!==n.mipmaps&&(s.mipmaps=n.mipmaps,s.minFilter=y),l===n.mipmapCount&&
(s.minFilter=g),void 0!==n.generateMipmaps&&
(s.generateMipmaps=n.generateMipmaps),s.needsUpdate=!0,e&&e(s,n)}),n,i),s)}class
pl extends ol{constructor(t){super(t)}load(t,e,n,i){const r=new bt,s=new
hl(this.manager);return
s.setCrossOrigin(this.crossOrigin),s.setPath(this.path),s.load(t,(function(n)

```

```

{r.image=n;const i=t.search(/\.\jpe?
g($|\?)/i)>0||0===t.search(/^data:image\/jpeg/);r.format=i?
T:E,r.needsUpdate=!0,void 0!==e&&e(r)}),n,i),r}}class ml{constructor()
{this.type="Curve",this.arcLengthDivisions=200}getPoint(){return
console.warn("THREE.Curve: .getPoint() not implemented."),null}getPointAt(t,e)
{const n=this.getUtoTmapping(t);return this.getPoint(n,e)}getPoints(t=5){const e=
[];for(let n=0;n<=t;n++)e.push(this.getPoint(n/t));return e}getSpacedPoints(t=5)
{const e=[];for(let n=0;n<=t;n++)e.push(this.getPointAt(n/t));return e}getLength()
{const t=this.getLengths();return t[t.length-
1]}getLengths(t=this.arcLengthDivisions)
{if(this.cacheArcLengths&&this.cacheArcLengths.length===t+1&&!this.needsUpdate)retu
rn this.cacheArcLengths;this.needsUpdate=!1;const e=[];let
n,i=this.getPoint(0),r=0;e.push(0);for(let
s=1;s<=t;s++)n=this.getPoint(s/t),r+=n.distanceTo(i),e.push(r),i=n;return
this.cacheArcLengths=e,e}updateArcLengths()
{this.needsUpdate=!0,this.getLengths()}getUtoTmapping(t,e){const
n=this.getLengths();let i=0;const r=n.length;let s;s=e||t*n[r-1];let a,o=0,l=r-
1;for(;o<=l;)if(i=Math.floor(o+(l-o)/2),a=n[i]-s,a<0)o=i+1;else{if(!(a>0))
{l=i;break}l=i-1}if(i=l,n[i]===s)return i/(r-1);const c=n[i];return(i+(s-
c)/(n[i+1]-c))/(r-1)}getTangent(t,e){const n=1e-4;let i=t-n,r=t+n;i<0&&(i=0),r>1&&
(r=1);const s=this.getPoint(i),a=this.getPoint(r),o=e||(s.isVector2?new vt:new
Lt);return o.copy(a).sub(s).normalize(),o}getTangentAt(t,e){const
n=this.getUtoTmapping(t);return this.getTangent(n,e)}computeFrenetFrames(t,e){const
n=new Lt,i=[],r=[],s=[],a=new Lt,o=new se;for(let e=0;e<=t;e++){const
n=e/t;i[e]=this.getTangentAt(n,new Lt),i[e].normalize()}r[0]=new Lt,s[0]=new Lt;let
l=Number.MAX_VALUE;const
c=Math.abs(i[0].x),h=Math.abs(i[0].y),u=Math.abs(i[0].z);c<=l&&
(l=c,n.set(1,0,0)),h<=l&&
(l=h,n.set(0,1,0)),u<=l&&n.set(0,0,1),a.crossVectors(i[0],n).normalize(),r[0].cross
Vectors(i[0],a),s[0].crossVectors(i[0],r[0]);for(let e=1;e<=t;e++){if(r[e]=r[e-
1].clone(),s[e]=s[e-1].clone(),a.crossVectors(i[e-
1],i[e]),a.length()>Number.EPSILON){a.normalize();const t=Math.acos(ht(i[e-
1].dot(i[e]),-1,1));r[e].applyMatrix4(o.makeRotationAxis(a,t))}s[e].crossVectors(i[
e],r[e])}if(!0===e){let
e=Math.acos(ht(r[0].dot(r[t]),-1,1));e/=t,i[0].dot(a.crossVectors(r[0],r[t]))>0&&
(e=-e);for(let
n=1;n<=t;n++)r[n].applyMatrix4(o.makeRotationAxis(i[n],e*n)),s[n].crossVectors(i[n]
,r[n])}return{tangents:i,normals:r,binormals:s}}clone(){return(new
this.constructor).copy(this)}copy(t){return
this.arcLengthDivisions=t.arcLengthDivisions,this}toJSON(){const t={metadata:
{version:4.5,type:"Curve",generator:"Curve.toJSON"}};return
t.arcLengthDivisions=this.arcLengthDivisions,t.type=this.type,t}fromJSON(t){return
this.arcLengthDivisions=t.arcLengthDivisions,this}}class fl extends
ml{constructor(t=0,e=0,n=1,i=1,r=0,s=2*Math.PI,a=!1,o=0)
{super(),this.type="EllipseCurve",this.aX=t,this.aY=e,this.xRadius=n,this.yRadius=i
,this.aStartAngle=r,this.aEndAngle=s,this.aClockwise=a,this.aRotation=o}getPoint(t,
e){const n=e||new vt,i=2*Math.PI;let r=this.aEndAngle-this.aStartAngle;const
s=Math.abs(r)<Number.EPSILON;for(;r<0;)r+=i;for(;r>i;)r-=i;r<Number.EPSILON&&(r=s?
0:i),!0!==this.aClockwise||s||(r===i?r=-i:r=-i);const a=this.aStartAngle+t*r;let
o=this.aX+this.xRadius*Math.cos(a),l=this.aY+this.yRadius*Math.sin(a);if(0!==this.a
Rotation){const t=Math.cos(this.aRotation),e=Math.sin(this.aRotation),n=o-
this.aX,i=l-this.aY;o=n*t-i*e+this.aX,l=n*e+i*t+this.aY}return n.set(o,l)}copy(t)
{return
super.copy(t),this.aX=t.aX,this.aY=t.aY,this.xRadius=t.xRadius,this.yRadius=t.yRadi
us,this.aStartAngle=t.aStartAngle,this.aEndAngle=t.aEndAngle,this.aClockwise=t.aClo
ckwise,this.aRotation=t.aRotation,this}toJSON(){const t=super.toJSON();return
t.aX=this.aX,t.aY=this.aY,t.xRadius=this.xRadius,t.yRadius=this.yRadius,t.aStartAng
le=this.aStartAngle,t.aEndAngle=this.aEndAngle,t.aClockwise=this.aClockwise,t.aRota

```

```

tion=this.aRotation,t}fromJSON(t){return
super.fromJSON(t),this.aX=t.aX,this.aY=t.aY,this.xRadius=t.xRadius,this.yRadius=t.y
Radius,this.aStartAngle=t.aStartAngle,this.aEndAngle=t.aEndAngle,this.aClockwise=t.
aClockwise,this.aRotation=t.aRotation,this}}fl.prototype.isEllipseCurve=!0;class gl
extends fl{constructor(t,e,n,i,r,s){
super(t,e,n,n,i,r,s),this.type="ArcCurve"}}function vl(){let
t=0,e=0,n=0,i=0;function r(r,s,a,o){t=r,e=a,n=-3*r+3*s-2*a-o,i=2*r-
2*s+a+o}return{initCatmullRom:function(t,e,n,i,s){r(e,n,s*(n-t),s*(i-
e))},initNonuniformCatmullRom:function(t,e,n,i,s,a,o){let l=(e-t)/s-(n-t)/(s+a)+(n-
e)/a,c=(n-e)/a-(i-e)/(a+o)+(i-n)/o;l*=a,c*=a,r(e,n,l,c)},calc:function(r){const
s=r*r;return t+e*r+n*s+i*(s*r)}}gl.prototype.isArcCurve=!0;const yl=new Lt,xl=new
vl,_l=new vl,wl=new vl;class bl extends ml{constructor(t=
[],e=!1,n="centripetal",i=.5){
super(),this.type="CatmullRomCurve3",this.points=t,this.closed=e,this.curveType=n,
this.tension=i}getPoint(t,e=new Lt){const n=e,i=this.points,r=i.length,s=(r-
(this.closed?0:1))*t;let a,o,l=Math.floor(s),c=s-l;this.closed?l+=l>0?0:
(Math.floor(Math.abs(l)/r)+1)*r:0===c&&l===r-l&&(l=r-2,c=1),this.closed||l>0?
a=i[(l-1)%r]:(yl.subVectors(i[0],i[1])).add(i[0]),a=yl;const
h=i[l%r],u=i[(l+1)%r];if(this.closed||l+2<r?o=i[(l+2)%r]:(yl.subVectors(i[r-1],i[r-
2])).add(i[r-1]),o=yl),"centripetal"===this.curveType||"chordal"===this.curveType)
{const t="chordal"===this.curveType?.5:.25;let
e=Math.pow(a.distanceToSquared(h),t),n=Math.pow(h.distanceToSquared(u),t),i=Math.po
w(u.distanceToSquared(o),t);n<1e-4&&(n=1),e<1e-4&&(e=n),i<1e-4&&
(i=n),xl.initNonuniformCatmullRom(a.x,h.x,u.x,o.x,e,n,i),_l.initNonuniformCatmullRo
m(a.y,h.y,u.y,o.y,e,n,i),wl.initNonuniformCatmullRom(a.z,h.z,u.z,o.z,e,n,i)}else"ca
tmullrom"===this.curveType&&
(xl.initCatmullRom(a.x,h.x,u.x,o.x,this.tension),_l.initCatmullRom(a.y,h.y,u.y,o.y,
this.tension),wl.initCatmullRom(a.z,h.z,u.z,o.z,this.tension));return
n.set(xl.calc(c),_l.calc(c),wl.calc(c)),n}copy(t){super.copy(t),this.points=
[];for(let e=0,n=t.points.length;e<n;e++){const
n=t.points[e];this.points.push(n.clone())}return
this.closed=t.closed,this.curveType=t.curveType,this.tension=t.tension,this}toJSON(
){const t=super.toJSON();t.points=[];for(let e=0,n=this.points.length;e<n;e++)
{const n=this.points[e];t.points.push(n.toArray())}return
t.closed=this.closed,t.curveType=this.curveType,t.tension=this.tension,t}fromJSON(t
){super.fromJSON(t),this.points=[];for(let e=0,n=t.points.length;e<n;e++){const
n=t.points[e];this.points.push((new Lt).fromArray(n))}return
this.closed=t.closed,this.curveType=t.curveType,this.tension=t.tension,this}}functi
on Ml(t,e,n,i,r){const s=.5*(i-e),a=.5*(r-n),o=t*t;return(2*n-2*i+s+a)*(t*o)+
(-3*n+3*i-2*s-a)*o+s*t+n}function Sl(t,e,n,i){return function(t,e){const n=1-
t;return n*n*e}(t,e)+function(t,e){return 2*(1-t)*t*e}(t,n)+function(t,e){return
t*t*e}(t,i)}function Tl(t,e,n,i,r){return function(t,e){const n=1-t;return n*n*n*e}
(t,e)+function(t,e){const n=1-t;return 3*n*n*t*e}(t,n)+function(t,e){return 3*(1-
t)*t*t*e}(t,i)+function(t,e){return t*t*t*e}
(t,r)}bl.prototype.isCatmullRomCurve3=!0;class El extends ml{constructor(t=new
vt,e=new vt,n=new vt,i=new vt){
super(),this.type="CubicBezierCurve",this.v0=t,this.v1=e,this.v2=n,this.v3=i}getPo
int(t,e=new vt){const n=e,i=this.v0,r=this.v1,s=this.v2,a=this.v3;return
n.set(Tl(t,i.x,r.x,s.x,a.x),Tl(t,i.y,r.y,s.y,a.y)),n}copy(t){return
super.copy(t),this.v0.copy(t.v0),this.v1.copy(t.v1),this.v2.copy(t.v2),this.v3.copy
(t.v3),this}toJSON(){const t=super.toJSON();return
t.v0=this.v0.toArray(),t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t.v3=this.v3.t
oArray(),t}fromJSON(t){return
super.fromJSON(t),this.v0.fromArray(t.v0),this.v1.fromArray(t.v1),this.v2.fromArray
(t.v2),this.v3.fromArray(t.v3),this}}El.prototype.isCubicBezierCurve=!0;class Al
extends ml{constructor(t=new Lt,e=new Lt,n=new Lt,i=new Lt){
super(),this.type="CubicBezierCurve3",this.v0=t,this.v1=e,this.v2=n,this.v3=i}getP
oint(t,e=new Lt){const n=e,i=this.v0,r=this.v1,s=this.v2,a=this.v3;return

```



```

n.set(Tl(t,i.x,r.x,s.x,a.x),Tl(t,i.y,r.y,s.y,a.y),Tl(t,i.z,r.z,s.z,a.z)),n}copy(t)
{return
super.copy(t),this.v0.copy(t.v0),this.v1.copy(t.v1),this.v2.copy(t.v2),this.v3.copy
(t.v3),this}toJSON(){const t=super.toJSON();return
t.v0=this.v0.toArray(),t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t.v3=this.v3.t
oArray(),t}fromJSON(t){return
super.fromJSON(t),this.v0.fromArray(t.v0),this.v1.fromArray(t.v1),this.v2.fromArray
(t.v2),this.v3.fromArray(t.v3),this}}Al.prototype.isCubicBezierCurve3=!0;class Ll
extends ml{constructor(t=new vt,e=new vt)
{super(),this.type="LineCurve",this.v1=t,this.v2=e}getPoint(t,e=new vt){const
n=e;return 1===t?n.copy(this.v2):
(n.copy(this.v2).sub(this.v1),n.multiplyScalar(t).add(this.v1)),n}getPointAt(t,e)
{return this.getPoint(t,e)}getTangent(t,e){const n=e|new vt;return
n.copy(this.v2).sub(this.v1).normalize(),n}copy(t){return
super.copy(t),this.v1.copy(t.v1),this.v2.copy(t.v2),this}toJSON(){const
t=super.toJSON();return t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t}fromJSON(t)
{return
super.fromJSON(t),this.v1.fromArray(t.v1),this.v2.fromArray(t.v2),this}}Ll.prototype
e.isLineCurve=!0;class Rl extends ml{constructor(t=new Lt,e=new Lt)
{super(),this.type="LineCurve3",this.isLineCurve3=!0,this.v1=t,this.v2=e}getPoint(t
,e=new Lt){const n=e;return 1===t?n.copy(this.v2):
(n.copy(this.v2).sub(this.v1),n.multiplyScalar(t).add(this.v1)),n}getPointAt(t,e)
{return this.getPoint(t,e)}copy(t){return
super.copy(t),this.v1.copy(t.v1),this.v2.copy(t.v2),this}toJSON(){const
t=super.toJSON();return t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t}fromJSON(t)
{return
super.fromJSON(t),this.v1.fromArray(t.v1),this.v2.fromArray(t.v2),this}}class Cl
extends ml{constructor(t=new vt,e=new vt,n=new vt)
{super(),this.type="QuadraticBezierCurve",this.v0=t,this.v1=e,this.v2=n}getPoint(t,
e=new vt){const n=e,i=this.v0,r=this.v1,s=this.v2;return
n.set(Sl(t,i.x,r.x,s.x),Sl(t,i.y,r.y,s.y)),n}copy(t){return
super.copy(t),this.v0.copy(t.v0),this.v1.copy(t.v1),this.v2.copy(t.v2),this}toJSON(
){const t=super.toJSON();return
t.v0=this.v0.toArray(),t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t}fromJSON(t)
{return
super.fromJSON(t),this.v0.fromArray(t.v0),this.v1.fromArray(t.v1),this.v2.fromArray
(t.v2),this}}Cl.prototype.isQuadraticBezierCurve=!0;class Pl extends
ml{constructor(t=new Lt,e=new Lt,n=new Lt)
{super(),this.type="QuadraticBezierCurve3",this.v0=t,this.v1=e,this.v2=n}getPoint(t
,e=new Lt){const n=e,i=this.v0,r=this.v1,s=this.v2;return
n.set(Sl(t,i.x,r.x,s.x),Sl(t,i.y,r.y,s.y),Sl(t,i.z,r.z,s.z)),n}copy(t){return
super.copy(t),this.v0.copy(t.v0),this.v1.copy(t.v1),this.v2.copy(t.v2),this}toJSON(
){const t=super.toJSON();return
t.v0=this.v0.toArray(),t.v1=this.v1.toArray(),t.v2=this.v2.toArray(),t}fromJSON(t)
{return
super.fromJSON(t),this.v0.fromArray(t.v0),this.v1.fromArray(t.v1),this.v2.fromArray
(t.v2),this}}Pl.prototype.isQuadraticBezierCurve3=!0;class Dl extends
ml{constructor(t=[]){super(),this.type="SplineCurve",this.points=t}getPoint(t,e=new
vt){const n=e,i=this.points,r=(i.length-1)*t,s=Math.floor(r),a=r-s,o=i[0===s?s:s-
1],l=i[s],c=i[s>i.length-2?i.length-1:s+1],h=i[s>i.length-3?i.length-1:s+2];return
n.set(Ml(a,o.x,l.x,c.x,h.x),Ml(a,o.y,l.y,c.y,h.y)),n}copy(t)
{super.copy(t),this.points=[];for(let e=0,n=t.points.length;e<n;e++){const
n=t.points[e];this.points.push(n.clone())}return this}toJSON(){const
t=super.toJSON();t.points=[];for(let e=0,n=t.points.length;e<n;e++){const
n=this.points[e];t.points.push(n.toArray())}return t}fromJSON(t)
{super.fromJSON(t),this.points=[];for(let e=0,n=t.points.length;e<n;e++){const
n=t.points[e];this.points.push((new vt).fromArray(n))}return
this}}Dl.prototype.isSplineCurve=!0;var

```

```

Il=Object.freeze({__proto__:null,ArcCurve:gl,CatmullRomCurve3:bl,CubicBezierCurve:E
l,CubicBezierCurve3:Al,EllipseCurve:fl,LineCurve:Ll,LineCurve3:Rl,QuadraticBezierCu
rve:Cl,QuadraticBezierCurve3:Pl,SplineCurve:Dl});class Nl extends ml{constructor()
{super(),this.type="CurvePath",this.curves=[],this.autoClose=!1}add(t)
{this.curves.push(t)}closePath(){const
t=this.curves[0].getPoint(0),e=this.curves[this.curves.length-
1].getPoint(1);t.equals(e)||this.curves.push(new Ll(e,t))}getPoint(t){const
e=t*this.getLength(),n=this.getCurveLengths();let i=0;for(;i<n.length;){if(n[i]>=e)
{const t=n[i]-e,r=this.curves[i],s=r.getLength(),a=0===s?0:1-t/s;return
r.getPointAt(a)}i++}return null}getLength(){const t=this.getCurveLengths();return
t[t.length-1]}updateArcLengths()
{this.needsUpdate=!0,this.cacheLengths=null,this.getCurveLengths()}getCurveLengths()
{if(this.cacheLengths&&this.cacheLengths.length===this.curves.length)return
this.cacheLengths;const t=[];let e=0;for(let
n=0,i=this.curves.length;n<i;n++)e+=this.curves[n].getLength(),t.push(e);return
this.cacheLengths=t,t}getSpacedPoints(t=40){const e=[];for(let
n=0;n<=t;n++)e.push(this.getPoint(n/t));return
this.autoClose&&e.push(e[0]),e}getPoints(t=12){const e=[];let n;for(let
i=0,r=this.curves;i<r.length;i++){const s=r[i],a=s&&s.isEllipseCurve?2*t:s&&
(s.isLineCurve||s.isLineCurve3)?1:s&&s.isSplineCurve?
t*s.points.length:t,o=s.getPoints(a);for(let t=0;t<o.length;t++){const
i=o[t];n&&n.equals(i)||e.push(i),n=i}}return
this.autoClose&&e.length>1&&!e[e.length-1].equals(e[0])&&e.push(e[0]),e}copy(t)
{super.copy(t),this.curves=[];for(let e=0,n=t.curves.length;e<n;e++){const
n=t.curves[e];this.curves.push(n.clone())}return
this.autoClose=t.autoClose,this}toJSON(){const
t=super.toJSON();t.autoClose=this.autoClose,t.curves=[];for(let
e=0,n=this.curves.length;e<n;e++){const
n=this.curves[e];t.curves.push(n.toJSON())}return t}fromJSON(t)
{super.fromJSON(t),this.autoClose=t.autoClose,this.curves=[];for(let
e=0,n=t.curves.length;e<n;e++){const n=t.curves[e];this.curves.push((new
Il[n.type]).fromJSON(n))}return this}}class Bl extends Nl{constructor(t)
{super(),this.type="Path",this.currentPoint=new
vt,t&&this.setFromPoints(t)}setFromPoints(t){this.moveTo(t[0].x,t[0].y);for(let
e=1,n=t.length;e<n;e++)this.lineTo(t[e].x,t[e].y);return this}moveTo(t,e){return
this.currentPoint.set(t,e),this}lineTo(t,e){const n=new
Ll(this.currentPoint.clone(),new vt(t,e));return
this.curves.push(n),this.currentPoint.set(t,e),this}quadraticCurveTo(t,e,n,i){const
r=new Cl(this.currentPoint.clone(),new vt(t,e),new vt(n,i));return
this.curves.push(r),this.currentPoint.set(n,i),this}bezierCurveTo(t,e,n,i,r,s)
{const a=new El(this.currentPoint.clone(),new vt(t,e),new vt(n,i),new
vt(r,s));return this.curves.push(a),this.currentPoint.set(r,s),this}splineThru(t)
{const e=[this.currentPoint.clone()].concat(t),n=new Dl(e);return
this.curves.push(n),this.currentPoint.copy(t[t.length-1]),this}arc(t,e,n,i,r,s)
{const a=this.currentPoint.x,o=this.currentPoint.y;return
this.absarc(t+a,e+o,n,i,r,s),this}absarc(t,e,n,i,r,s){return
this.absellipse(t,e,n,n,i,r,s),this}ellipse(t,e,n,i,r,s,a,o){const
l=this.currentPoint.x,c=this.currentPoint.y;return
this.absellipse(t+l,e+c,n,i,r,s,a,o),this}absellipse(t,e,n,i,r,s,a,o){const l=new
fl(t,e,n,i,r,s,a,o);if(this.curves.length>0){const
t=l.getPoint(0);t.equals(this.currentPoint)||this.lineTo(t.x,t.y)}this.curves.push(
l);const c=l.getPoint(1);return this.currentPoint.copy(c),this}copy(t){return
super.copy(t),this.currentPoint.copy(t.currentPoint),this}toJSON(){const
t=super.toJSON();return t.currentPoint=this.currentPoint.toArray(),t}fromJSON(t)
{return super.fromJSON(t),this.currentPoint.fromArray(t.currentPoint),this}}class
zl extends Bl{constructor(t){super(t),this.uuid=ct(),this.type="Shape",this.holes=
[]}getPointsHoles(t){const e=[];for(let
n=0,i=this.holes.length;n<i;n++)e[n]=this.holes[n].getPoints(t);return

```

```

e}extractPoints(t)
{return{shape:this.getPoints(t),holes:this.getPointsHoles(t)}}copy(t)
{super.copy(t),this.holes=[];for(let e=0,n=t.holes.length;e<n;e++){const
n=t.holes[e];this.holes.push(n.clone())}return this}toJSON(){const
t=super.toJSON();t.uuid=this.uuid,t.holes=[];for(let
e=0,n=this.holes.length;e<n;e++){const
n=this.holes[e];t.holes.push(n.toJSON())}return t}fromJSON(t)
{super.fromJSON(t),this.uuid=t.uuid,this.holes=[];for(let
e=0,n=t.holes.length;e<n;e++){const n=t.holes[e];this.holes.push((new
Bl).fromJSON(n))}return this}}class Fl extends Ce{constructor(t,e=1)
{super(),this.type="Light",this.color=new tn(t),this.intensity=e}dispose(){}copy(t)
{return
super.copy(t),this.color.copy(t.color),this.intensity=t.intensity,this}toJSON(t)
{const e=super.toJSON(t);return
e.object.color=this.color.getHex(),e.object.intensity=this.intensity,void
0!==thisgroundColor&&(e.objectgroundColor=thisgroundColor.getHex()),void
0!==this.distance&&(e.object.distance=this.distance),void 0!==this.angle&&
(e.object.angle=this.angle),void 0!==this.decay&&(e.object.decay=this.decay),void
0!==this.penumbra&&(e.object.penumbra=this.penumbra),void 0!==this.shadow&&
(e.object.shadow=this.shadow.toJSON()),e}}Fl.prototype.isLight=!0;class Ol extends
Fl{constructor(t,e,n)
{super(t,n),this.type="HemisphereLight",this.position.copy(Ce.DefaultUp),this.updat
eMatrix(),thisgroundColor=new tn(e)}copy(t){return
Fl.prototype.copy.call(this,t),thisgroundColor.copy(tgroundColor),this}}Ol.protot
ype.isHemisphereLight=!0;const Hl=new se,Gl=new Lt,Ul=new Lt;class
kl{constructor(t)
{this.camera=t,this.bias=0,this.normalBias=0,this.radius=1,this.mapSize=new
vt(512,512),this.map=null,this.mapPass=null,this.matrix=new
se,this.autoUpdate=!0,this.needsUpdate=!1,this._frustum=new
ai,this._frameExtents=new vt(1,1),this._viewportCount=1,this._viewports=[new
St(0,0,1,1)]}getViewportCount(){return this._viewportCount}getFrustum(){return
this._frustum}updateMatrices(t){const
e=this.camera,n=this.matrix;Gl.setFromMatrixPosition(t.matrixWorld),e.position.copy
(Gl),Ul.setFromMatrixPosition(t.target.matrixWorld),e.lookAt(Ul),e.updateMatrixWorl
d(),Hl.multiplyMatrices(e.projectionMatrix,e.matrixWorldInverse),this._frustum.setF
romProjectionMatrix(Hl),n.set(.5,0,0,.5,0,.5,0,.5,0,0,.5,.5,0,0,0,1),n.multiply(e.p
rojectionMatrix),n.multiply(e.matrixWorldInverse)}getViewport(t){return
this._viewports[t]}getFrameExtents(){return this._frameExtents}dispose()
{this.map&&this.map.dispose(),this.mapPass&&this.mapPass.dispose()}copy(t){return
this.camera=t.camera.clone(),this.bias=t.bias,this.radius=t.radius,this.mapSize.cop
y(t.mapSize),this}clone(){return(new this.constructor).copy(this)}toJSON(){const t=
{};return 0!==this.bias&&(t.bias=this.bias),0!==this.normalBias&&
(t.normalBias=this.normalBias),1!==this.radius&&
(t.radius=this.radius),512===this.mapSize.x&&512===this.mapSize.y||
(t.mapSize=this.mapSize.toArray()),t.camera=this.camera.toJSON(!1).object,delete
t.camera.matrix,t}}class Vl extends kl{constructor(){super(new
Kn(50,1,.5,500)),this.focus=1}updateMatrices(t){const
e=this.camera,n=2*lt*t.angle*this.focus,i=this.mapSize.width/this.mapSize.height,r=
t.distance||e.far;n===e.fov&&i===e.aspect&&r===e.far||
(e.fov=n,e.aspect=i,e.far=r,e.updateProjectionMatrix()),super.updateMatrices(t)}cop
y(t){return
super.copy(t),this.focus=t.focus,this}}Vl.prototype.isSpotLightShadow=!0;class Wl
extends Fl{constructor(t,e,n=0,i=Math.PI/3,r=0,s=1)
{super(t,e),this.type="SpotLight",this.position.copy(Ce.DefaultUp),this.updateMatri
x(),this.target=new
Ce,this.distance=n,this.angle=i,this.penumbra=r,this.decay=s,this.shadow=new Vl}get
power(){return this.intensity*Math.PI}set power(t)
{this.intensity=t/Math.PI}dispose(){this.shadow.dispose()}copy(t){return

```

```

super.copy(t),this.distance=t.distance,this.angle=t.angle,this.penumbra=t.penumbra,
this.decay=t.decay,this.target=t.target.clone(),this.shadow=t.shadow.clone(),this}}
Wl.prototype.isSpotLight=!0;const jl=new se,ql=new Lt,Xl=new Lt;class Yl extends
kl{constructor(){super(new Kn(90,1,.5,500)),this._frameExtents=new
vt(4,2),this._viewportCount=6,this._viewports=[new St(2,1,1,1),new St(0,1,1,1),new
St(3,1,1,1),new St(1,1,1,1),new St(3,0,1,1),new St(1,0,1,1)],this._cubeDirections=
[new Lt(1,0,0),new Lt(-1,0,0),new Lt(0,0,1),new Lt(0,0,-1),new Lt(0,1,0),new
Lt(0,-1,0)],this._cubeUps=[new Lt(0,1,0),new Lt(0,1,0),new Lt(0,1,0),new
Lt(0,1,0),new Lt(0,0,1),new Lt(0,0,-1)]}updateMatrices(t,e=0){const
n=this.camera,i=this.matrix,r=t.distance||n.far;r!:=n.far&&
(n.far=r,n.updateProjectionMatrix()),ql.setFromMatrixPosition(t.matrixWorld),n.posi
tion.copy(ql),Xl.copy(n.position),Xl.add(this._cubeDirections[e]),n.up.copy(this._c
ubeUps[e]),n.lookAt(Xl),n.updateMatrixWorld(),i.makeTranslation(-ql.x,-ql.y,-
ql.z),jl.multiplyMatrices(n.projectionMatrix,n.matrixWorldInverse),this._frustum.se
tFromProjectionMatrix(jl)}}Yl.prototype.isPointLightShadow=!0;class Zl extends
Fl{constructor(t,e,n=0,i=1)
{super(t,e),this.type="PointLight",this.distance=n,this.decay=i,this.shadow=new
Yl}get power(){return 4*this.intensity*Math.PI}set power(t)
{this.intensity=t/(4*Math.PI)}dispose(){this.shadow.dispose()}copy(t){return
super.copy(t),this.distance=t.distance,this.decay=t.decay,this.shadow=t.shadow.clon
e(),this}}Zl.prototype.isPointLight=!0;class Jl extends
Qn{constructor(t=-1,e=1,n=1,i=-1,r=.1,s=2e3)
{super(),this.type="OrthographicCamera",this.zoom=1,this.view=null,this.left=t,this
.right=e,this.top=n,this.bottom=i,this.near=r,this.far=s,this.updateProjectionMatri
x()}copy(t,e){return
super.copy(t,e),this.left=t.left,this.right=t.right,this.top=t.top,this.bottom=t.bo
ttom,this.near=t.near,this.far=t.far,this.zoom=t.zoom,this.view=null===t.view?
null:Object.assign({},t.view),this}setViewOffset(t,e,n,i,r,s){null===this.view&&
(this.view=
{enabled:!0,fullWidth:1,fullHeight:1,offsetX:0,offsetY:0,width:1,height:1}),this.vi
ew.enabled=!0,this.view.fullWidth=t,this.view.fullHeight=e,this.view.offsetX=n,this
.view.offsetY=i,this.view.width=r,this.view.height=s,this.updateProjectionMatrix()}
clearViewOffset(){null!==this.view&&
(this.view.enabled=!1),this.updateProjectionMatrix()}updateProjectionMatrix(){const
t=(this.right-this.left)/(2*this.zoom),e=(this.top-this.bottom)/(2*this.zoom),n=
(this.right+this.left)/2,i=(this.top+this.bottom)/2;let r=n-t,s=n+t,a=i+e,o=i-
e;if(null!==this.view&&this.view.enabled){const t=(this.right-
this.left)/this.view.fullWidth/this.zoom,e=(this.top-
this.bottom)/this.view.fullHeight/this.zoom;r+=t*this.view.offsetX,s=r+t*this.view.
width,a-=e*this.view.offsetY,o=a-
e*this.view.height}this.projectionMatrix.makeOrthographic(r,s,a,o,this.near,this.far),this.projectionMatrixInverse.copy(this.projectionMatrix).invert()}toJSON(t)
{const e=super.toJSON(t);return
e.object.zoom=this.zoom,e.object.left=this.left,e.object.right=this.right,e.object.
top=this.top,e.object.bottom=this.bottom,e.object.near=this.near,e.object.far=this.
far,null!==this.view&&
(e.object.view=Object.assign({},this.view)),e}}Jl.prototype.isOrthographicCamera=!0
;class Ql extends kl{constructor(){super(new
Jl(-5,5,5,-5,.5,500))}}Ql.prototype.isDirectionalLightShadow=!0;class Kl extends
Fl{constructor(t,e)
{super(t,e),this.type="DirectionalLight",this.position.copy(Ce.DefaultUp),this.upda
teMatrix(),this.target=new Ce,this.shadow=new Ql}dispose()
{this.shadow.dispose()}copy(t){return
super.copy(t),this.target=t.target.clone(),this.shadow=t.shadow.clone(),this}}Kl.pr
ototype.isDirectionalLight=!0;class $l extends Fl{constructor(t,e)
{super(t,e),this.type="AmbientLight"}}$l.prototype.isAmbientLight=!0;class tc
extends Fl{constructor(t,e,n=10,i=10)
{super(t,e),this.type="RectAreaLight",this.width=n,this.height=i}copy(t){return

```

```

super.copy(t),this.width=t.width,this.height=t.height,this}toJSON(t){const
e=super.toJSON(t);return
e.object.width=this.width,e.object.height=this.height,e}}tc.prototype.isRectAreaLig
ht=!0;class ec{constructor(){this.coefficients=[];for(let
t=0;t<9;t++)this.coefficients.push(new Lt)}set(t){for(let
e=0;e<9;e++)this.coefficients[e].copy(t[e]);return this}zero(){for(let
t=0;t<9;t++)this.coefficients[t].set(0,0,0);return this}getAt(t,e){const
n=t.x,i=t.y,r=t.z,s=this.coefficients;return
e.copy(s[0]).multiplyScalar(.282095),e.addScaledVector(s[1],.488603*i),e.addScaledV
ector(s[2],.488603*r),e.addScaledVector(s[3],.488603*n),e.addScaledVector(s[4],n*i*
1.092548),e.addScaledVector(s[5],i*r*1.092548),e.addScaledVector(s[6],.315392*
(3*r*r-1)),e.addScaledVector(s[7],n*r*1.092548),e.addScaledVector(s[8],.546274*
(n*n-i*i)),e}getIrradianceAt(t,e){const
n=t.x,i=t.y,r=t.z,s=this.coefficients;return
e.copy(s[0]).multiplyScalar(.886227),e.addScaledVector(s[1],1.023328*i),e.addScaled
Vector(s[2],1.023328*r),e.addScaledVector(s[3],1.023328*n),e.addScaledVector(s[4],.
858086*n*i),e.addScaledVector(s[5],.858086*i*r),e.addScaledVector(s[6],.743125*r*r-
.247708),e.addScaledVector(s[7],.858086*n*r),e.addScaledVector(s[8],.429043*(n*n-
i*i)),e}add(t){for(let
e=0;e<9;e++)this.coefficients[e].add(t.coefficients[e]);return
this}addScaledSH(t,e){for(let
n=0;n<9;n++)this.coefficients[n].addScaledVector(t.coefficients[n],e);return
this}scale(t){for(let e=0;e<9;e++)this.coefficients[e].multiplyScalar(t);return
this}lerp(t,e){for(let
n=0;n<9;n++)this.coefficients[n].lerp(t.coefficients[n],e);return this}equals(t)
{for(let
e=0;e<9;e++)if(!this.coefficients[e].equals(t.coefficients[e]))return!1;return!0}co
py(t){return this.set(t.coefficients)}clone(){return(new
this.constructor).copy(this)}fromArray(t,e=0){const n=this.coefficients;for(let
i=0;i<9;i++)n[i].fromArray(t,e+3*i);return this}toArray(t=[],e=0){const
n=this.coefficients;for(let i=0;i<9;i++)n[i].toArray(t,e+3*i);return t}static
getBasisAt(t,e){const
n=t.x,i=t.y,r=t.z;e[0]=.282095,e[1]=.488603*i,e[2]=.488603*r,e[3]=.488603*n,e[4]=1.
092548*n*i,e[5]=1.092548*i*r,e[6]=.315392*(3*r*r-1),e[7]=1.092548*n*r,e[8]=.546274*
(n*n-i*i)}}ec.prototype.isSphericalHarmonics3=!0;class nc extends
Fl{constructor(t=new ec,e=1){super(void 0,e),this.sh=t}copy(t){return
super.copy(t),this.sh.copy(t.sh),this}fromJSON(t){return
this.intensity=t.intensity,this.sh.fromArray(t.sh),this}toJSON(t){const
e=super.toJSON(t);return
e.object.sh=this.sh.toArray(),e}}nc.prototype.isLightProbe=!0;class ic extends
ol{constructor(t){super(t),this.textures={}}load(t,e,n,i){const r=this,s=new
cl(r.manager);s.setPath(r.path),s.setRequestHeader(r.requestHeader),s.setWithCreden
tials(r.withCredentials),s.load(t,(function(n)
{try{e(r.parse(JSON.parse(n)))}}catch(e){i?
i(e):console.error(e),r.manager.itemError(t)})),n,i)}parse(t){const
e=this.textures;function n(t){return void
0===e[t]&&console.warn("THREE.MaterialLoader: Undefined texture",t),e[t]}const
i=new ko[t.type];if(void 0!==t.uuid&&(i.uuid=t.uuid),void 0!==t.name&&
(i.name=t.name),void 0!==t.color&&void 0!==i.color&&i.color.setHex(t.color),void
0!==t.roughness&&(i.roughness=t.roughness),void 0!==t.metalness&&
(i.metalness=t.metalness),void 0!==t.shininess&&(i.shininess=(new tn).setHex(t.shininess)),void
0!==t.emissive&&void 0!==i.emissive&&i.emissive.setHex(t.emissive),void
0!==t.specular&&void 0!==i.specular&&i.specular.setHex(t.specular),void
0!==t.shininess&&(i.shininess=t.shininess),void 0!==t.clearcoat&&
(i.clearcoat=t.clearcoat),void 0!==t.clearcoatRoughness&&
(i.clearcoatRoughness=t.clearcoatRoughness),void 0!==t.fog&&(i.fog=t.fog),void
0!==t.flatShading&&(i.flatShading=t.flatShading),void 0!==t.blending&&
(i.blending=t.blending),void 0!==t.combine&&(i.combine=t.combine),void 0!==t.side&&

```

```

(i.side=t.side),void 0!==t.shadowSide&&(i.shadowSide=t.shadowSide),void
0!==t.opacity&&(i.opacity=t.opacity),void 0!==t.transparent&&
(i.transparent=t.transparent),void 0!==t.alphaTest&&(i.alphaTest=t.alphaTest),void
0!==t.depthTest&&(i.depthTest=t.depthTest),void 0!==t.depthWrite&&
(i.depthWrite=t.depthWrite),void 0!==t.colorWrite&&(i.colorWrite=t.colorWrite),void
0!==t.stencilWrite&&(i.stencilWrite=t.stencilWrite),void 0!==t.stencilWriteMask&&
(i.stencilWriteMask=t.stencilWriteMask),void 0!==t.stencilFunc&&
(i.stencilFunc=t.stencilFunc),void 0!==t.stencilRef&&
(i.stencilRef=t.stencilRef),void 0!==t.stencilFuncMask&&
(i.stencilFuncMask=t.stencilFuncMask),void 0!==t.stencilFail&&
(i.stencilFail=t.stencilFail),void 0!==t.stencilZFail&&
(i.stencilZFail=t.stencilZFail),void 0!==t.stencilZPass&&
(i.stencilZPass=t.stencilZPass),void 0!==t.wireframe&&
(i.wireframe=t.wireframe),void 0!==t.wireframeLinewidth&&
(i.wireframeLinewidth=t.wireframeLinewidth),void 0!==t.wireframeLinecap&&
(i.wireframeLinecap=t.wireframeLinecap),void 0!==t.wireframeLinejoin&&
(i.wireframeLinejoin=t.wireframeLinejoin),void 0!==t.rotation&&
(i.rotation=t.rotation),1!==t.linewidth&&(i.linewidth=t.linewidth),void
0!==t.dashSize&&(i.dashSize=t.dashSize),void 0!==t.gapSize&&
(i.gapSize=t.gapSize),void 0!==t.scale&&(i.scale=t.scale),void
0!==t.polygonOffset&&(i.polygonOffset=t.polygonOffset),void
0!==t.polygonOffsetFactor&&(i.polygonOffsetFactor=t.polygonOffsetFactor),void
0!==t.polygonOffsetUnits&&(i.polygonOffsetUnits=t.polygonOffsetUnits),void
0!==t.skinning&&(i.skinning=t.skinning),void 0!==t.morphTargets&&
(i.morphTargets=t.morphTargets),void 0!==t.morphNormals&&
(i.morphNormals=t.morphNormals),void 0!==t.dithering&&
(i.dithering=t.dithering),void 0!==t.alphaToCoverage&&
(i.alphaToCoverage=t.alphaToCoverage),void 0!==t.premultipliedAlpha&&
(i.premultipliedAlpha=t.premultipliedAlpha),void 0!==t.vertexTangents&&
(i.vertexTangents=t.vertexTangents),void 0!==t.visible&&(i.visible=t.visible),void
0!==t.toneMapped&&(i.toneMapped=t.toneMapped),void 0!==t.userData&&
(i.userData=t.userData),void 0!==t.vertexColors&&("number"===typeof t.vertexColors?
i.vertexColors=t.vertexColors>0:i.vertexColors=t.vertexColors),void
0!==t.uniforms)for(const e in t.uniforms){const
r=t.uniforms[e];switch(i.uniforms[e]={},r.type)
{case"t":i.uniforms[e].value=n(r.value);break;case"c":i.uniforms[e].value=(new
tn).setHex(r.value);break;case"v2":i.uniforms[e].value=(new
vt).fromArray(r.value);break;case"v3":i.uniforms[e].value=(new
Lt).fromArray(r.value);break;case"v4":i.uniforms[e].value=(new
St).fromArray(r.value);break;case"m3":i.uniforms[e].value=(new
yt).fromArray(r.value);break;case"m4":i.uniforms[e].value=(new
se).fromArray(r.value);break;default:i.uniforms[e].value=r.value}}if(void
0!==t.defines&&(i.defines=t.defines),void 0!==t.vertexShader&&
(i.vertexShader=t.vertexShader),void 0!==t.fragmentShader&&
(i.fragmentShader=t.fragmentShader),void 0!==t.extensions)for(const e in
t.extensions)i.extensions[e]=t.extensions[e];if(void 0!==t.shading&&
(i.flatShading=1===t.shading),void 0!==t.size&&(i.size=t.size),void
0!==t.sizeAttenuation&&(i.sizeAttenuation=t.sizeAttenuation),void 0!==t.map&&
(i.map=n(t.map)),void 0!==t.matcap&&(i.matcap=n(t.matcap)),void 0!==t.alphaMap&&
(i.alphaMap=n(t.alphaMap)),void 0!==t.bumpMap&&(i.bumpMap=n(t.bumpMap)),void
0!==t.bumpScale&&(i.bumpScale=t.bumpScale),void 0!==t.normalMap&&
(i.normalMap=n(t.normalMap)),void 0!==t.normalMapType&&
(i.normalMapType=t.normalMapType),void 0!==t.normalScale){let
e=t.normalScale;1===Array.isArray(e)&&(e=[e,e]),i.normalScale=(new
vt).fromArray(e)}return void 0!==t.displacementMap&&
(i.displacementMap=n(t.displacementMap)),void 0!==t.displacementScale&&
(i.displacementScale=t.displacementScale),void 0!==t.displacementBias&&
(i.displacementBias=t.displacementBias),void 0!==t.roughnessMap&&

```

```

(i.roughnessMap=n(t.roughnessMap)),void 0!==t.metalnessMap&&
(i.metalnessMap=n(t.metalnessMap)),void 0!==t.emissiveMap&&
(i.emissiveMap=n(t.emissiveMap)),void 0!==t.emissiveIntensity&&
(i.emissiveIntensity=t.emissiveIntensity),void 0!==t.specularMap&&
(i.specularMap=n(t.specularMap)),void 0!==t.envMap&&(i.envMap=n(t.envMap)),void
0!==t.envMapIntensity&&(i.envMapIntensity=t.envMapIntensity),void
0!==t.reflectivity&&(i.reflectivity=t.reflectivity),void 0!==t.refractionRatio&&
(i.refractionRatio=t.refractionRatio),void 0!==t.lightMap&&
(i.lightMap=n(t.lightMap)),void 0!==t.lightMapIntensity&&
(i.lightMapIntensity=t.lightMapIntensity),void 0!==t.aoMap&&
(i.aoMap=n(t.aoMap)),void 0!==t.aoMapIntensity&&
(i.aoMapIntensity=t.aoMapIntensity),void 0!==t.gradientMap&&
(i.gradientMap=n(t.gradientMap)),void 0!==t.clearcoatMap&&
(i.clearcoatMap=n(t.clearcoatMap)),void 0!==t.clearcoatRoughnessMap&&
(i.clearcoatRoughnessMap=n(t.clearcoatRoughnessMap)),void
0!==t.clearcoatNormalMap&&(i.clearcoatNormalMap=n(t.clearcoatNormalMap)),void
0!==t.clearcoatNormalScale&&(i.clearcoatNormalScale=(new
vt).fromArray(t.clearcoatNormalScale)),void 0!==t.transmission&&
(i.transmission=t.transmission),void 0!==t.transmissionMap&&
(i.transmissionMap=n(t.transmissionMap)),i.setTextures(t){return
this.textures=t,this}}class rc{static decodeText(t){if("undefined"!==typeof
TextDecoder)return(new TextDecoder).decode(t);let e="";for(let
n=0,i=t.length;n<i;n++)e+=String.fromCharCode(t[n]);try{return
decodeURIComponent(escape(e))}catch(t){return e}}static extractUrlBase(t){const
e=t.lastIndexOf("/");return-1===e?".":"/":t.substr(0,e+1)}}class sc extends
En{constructor()
{super(),this.type="InstancedBufferGeometry",this.instanceCount=1/0}copy(t){return
super.copy(t),this.instanceCount=t.instanceCount,this}clone(){return(new
this.constructor).copy(this)}toJSON(){const t=super.toJSON(this);return
t.instanceCount=this.instanceCount,t.isInstancedBufferGeometry=!0,t}}sc.prototype.i
sInstancedBufferGeometry=!0;class ac extends sn{constructor(t,e,n,i)
{"number"===typeof n&&(i=n,n=!1,console.error("THREE.InstancedBufferAttribute: The
constructor now expects normalized as the third
argument.")),super(t,e,n),this.meshPerAttribute=i||1}copy(t){return
super.copy(t),this.meshPerAttribute=t.meshPerAttribute,this}toJSON(){const
t=super.toJSON();return
t.meshPerAttribute=this.meshPerAttribute,t.isInstancedBufferAttribute=!0,t}}ac.prot
otype.isInstancedBufferAttribute=!0;class oc extends ol{constructor(t)
{super(t)}load(t,e,n,i){const r=this,s=new
cl(r.manager);s.setPath(r.path),s.setRequestHeader(r.requestHeader),s.setWithCreden
tials(r.withCredentials),s.load(t,(function(n)
{try{e(r.parse(JSON.parse(n)))}catch(e){i?
i(e):console.error(e),r.manager.itemError(t)}}),n,i)}parse(t){const e={},n=
{};function i(t,i){if(void 0!==e[i])return e[i];const
r=t.interleavedBuffers[i],s=function(t,e){if(void 0!==n[e])return n[e];const
i=t.arrayBuffers[e],r=new Uint32Array(i).buffer;return n[e]=r,r}
(t,r.buffer),a=yn(r.type,s),o=new Es(a,r.stride);return
o.uuid=r.uuid,e[i]=o,o}const r=t.isInstancedBufferGeometry?new sc:new
En,s=t.data.index;if(void 0!==s){const t=yn(s.type,s.array);r.setIndex(new
sn(t,1))}const a=t.data.attributes;for(const e in a){const n=a[e];let
s;if(n.isInterleavedBufferAttribute){const e=i(t.data,n.data);s=new
ls(e,n.itemSize,n.offset,n.normalized)}else{const
t=yn(n.type,n.array);s=new(n.isInstancedBufferAttribute?ac:sn)
(t,n.itemSize,n.normalized)}void 0!==n.name&&(s.name=n.name),void
0!==n.usage&&s.setUsage(n.usage),void 0!==n.updateRange&&
(s.updateRange.offset=n.updateRange.offset,s.updateRange.count=n.updateRange.count)
,r.setAttribute(e,s)}const o=t.data.morphAttributes;if(o)for(const e in o){const
n=o[e],s=[];for(let e=0,r=n.length;e<r;e++){const r=n[e];let

```

```

a;if(r.isInterleavedBufferAttribute){const e=i(t.data,r.data);a=new
Ls(e,r.itemSize,r.offset,r.normalized)}else{const t=yn(r.type,r.array);a=new
sn(t,r.itemSize,r.normalized)}void 0!==r.name&&
(a.name=r.name),s.push(a)}r.morphAttributes[e]=s}t.data.morphTargetsRelative&&
(r.morphTargetsRelative=!0);const
l=t.data.groups||t.data.drawcalls||t.data.offsets;if(void 0!==l)for(let
t=0,e=l.length;t!==e;++t){const
e=l[t];r.addGroup(e.start,e.count,e.materialIndex)}const
c=t.data.boundingSphere;if(void 0!==c){const t=new Lt;void
0!==c.center&&t.fromArray(c.center),r.boundingSphere=new Jt(t,c.radius)}return
t.name&&(r.name=t.name),t.userData&&(r.userData=t.userData),r}}const lc=
{UVMMapping:i,CubeReflectionMapping:r,CubeRefractionMapping:s,EquirectangularReflect
ionMapping:a,EquirectangularRefractionMapping:o,CubeUVReflectionMapping:l,CubeUVRef
ractionMapping:c},cc=
{RepeatWrapping:h,ClampToEdgeWrapping:u,MirroredRepeatWrapping:d},hc=
{NearestFilter:p,NearestMipmapNearestFilter:m,NearestMipmapLinearFilter:f,LinearFil
ter:g,LinearMipmapNearestFilter:v,LinearMipmapLinearFilter:y};class uc extends
ol{constructor(t){super(t),"undefined"===typeof
createImageBitmap&&console.warn("THREE.ImageBitmapLoader: createImageBitmap() not
supported."),"undefined"===typeof fetch&&console.warn("THREE.ImageBitmapLoader:
fetch() not supported."),this.options={premultiplyAlpha:"none"}}setOptions(t)
{return this.options=t,this}load(t,e,n,i){void 0===t&&(t=""),void 0!==this.path&&
(t=this.path+t),t=this.manager.resolveURL(t);const r=this,s=rl.get(t);if(void
0!==s)return r.manager.itemStart(t),setTimeout((function()
{e&&e(s),r.manager.itemEnd(t)}),0),s;const a=
{};a.credentials="anonymous"===this.crossOrigin?"same-
origin":"include",a.headers=this.requestHeader,fetch(t,a).then((function(t){return
t.blob()})).then((function(t){return createImageBitmap(t,Object.assign(r.options,
{colorSpaceConversion:"none"}))})).then((function(n)
{rl.add(t,n),e&&e(n),r.manager.itemEnd(t)})).catch((function(e)
{i&&i(e),r.manager.itemError(t),r.manager.itemEnd(t)})),r.manager.itemStart(t)}uc.
prototype.isImageBitmapLoader=!0;class dc{constructor()
{this.type="ShapePath",this.color=new tn,this.subPaths=
[],this.currentPath=null}moveTo(t,e){return this.currentPath=new
Bl,this.subPaths.push(this.currentPath),this.currentPath.moveTo(t,e),this}lineTo(t,
e){return this.currentPath.lineTo(t,e),this}quadraticCurveTo(t,e,n,i){return
this.currentPath.quadraticCurveTo(t,e,n,i),this}bezierCurveTo(t,e,n,i,r,s){return
this.currentPath.bezierCurveTo(t,e,n,i,r,s),this}splineThru(t){return
this.currentPath.splineThru(t),this}toShapes(t,e){function n(t){const e=[];for(let
n=0,i=t.length;n<i;n++){const i=t[n],r=new zl;r.curves=i.curves,e.push(r)}return
e}function i(t,e){const n=e.length;let i=!1;for(let r=n-1,s=0;s<n;r=s++){let
n=e[r],a=e[s],o=a.x-n.x,l=a.y-n.y;if(Math.abs(l)>Number.EPSILON){if(l<0&&
(n=e[s],o=-o,a=e[r],l=-l),t.y<n.y||t.y>a.y)continue;if(t.y===n.y)
{if(t.x===n.x)return!0}else{const e=l*(t.x-n.x)-o*(t.y-
n.y);if(0===e)return!0;if(e<0)continue;i=!i}}else{if(t.y!==n.y)continue;if(a.x<=t.x
&&t.x<=n.x||n.x<=t.x&&t.x<=a.x)return!0}}return i}const
r=po.isClockwise,s=this.subPaths;if(0===s.length)return[];if(!0===e)return n(s);let
a,o,l;const c=[];if(l===s.length)return o=s[0],l=new
zl,l.curves=o.curves,c.push(l),c;let h=!r(s[0].getPoints());h=t?!h:h;const u=[],d=
[];let p,m,f=[],g=0;d[g]=void 0,f[g]=[];for(let
e=0,n=s.length;e<n;e++)o=s[e],p=o.getPoints(),a=r(p),a=t?!a:a,a?
(!h&&d[g]&&g++,d[g]={s:new zl,p:p},d[g].s.curves=o.curves,h&&g++,f[g]=
[]):f[g].push({h:o,p:p[0]});if(!d[0])return n(s);if(d.length>1){let t=!1;const e=
[];for(let t=0,e=d.length;t<e;t++)u[t]=[];for(let n=0,r=d.length;n<r;n++){const
r=f[n];for(let s=0;s<r.length;s++){const a=r[s];let o=!0;for(let
r=0;r<d.length;r++)i(a.p,d[r].p)&&(n!==r&&e.push({froms:n,tos:r,hole:s}),o?
(o=!1,u[r].push(a)):t=!0);o&&u[n].push(a)}}e.length>0&&(t||(f=u))}for(let
t=0,e=d.length;t<e;t++){l=d[t].s,c.push(l),m=f[t];for(let

```



```

t=0,e=m.length;t<e;t++)l.holes.push(m[t].h)}return c}}class pc{constructor(t)
{this.type="Font",this.data=t}generateShapes(t,e=100){const n=[],i=function(t,e,n)
{const i=Array.from(t),r=e/n.resolution,s=(n.boundingBox.yMax-
n.boundingBox.yMin+n.underlineThickness)*r,a=[];let o=0,l=0;for(let
t=0;t<i.length;t++){const e=i[t];if("\n"===e)o=0,l-=s;else{const
t=mc(e,r,o,l,n);o+=t.offsetX,a.push(t.path)}}return a}(t,e,this.data);for(let
t=0,e=i.length;t<e;t++)Array.prototype.push.apply(n,i[t].toShapes());return
n}}function mc(t,e,n,i,r){const s=r.glyphs[t]||r.glyphs["?"];if(!s)return void
console.error('THREE.Font: character "'+t+'" does not exists in font family
'+r.familyName+'.');const a=new dc;let o,l,c,h,u,d,p,m;if(s.o){const
t=s._cachedOutline||(s._cachedOutline=s.o.split(" "));for(let r=0,s=t.length;r<s;)
{switch(t[r++])
{case"m":o=t[r++]*e+n,l=t[r++]*e+i,a.moveTo(o,l);break;case"l":o=t[r++]*e+n,l=t[r++]*e+i,a.lineTo(o,l);break;case"q":c=t[r++]*e+n,h=t[r++]*e+i,u=t[r++]*e+n,d=t[r++]*e+i,a.quadraticCurveTo(u,d,c,h);break;case"b":c=t[r++]*e+n,h=t[r++]*e+i,u=t[r++]*e+n,d=t[r++]*e+i,p=t[r++]*e+n,m=t[r++]*e+i,a.bezierCurveTo(u,d,p,m,c,h)}}return{offsetX:s.ha*e,path:a}}pc.prototype.isFont=!0;let fc;const gc={getContext:function()
{return void 0===fc&&
(fc=new(window.AudioContext||window.webkitAudioContext)),fc},setContext:function(t)
{fc=t}};class vc extends ol{constructor(t){super(t)}load(t,e,n,i){const
r=this,s=new
cl(this.manager);s.setResponseType("arraybuffer"),s.setPath(this.path),s.setRequest
Header(this.requestHeader),s.setWithCredentials(this.withCredentials),s.load(t,
(function(n){try{const t=n.slice(0);gc.getContext().decodeAudioData(t,(function(t)
{e(t)}))}catch(e){i?i(e):console.error(e),r.manager.itemError(t)}}),n,i)}}class yc
extends nc{constructor(t,e,n=1){super(void 0,n);const i=(new tn).set(t),r=(new
tn).set(e),s=new Lt(i.r,i.g,i.b),a=new
Lt(r.r,r.g,r.b),o=Math.sqrt(Math.PI),l=o*Math.sqrt(.75);this.sh.coefficients[0].cop
y(s).add(a).multiplyScalar(o),this.sh.coefficients[1].copy(s).sub(a).multiplyScalar
(l)}yc.prototype.isHemisphereLightProbe=!0;class xc extends nc{constructor(t,e=1)
{super(void 0,e);const n=(new
tn).set(t);this.sh.coefficients[0].set(n.r,n.g,n.b).multiplyScalar(2*Math.sqrt(Math
.PI))}xc.prototype.isAmbientLightProbe=!0;const _c=new se,wc=new se;class
bc{constructor(t=!0)
{this.autoStart=t,this.startTime=0,this.oldTime=0,this.elapsedTime=0,this.running=!
1}start()
{this.startTime=Mc(),this.oldTime=this.startTime,this.elapsedTime=0,this.running=!0
}stop(){this.getElapsedTime(),this.running=!1,this.autoStart=!1}getElapsedTime()
{return this.getDelta(),this.elapsedTime}getDelta(){let
t=0;if(this.autoStart&&!this.running)return this.start(),0;if(this.running){const
e=Mc();t=(e-this.oldTime)/1e3,this.oldTime=e,this.elapsedTime+=t}return t}}function
Mc(){return("undefined"==typeof performance?Date:performance).now()}const Sc=new
Lt,Tc=new At,Ec=new Lt,Ac=new Lt;class Lc extends Ce{constructor(t)
{super(),this.type="Audio",this.listener=t,this.context=t.context,this.gain=this.co
ntext.createGain(),this.gain.connect(t.getInput()),this.autoplay=!1,this.buffer=nul
l,this.detune=0,this.loop=!1,this.loopStart=0,this.loopEnd=0,this.offset=0,this.dur
ation=void
0,this.playbackRate=1,this.isPlaying=!1,this.hasPlaybackControl=!0,this.source=null
,this.sourceType="empty",this._startedAt=0,this._progress=0,this._connected=!1,this
.filters=[]}getOutput(){return this.gain}setNodeSource(t){return
this.hasPlaybackControl=!1,this.sourceType="audioNode",this.source=t,this.connect()
,this}setMediaElementSource(t){return
this.hasPlaybackControl=!1,this.sourceType="mediaNode",this.source=this.context.cre
ateMediaElementSource(t),this.connect(),this}setMediaStreamSource(t){return
this.hasPlaybackControl=!1,this.sourceType="mediaStreamNode",this.source=this.conte
xt.createMediaStreamSource(t),this.connect(),this}setBuffer(t){return
this.buffer=t,this.sourceType="buffer",this.autoplay&&this.play(),this}play(t=0)
{if(!0===this.isPlaying)return void console.warn("THREE.Audio: Audio is already

```

```

playing.");if(!1===this.hasPlaybackControl)return void console.warn("THREE.Audio:
this Audio has no playback
control.");this._startedAt=this.context.currentTime+t;const
e=this.context.createBufferSource();return
e.buffer=this.buffer,e.loop=this.loop,e.loopStart=this.loopStart,e.loopEnd=this.loo
pEnd,e.onended=this.onEnded.bind(this),e.start(this._startedAt,this._progress+this.
offset,this.duration),this.isPlaying=!0,this.source=e,this.setDetune(this.detune),t
his.setPlaybackRate(this.playbackRate),this.connect()}pause()
{if(!1!==this.hasPlaybackControl)return!0===this.isPlaying&&
(this._progress+=Math.max(this.context.currentTime-
this._startedAt,0)*this.playbackRate,!0===this.loop&&
(this._progress=this._progress%
(this.duration||this.buffer.duration)),this.source.stop(),this.source.onended=null,
this.isPlaying=!1),this;console.warn("THREE.Audio: this Audio has no playback
control.")}stop(){if(!1!==this.hasPlaybackControl)return
this._progress=0,this.source.stop(),this.source.onended=null,this.isPlaying=!1,this
;console.warn("THREE.Audio: this Audio has no playback control.")}connect()
{if(this.filters.length>0){this.source.connect(this.filters[0]);for(let
t=1,e=this.filters.length;t<e;t++)this.filters[t-
1].connect(this.filters[t]);this.filters[this.filters.length-
1].connect(this.getOutput())}else this.source.connect(this.getOutput());return
this._connected=!0,this}disconnect(){if(this.filters.length>0)
{this.source.disconnect(this.filters[0]);for(let
t=1,e=this.filters.length;t<e;t++)this.filters[t-
1].disconnect(this.filters[t]);this.filters[this.filters.length-
1].disconnect(this.getOutput())}else
this.source.disconnect(this.getOutput());return
this._connected=!1,this}getFilters(){return this.filters}setFilters(t){return t||
(t=[]),!0===this._connected?
(this.disconnect(),this.filters=t.slice(),this.connect()):this.filters=t.slice(),th
is}setDetune(t){if(this.detune=t,void
0!==this.source.detune)return!0===this.isPlaying&&this.source.detune.setTargetAtTim
e(this.detune,this.context.currentTime,.01),this}getDetune(){return
this.detune}getFilter(){return this.getFilters()[0]}setFilter(t){return
this.setFilters(t?[t]:[])}setPlaybackRate(t){if(!1!==this.hasPlaybackControl)return
this.playbackRate=t,!0===this.isPlaying&&this.source.playbackRate.setTargetAtTime(t
his.playbackRate,this.context.currentTime,.01),this;console.warn("THREE.Audio: this
Audio has no playback control.")}getPlaybackRate(){return
this.playbackRate}onEnded(){this.isPlaying=!1}getLoop()
{return!1===this.hasPlaybackControl?(console.warn("THREE.Audio: this Audio has no
playback control."),!1):this.loop}setLoop(t){if(!1!==this.hasPlaybackControl)return
this.loop=t,!0===this.isPlaying&&
(this.source.loop=this.loop),this;console.warn("THREE.Audio: this Audio has no
playback control.")}setLoopStart(t){return this.loopStart=t,this}setLoopEnd(t)
{return this.loopEnd=t,this}getVolume(){return this.gain.gain.value}setVolume(t)
{return this.gain.gain.setTargetAtTime(t,this.context.currentTime,.01),this}}const
Rc=new Lt,Cc=new At,Pc=new Lt,Dc=new Lt;class Ic{constructor(t,e=2048)
{this.analyser=t.context.createAnalyser(),this.analyser.fftSize=e,this.data=new
Uint8Array(this.analyser.frequencyBinCount),t.getOutput().connect(this.analyser)}ge
tFrequencyData(){return
this.analyser.getBytesFrequencyData(this.data),this.data}getAverageFrequency(){let
t=0;const e=this.getFrequencyData();for(let n=0;n<e.length;n++)t+=e[n];return
t/e.length}}class Nc{constructor(t,e,n){let
i,r,s;switch(this.binding=t,this.valueSize=n,e)
{case"quaternion":i=this._slerp,r=this._slerpAdditive,s=this._setAdditiveIdentityQu
aternion,this.buffer=new
Float64Array(6*n),this._workIndex=5;break;case"string":case"bool":i=this._select,r=
this._select,s=this._setAdditiveIdentityOther,this.buffer=new

```

```

Array(5*n);break;default:i=this._lerp,r=this._lerpAdditive,s=this._setAdditiveIdentityNumeric,this.buffer=new
Float64Array(5*n)}this._mixBufferRegion=i,this._mixBufferRegionAdditive=r,this._setIdentity=s,this._origIndex=3,this._addIndex=4,this.cumulativeWeight=0,this.cumulativeWeightAdditive=0,this.useCount=0,this.referenceCount=0}accumulate(t,e){const n=this.buffer,i=this.valueSize,r=t*i+i;let s=this.cumulativeWeight;if(0===s){for(let t=0;t!==i;++t)n[r+t]=n[t];s=e}else{s+=e;const t=e/s;this._mixBufferRegion(n,r,0,t,i)}this.cumulativeWeight=s}accumulateAdditive(t){const e=this.buffer,n=this.valueSize,i=n*this._addIndex;0===this.cumulativeWeightAdditive&&this._setIdentity(),this._mixBufferRegionAdditive(e,i,0,t,n),this.cumulativeWeightAdditive+=t}apply(t){const e=this.valueSize,n=this.buffer,i=t*e,e,r=this.cumulativeWeight,s=this.cumulativeWeightAdditive,a=this.binding;if(this.cumulativeWeight=0,this.cumulativeWeightAdditive=0,r<1){const t=e*this._origIndex;this._mixBufferRegion(n,i,t,1-r,e)}s>0&&this._mixBufferRegionAdditive(n,i,this._addIndex*e,1,e);for(let t=e,r=e+e;t!==r;++t)if(n[t]!==n[t+e]){a.setValue(n,i);break}}saveOriginalState(){const t=this.binding,e=this.buffer,n=this.valueSize,i=n*this._origIndex;t.getValue(e,i);for(let t=n,r=i;t!==r;++t)e[t]=e[i+t%n];this._setIdentity(),this.cumulativeWeight=0,this.cumulativeWeightAdditive=0}restoreOriginalState(){const t=3*this.valueSize;this.binding.setValue(this.buffer,t)}_setAdditiveIdentityNumeric(){const t=this._addIndex*this.valueSize,e=t+this.valueSize;for(let n=t;n<e;n++)this.buffer[n]=0}_setAdditiveIdentityQuaternion(){this._setAdditiveIdentityNumeric(),this.buffer[this._addIndex*this.valueSize+3]=1}_setAdditiveIdentityOther(){const t=this._origIndex*this.valueSize,e=this._addIndex*this.valueSize;for(let n=0;n<this.valueSize;n++)this.buffer[e+n]=this.buffer[t+n]}_select(t,e,n,i,r){if(i>=.5)for(let i=0;i!==r;++i)t[e+i]=t[n+i]}_slerp(t,e,n,i){At.slerpFlat(t,e,t,e,t,n,i)}_slerpAdditive(t,e,n,i,r){const s=this._workIndex*r;At.multiplyQuaternionsFlat(t,s,t,e,t,n),At.slerpFlat(t,e,t,e,t,s,i)}_lerp(t,e,n,i,r){const s=1-i;for(let a=0;a!==r;++a){const r=e+a;t[r]=t[r]*s+t[n+a]*i}_lerpAdditive(t,e,n,i,r){for(let s=0;s!==r;++s){const r=e+s;t[r]=t[r]+t[n+s]*i}}const Bc="\\[\\]\\.:\\/","zc=new RegExp("\\[\\]\\.:\\/","g"),Fc="^[\\[\\]\\.:\\/","Oc="^[^"+Bc.replace("\\.", "")+"",Hc="/((?:WC+[/\\:]))*/.source.replace("WC",Fc),Gc="/(WCOD+)?/.source.replace("WCOD",Oc),Uc="/(?:\\. (WC+)(?:\\[ (\\+\\)\\)\\)?)/.source.replace("WC",Fc),kc="/\\. (WC+)(?:\\[ (\\+\\)\\)\\)?)/.source.replace("WC",Fc),Vc=new RegExp("^"+Hc+Gc+Uc+kc+"$"),Wc=["material","materials","bones"];class jc{constructor(t,e,n){this.path=e,this.parsedPath=n||jc.parseTrackName(e),this.node=jc.findNode(t,this.parsedPath.nodeName)||t,this.rootNode=t,this.getValue=this._getValue_unbound,this.setValue=this._setValue_unbound}static create(t,e,n){return t&&t.isAnimationObjectGroup?new jc.Composite(t,e,n):new jc(t,e,n)}static sanitizeNodeName(t){return t.replace(/\\/s/g,"_").replace(zc,"")}static parseTrackName(t){const e=Vc.exec(t);if(!e)throw new Error("PropertyBinding: Cannot parse trackName: "+t);const n={nodeName:e[2],objectName:e[3],objectIndex:e[4],propertyName:e[5],propertyIndex:e[6]},i=n.nodeName&&n.nodeName.lastIndexOf(".");if(void 0!==i&&-1!==i){const t=n.nodeName.substring(i+1);-1!==Wc.indexOf(t)&&(n.nodeName=n.nodeName.substring(0,i),n.objectName=t)}if(null===n.propertyName||0===n.propertyName.length)throw new Error("PropertyBinding: can not parse propertyName from trackName: "+t);return n}static findNode(t,e){if(!e||""===e||"."===e||-1===e||e===t.name||e===t.uuid)return t;if(t.skeleton){const n=t.skeleton.getBoneByName(e);if(void 0!==n)return n}if(t.children){const n=function(t){for(let i=0;i<t.length;i++){const r=t[i];if(r.name===e||r.uuid===e)return r;const s=n(r.children);if(s)return

```

```

s{return null},i=n(t.children);if(i)return i{return null}_getValue_unavailable()
{}_setValue_unavailable(){}_getValue_direct(t,e)
{t[e]=this.node[this.propertyName]}_getValue_array(t,e){const
n=this.resolvedProperty;for(let
i=0,r=n.length;i!==r;++i)t[e++]=n[i]}_getValue_arrayElement(t,e)
{t[e]=this.resolvedProperty[this.propertyIndex]}_getValue_toArray(t,e)
{this.resolvedProperty.toArray(t,e)}_setValue_direct(t,e)
{this.targetObject[this.propertyName]=t[e]}_setValue_direct_setNeedsUpdate(t,e)
{this.targetObject[this.propertyName]=t[e],this.targetObject.needsUpdate=!0}_setVal
ue_direct_setMatrixWorldNeedsUpdate(t,e)
{this.targetObject[this.propertyName]=t[e],this.targetObject.matrixWorldNeedsUpdate
=!0}_setValue_array(t,e){const n=this.resolvedProperty;for(let
i=0,r=n.length;i!==r;++i)n[i]=t[e++]}_setValue_array_setNeedsUpdate(t,e){const
n=this.resolvedProperty;for(let
i=0,r=n.length;i!==r;++i)n[i]=t[e++];this.targetObject.needsUpdate=!0}_setValue_arr
ay_setMatrixWorldNeedsUpdate(t,e){const n=this.resolvedProperty;for(let
i=0,r=n.length;i!==r;++i)n[i]=t[e++];this.targetObject.matrixWorldNeedsUpdate=!0}_s
etValue_arrayElement(t,e)
{this.resolvedProperty[this.propertyIndex]=t[e]}_setValue_arrayElement_setNeedsUpda
te(t,e)
{this.resolvedProperty[this.propertyIndex]=t[e],this.targetObject.needsUpdate=!0}_s
etValue_arrayElement_setMatrixWorldNeedsUpdate(t,e)
{this.resolvedProperty[this.propertyIndex]=t[e],this.targetObject.matrixWorldNeedsU
pdate=!0}_setValue_fromArray(t,e)
{this.resolvedProperty.fromArray(t,e)}_setValue_fromArray_setNeedsUpdate(t,e)
{this.resolvedProperty.fromArray(t,e),this.targetObject.needsUpdate=!0}_setValue_fr
omArray_setMatrixWorldNeedsUpdate(t,e)
{this.resolvedProperty.fromArray(t,e),this.targetObject.matrixWorldNeedsUpdate=!0}_
getValue_unbound(t,e){this.bind(),this.getValue(t,e)}_setValue_unbound(t,e)
{this.bind(),this.setValue(t,e)}bind(){let t=this.node;const
e=this.parsedPath,n=e.objectName,i=e.propertyName;let r=e.propertyIndex;if(t||
(t=jc.findNode(this.rootNode,e.nodeName)||this.rootNode,this.node=t),this.getValue=
this._getValue_unavailable,this.setValue=this._setValue_unavailable,!t)return void
console.error("THREE.PropertyBinding: Trying to update node for track:
"+this.path+" but it wasn't found.");if(n){let i=e.objectIndex;switch(n)
{case"materials":if(!t.material)return void console.error("THREE.PropertyBinding:
Can not bind to material as node does not have a
material.",this);if(!t.material.materials)return void
console.error("THREE.PropertyBinding: Can not bind to material.materials as
node.material does not have a materials
array.",this);t=t.material.materials;break;case"bones":if(!t.skeleton)return void
console.error("THREE.PropertyBinding: Can not bind to bones as node does not have a
skeleton.",this);t=t.skeleton.bones;for(let e=0;e<t.length;e++)if(t[e].name===i)
{i=e;break}break;default:if(void 0===t[n])return void
console.error("THREE.PropertyBinding: Can not bind to objectName of node
undefined.",this);t=t[n]}if(void 0!==i){if(void 0===t[i])return void
console.error("THREE.PropertyBinding: Trying to bind to objectIndex of objectName,
but is undefined.",this,t);t=t[i]}const s=t[i];if(void 0===s){const
n=e.nodeName;return void console.error("THREE.PropertyBinding: Trying to update
property for track: "+n+"."+i+" but it wasn't found.",t)}let
a=this.Versioning.None;this.targetObject=t,void 0!==t.needsUpdate?
a=this.Versioning.NeedsUpdate:void 0!==t.matrixWorldNeedsUpdate&&
(a=this.Versioning.MatrixWorldNeedsUpdate);let o=this.BindingType.Direct;if(void
0!==r){if("morphTargetInfluences"===i){if(!t.geometry)return void
console.error("THREE.PropertyBinding: Can not bind to morphTargetInfluences because
node does not have a geometry.",this);if(!t.geometry.isBufferGeometry)return void
console.error("THREE.PropertyBinding: Can not bind to morphTargetInfluences on
THREE.Geometry. Use THREE.BufferGeometry

```

```

instead.",this);if(!t.geometry.morphAttributes)return void
console.error("THREE.PropertyBinding: Can not bind to morphTargetInfluences because
node does not have a geometry.morphAttributes.",this);void
0!==(t.morphTargetDictionary[r]&&
(r=t.morphTargetDictionary[r]))o=this.BindingType.ArrayElement,this.resolvedPropert
y=s,this.propertyIndex=r}else void 0!==(s.fromArray&&void 0!==(s.toArray?
(o=this.BindingType.HasFromToArray,this.resolvedProperty=s):Array.isArray(s)?
(o=this.BindingType.EntireArray,this.resolvedProperty=s):this.propertyName=i;this.g
etValue=this.GetterByBindingType[o],this.setValue=this.SetterByBindingTypeAndVersio
ning[o][a])unbind()
{this.node=null,this.getValue=this._getValue_unbound,this.setValue=this._setValue_u
nbound}}jc.Composite=class{constructor(t,e,n){const
i=n||jc.parseTrackName(e);this._targetGroup=t,this._bindings=t.subscribe_(e,i)}getV
alue(t,e){this.bind();const
n=this._targetGroup.nCachedObjects_,i=this._bindings[n];void
0!==(i&&i.getValue(t,e))setValue(t,e){const n=this._bindings;for(let
i=this._targetGroup.nCachedObjects_,r=n.length;i!=r;++i)n[i].setValue(t,e)}bind()
{const t=this._bindings;for(let
e=this._targetGroup.nCachedObjects_,n=t.length;e!=n;++e)t[e].bind()}unbind(){const
t=this._bindings;for(let
e=this._targetGroup.nCachedObjects_,n=t.length;e!=n;++e)t[e].unbind()}}jc.prototy
pe.BindingType=
{Direct:0,EntireArray:1,ArrayElement:2,HasFromToArray:3},jc.prototype.Versioning=
{None:0,NeedsUpdate:1,MatrixWorldNeedsUpdate:2},jc.prototype.GetterByBindingType=
[jc.prototype._getValue_direct,jc.prototype._getValue_array,jc.prototype._getValue_
arrayElement,jc.prototype._getValue_toArray],jc.prototype.SetterByBindingTypeAndVer
sioning=
[[jc.prototype._setValue_direct,jc.prototype._setValue_direct_setNeedsUpdate,jc.pro
totype._setValue_direct_setMatrixWorldNeedsUpdate],
[jc.prototype._setValue_array,jc.prototype._setValue_array_setNeedsUpdate,jc.protot
ype._setValue_array_setMatrixWorldNeedsUpdate],
[jc.prototype._setValue_arrayElement,jc.prototype._setValue_arrayElement_setNeedsUp
date,jc.prototype._setValue_arrayElement_setMatrixWorldNeedsUpdate],
[jc.prototype._setValue_fromArray,jc.prototype._setValue_fromArray_setNeedsUpdate,j
c.prototype._setValue_fromArray_setMatrixWorldNeedsUpdate]];class qc{constructor()
{this.uuid=ct(),this._objects=Array.prototype.slice.call(arguments),this.nCachedObj
ects_=0;const t={};this._indicesByUUID=t;for(let
e=0,n=arguments.length;e!=n;++e)t[arguments[e].uuid]=e;this._paths=
[],this._parsedPaths=[],this._bindings=[],this._bindingsIndicesByPath={};const
e=this;this.stats={objects:{get total(){return e._objects.length},get inUse()
{return this.total-e.nCachedObjects_}},get bindingsPerObject(){return
e._bindings.length}}add(){const
t=this._objects,e=this._indicesByUUID,n=this._paths,i=this._parsedPaths,r=this._bin
dings,s=r.length;let a,o=t.length,l=this.nCachedObjects_;for(let
c=0,h=arguments.length;c!=h;++c){const h=arguments[c],u=h.uuid;let d=e[u];if(void
0===d){d=o++,e[u]=d,t.push(h);for(let t=0,e=s;t!=e;++t)r[t].push(new
jc(h,n[t],i[t]))}else if(d<l){a=t[d];const o--
l,c=t[o];e[c.uuid]=d,t[d]=c,e[u]=o,t[o]=h;for(let t=0,e=s;t!=e;++t){const
e=r[t],s=e[o];let a=e[d];e[d]=s,void 0===a&&(a=new jc(h,n[t],i[t])),e[o]=a}else
t[d]!==a&&console.error("THREE.AnimationObjectGroup: Different objects with the
same UUID detected. Clean the caches or recreate your infrastructure when reloading
scenes.")}this.nCachedObjects_=l}remove(){const
t=this._objects,e=this._indicesByUUID,n=this._bindings,i=n.length;let
r=this.nCachedObjects_;for(let s=0,a=arguments.length;s!=a;++s){const
a=arguments[s],o=a.uuid,l=e[o];if(void 0!==(l&&l>=r){const
s=r++,c=t[s];e[c.uuid]=l,t[l]=c,e[o]=s,t[s]=a;for(let t=0,e=i;t!=e;++t){const
e=n[t],i=e[s],r=e[l];e[l]=i,e[s]=r}}this.nCachedObjects_=r}uncache(){const
t=this._objects,e=this._indicesByUUID,n=this._bindings,i=n.length;let

```

```

r=this.nCachedObjects_,s=t.length;for(let a=0,o=arguments.length;a!==(o+1);++a){const
o=arguments[a].uuid,l=e[o];if(void 0!==(l))if(delete e[o],l<r){const a=--r,o=t[a],c=
-s,h=t[c];e[o.uuid]=l,t[l]=o,e[h.uuid]=a,t[a]=h,t.pop();for(let t=0,e=i;t!==(e+1);++t)
{const e=n[t],i=e[a],r=e[c];e[l]=i,e[a]=r,e.pop()}}else{const r=--s,a=t[r];r>0&&
(e[a.uuid]=l),t[l]=a,t.pop();for(let t=0,e=i;t!==(e+1);++t){const
e=n[t];e[l]=e[r],e.pop()}}this.nCachedObjects_=r}subscribe_(t,e){const
n=this._bindingsIndicesByPath;let i=n[t];const r=this._bindings;if(void
0!==(i))return r[i];const
s=this._paths,a=this._parsedPaths,o=this._objects,l=o.length,c=this.nCachedObjects_
,h=new Array(l);i=r.length,n[t]=i,s.push(t),a.push(e),r.push(h);for(let
n=c,i=o.length;n!==(i+1);++n){const i=o[n];h[n]=new jc(i,t,e)}return h}unsubscribe_(t)
{const e=this._bindingsIndicesByPath,n=e[t];if(void 0!==(n)){const
i=this._paths,r=this._parsedPaths,s=this._bindings,a=s.length-
1,o=s[a];e[t[a]]=n,s[n]=o,s.pop(),r[n]=r[a],r.pop(),i[n]=i[a],i.pop()}}qc.prototype
e.isAnimationObjectGroup=!0;class Xc{constructor(t,e,n=null,i=e.blendMode)
{this._mixer=t,this._clip=e,this._localRoot=n,this.blendMode=i;const
r=e.tracks,s=r.length,a=new Array(s),o={endingStart:k,endingEnd:k};for(let
t=0;t!==(s+1);++t){const
e=r[t].createInterpolant(null);a[t]=e,e.settings=o}this._interpolantSettings=o,this
._interpolants=a,this._propertyBindings=new
Array(s),this._cacheIndex=null,this._byClipCacheIndex=null,this._timeScaleInterpolan
t=null,this._weightInterpolant=null,this.loop=2201,this._loopCount=-1,this._startT
ime=null,this.time=0,this.timeScale=1,this._effectiveTimeScale=1,this.weight=1,this
._effectiveWeight=1,this.repetitions=1/0,this.paused=!1,this.enabled=!0,this.clampW
henFinished=!1,this.zeroSlopeAtStart=!0,this.zeroSlopeAtEnd=!0}play(){return
this._mixer._activateAction(this),this}stop(){return
this._mixer._deactivateAction(this),this.reset()}reset(){return
this.paused=!1,this.enabled=!0,this.time=0,this._loopCount=-1,this._startTime=null,
this.stopFading().stopWarping()}isRunning(){return
this.enabled&&!this.paused&&0!==(this.timeScale&&null===this._startTime&&this._mixer
._isActiveAction(this))isScheduled(){return
this._mixer._isActiveAction(this)}startAt(t){return
this._startTime=t,this}setLoop(t,e){return
this.loop=t,this.repetitions=e,this}setEffectiveWeight(t){return
this.weight=t,this._effectiveWeight=this.enabled?
t:0,this.stopFading()}getEffectiveWeight(){return this._effectiveWeight}fadeIn(t)
{return this._scheduleFading(t,0,1)}fadeOut(t){return
this._scheduleFading(t,1,0)}crossFadeFrom(t,e,n){if(t.fadeOut(e),this.fadeIn(e),n)
{const
n=this._clip.duration,i=t._clip.duration,r=i/n,s=n/i;t.warp(1,r,e),this.warp(s,1,e)
}return this}crossFadeTo(t,e,n){return t.crossFadeFrom(this,e,n)}stopFading(){const
t=this._weightInterpolant;return null!==(t&&
(this._weightInterpolant=null,this._mixer._takeBackControlInterpolant(t)),this)setE
ffectiveTimeScale(t){return this.timeScale=t,this._effectiveTimeScale=this.paused?
0:t,this.stopWarping()}getEffectiveTimeScale(){return
this._effectiveTimeScale}setDuration(t){return
this.timeScale=this._clip.duration/t,this.stopWarping()}syncWith(t){return
this.time=t.time,this.timeScale=t.timeScale,this.stopWarping()}halt(t){return
this.warp(this._effectiveTimeScale,0,t)}warp(t,e,n){const
i=this._mixer,r=i.time,s=this.timeScale;let a=this._timeScaleInterpolant;null===a&&
(a=i._lendControlInterpolant(),this._timeScaleInterpolant=a);const
o=a.parameterPositions,l=a.sampleValues;return
o[0]=r,o[1]=r+n,l[0]=t/s,l[1]=e/s,this}stopWarping(){const
t=this._timeScaleInterpolant;return null!==(t&&
(this._timeScaleInterpolant=null,this._mixer._takeBackControlInterpolant(t)),this)g
etMixer(){return this._mixer}getClip(){return this._clip}getRoot(){return
this._localRoot||this._mixer._root}update(t,e,n,i){if(!this.enabled)return void
this._updateWeight(t);const r=this._startTime;if(null!==(r)){const i=(t-

```

```

r)*n;if(i<0||0===n)return;this._startTime=null,e=n*i}e*=this._updateTimeScale(t);co
nst s=this._updateTime(e),a=this._updateWeight(t);if(a>0){const
t=this._interpolants,e=this._propertyBindings;switch(this.blendMode){case q:for(let
n=0,i=t.length;n!==i;++n)t[n].evaluate(s),e[n].accumulateAdditive(a);break;case
j:default:for(let
n=0,r=t.length;n!==r;++n)t[n].evaluate(s),e[n].accumulate(i,a)}}}_updateWeight(t)
{let e=0;if(this.enabled){e=this.weight;const
n=this._weightInterpolant;if(null!==n){const i=n.evaluate(t)
[0];e*=i,t>n.parameterPositions[1]&&(this.stopFading(),0===i&&
(this.enabled=!1))}}return this._effectiveWeight=e,e}_updateTimeScale(t){let
e=0;if(!this.paused){e=this.timeScale;const
n=this._timeScaleInterpolant;if(null!==n){e*=n.evaluate(t)
[0],t>n.parameterPositions[1]&&(this.stopWarping(),0===e?
this.paused=!0:this.timeScale=e)}}return
this._effectiveTimeScale=e,e}_updateTime(t){const
e=this._clip.duration,n=this.loop;let i=this.time+t,r=this._loopCount;const
s=2202===n;if(0===t)return-1===r?i:s&&1==(1&r)?e-i:i;if(2200===n){-1===r&&
(this._loopCount=0,this._setEndings(!0,!0,!1));t:{if(i>=e)i=e;else{if(!(i<0))
{this.time=i;break t}i=0}this.clampWhenFinished?
this.paused=!0:this.enabled=!1,this.time=i,this._mixer.dispatchEvent({type:"finishe
d",action:this,direction:t<0?-1:1})}}else{if(-1===r&&(t>=0?
(r=0,this._setEndings(!0,0===this.repetitions,s)):this._setEndings(0===this.repetit
ions,!0,s)),i>=e||i<0){const n=Math.floor(i/e);i-=e*n,r+=Math.abs(n);const
a=this.repetitions-r;if(a<=0)this.clampWhenFinished?
this.paused=!0:this.enabled=!1,i=t>0?
e:0,this.time=i,this._mixer.dispatchEvent({type:"finished",action:this,direction:t>
0?1:-1});else{if(1===a){const e=t<0;this._setEndings(e,!e,s)}else
this._setEndings(!1,!1,s);this._loopCount=r,this.time=i,this._mixer.dispatchEvent({
type:"loop",action:this,loopDelta:n})}}else this.time=i;if(s&&1==(1&r))return e-
i}return i}_setEndings(t,e,n){const i=this._interpolantSettings;n?
(i.endingStart=V,i.endingEnd=V):(i.endingStart=t?this.zeroSlopeAtStart?
V:k:W,i.endingEnd=e?this.zeroSlopeAtEnd?V:k:W)}_scheduleFading(t,e,n){const
i=this._mixer,r=i.time;let s=this._weightInterpolant;null===s&&
(s=i._lendControlInterpolant(),this._weightInterpolant=s);const
a=s.parameterPositions,o=s.sampleValues;return
a[0]=r,o[0]=e,a[1]=r+t,o[1]=n,this}}class Yc extends rt{constructor(t)
{super(),this._root=t,this._initMemoryManager(),this._accuIndex=0,this.time=0,this.
timeScale=1}_bindAction(t,e){const
n=t._localRoot||this._root,i=t._clip.tracks,r=i.length,s=t._propertyBindings,a=t._i
nterpolants,o=n.uuid,l=this._bindingsByRootAndName;let c=l[o];void 0===c&&(c=
{},l[o]=c);for(let t=0;t!==r;++t){const r=i[t],l=r.name;let h=c[l];if(void
0!==h)s[t]=h;else{if(h=s[t],void 0!==h){null===h._cacheIndex&&
(++h.referenceCount,this._addInactiveBinding(h,o,l));continue}const
i=e&&e._propertyBindings[t].binding.parsedPath;h=new
Nc(jc.create(n,l,i),r.ValueTypeName,r.getValueSize()),++h.referenceCount,this._addI
nactiveBinding(h,o,l),s[t]=h}a[t].resultBuffer=h.buffer}}_activateAction(t)
{if(!this._isActiveAction(t)){if(null===t._cacheIndex){const e=
(t._localRoot||this._root).uuid,n=t._clip.uuid,i=this._actionsByClip[n];this._bindA
ction(t,i&&i.knownActions[0]),this._addInactiveAction(t,n,e)}const
e=t._propertyBindings;for(let t=0,n=e.length;t!==n;++t){const
n=e[t];0===n.useCount++&&
(this._lendBinding(n),n.saveOriginalState())}this._lendAction(t)}}_deactivateAction
(t){if(this._isActiveAction(t)){const e=t._propertyBindings;for(let
t=0,n=e.length;t!==n;++t){const n=e[t];0===--n.useCount&&
(n.restoreOriginalState(),this._takeBackBinding(n))}this._takeBackAction(t)}}_initM
emoryManager(){this._actions=[],this._nActiveActions=0,this._actionsByClip=
{},this._bindings=[],this._nActiveBindings=0,this._bindingsByRootAndName=
{},this._controlInterpolants=[],this._nActiveControlInterpolants=0;const

```

```

t=this;this.stats={actions:{get total(){return t._actions.length},get inUse()
{return t._activeActions}},bindings:{get total(){return t._bindings.length},get
inUse(){return t._activeBindings}},controlInterpolants:{get total(){return
t._controlInterpolants.length},get inUse(){return
t._activeControlInterpolants}}}_isActiveAction(t){const e=t._cacheIndex;return
null!=e&&e<this._activeActions}_addInactiveAction(t,e,n){const
i=this._actions,r=this._actionsByClip;let s=r[e];if(void 0===s)s={knownActions:
[t],actionByRoot:{}},t._byClipCacheIndex=0,r[e]=s;else{const
e=s.knownActions;t._byClipCacheIndex=e.length,e.push(t)}t._cacheIndex=i.length,i.pu
sh(t),s.actionByRoot[n]=t}_removeInactiveAction(t){const
e=this._actions,n=e[e.length-
1],i=t._cacheIndex;n._cacheIndex=i,e[i]=n,e.pop(),t._cacheIndex=null;const
r=t._clip.uuid,s=this._actionsByClip,a=s[r],o=a.knownActions,l=o[o.length-
1],c=t._byClipCacheIndex;l._byClipCacheIndex=c,o[c]=l,o.pop(),t._byClipCacheIndex=n
1;delete a.actionByRoot[(t._localRoot||this._root).uuid],0===o.length&&delete
s[r],this._removeInactiveBindingsForAction(t)}_removeInactiveBindingsForAction(t)
{const e=t._propertyBindings;for(let t=0,n=e.length;t!=n;++t){const n=e[t];0===--
n.referenceCount&&this._removeInactiveBinding(n)}_lendAction(t){const
e=this._actions,n=t._cacheIndex,i=this._activeActions++,r=e[i];t._cacheIndex=i,e[i
]=t,r._cacheIndex=n,e[n]=r}_takeBackAction(t){const
e=this._actions,n=t._cacheIndex,i=--
this._activeActions,r=e[i];t._cacheIndex=i,e[i]=t,r._cacheIndex=n,e[n]=r}_addInact
iveBinding(t,e,n){const i=this._bindingsByRootAndName,r=this._bindings;let
s=i[e];void 0===s&&(s=
{}),i[e]=s,s[n]=t,t._cacheIndex=r.length,r.push(t)}_removeInactiveBinding(t){const
e=this._bindings,n=t.binding,i=n.rootNode.uuid,r=n.path,s=this._bindingsByRootAndNa
me,a=s[i],o=e[e.length-1],l=t._cacheIndex;o._cacheIndex=l,e[l]=o,e.pop(),delete
a[r],0===Object.keys(a).length&&delete s[i]}_lendBinding(t){const
e=this._bindings,n=t._cacheIndex,i=this._activeBindings++,r=e[i];t._cacheIndex=i,e
[i]=t,r._cacheIndex=n,e[n]=r}_takeBackBinding(t){const
e=this._bindings,n=t._cacheIndex,i=--
this._activeBindings,r=e[i];t._cacheIndex=i,e[i]=t,r._cacheIndex=n,e[n]=r}_lendCon
trolInterpolant(){const
t=this._controlInterpolants,e=this._activeControlInterpolants++;let n=t[e];return
void 0===n&&(n=new qo(new Float32Array(2),new
Float32Array(2),1,this._controlInterpolantsResultBuffer),n.__cacheIndex=e,t[e]=n),n
}_takeBackControlInterpolant(t){const
e=this._controlInterpolants,n=t.__cacheIndex,i=--
this._activeControlInterpolants,r=e[i];t.__cacheIndex=i,e[i]=t,r.__cacheIndex=n,e[
n]=r}_clipAction(t,e,n){const i=e||this._root,r=i.uuid;let s="string"==typeof t?
nl.findByName(i,t):t;const a=null!=s?s.uuid:t,o=this._actionsByClip[a];let
l=null;if(void 0===n&&(n=null!=s?s.blendMode:j),void 0!=o){const
t=o.actionByRoot[r];if(void 0!=t&&t.blendMode===n)return
t;l=o.knownActions[0],null===s&&(s=l._clip)}if(null===s)return null;const c=new
Xc(this,s,e,n);return
this._bindAction(c,l),this._addInactiveAction(c,a,r),c}_existingAction(t,e){const
n=e||this._root,i=n.uuid,r="string"==typeof t?nl.findByName(n,t):t,s=r?
r.uuid:t,a=this._actionsByClip[s];return void
0!=a&&a.actionByRoot[i]||null}_stopAllAction(){const t=this._actions;for(let
e=this._activeActions-1;e>=0;--e)t[e].stop();return this}_update(t)
{t*=this.timeScale;const
e=this._actions,n=this._activeActions,i=this.time+=t,r=Math.sign(t),s=this._accuIn
dex^=1;for(let a=0;a!=n;++a){e[a]._update(i,t,r,s)}const
a=this._bindings,o=this._activeBindings;for(let t=0;t!=o;++t)a[t].apply(s);return
this}_setTime(t){this.time=0;for(let
t=0;t<this._actions.length;t++)this._actions[t].time=0;return
this}_update(t){getRoot(){return this._root}_uncacheClip(t){const
e=this._actions,n=t.uuid,i=this._actionsByClip,r=i[n];if(void 0!=r){const

```



```

t=r.knownActions;for(let n=0,i=t.length;n!==i;++n){const
i=t[n];this._deactivateAction(i);const r=i._cacheIndex,s=e[e.length-
1];i._cacheIndex=null,i._byClipCacheIndex=null,s._cacheIndex=r,e[r]=s,e.pop(),this.
_removeInactiveBindingsForAction(i)}delete i[n]}uncacheRoot(t){const
e=t.uuid,n=this._actionsByClip;for(const t in n){const i=n[t].actionByRoot[e];void
0!==i&&(this._deactivateAction(i),this._removeInactiveAction(i))}const
i=this._bindingsByRootAndName[e];if(void 0!==i)for(const t in i){const
e=i[t];e.restoreOriginalState(),this._removeInactiveBinding(e)}uncacheAction(t,e)
{const n=this.existingAction(t,e);null!==n&&
(this._deactivateAction(n),this._removeInactiveAction(n))}Yc.prototype._controlInt
erpolantsResultBuffer=new Float32Array(1);class Zc{constructor(t){"string"==typeof
t&&(console.warn("THREE.Uniform: Type parameter is no longer
needed."),t=arguments[1]),this.value=t}clone(){return new Zc(void
0===this.value.clone?this.value:this.value.clone())}class Jc extends
Es{constructor(t,e,n=1){super(t,e),this.meshPerAttribute=n||1}copy(t){return
super.copy(t),this.meshPerAttribute=t.meshPerAttribute,this.clone(t){const
e=super.clone(t);return e.meshPerAttribute=this.meshPerAttribute,e}toJSON(t){const
e=super.toJSON(t);return
e.isInstancedInterleavedBuffer=!0,e.meshPerAttribute=this.meshPerAttribute,e}}Jc.pr
ototype.isInstancedInterleavedBuffer=!0;class Qc{constructor(t,e,n,i,r)
{this.buffer=t,this.type=e,this.itemSize=n,this.elementSize=i,this.count=r,this.ver
sion=0}set needsUpdate(t){!0===t&&this.version++}setBuffer(t){return
this.buffer=t,this}setType(t,e){return
this.type=t,this.elementSize=e,this}setItemSize(t){return
this.itemSize=t,this}setCount(t){return
this.count=t,this}}Qc.prototype.isGLBufferAttribute=!0;function Kc(t,e){return
t.distance-e.distance}function $c(t,e,n,i)
{if(t.layers.test(e.layers)&&t.raycast(e,n,!0===i){const i=t.children;for(let
t=0,r=i.length;t<r;t++)$c(i[t],e,n,!0)}const th=new vt;class eh{constructor(t=new
vt(1/0,1/0),e=new vt(-1/0,-1/0)){this.min=t,this.max=e}set(t,e){return
this.min.copy(t),this.max.copy(e),this}setFromPoints(t){this.makeEmpty();for(let
e=0,n=t.length;e<n;e++)this.expandByPoint(t[e]);return
this}setFromCenterAndSize(t,e){const n=th.copy(e).multiplyScalar(.5);return
this.min.copy(t).sub(n),this.max.copy(t).add(n),this}clone(){return(new
this.constructor).copy(this)}copy(t){return
this.min.copy(t.min),this.max.copy(t.max),this}makeEmpty(){return
this.min.x=this.min.y=1/0,this.max.x=this.max.y=-1/0,this}isEmpty(){return
this.max.x<this.min.x||this.max.y<this.min.y}getCenter(t){return void 0===t&&
(console.warn("THREE.Box2: .getCenter() target is now required"),t=new
vt),this.isEmpty()?
t.set(0,0):t.addVectors(this.min,this.max).multiplyScalar(.5)}getSize(t){return
void 0===t&&(console.warn("THREE.Box2: .getSize() target is now required"),t=new
vt),this.isEmpty()?t.set(0,0):t.subVectors(this.max,this.min)}expandByPoint(t)
{return this.min.min(t),this.max.max(t),this}expandByVector(t){return
this.min.sub(t),this.max.add(t),this}expandByScalar(t){return this.min.addScalar(-
t),this.max.addScalar(t),this}containsPoint(t){return!
(t.x<this.min.x||t.x>this.max.x||t.y<this.min.y||t.y>this.max.y)}containsBox(t)
{return
this.min.x<=t.min.x&&t.max.x<=this.max.x&&t.min.y<=t.min.y&&t.max.y<=this.max.y}
getParameter(t,e){return void 0===e&&(console.warn("THREE.Box2: .getParameter()
target is now required"),e=new vt),e.set((t.x-this.min.x)/(this.max.x-this.min.x),
(t.y-this.min.y)/(this.max.y-this.min.y))}intersectsBox(t){return!
(t.max.x<this.min.x||t.min.x>this.max.x||t.max.y<this.min.y||t.min.y>this.max.y)}cl
ampPoint(t,e){return void 0===e&&(console.warn("THREE.Box2: .clampPoint() target is
now required"),e=new vt),e.copy(t).clamp(this.min,this.max)}distanceToPoint(t)
{return th.copy(t).clamp(this.min,this.max).sub(t).length()}intersect(t){return
this.min.max(t.min),this.max.min(t.max),this}union(t){return
this.min.min(t.min),this.max.max(t.max),this}translate(t){return

```

```

this.min.add(t),this.max.add(t),this}equals(t){return
t.min.equals(this.min)&&t.max.equals(this.max)}}eh.prototype.isBox2=!0;const nh=new
Lt,ih=new Lt;class rh{constructor(t=new Lt,e=new Lt)
{this.start=t,this.end=e}set(t,e){return
this.start.copy(t),this.end.copy(e),this}copy(t){return
this.start.copy(t.start),this.end.copy(t.end),this}getCenter(t){return void 0===t&&
(console.warn("THREE.Line3: .getCenter() target is now required"),t=new
Lt),t.addVectors(this.start,this.end).multiplyScalar(.5)}delta(t){return void
0===t&&(console.warn("THREE.Line3: .delta() target is now required"),t=new
Lt),t.subVectors(this.end,this.start)}distanceSq(){return
this.start.distanceToSquared(this.end)}distance(){return
this.start.distanceTo(this.end)}at(t,e){return void 0===e&&
(console.warn("THREE.Line3: .at() target is now required"),e=new
Lt),this.delta(e).multiplyScalar(t).add(this.start)}closestPointToPointParameter(t,
e){nh.subVectors(t,this.start),ih.subVectors(this.end,this.start);const
n=ih.dot(ih);let i=ih.dot(nh)/n;return e&&
(i=ht(i,0,1)),i}closestPointToPoint(t,e,n){const
i=this.closestPointToPointParameter(t,e);return void 0===n&&
(console.warn("THREE.Line3: .closestPointToPoint() target is now required"),n=new
Lt),this.delta(n).multiplyScalar(i).add(this.start)}applyMatrix4(t){return
this.start.applyMatrix4(t),this.end.applyMatrix4(t),this}equals(t){return
t.start.equals(this.start)&&t.end.equals(this.end)}clone(){return(new
this.constructor).copy(this)}}class sh extends Ce{constructor(t)
{super(),this.material=t,this.render=function()
{}},this.hasPositions=!1,this.hasNormals=!1,this.hasColors=!1,this.hasUvs=!1,this.po
sitionArray=null,this.normalArray=null,this.colorArray=null,this.uvArray=null,this.
count=0}}sh.prototype.isImmediateRenderObject=!0;const ah=new Lt;const oh=new
Lt,lh=new se,ch=new se;class hh extends ya{constructor(t){const e=uh(t),n=new En,i=
[],r=[],s=new tn(0,0,1),a=new tn(0,1,0);for(let t=0;t<e.length;t++){const
n=e[t];n.parent&&n.parent.isBone&&
(i.push(0,0,0),i.push(0,0,0),r.push(s.r,s.g,s.b),r.push(a.r,a.g,a.b))}n.setAttribut
e("position",new mn(i,3)),n.setAttribute("color",new mn(r,3));super(n,new
ca({vertexColors:!0,depthTest:!1,depthWrite:!1,toneMapped:!1,transparent:!0})),this
.type="SkeletonHelper",this.isSkeletonHelper=!0,this.root=t,this.bones=e,this.matri
x=t.matrixWorld,this.matrixAutoUpdate=!1}updateMatrixWorld(t){const
e=this.bones,n=this.geometry,i=n.getAttribute("position");ch.copy(this.root.matrixW
orld).invert();for(let t=0,n=0;t<e.length;t++){const
r=e[t];r.parent&&r.parent.isBone&&
(lh.multiplyMatrices(ch,r.matrixWorld),oh.setFromMatrixPosition(lh),i.setXYZ(n,oh.x
,oh.y,oh.z),lh.multiplyMatrices(ch,r.parent.matrixWorld),oh.setFromMatrixPosition(l
h),i.setXYZ(n+1,oh.x,oh.y,oh.z),n+=2)}n.getAttribute("position").needsUpdate=!0,sup
er.updateMatrixWorld(t)}}function uh(t){const e=[];t&&t.isBone&&e.push(t);for(let
n=0;n<t.children.length;n++)e.push.apply(e,uh(t.children[n]));return e}const dh=new
Lt,ph=new tn,mh=new tn;class fh extends
ya{constructor(t=10,e=10,n=4473924,i=8947848){n=new tn(n),i=new tn(i);const
r=e/2,s=t/e,a=t/2,o=[],l=[];for(let t=0,c=0,h=-a;t<=e;t++,h+=s){o.push(-
a,0,h,a,0,h),o.push(h,0,-a,h,0,a);const e=t===r?
n:i;e.toArray(1,c),c+=3,e.toArray(1,c),c+=3,e.toArray(1,c),c+=3,e.toArray(1,c),c+=3
}const c=new En;c.setAttribute("position",new mn(o,3)),c.setAttribute("color",new
mn(1,3));super(c,new
ca({vertexColors:!0,toneMapped:!1})),this.type="GridHelper"}}const gh=new Lt,vh=new
Lt,yh=new Lt;const xh=new Lt,_h=new Qn;function wh(t,e,n,i,r,s,a)
{XH.set(r,s,a).unproject(i);const o=e[t];if(void 0!==o){const
t=n.getAttribute("position");for(let
e=0,n=o.length;e<n;e++)t.setXYZ(o[e],xh.x,xh.y,xh.z)}}const bh=new Pt;class Mh
extends ya{constructor(t,e=16776960){const n=new
Uint16Array([0,1,1,2,2,3,3,0,4,5,5,6,6,7,7,4,0,4,1,5,2,6,3,7]),i=new
Float32Array(24),r=new En;r.setIndex(new sn(n,1)),r.setAttribute("position",new

```

```

sn(i,3)),super(r,new
ca({color:e,toneMapped:!1})),this.object=t,this.type="BoxHelper",this.matrixAutoUpd
ate=!1,this.update())update(t){if(void 0!==t&&console.warn("THREE.BoxHelper:
.update() has no longer arguments."),void
0!==this.object&&bh.setFromObject(this.object),bh.isEmpty())return;const
e=bh.min,n=bh.max,i=this.geometry.attributes.position,r=i.array;r[0]=n.x,r[1]=n.y,r
[2]=n.z,r[3]=e.x,r[4]=n.y,r[5]=n.z,r[6]=e.x,r[7]=e.y,r[8]=n.z,r[9]=n.x,r[10]=e.y,r[
11]=n.z,r[12]=n.x,r[13]=n.y,r[14]=e.z,r[15]=e.x,r[16]=n.y,r[17]=e.z,r[18]=e.x,r[19]
=e.y,r[20]=e.z,r[21]=n.x,r[22]=e.y,r[23]=e.z,i.needsUpdate=!0,this.geometry.compute
BoundingSphere())setFromObject(t){return this.object=t,this.update(),this.copy(t)
{return ya.prototype.call(this,t),this.object=t,this}}const Sh=new
Lt;let Th,Eh;class Ah extends ya{constructor(t=1){const e=
[0,0,0,t,0,0,0,0,0,0,t,0,0,0,0,0,t],n=new En;n.setAttribute("position",new
mn(e,3)),n.setAttribute("color",new
mn([1,0,0,1,.6,0,0,1,0,.6,1,0,0,0,1,0,.6,1],3));super(n,new
ca({vertexColors:!0,toneMapped:!1})),this.type="AxesHelper"}dispose()
{this.geometry.dispose(),this.material.dispose()}const Lh=new
Float32Array(1),Rh=new Int32Array(Lh.buffer);const Ch=Math.pow(2,8),Ph=
[.125,.215,.35,.446,.526,.582],Dh=5+Ph.length,Ih=20,Nh=[{X:0,[Y]:1,[J]:2,[Q]:3,
[K]:4,[S]:5,[Z]:6},Bh=new en({side:1,depthWrite:!1,depthTest:!1}),zh=new Wn(new
qn,Bh),Fh=new J1,{_lodPlanes:Oh,_sizeLods:Hh,_sigmas:Gh}=Yh(),Uh=new tn;let
kh=null;const Vh=(1+Math.sqrt(5))/2,Wh=1/Vh,jh=[new Lt(1,1,1),new Lt(-1,1,1),new
Lt(1,1,-1),new Lt(-1,1,-1),new Lt(0,Vh,Wh),new Lt(0,Vh,-Wh),new Lt(Wh,0,Vh),new
Lt(-Wh,0,Vh),new Lt(Vh,Wh,0),new Lt(-Vh,Wh,0)];function qh(t){const
e=Math.max(t.r,t.g,t.b),n=Math.min(Math.max(Math.ceil(Math.log2(e)),-128),127);t.mu
ltiplyScalar(Math.pow(2,-n));return(n+128)/255}function Xh(t){return void
0!==t&t.type===x&(t.encoding===X|t.encoding===Y|t.encoding===Z)}function Yh()
{const t=[],e=[],n=[];let i=8;for(let r=0;r<Dh;r++){const
s=Math.pow(2,i);e.push(s);let a=1/s;r>4?a=Ph[r-8+4-1]:0==r&(a=0),n.push(a);const
o=1/(s-1),l=-o/2,c=1+o/2,h=[1,l,c,l,c,c,l,l,c,c,l,c],u=6,d=6,p=3,m=2,f=1,g=new
Float32Array(p*d*u),v=new Float32Array(m*d*u),y=new Float32Array(f*d*u);for(let
t=0;t<u;t++){const e=t%3*2/3-1,n=t>2?0:-1,i=
[e,n,0,e+2/3,n,0,e+2/3,n+1,0,e,n,0,e+2/3,n+1,0,e,n+1,0];g.set(i,p*d*t),v.set(h,m*d*
t);const r=[t,t,t,t,t];y.set(r,f*d*t)}const x=new
En;x.setAttribute("position",new sn(g,p)),x.setAttribute("uv",new
sn(v,m)),x.setAttribute("faceIndex",new sn(y,f)),t.push(x),i>4&i--
}return{_lodPlanes:t,_sizeLods:e,_sigmas:n}}function Zh(t){const e=new
Tt(3*Ch,3*Ch,t);return
e.texture.mapping=1,e.texture.name="PMREM.cubeUv",e.scissorTest=!0,e}function
Jh(t,e,n,i,r){t.viewports.set(e,n,i,r),t.scissors.set(e,n,i,r)}function Qh(){const
t=new vt(1,1);return new Io({name:"EquirectangularToCubeUV",uniforms:{envMap:
{value:null},texelSize:{value:t},inputEncoding:{value:Nh[3e3]},outputEncoding:
{value:Nh[3e3]}},vertexShader:$h(),fragmentShader:`\n\n\t\t\tprecision mediump
float;\n\n\t\t\t\tprecision mediump int;\n\n\t\t\t\t\tvarying vec3
vOutputDirection;\n\n\t\t\t\t\tuniform sampler2D envMap;\n\n\t\t\t\t\tuniform vec2
texelSize;\n\n\t\t\t\t\t\t$tu()\n\n\t\t\t\t\t#include <common>\n\n\t\t\t\t\t\tvoid main()
{\n\n\t\t\t\t\t\t\tgl_FragColor = vec4( 0.0, 0.0, 0.0, 1.0 );\n\n\t\t\t\t\t\t\tvec3
outputDirection = normalize( vOutputDirection );\n\n\t\t\t\t\t\t\tvec2 uv = equirectUv(
outputDirection );\n\n\t\t\t\t\t\t\tvec2 f = fract( uv / texelSize - 0.5 );\n\n\t\t\t\t\t\t\ttuv
= f * texelSize;\n\n\t\t\t\t\t\t\tvec3 tl = envMapTexelToLinear( texture2D ( envMap, uv )
).rgb;\n\n\t\t\t\t\t\t\ttuv.x += texelSize.x;\n\n\t\t\t\t\t\t\tvec3 tr = envMapTexelToLinear(
texture2D ( envMap, uv ) ).rgb;\n\n\t\t\t\t\t\t\ttuv.y += texelSize.y;\n\n\t\t\t\t\t\t\tvec3 br =
envMapTexelToLinear( texture2D ( envMap, uv ) ).rgb;\n\n\t\t\t\t\t\t\ttuv.x -=
texelSize.x;\n\n\t\t\t\t\t\t\tvec3 bl = envMapTexelToLinear( texture2D ( envMap, uv )
).rgb;\n\n\t\t\t\t\t\t\ttm = mix( tl, tr, f.x );\n\n\t\t\t\t\t\t\tbm = mix( bl, br, f.x
);\n\n\t\t\t\t\t\t\tgl_FragColor.rgb = mix( tm, bm, f.y );\n\n\t\t\t\t\t\t\tgl_FragColor =
linearToOutputTexel( gl_FragColor
);\n\n\t\t\t\t\t\t\t\tblending:0,depthTest:!1,depthWrite:!1)}}function Kh(){return

```

[illegible]

```

{add:function(){console.error("THREE.Loader: Handlers.add() has been removed. Use
LoadingManager.addHandler() instead.")},get:function(){console.error("THREE.Loader:
Handlers.get() has been removed. Use LoadingManager.getHandler()
instead.")}},eh.prototype.center=function(t){return console.warn("THREE.Box2:
.center() has been renamed to
.getCenter()."),this.getCenter(t)},eh.prototype.empty=function(){return
console.warn("THREE.Box2: .empty() has been renamed to
.isEmpty()."),this.isEmpty()},eh.prototype.isIntersectionBox=function(t){return
console.warn("THREE.Box2: .isIntersectionBox() has been renamed to
.intersectsBox()."),this.intersectsBox(t)},eh.prototype.size=function(t){return
console.warn("THREE.Box2: .size() has been renamed to
.getSize()."),this.getSize(t)},Pt.prototype.center=function(t){return
console.warn("THREE.Box3: .center() has been renamed to
.getCenter()."),this.getCenter(t)},Pt.prototype.empty=function(){return
console.warn("THREE.Box3: .empty() has been renamed to
.isEmpty()."),this.isEmpty()},Pt.prototype.isIntersectionBox=function(t){return
console.warn("THREE.Box3: .isIntersectionBox() has been renamed to
.intersectsBox()."),this.intersectsBox(t)},Pt.prototype.isIntersectionSphere=functi
on(t){return console.warn("THREE.Box3: .isIntersectionSphere() has been renamed to
.intersectsSphere()."),this.intersectsSphere(t)},Pt.prototype.size=function(t)
{return console.warn("THREE.Box3: .size() has been renamed to
.getSize()."),this.getSize(t)},Jt.prototype.empty=function(){return
console.warn("THREE.Sphere: .empty() has been renamed to
.isEmpty()."),this.isEmpty()},ai.prototype.setFromMatrix=function(t){return
console.warn("THREE.Frustum: .setFromMatrix() has been renamed to
.setFromProjectionMatrix()."),this.setFromProjectionMatrix(t)},rh.prototype.center=
function(t){return console.warn("THREE.Line3: .center() has been renamed to
.getCenter()."),this.getCenter(t)},yt.prototype.flattenToArrayOffset=function(t,e)
{return console.warn("THREE.Matrix3: .flattenToArrayOffset() has been deprecated.
Use .toArray()
instead."),this.toArray(t,e)},yt.prototype.multiplyVector3=function(t){return
console.warn("THREE.Matrix3: .multiplyVector3() has been removed. Use
vector.applyMatrix3( matrix )
instead."),t.applyMatrix3(this)},yt.prototype.multiplyVector3Array=function()
{console.error("THREE.Matrix3: .multiplyVector3Array() has been
removed.")},yt.prototype.applyToBufferAttribute=function(t){return
console.warn("THREE.Matrix3: .applyToBufferAttribute() has been removed. Use
attribute.applyMatrix3( matrix )
instead."),t.applyMatrix3(this)},yt.prototype.applyToVector3Array=function()
{console.error("THREE.Matrix3: .applyToVector3Array() has been
removed.")},yt.prototype.getInverse=function(t){return console.warn("THREE.Matrix3:
.getInverse() has been removed. Use matrixInv.copy( matrix ).invert();
instead."),this.copy(t).invert()},se.prototype.extractPosition=function(t){return
console.warn("THREE.Matrix4: .extractPosition() has been renamed to
.copyPosition()."),this.copyPosition(t)},se.prototype.flattenToArrayOffset=function
(t,e){return console.warn("THREE.Matrix4: .flattenToArrayOffset() has been
deprecated. Use .toArray()
instead."),this.toArray(t,e)},se.prototype.getPosition=function(){return
console.warn("THREE.Matrix4: .getPosition() has been removed. Use
Vector3.setFromMatrixPosition( matrix ) instead."),(new
Lt).setFromMatrixColumn(this,3)},se.prototype.setRotationFromQuaternion=function(t)
{return console.warn("THREE.Matrix4: .setRotationFromQuaternion() has been renamed
to
.makeRotationFromQuaternion()."),this.makeRotationFromQuaternion(t)},se.prototype.m
ultiplyToArray=function(){console.warn("THREE.Matrix4: .multiplyToArray() has been
removed.")},se.prototype.multiplyVector3=function(t){return
console.warn("THREE.Matrix4: .multiplyVector3() has been removed. Use
vector.applyMatrix4( matrix )

```

```

instead."),t.applyMatrix4(this)},se.prototype.multiplyVector4=function(t){return
console.warn("THREE.Matrix4: .multiplyVector4() has been removed. Use
vector.applyMatrix4( matrix )
instead."),t.applyMatrix4(this)},se.prototype.multiplyVector3Array=function()
{console.error("THREE.Matrix4: .multiplyVector3Array() has been
removed.")},se.prototype.rotateAxis=function(t){console.warn("THREE.Matrix4:
.rotateAxis() has been removed. Use Vector3.transformDirection( matrix )
instead."),t.transformDirection(this)},se.prototype.crossVector=function(t){return
console.warn("THREE.Matrix4: .crossVector() has been removed. Use
vector.applyMatrix4( matrix )
instead."),t.applyMatrix4(this)},se.prototype.translate=function()
{console.error("THREE.Matrix4: .translate() has been
removed.")},se.prototype.rotateX=function(){console.error("THREE.Matrix4:
.rotateX() has been removed.")},se.prototype.rotateY=function()
{console.error("THREE.Matrix4: .rotateY() has been
removed.")},se.prototype.rotateZ=function(){console.error("THREE.Matrix4:
.rotateZ() has been removed.")},se.prototype.rotateByAxis=function()
{console.error("THREE.Matrix4: .rotateByAxis() has been
removed.")},se.prototype.applyToBufferAttribute=function(t){return
console.warn("THREE.Matrix4: .applyToBufferAttribute() has been removed. Use
attribute.applyMatrix4( matrix )
instead."),t.applyMatrix4(this)},se.prototype.applyToVector3Array=function()
{console.error("THREE.Matrix4: .applyToVector3Array() has been
removed.")},se.prototype.makeFrustum=function(t,e,n,i,r,s){return
console.warn("THREE.Matrix4: .makeFrustum() has been removed. Use .makePerspective(
left, right, top, bottom, near, far )
instead."),this.makePerspective(t,e,i,n,r,s)},se.prototype.getInverse=function(t)
{return console.warn("THREE.Matrix4: .getInverse() has been removed. Use
matrixInv.copy( matrix ).invert();
instead."),this.copy(t).invert()},Ne.prototype.isIntersectionLine=function(t)
{return console.warn("THREE.Plane: .isIntersectionLine() has been renamed to
.intersectsLine()."),this.intersectsLine(t)},At.prototype.multiplyVector3=function(
t){return console.warn("THREE.Quaternion: .multiplyVector3() has been removed. Use
is now vector.applyQuaternion( quaternion )
instead."),t.applyQuaternion(this)},At.prototype.inverse=function(){return
console.warn("THREE.Quaternion: .inverse() has been renamed to
invert()."),this.invert()},re.prototype.isIntersectionBox=function(t){return
console.warn("THREE.Ray: .isIntersectionBox() has been renamed to
.intersectsBox()."),this.intersectsBox(t)},re.prototype.isIntersectionPlane=functio
n(t){return console.warn("THREE.Ray: .isIntersectionPlane() has been renamed to
.intersectsPlane()."),this.intersectsPlane(t)},re.prototype.isIntersectionSphere=fu
nction(t){return console.warn("THREE.Ray: .isIntersectionSphere() has been renamed
to .intersectsSphere()."),this.intersectsSphere(t)},je.prototype.area=function()
{return console.warn("THREE.Triangle: .area() has been renamed to
.getArea()."),this.getArea()},je.prototype.barycoordFromPoint=function(t,e){return
console.warn("THREE.Triangle: .barycoordFromPoint() has been renamed to
.getBarycoord()."),this.getBarycoord(t,e)},je.prototype.midpoint=function(t){return
console.warn("THREE.Triangle: .midpoint() has been renamed to
.getMidpoint()."),this.getMidpoint(t)},je.prototype.normal=function(t){return
console.warn("THREE.Triangle: .normal() has been renamed to
.getNormal()."),this.getNormal(t)},je.prototype.plane=function(t){return
console.warn("THREE.Triangle: .plane() has been renamed to
.getPlane()."),this.getPlane(t)},je.barycoordFromPoint=function(t,e,n,i,r){return
console.warn("THREE.Triangle: .barycoordFromPoint() has been renamed to
.getBarycoord()."),je.getBarycoord(t,e,n,i,r)},je.normal=function(t,e,n,i){return
console.warn("THREE.Triangle: .normal() has been renamed to
.getNormal()."),je.getNormal(t,e,n,i)},zl.prototype.extractAllPoints=function(t)
{return console.warn("THREE.Shape: .extractAllPoints() has been removed. Use

```

```

.extractPoints() instead."),this.extractPoints(t)},zl.prototype.extrude=function(t)
{return console.warn("THREE.Shape: .extrude() has been removed. Use
ExtrudeGeometry() instead."),new go(this,t)},zl.prototype.makeGeometry=function(t)
{return console.warn("THREE.Shape: .makeGeometry() has been removed. Use
ShapeGeometry() instead."),new
Mo(this,t)},vt.prototype.fromAttribute=function(t,e,n){return
console.warn("THREE.Vector2: .fromAttribute() has been renamed to
.fromBufferAttribute()."),this.fromBufferAttribute(t,e,n)},vt.prototype.distanceToM
anhattan=function(t){return console.warn("THREE.Vector2: .distanceToManhattan() has
been renamed to
.manhattanDistanceTo()."),this.manhattanDistanceTo(t)},vt.prototype.lengthManhattan
=function(){return console.warn("THREE.Vector2: .lengthManhattan() has been renamed
to
.manhattanLength()."),this.manhattanLength()},Lt.prototype.setEulerFromRotationMatr
ix=function(){console.error("THREE.Vector3: .setEulerFromRotationMatrix() has been
removed. Use Euler.setFromRotationMatrix()
instead.")},Lt.prototype.setEulerFromQuaternion=function()
{console.error("THREE.Vector3: .setEulerFromQuaternion() has been removed. Use
Euler.setFromQuaternion()
instead.")},Lt.prototype.getPositionFromMatrix=function(t){return
console.warn("THREE.Vector3: .getPositionFromMatrix() has been renamed to
.setFromMatrixPosition()."),this.setFromMatrixPosition(t)},Lt.prototype.getScaleFro
mMatrix=function(t){return console.warn("THREE.Vector3: .getScaleFromMatrix() has
been renamed to
.setFromMatrixScale()."),this.setFromMatrixScale(t)},Lt.prototype.getColumnFromMatr
ix=function(t,e){return console.warn("THREE.Vector3: .getColumnFromMatrix() has
been renamed to
.setFromMatrixColumn()."),this.setFromMatrixColumn(e,t)},Lt.prototype.applyProjecti
on=function(t){return console.warn("THREE.Vector3: .applyProjection() has been
removed. Use .applyMatrix4( m )
instead."),this.applyMatrix4(t)},Lt.prototype.fromAttribute=function(t,e,n){return
console.warn("THREE.Vector3: .fromAttribute() has been renamed to
.fromBufferAttribute()."),this.fromBufferAttribute(t,e,n)},Lt.prototype.distanceToM
anhattan=function(t){return console.warn("THREE.Vector3: .distanceToManhattan() has
been renamed to
.manhattanDistanceTo()."),this.manhattanDistanceTo(t)},Lt.prototype.lengthManhattan
=function(){return console.warn("THREE.Vector3: .lengthManhattan() has been renamed
to
.manhattanLength()."),this.manhattanLength()},St.prototype.fromAttribute=function(t
,e,n){return console.warn("THREE.Vector4: .fromAttribute() has been renamed to
.fromBufferAttribute()."),this.fromBufferAttribute(t,e,n)},St.prototype.lengthManha
ttan=function(){return console.warn("THREE.Vector4: .lengthManhattan() has been
renamed to
.manhattanLength()."),this.manhattanLength()},Ce.prototype.getChildByName=function(
t){return console.warn("THREE.Object3D: .getChildByName() has been renamed to
.getObjectByName()."),this.getObjectByName(t)},Ce.prototype.renderDepth=function()
{console.warn("THREE.Object3D: .renderDepth has been removed. Use .renderOrder,
instead.")},Ce.prototype.translate=function(t,e){return
console.warn("THREE.Object3D: .translate() has been removed. Use .translateOnAxis(
axis, distance )
instead."),this.translateOnAxis(e,t)},Ce.prototype.getWorldRotation=function()
{console.error("THREE.Object3D: .getWorldRotation() has been removed. Use
THREE.Object3D.getWorldQuaternion( target )
instead.")},Ce.prototype.applyMatrix=function(t){return
console.warn("THREE.Object3D: .applyMatrix() has been renamed to
.applyMatrix4()."),this.applyMatrix4(t)},Object.defineProperties(Ce.prototype,
{eulerOrder:{get:function(){return console.warn("THREE.Object3D: .eulerOrder is now
.rotation.order."),this.rotation.order},set:function(t)

```

```

{console.warn("THREE.Object3D: .eulerOrder is now
.rotation.order."),this.rotation.order=t}},useQuaternion:{get:function()
{console.warn("THREE.Object3D: .useQuaternion has been removed. The library now
uses quaternions by default.")},set:function(){console.warn("THREE.Object3D:
.useQuaternion has been removed. The library now uses quaternions by
default.")}}}},Wn.prototype.setDrawMode=function(){console.error("THREE.Mesh:
.setDrawMode() has been removed. The renderer now always assumes
THREE.TrianglesDrawMode. Transform your geometry via
BufferGeometryUtils.toTrianglesDrawMode() if
necessary.")},Object.defineProperties(Wn.prototype,{drawMode:{get:function(){return
console.error("THREE.Mesh: .drawMode has been removed. The renderer now always
assumes THREE.TrianglesDrawMode."),0},set:function(){console.error("THREE.Mesh:
.drawMode has been removed. The renderer now always assumes
THREE.TrianglesDrawMode. Transform your geometry via
BufferGeometryUtils.toTrianglesDrawMode() if
necessary.")}}}},$s.prototype.initBones=function()
{console.error("THREE.SkinnedMesh: initBones() has been
removed.")},Kn.prototype.setLens=function(t,e)
{console.warn("THREE.PerspectiveCamera.setLens is deprecated. Use .setFocalLength
and .filmGauge for a photographic setup."),void 0!==e&&
(this.filmGauge=e),this.setFocalLength(t)},Object.defineProperties(Fl.prototype,
{onlyShadow:{set:function(){console.warn("THREE.Light: .onlyShadow has been
removed.")}},shadowCameraFov:{set:function(t){console.warn("THREE.Light:
.shadowCameraFov is now
.shadow.camera.fov."),this.shadow.camera.fov=t}},shadowCameraLeft:{set:function(t)
{console.warn("THREE.Light: .shadowCameraLeft is now
.shadow.camera.left."),this.shadow.camera.left=t}},shadowCameraRight:
{set:function(t){console.warn("THREE.Light: .shadowCameraRight is now
.shadow.camera.right."),this.shadow.camera.right=t}},shadowCameraTop:
{set:function(t){console.warn("THREE.Light: .shadowCameraTop is now
.shadow.camera.top."),this.shadow.camera.top=t}},shadowCameraBottom:
{set:function(t){console.warn("THREE.Light: .shadowCameraBottom is now
.shadow.camera.bottom."),this.shadow.camera.bottom=t}},shadowCameraNear:
{set:function(t){console.warn("THREE.Light: .shadowCameraNear is now
.shadow.camera.near."),this.shadow.camera.near=t}},shadowCameraFar:{set:function(t)
{console.warn("THREE.Light: .shadowCameraFar is now
.shadow.camera.far."),this.shadow.camera.far=t}},shadowCameraVisible:
{set:function(){console.warn("THREE.Light: .shadowCameraVisible has been removed.
Use new THREE.CameraHelper( light.shadow.camera ) instead.")}},shadowBias:
{set:function(t){console.warn("THREE.Light: .shadowBias is now
.shadow.bias."),this.shadow.bias=t}},shadowDarkness:{set:function()
{console.warn("THREE.Light: .shadowDarkness has been removed.")}},shadowMapWidth:
{set:function(t){console.warn("THREE.Light: .shadowMapWidth is now
.shadow.mapSize.width."),this.shadow.mapSize.width=t}},shadowMapHeight:
{set:function(t){console.warn("THREE.Light: .shadowMapHeight is now
.shadow.mapSize.height."),this.shadow.mapSize.height=t}},Object.defineProperties(
sn.prototype,{length:{get:function(){return console.warn("THREE.BufferAttribute:
.length has been deprecated. Use .count instead."),this.array.length}},dynamic:
{get:function(){return console.warn("THREE.BufferAttribute: .dynamic has been
deprecated. Use .usage instead."),this.usage===nt},set:function()
{console.warn("THREE.BufferAttribute: .dynamic has been deprecated. Use .usage
instead."),this.setUsage(nt)}}}},sn.prototype.setDynamic=function(t){return
console.warn("THREE.BufferAttribute: .setDynamic() has been deprecated. Use
.setUsage() instead."),this.setUsage(!0===t?
nt:et),this},sn.prototype.copyIndicesArray=function()
{console.error("THREE.BufferAttribute: .copyIndicesArray() has been
removed.")},sn.prototype.setArray=function(){console.error("THREE.BufferAttribute:
.setArray has been removed. Use BufferGeometry .setAttribute to replace/resize

```



```

attribute buffers"}},En.prototype.addIndex=function(t)
{console.warn("THREE.BufferGeometry: .addIndex() has been renamed to
.setIndex()."),this.setIndex(t)},En.prototype.addAttribute=function(t,e){return
console.warn("THREE.BufferGeometry: .addAttribute() has been renamed to
.setAttribute()."),e&&e.isBufferAttribute||e&&e.isInterleavedBufferAttribute?"index
===t?(console.warn("THREE.BufferGeometry.addAttribute: Use .setIndex() for index
attribute."),this.setIndex(e),this):this.setAttribute(t,e):
(console.warn("THREE.BufferGeometry: .addAttribute() now expects ( name, attribute
)."),this.setAttribute(t,new
sn(arguments[1],arguments[2])))},En.prototype.addDrawCall=function(t,e,n){void
0!==(n&&console.warn("THREE.BufferGeometry: .addDrawCall() no longer supports
indexOffset."),console.warn("THREE.BufferGeometry: .addDrawCall() is now
.addGroup()."),this.addGroup(t,e)},En.prototype.clearDrawCalls=function()
{console.warn("THREE.BufferGeometry: .clearDrawCalls() is now
.clearGroups()."),this.clearGroups()},En.prototype.computeOffsets=function()
{console.warn("THREE.BufferGeometry: .computeOffsets() has been
removed.")},En.prototype.removeAttribute=function(t){return
console.warn("THREE.BufferGeometry: .removeAttribute() has been renamed to
.deleteAttribute()."),this.deleteAttribute(t)},En.prototype.applyMatrix=function(t)
{return console.warn("THREE.BufferGeometry: .applyMatrix() has been renamed to
.applyMatrix4()."),this.applyMatrix4(t)},Object.defineProperties(En.prototype,
{drawcalls:{get:function(){return console.error("THREE.BufferGeometry: .drawcalls
has been renamed to .groups."),this.groups}},offsets:{get:function(){return
console.warn("THREE.BufferGeometry: .offsets has been renamed to
.groups."),this.groups}}}),Es.prototype.setDynamic=function(t){return
console.warn("THREE.InterleavedBuffer: .setDynamic() has been deprecated. Use
.setUsage() instead."),this.setUsage(!0===t?
nt:et),this},Es.prototype.setArray=function()
{console.error("THREE.InterleavedBuffer: .setArray has been removed. Use
BufferGeometry .setAttribute to replace/resize attribute
buffers")},go.prototype.getArrays=function(){console.error("THREE.ExtrudeGeometry:
.getArrays() has been removed.")},go.prototype.addShapeList=function()
{console.error("THREE.ExtrudeGeometry: .addShapeList() has been
removed.")},go.prototype.addShape=function(){console.error("THREE.ExtrudeGeometry:
.addShape() has been removed.")},Ts.prototype.dispose=function()
{console.error("THREE.Scene: .dispose() has been
removed.")},Zc.prototype.onUpdate=function(){return console.warn("THREE.Uniform:
.onUpdate() has been removed. Use object.onBeforeRender()
instead."),this},Object.defineProperties(Xe.prototype,{wrapAround:{get:function()
{console.warn("THREE.Material: .wrapAround has been removed.")},set:function()
{console.warn("THREE.Material: .wrapAround has been removed.")}},overdraw:
{get:function(){console.warn("THREE.Material: .overdraw has been
removed.")},set:function(){console.warn("THREE.Material: .overdraw has been
removed.")}},wrapRGB:{get:function(){return console.warn("THREE.Material: .wrapRGB
has been removed."),new tn}},shading:{get:function()
{console.error("THREE."+this.type+": .shading has been removed. Use the boolean
.flatShading instead.")},set:function(t){console.warn("THREE."+this.type+":
.shading has been removed. Use the boolean .flatShading
instead."),this.flatShading=1===t}},stencilMask:{get:function(){return
console.warn("THREE."+this.type+": .stencilMask has been removed. Use
.stencilFuncMask instead."),this.stencilFuncMask},set:function(t)
{console.warn("THREE."+this.type+": .stencilMask has been removed. Use
.stencilFuncMask
instead."),this.stencilFuncMask=t}}}),Object.defineProperties(Jn.prototype,
{derivatives:{get:function(){return console.warn("THREE.ShaderMaterial:
.derivatives has been moved to
.extensions.derivatives."),this.extensions.derivatives},set:function(t)
{console.warn("THREE. ShaderMaterial: .derivatives has been moved to

```

```

.extensions.derivatives."),this.extensions.derivatives=t}}}),ws.prototype.clearTarget=function(t,e,n,i){console.warn("THREE.WebGLRenderer: .clearTarget() has been deprecated. Use .setRenderTarget() and .clear() instead."),this.setRenderTarget(t),this.clear(e,n,i)},ws.prototype.animate=function(t){console.warn("THREE.WebGLRenderer: .animate() is now .setAnimationLoop()."),this.setAnimationLoop(t)},ws.prototype.getCurrentRenderTarget=function(){return console.warn("THREE.WebGLRenderer: .getCurrentRenderTarget() is now .getRenderTarget()."),this.getRenderTarget()},ws.prototype.getMaxAnisotropy=function(){return console.warn("THREE.WebGLRenderer: .getMaxAnisotropy() is now .capabilities.getMaxAnisotropy()."),this.capabilities.getMaxAnisotropy()},ws.prototype.getPrecision=function(){return console.warn("THREE.WebGLRenderer: .getPrecision() is now .capabilities.precision."),this.capabilities.precision},ws.prototype.resetGLState=function(){return console.warn("THREE.WebGLRenderer: .resetGLState() is now .state.reset()."),this.state.reset()},ws.prototype.supportsFloatTextures=function(){return console.warn("THREE.WebGLRenderer: .supportsFloatTextures() is now .extensions.get( 'OES_texture_float' )."),this.extensions.get("OES_texture_float")},ws.prototype.supportsHalfFloatTextures=function(){return console.warn("THREE.WebGLRenderer: .supportsHalfFloatTextures() is now .extensions.get( 'OES_texture_half_float' )."),this.extensions.get("OES_texture_half_float")},ws.prototype.supportsStandardDerivatives=function(){return console.warn("THREE.WebGLRenderer: .supportsStandardDerivatives() is now .extensions.get( 'OES_standard_derivatives' )."),this.extensions.get("OES_standard_derivatives")},ws.prototype.supportsCompressedTextureS3TC=function(){return console.warn("THREE.WebGLRenderer: .supportsCompressedTextureS3TC() is now .extensions.get( 'WEBGL_compressed_texture_s3tc' )."),this.extensions.get("WEBGL_compressed_texture_s3tc")},ws.prototype.supportsCompressedTexturePVRTC=function(){return console.warn("THREE.WebGLRenderer: .supportsCompressedTexturePVRTC() is now .extensions.get( 'WEBGL_compressed_texture_pvrtc' )."),this.extensions.get("WEBGL_compressed_texture_pvrtc")},ws.prototype.supportsBlendMinMax=function(){return console.warn("THREE.WebGLRenderer: .supportsBlendMinMax() is now .extensions.get( 'EXT_blend_minmax' )."),this.extensions.get("EXT_blend_minmax")},ws.prototype.supportsVertexTextures=function(){return console.warn("THREE.WebGLRenderer: .supportsVertexTextures() is now .capabilities.vertexTextures."),this.capabilities.vertexTextures},ws.prototype.supportsInstancedArrays=function(){return console.warn("THREE.WebGLRenderer: .supportsInstancedArrays() is now .extensions.get( 'ANGLE_instanced_arrays' )."),this.extensions.get("ANGLE_instanced_arrays")},ws.prototype.enableScissorTest=function(t){console.warn("THREE.WebGLRenderer: .enableScissorTest() is now .setScissorTest()."),this.setScissorTest(t)},ws.prototype.initMaterial=function(){console.warn("THREE.WebGLRenderer: .initMaterial() has been removed.")},ws.prototype.addPrePlugin=function(){console.warn("THREE.WebGLRenderer: .addPrePlugin() has been removed.")},ws.prototype.addPostPlugin=function(){console.warn("THREE.WebGLRenderer: .addPostPlugin() has been removed.")},ws.prototype.updateShadowMap=function(){console.warn("THREE.WebGLRenderer: .updateShadowMap() has been removed.")},ws.prototype.setFaceCulling=function(){console.warn("THREE.WebGLRenderer: .setFaceCulling() has been removed.")},ws.prototype.allocTextureUnit=function(){console.warn("THREE.WebGLRenderer: .allocTextureUnit() has been removed.")},ws.prototype.setTexture=function(){console.warn("THREE.WebGLRenderer: .setTexture() has been removed.")},ws.prototype.setTexture2D=function(){console.warn("THREE.WebGLRenderer: .setTexture2D() has been removed.")},ws.prototype.setTextureCube=function(){

```

```

{console.warn("THREE.WebGLRenderer: .setTextureCube() has been
removed.")},ws.prototype.getActiveMipMapLevel=function(){return
console.warn("THREE.WebGLRenderer: .getActiveMipMapLevel() is now
.getActiveMipmapLevel()."),this.getActiveMipmapLevel()},Object.defineProperties(ws.
prototype,{shadowMapEnabled:{get:function(){return
this.shadowMap.enabled},set:function(t){console.warn("THREE.WebGLRenderer:
.shadowMapEnabled is now
.shadowMap.enabled."),this.shadowMap.enabled=t}},shadowMapType:{get:function()
{return this.shadowMap.type},set:function(t){console.warn("THREE.WebGLRenderer:
.shadowMapType is now .shadowMap.type."),this.shadowMap.type=t}},shadowMapCullFace:
{get:function(){console.warn("THREE.WebGLRenderer: .shadowMapCullFace has been
removed. Set Material.shadowSide instead.")},set:function()
{console.warn("THREE.WebGLRenderer: .shadowMapCullFace has been removed. Set
Material.shadowSide instead.")}},context:{get:function(){return
console.warn("THREE.WebGLRenderer: .context has been removed. Use .getContext()
instead."),this.getContext()}},vr:{get:function(){return
console.warn("THREE.WebGLRenderer: .vr has been renamed to
.xr"),this.xr}},gammaInput:{get:function(){return
console.warn("THREE.WebGLRenderer: .gammaInput has been removed. Set the encoding
for textures via Texture.encoding instead."),!1},set:function()
{console.warn("THREE.WebGLRenderer: .gammaInput has been removed. Set the encoding
for textures via Texture.encoding instead.")}},gammaOutput:{get:function(){return
console.warn("THREE.WebGLRenderer: .gammaOutput has been removed. Set
WebGLRenderer.outputEncoding instead."),!1},set:function(t)
{console.warn("THREE.WebGLRenderer: .gammaOutput has been removed. Set
WebGLRenderer.outputEncoding instead."),this.outputEncoding=!0===t?
Y:X}},toneMappingWhitePoint:{get:function(){return
console.warn("THREE.WebGLRenderer: .toneMappingWhitePoint has been
removed."),1},set:function(){console.warn("THREE.WebGLRenderer:
.toneMappingWhitePoint has been
removed.")}},Object.defineProperties(us.prototype,{cullFace:{get:function()
{console.warn("THREE.WebGLRenderer: .shadowMap.cullFace has been removed. Set
Material.shadowSide instead.")},set:function(){console.warn("THREE.WebGLRenderer:
.shadowMap.cullFace has been removed. Set Material.shadowSide
instead.")}},renderReverseSided:{get:function(){console.warn("THREE.WebGLRenderer:
.shadowMap.renderReverseSided has been removed. Set Material.shadowSide
instead.")},set:function(){console.warn("THREE.WebGLRenderer:
.shadowMap.renderReverseSided has been removed. Set Material.shadowSide
instead.")}},renderSingleSided:{get:function(){console.warn("THREE.WebGLRenderer:
.shadowMap.renderSingleSided has been removed. Set Material.shadowSide
instead.")},set:function(){console.warn("THREE.WebGLRenderer:
.shadowMap.renderSingleSided has been removed. Set Material.shadowSide
instead.")}}}),Object.defineProperties(Tt.prototype,{wrapS:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .wrapS is now
.texture.wrapS."),this.texture.wrapS},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .wrapS is now
.texture.wrapS."),this.texture.wrapS=t}},wrapT:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .wrapT is now
.texture.wrapT."),this.texture.wrapT},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .wrapT is now
.texture.wrapT."),this.texture.wrapT=t}},magFilter:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .magFilter is now
.texture.magFilter."),this.texture.magFilter},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .magFilter is now
.texture.magFilter."),this.texture.magFilter=t}},minFilter:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .minFilter is now
.texture.minFilter."),this.texture.minFilter},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .minFilter is now

```

```

.texture.minFilter."),this.texture.minFilter=t}},anisotropy:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .anisotropy is now
.texture.anisotropy."),this.texture.anisotropy},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .anisotropy is now
.texture.anisotropy."),this.texture.anisotropy=t}},offset:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .offset is now
.texture.offset."),this.texture.offset},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .offset is now
.texture.offset."),this.texture.offset=t}},repeat:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .repeat is now
.texture.repeat."),this.texture.repeat},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .repeat is now
.texture.repeat."),this.texture.repeat=t}},format:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .format is now
.texture.format."),this.texture.format},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .format is now
.texture.format."),this.texture.format=t}},type:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .type is now
.texture.type."),this.texture.type},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .type is now
.texture.type."),this.texture.type=t}},generateMipmaps:{get:function(){return
console.warn("THREE.WebGLRenderTarget: .generateMipmaps is now
.texture.generateMipmaps."),this.texture.generateMipmaps},set:function(t)
{console.warn("THREE.WebGLRenderTarget: .generateMipmaps is now
.texture.generateMipmaps."),this.texture.generateMipmaps=t}}}),Lc.prototype.load=fu
nction(t){console.warn("THREE.Audio: .load has been deprecated. Use
THREE.AudioLoader instead.");const e=this;return(new vc).load(t,(function(t)
{e.setBuffer(t)})),this},Ic.prototype.getData=function(){return
console.warn("THREE.AudioAnalyser: .getData() is now
.getFrequencyData()."),this.getFrequencyData()},ti.prototype.updateCubeMap=function
(t,e){return console.warn("THREE.CubeCamera: .updateCubeMap() is now
.update()."),this.update(t,e)},ti.prototype.clear=function(t,e,n,i){return
console.warn("THREE.CubeCamera: .clear() is now
.renderTarget.clear()."),this.renderTarget.clear(t,e,n,i)},_t.crossOrigin=void
0,_t.loadTexture=function(t,e,n,i){console.warn("THREE.ImageUtils.loadTexture has
been deprecated. Use THREE.TextureLoader() instead.");const r=new
pl;r.setCrossOrigin(this.crossOrigin);const s=r.load(t,n,void 0,i);return e&&
(s.mapping=e),s},_t.loadTextureCube=function(t,e,n,i)
{console.warn("THREE.ImageUtils.loadTextureCube has been deprecated. Use
THREE.CubeTextureLoader() instead.");const r=new
ul;r.setCrossOrigin(this.crossOrigin);const s=r.load(t,n,void 0,i);return e&&
(s.mapping=e),s},_t.loadCompressedTexture=function()
{console.error("THREE.ImageUtils.loadCompressedTexture has been removed. Use
THREE.DDSLoader instead.")},_t.loadCompressedTextureCube=function()
{console.error("THREE.ImageUtils.loadCompressedTextureCube has been removed. Use
THREE.DDSLoader instead.")};const eu={createMultiMaterialObject:function()
{console.error("THREE.SceneUtils has been moved to
/examples/jsm/utils/SceneUtils.js")},detach:function()
{console.error("THREE.SceneUtils has been moved to
/examples/jsm/utils/SceneUtils.js")},attach:function()
{console.error("THREE.SceneUtils has been moved to
/examples/jsm/utils/SceneUtils.js")}};"undefined"!==typeof
__THREE_DEVTOOLS__&&__THREE_DEVTOOLS__.dispatchEvent(new CustomEvent("register",
{detail:{revision:e}})), "undefined"!==typeof window&&(window.__THREE__?
console.warn("WARNING: Multiple instances of Three.js being
imported."):window.__THREE__=e),t.ACESFilmicToneMapping=4,t.AddEquation=n,t.AddOper
ation=2,t.AdditiveAnimationBlendMode=q,t.AdditiveBlending=2,t.AlphaFormat=1021,t.Al
waysDepth=1,t.AlwaysStencilFunc=519,t.AmbientLight=$l,t.AmbientLightProbe=xc,t.Anim

```

```

ationClip=nl,t.AnimationLoader=class extends ol{constructor(t)
{super(t)}load(t,e,n,i){const r=this,s=new
cl(this.manager);s.setPath(this.path),s.setRequestHeader(this.requestHeader),s.setW
ithCredentials(this.withCredentials),s.load(t,(function(n)
{try{e(r.parse(JSON.parse(n)))}catch(e){i?
i(e):console.error(e),r.manager.itemError(t)}}),n,i)}parse(t){const e=[];for(let
n=0;n<t.length;n++){const i=nl.parse(t[n]);e.push(i)}return
e}},t.AnimationMixer=Yc,t.AnimationObjectGroup=qc,t.AnimationUtils=Vo,t.ArcCurve=gl
,t.ArrayCamera=fs,t.ArrowHelper=class extends Ce{constructor(t=new Lt(0,0,1),e=new
Lt(0,0,0),n=1,i=16776960,r=.2*n,s=.2*r){super(),this.type="ArrowHelper",void
0===Th&&(Th=new En,Th.setAttribute("position",new mn([0,0,0,0,1,0],3)),Eh=new
Da(0,.5,1,5,1),Eh.translate(0,-.5,0)),this.position.copy(e),this.line=new fa(Th,new
ca({color:i,toneMapped:!1})),this.line.matrixAutoUpdate=!1,this.add(this.line),this
.cone=new Wn(Eh,new
en({color:i,toneMapped:!1})),this.cone.matrixAutoUpdate=!1,this.add(this.cone),this
.setDirection(t),this.setLength(n,r,s)}setDirection(t)
{if(t.y>.99999)this.quaternion.set(0,0,0,1);else
if(t.y<-.99999)this.quaternion.set(1,0,0,0);else{Sh.set(t.z,0,-
t.x).normalize();const
e=Math.acos(t.y);this.quaternion.setFromAxisAngle(Sh,e)}}setLength(t,e=.2*t,n=.2*e)
{this.line.scale.set(1,Math.max(1e-4,t-
e),1),this.line.updateMatrix(),this.cone.scale.set(n,e,n),this.cone.position.y=t,th
is.cone.updateMatrix()}setColor(t)
{this.line.material.color.set(t),this.cone.material.color.set(t)}copy(t){return
super.copy(t,!1),this.line.copy(t.line),this.cone.copy(t.cone),this}},t.Audio=Lc,t.
AudioAnalyser=Ic,t.AudioContext=gc,t.AudioListener=class extends Ce{constructor()
{super(),this.type="AudioListener",this.context=gc.getContext(),this.gain=this.cont
ext.createGain(),this.gain.connect(this.context.destination),this.filter=null,this.
timeDelta=0,this._clock=new bc}getInput(){return this.gain}removeFilter(){return
null!==(this.filter&&
(this.gain.disconnect(this.filter),this.filter.disconnect(this.context.destination)
,this.gain.connect(this.context.destination),this.filter=null),this)}getFilter()
{return this.filter}setFilter(t){return null!==(this.filter?
(this.gain.disconnect(this.filter),this.filter.disconnect(this.context.destination)
):this.gain.disconnect(this.context.destination),this.filter=t,this.gain.connect(th
is.filter),this.filter.connect(this.context.destination),this)}getMasterVolume()
{return this.gain.gain.value}setMasterVolume(t){return
this.gain.gain.setTargetAtTime(t,this.context.currentTime,.01),this}updateMatrixWor
ld(t){super.updateMatrixWorld(t);const
e=this.context.listener,n=this.up;if(this.timeDelta=this._clock.getDelta(),this.mat
rixWorld.decompose(Sc,Tc,Ec),Ac.set(0,0,-1).applyQuaternion(Tc),e.positionX){const
t=this.context.currentTime+this.timeDelta;e.positionX.linearRampToValueAtTime(Sc.x,
t),e.positionY.linearRampToValueAtTime(Sc.y,t),e.positionZ.linearRampToValueAtTime(
Sc.z,t),e.forwardX.linearRampToValueAtTime(Ac.x,t),e.forwardY.linearRampToValueAtTi
me(Ac.y,t),e.forwardZ.linearRampToValueAtTime(Ac.z,t),e.upX.linearRampToValueAtTime
(n.x,t),e.upY.linearRampToValueAtTime(n.y,t),e.upZ.linearRampToValueAtTime(n.z,t)}e
lse
e.setPosition(Sc.x,Sc.y,Sc.z),e.setOrientation(Ac.x,Ac.y,Ac.z,n.x,n.y,n.z)}},t.Audi
oLoader=vc,t.AxesHelper=Ah,t.AxisHelper=function(t){return
console.warn("THREE.AxisHelper has been renamed to THREE.AxesHelper."),new
Ah(t)},t.BackSide=1,t.BasicDepthPacking=3200,t.BasicShadowMap=0,t.BinaryTextureLoad
er=function(t){return console.warn("THREE.BinaryTextureLoader has been renamed to
THREE.DataTextureLoader."),new
dl(t)},t.Bone=ta,t.BooleanKeyframeTrack=Zo,t.BoundingBoxHelper=function(t,e){return
console.warn("THREE.BoundingBoxHelper has been deprecated. Creating a
THREE.BoxHelper instead."),new Mh(t,e)},t.Box2=eh,t.Box3=Pt,t.Box3Helper=class
extends ya{constructor(t,e=16776960){const n=new
Uint16Array([0,1,1,2,2,3,3,0,4,5,5,6,6,7,7,4,0,4,1,5,2,6,3,7]),i=new

```

```
En,i.setIndex(new sn(n,1)),i.setAttribute("position",new
mn([1,1,1,-1,1,1,-1,-1,1,1,-1,1,1,1,-1,-1,1,-1,-1,-1,1,-1,-1],3)),super(i,new
ca({color:e,toneMapped:!1})),this.box=t,this.type="Box3Helper",this.geometry.compute
eBoundingSphere()}updateMatrixWorld(t){const e=this.box;e.isEmpty()||
(e.getCenter(this.position),e.getSize(this.scale),this.scale.multiplyScalar(.5),sup
er.updateMatrixWorld(t))},t.BoxBufferGeometry=qn,t.BoxGeometry=qn,t.BoxHelper=Mh,t
.BufferAttribute=sn,t.BufferGeometry=En,t.BufferGeometryLoader=oc,t.ByteType=1010,t
.Cache=rl,t.Camera=Qn,t.CameraHelper=class extends ya{constructor(t){const e=new
En,n=new ca({color:16777215,vertexColors:!0,toneMapped:!1}),i=[],r=[],s={},a=new
tn(16755200),o=new tn(16711680),l=new tn(43775),c=new tn(16777215),h=new
tn(3355443);function u(t,e,n){d(t,n),d(e,n)}function d(t,e)
{i.push(0,0,0),r.push(e.r,e.g,e.b),void 0===s[t]&&(s[t]=[]),s[t].push(i.length/3-
1)}u("n1","n2",a),u("n2","n4",a),u("n4","n3",a),u("n3","n1",a),u("f1","f2",a),u("f2
","f4",a),u("f4","f3",a),u("f3","f1",a),u("n1","f1",a),u("n2","f2",a),u("n3","f3",a
),u("n4","f4",a),u("p","n1",o),u("p","n2",o),u("p","n3",o),u("p","n4",o),u("u1","u2
",l),u("u2","u3",l),u("u3","u1",l),u("c","t",c),u("p","c",h),u("cn1","cn2",h),u("cn
3","cn4",h),u("cf1","cf2",h),u("cf3","cf4",h),e.setAttribute("position",new
mn(i,3)),e.setAttribute("color",new
mn(r,3)),super(e,n),this.type="CameraHelper",this.camera=t,this.camera.updateProjec
tionMatrix&&this.camera.updateProjectionMatrix(),this.matrix=t.matrixWorld,this.mat
rixAutoUpdate=!1,this.pointMap=s,this.update()}update(){const
t=this.geometry,e=this.pointMap;_h.projectionMatrixInverse.copy(this.camera.project
ionMatrixInverse),wh("c",e,t,_h,0,0,-1),wh("t",e,t,_h,0,0,1),wh("n1",e,t,_h,-1,-1,-
1),wh("n2",e,t,_h,1,-1,-1),wh("n3",e,t,_h,-1,1,-1),wh("n4",e,t,_h,1,1,-1),wh("f1",e
,t,_h,-1,-1,1),wh("f2",e,t,_h,1,-1,1),wh("f3",e,t,_h,-1,1,1),wh("f4",e,t,_h,1,1,1),
wh("u1",e,t,_h,.7,1,1,-1),wh("u2",e,t,_h,-.7,1,1,-1),wh("u3",e,t,_h,0,2,-1),wh("cf1
",e,t,_h,-1,0,1),wh("cf2",e,t,_h,1,0,1),wh("cf3",e,t,_h,0,-1,1),wh("cf4",e,t,_h,0,1
,1),wh("cn1",e,t,_h,-1,0,-1),wh("cn2",e,t,_h,1,0,-1),wh("cn3",e,t,_h,0,-1,-1),wh("c
n4",e,t,_h,0,1,-1),t.getAttribute("position").needsUpdate=!0}dispose())
{this.geometry.dispose(),this.material.dispose()}},t.CanvasRenderer=function()
{console.error("THREE.CanvasRenderer has been
removed")},t.CanvasTexture=Ra,t.CatmullRomCurve3=bl,t.CineonToneMapping=3,t.CircleB
ufferGeometry=Pa,t.CircleGeometry=Pa,t.ClampToEdgeWrapping=u,t.Clock=bc,t.Color=tn,
t.ColorKeyframeTrack=Jo,t.CompressedTexture=La,t.CompressedTextureLoader=class
extends ol{constructor(t){super(t)}load(t,e,n,i){const r=this,s=[],a=new La,o=new
cl(this.manager);o.setPath(this.path),o.setResponseType("arraybuffer"),o.setRequestHeader(this.requestHeader),o.setWithCredentials(r.withCredentials);let l=0;function
c(c){o.load(t[c],(function(t){const n=r.parse(t,!0);s[c]=
{width:n.width,height:n.height,format:n.format,mipmaps:n.mipmaps},l+=1,6===l&&
(1===n.mipmapCount&&
(a.minFilter=g),a.image=s,a.format=n.format,a.needsUpdate=!0,e&&e(a)}),n,i)}if(Arr
ay.isArray(t))for(let e=0,n=t.length;e<n;++e)c(e);else o.load(t,(function(t){const
n=r.parse(t,!0);if(n.isCubemap){const t=n.mipmaps.length/n.mipmapCount;for(let
e=0;e<t;e++){s[e]={mipmaps:[]};for(let
t=0;t<n.mipmapCount;t++)s[e].mipmaps.push(n.mipmaps[e*n.mipmapCount+t]),s[e].format
=n.format,s[e].width=n.width,s[e].height=n.height}a.image=s}else
a.image.width=n.width,a.image.height=n.height,a.mipmaps=n.mipmaps;1===n.mipmapCount
&&(a.minFilter=g),a.format=n.format,a.needsUpdate=!0,e&&e(a)}),n,i);return
a}},t.ConeBufferGeometry=Ia,t.ConeGeometry=Ia,t.CubeCamera=ti,t.CubeReflectionMappi
ng=r,t.CubeRefractionMapping=s,t.CubeTexture=ei,t.CubeTextureLoader=ul,t.CubeUVRefl
ectionMapping=l,t.CubeUVRefractionMapping=c,t.CubicBezierCurve=El,t.CubicBezierCurv
e3=Al,t.CubicInterpolant=jo,t.CullFaceBack=1,t.CullFaceFront=2,t.CullFaceFrontBack=
3,t.CullFaceNone=0,t.Curve=ml,t.CurvePath=Nl,t.CustomBlending=5,t.CustomToneMapping
=5,t.CylinderBufferGeometry=Da,t.CylinderGeometry=Da,t.Cylindrical=class{constructo
r(t=1,e=0,n=0){return this.radius=t,this.theta=e,this.y=n,this}set(t,e,n){return
this.radius=t,this.theta=e,this.y=n,this}copy(t){return
this.radius=t.radius,this.theta=t.theta,this.y=t.y,this}setFromVector3(t){return
this.setFromCartesianCoords(t.x,t.y,t.z)}setFromCartesianCoords(t,e,n){return
```

```

this.radius=Math.sqrt(t*t+n*n),this.theta=Math.atan2(t,n),this.y=e,this.clone()
{return(new
this.constructor).copy(this)},t.DataTexture=ii,t.DataTexture2DArray=Ai,t.DataTextu
re3D=Li,t.DataTextureLoader=dl,t.DataUtils=class{static toHalfFloat(t)
{Lh[0]=t;const e=Rh[0];let n=e>>16&32768,i=e>>12&2047;const r=e>>23&255;return
r<103?n:r>142?(n|=31744,n|=(255==r?0:1)&&8388607&e,n):r<113?(i|=2048,n|=(i>>114-r)+
(i>>113-r&1),n):(n|=r-
112<<10|i>>1,n+=1&i,n)}},t.DecrementStencilOp=7683,t.DecrementWrapStencilOp=34056,t
.DefaultLoadingManager=al,t.DepthFormat=A,t.DepthStencilFormat=L,t.DepthTexture=Ca,
t.DirectionallLight=Kl,t.DirectionallLightHelper=class extends Ce{constructor(t,e,n)
{super(),this.light=t,this.light.updateMatrixWorld(),this.matrix=t.matrixWorld,this
.matrixAutoUpdate=!1,this.color=n,void 0===e&&(e=1);let i=new
En;i.setAttribute("position",new mn([-e,e,0,e,e,0,e,-e,0,-e,-e,0,-e,e,0],3));const
r=new ca({fog:!1,toneMapped:!1});this.lightPlane=new
fa(i,r),this.add(this.lightPlane),i=new En,i.setAttribute("position",new
mn([0,0,0,0,0,1],3)),this.targetLine=new
fa(i,r),this.add(this.targetLine),this.update()}dispose()
{this.lightPlane.geometry.dispose(),this.lightPlane.material.dispose(),this.targetL
ine.geometry.dispose(),this.targetLine.material.dispose()}update()
{gh.setFromMatrixPosition(this.light.matrixWorld),vh.setFromMatrixPosition(this.lig
ht.target.matrixWorld),yh.subVectors(vh,gh),this.lightPlane.lookAt(vh),void
0!==this.color?
(this.lightPlane.material.color.set(this.color),this.targetLine.material.color.set(
this.color)):
(this.lightPlane.material.color.copy(this.light.color),this.targetLine.material.col
or.copy(this.light.color),this.targetLine.lookAt(vh),this.targetLine.scale.z=yh.le
ngth())},t.DiscreteInterpolant=Xo,t.DodecahedronBufferGeometry=Ba,t.DodecahedronGeo
metry=Ba,t.DoubleSide=2,t.DstAlphaFactor=206,t.DstColorFactor=208,t.DynamicBufferAt
tribute=function(t,e){return console.warn("THREE.DynamicBufferAttribute has been
removed. Use new THREE.BufferAttribute().setUsage( THREE.DynamicDrawUsage )
instead."),new
sn(t,e).setUsage(nt)},t.DynamicCopyUsage=35050,t.DynamicDrawUsage=nt,t.DynamicReadU
sage=35049,t.EdgesGeometry=Ga,t.EdgesHelper=function(t,e){return
console.warn("THREE.EdgesHelper has been removed. Use THREE.EdgesGeometry
instead."),new ya(new Ga(t.geometry),new ca({color:void 0!==e?
e:16777215}))},t.EllipseCurve=fl,t.EqualDepth=4,t.EqualStencilFunc=514,t.Equirectan
gularReflectionMapping=a,t.EquirectangularRefractionMapping=o,t.Euler=fe,t.EventDis
patcher=rt,t.ExtrudeBufferGeometry=go,t.ExtrudeGeometry=go,t.FaceColors=1,t.FileLoa
der=cl,t.FlatShading=1,t.Float16BufferAttribute=pn,t.Float32Attribute=function(t,e)
{return console.warn("THREE.Float32Attribute has been removed. Use new
THREE.Float32BufferAttribute() instead."),new
mn(t,e)},t.Float32BufferAttribute=mn,t.Float64Attribute=function(t,e){return
console.warn("THREE.Float64Attribute has been removed. Use new
THREE.Float64BufferAttribute() instead."),new
fn(t,e)},t.Float64BufferAttribute=fn,t.FloatType=b,t.Fog=Ss,t.FogExp2=Ms,t.Font=pc,
t.FontLoader=class extends ol{constructor(t){super(t)}load(t,e,n,i){const
r=this,s=new
cl(this.manager);s.setPath(this.path),s.setRequestHeader(this.requestHeader),s.setW
ithCredentials(r.withCredentials),s.load(t,(function(t){let
n;try{n=JSON.parse(t)}catch(e){console.warn("THREE.FontLoader: typeface.js support
is being deprecated. Use typeface.json
instead."),n=JSON.parse(t.substring(65,t.length-2))})const
i=r.parse(n);e&&e(i)},n,i)}parse(t){return new
pc(t)}},t.FrontSide=0,t.Frustum=ai,t.GLBufferAttribute=Qc,t.GLSL1="100",t.GLSL3=it,
t.GammaEncoding=Z,t.GreaterDepth=6,t.GreaterEqualDepth=5,t.GreaterEqualStencilFunc=
518,t.GreaterStencilFunc=516,t.GridHelper=fh,t.Group=gs,t.HalfFloatType=M,t.Hemisph
ereLight=Ol,t.HemisphereLightHelper=class extends Ce{constructor(t,e,n)
{super(),this.light=t,this.light.updateMatrixWorld(),this.matrix=t.matrixWorld,this

```

```

.matrixAutoUpdate=!1,this.color=n;const i=new
_o(e);i.rotateY(.5*Math.PI),this.material=new
en({wireframe:!0,fog:!1,toneMapped:!1}),void 0===this.color&&
(this.material.vertexColors=!0);const r=i.getAttribute("position"),s=new
Float32Array(3*r.count);i.setAttribute("color",new sn(s,3)),this.add(new
Wn(i,this.material)),this.update())dispose()
{this.children[0].geometry.dispose(),this.children[0].material.dispose()}update()
{const t=this.children[0];if(void
0!==this.color)this.material.color.set(this.color);else{const
e=t.geometry.getAttribute("color");ph.copy(this.light.color),mh.copy(this.light.gro
undColor);for(let t=0,n=e.count;t<n;t++){const i=t<n/2?
ph:mh;e.setXYZ(t,i.r,i.g,i.b)}e.needsUpdate=!0}t.lookAt(dh.setFromMatrixPosition(th
is.light.matrixWorld).negate()))},t.HemisphereLightProbe=yc,t.IcosahedronBufferGeom
etry=yo,t.IcosahedronGeometry=yo,t.ImageBitmapLoader=uc,t.ImageLoader=hl,t.ImageUtil
s=_t,t.ImmediateRenderObject=sh,t.IncrementStencilOp=7682,t.IncrementWrapStencilOp
=34055,t.InstancedBufferAttribute=ac,t.InstancedBufferGeometry=sc,t.InstancedInterl
eavedBuffer=Jc,t.InstancedMesh=la,t.Int16Attribute=function(t,e){return
console.warn("THREE.Int16Attribute has been removed. Use new
THREE.Int16BufferAttribute() instead."),new
cn(t,e)},t.Int16BufferAttribute=cn,t.Int32Attribute=function(t,e){return
console.warn("THREE.Int32Attribute has been removed. Use new
THREE.Int32BufferAttribute() instead."),new
un(t,e)},t.Int32BufferAttribute=un,t.Int8Attribute=function(t,e){return
console.warn("THREE.Int8Attribute has been removed. Use new
THREE.Int8BufferAttribute() instead."),new
an(t,e)},t.Int8BufferAttribute=an,t.IntType=1013,t.InterleavedBuffer=Es,t.Interleav
edBufferAttribute=Ls,t.Interpolant=Wo,t.InterpolateDiscrete=H,t.InterpolateLinear=G
,t.InterpolateSmooth=U,t.InvertStencilOp=5386,t.JSONLoader=function()
{console.error("THREE.JSONLoader has been
removed.")},t.KeepStencilOp=tt,t.KeyframeTrack=Yo,t.LOD=Xs,t.LatheBufferGeometry=xo
,t.LatheGeometry=xo,t.Layers=ge,t.LensFlare=function()
{console.error("THREE.LensFlare has been moved to
/examples/jsm/objects/Lensflare.js")},t.LessDepth=2,t.LessEqualDepth=3,t.LessEqualS
tencilFunc=515,t.LessStencilFunc=513,t.Light=Fl,t.LightProbe=nc,t.Line=fa,t.Line3=r
h,t.LineBasicMaterial=ca,t.LineCurve=Ll,t.LineCurve3=Rl,t.LineDashedMaterial=Uo,t.L
ineLoop=xa,t.LinePieces=1,t.LineSegments=ya,t.LineStrip=0,t.LinearEncoding=X,t.Line
arFilter=g,t.LinearInterpolant=qo,t.LinearMipMapLinearFilter=1008,t.LinearMipMapNea
restFilter=1007,t.LinearMipmapLinearFilter=y,t.LinearMipmapNearestFilter=v,t.Linear
ToneMapping=1,t.Loader=ol,t.LoaderUtils=rc,t.LoadingManager=sl,t.LogLuvEncoding=300
3,t.LoopOnce=2200,t.LoopPingPong=2202,t.LoopRepeat=2201,t.LuminanceAlphaFormat=1025
,t.LuminanceFormat=1024,t.MOUSE=
{LEFT:0,MIDDLE:1,RIGHT:2,ROTATE:0,DOLLY:1,PAN:2},t.Material=Xe,t.MaterialLoader=ic,
t.Math=gt,t.MathUtils=gt,t.Matrix3=yt,t.Matrix4=se,t.MaxEquation=104,t.Mesh=Wn,t.Me
shBasicMaterial=en,t.MeshDepthMaterial=cs,t.MeshDistanceMaterial=hs,t.MeshFaceMater
ial=function(t){return console.warn("THREE.MeshFaceMaterial has been removed. Use
an Array
instead."),t},t.MeshLambertMaterial=Ho,t.MeshMatcapMaterial=Go,t.MeshNormalMaterial
=Oo,t.MeshPhongMaterial=zo,t.MeshPhysicalMaterial=Bo,t.MeshStandardMaterial=No,t.Me
shToonMaterial=Fo,t.MinEquation=103,t.MirroredRepeatWrapping=d,t.MixOperation=1,t.M
ultiMaterial=function(t=[]){return console.warn("THREE.MultiMaterial has been
removed. Use an Array
instead."),t.isMultiMaterial=!0,t.materials=t,t.clone=function(){return
t.slice()},t},t.MultiplyBlending=4,t.MultiplyOperation=0,t.NearestFilter=p,t.Neares
tMipMapLinearFilter=1005,t.NearestMipMapNearestFilter=1004,t.NearestMipmapLinearFil
ter=f,t.NearestMipmapNearestFilter=m,t.NeverDepth=0,t.NeverStencilFunc=512,t.NoBlen
ding=0,t.NoColors=0,t.NoToneMapping=0,t.NormalAnimationBlendMode=j,t.NormalBlending
=1,t.NotEqualDepth=7,t.NotEqualStencilFunc=517,t.NumberKeyframeTrack=Qo,t.Object3D=
Ce,t.ObjectLoader=class extends ol{constructor(t){super(t)}load(t,e,n,i){const

```



```

r=this,s=""===this.path?
rc.extractUrlBase(t):this.path;this.resourcePath=this.resourcePath||s;const a=new
cl(this.manager);a.setPath(this.path),a.setRequestHeader(this.requestHeader),a.setW
ithCredentials(this.withCredentials),a.load(t,(function(n){let
s=null;try{s=JSON.parse(n)}catch(e){return void 0!==i&&i(e),void
console.error("THREE:ObjectLoader: Can't parse "+t+".",e.message)}const
a=s.metadata;void 0!==a&&void 0!==a.type&&"geometry"!==a.type.toLowerCase()?
r.parse(s,e):console.error("THREE.ObjectLoader: Can't load "+t)}),n,i)}parse(t,e)
{const
n=this.parseAnimations(t.animations),i=this.parseShapes(t.shapes),r=this.parseGeome
tries(t.geometries,i),s=this.parseImages(t.images,(function(){void
0!==e&&e(l)})),a=this.parseTextures(t.textures,s),o=this.parseMaterials(t.materials
,a),l=this.parseObject(t.object,r,o,n),c=this.parseSkeltons(t.skeltons,l);if(this
.bindSkeltons(l,c),void 0!==e){let t=!1;for(const e in s)if(s[e]instanceof
HTMLImageElement){t=!0;break}!l===t&&e(l)}return l}parseShapes(t){const e=
{};if(void 0!==t)for(let n=0,i=t.length;n<i;n++){const i=(new
zl).fromJSON(t[n]);e[i.uuid]=i}return e}parseSkeltons(t,e){const n={},i=
{};if(e.traverse((function(t){t.isBone&&(i[t.uuid]=t)})),void 0!==t)for(let
e=0,r=t.length;e<r;e++){const r=(new ia).fromJSON(t[e],i);n[r.uuid]=r}return
n}parseGeometries(t,e){const n={};let i;if(void 0!==t){const r=new oc;for(let
s=0,a=t.length;s<a;s++){let a;const o=t[s];switch(o.type)
{case"PlaneGeometry":case"PlaneBufferGeometry":a=new Po[o.type]
(o.width,o.height,o.widthSegments,o.heightSegments);break;case"BoxGeometry":case"Bo
xBufferGeometry":a=new Po[o.type]
(o.width,o.height,o.depth,o.widthSegments,o.heightSegments,o.depthSegments);break;c
ase"CircleGeometry":case"CircleBufferGeometry":a=new Po[o.type]
(o.radius,o.segments,o.thetaStart,o.thetaLength);break;case"CylinderGeometry":case"
CylinderBufferGeometry":a=new Po[o.type]
(o.radiusTop,o.radiusBottom,o.height,o.radialSegments,o.heightSegments,o.openEnded,
o.thetaStart,o.thetaLength);break;case"ConeGeometry":case"ConeBufferGeometry":a=new
Po[o.type]
(o.radius,o.height,o.radialSegments,o.heightSegments,o.openEnded,o.thetaStart,o.the
taLength);break;case"SphereGeometry":case"SphereBufferGeometry":a=new Po[o.type]
(o.radius,o.widthSegments,o.heightSegments,o.phiStart,o.phiLength,o.thetaStart,o.th
etaLength);break;case"DodecahedronGeometry":case"DodecahedronBufferGeometry":case"I
cosahedronGeometry":case"IcosahedronBufferGeometry":case"OctahedronGeometry":case"O
ctahedronBufferGeometry":case"TetrahedronGeometry":case"TetrahedronBufferGeometry":
a=new Po[o.type]
(o.radius,o.detail);break;case"RingGeometry":case"RingBufferGeometry":a=new
Po[o.type]
(o.innerRadius,o.outerRadius,o.thetaSegments,o.phiSegments,o.thetaStart,o.thetaLeng
th);break;case"TorusGeometry":case"TorusBufferGeometry":a=new Po[o.type]
(o.radius,o.tube,o.radialSegments,o.tubularSegments,o.arc);break;case"TorusKnotGeom
etry":case"TorusKnotBufferGeometry":a=new Po[o.type]
(o.radius,o.tube,o.tubularSegments,o.radialSegments,o.p,o.q);break;case"TubeGeometr
y":case"TubeBufferGeometry":a=new Po[o.type]((new
Il[o.path.type]).fromJSON(o.path),o.tubularSegments,o.radius,o.radialSegments,o.clo
sed);break;case"LatheGeometry":case"LatheBufferGeometry":a=new Po[o.type]
(o.points,o.segments,o.phiStart,o.phiLength);break;case"PolyhedronGeometry":case"Po
lyhedronBufferGeometry":a=new Po[o.type]
(o.vertices,o.indices,o.radius,o.details);break;case"ShapeGeometry":case"ShapeBuffe
rGeometry":i=[];for(let t=0,n=o.shapes.length;t<n;t++){const
n=e[o.shapes[t]];i.push(n)}a=new Po[o.type]
(i,o.curveSegments);break;case"ExtrudeGeometry":case"ExtrudeBufferGeometry":i=
[];for(let t=0,n=o.shapes.length;t<n;t++){const n=e[o.shapes[t]];i.push(n)}const
t=o.options.extrudePath;void 0!==t&&(o.options.extrudePath=(new
Il[t.type]).fromJSON(t)),a=new Po[o.type]
(i,o.options);break;case"BufferGeometry":case"InstancedBufferGeometry":a=r.parse(o)

```

```

;break;case"Geometry":console.error('THREE.ObjectLoader: Loading "Geometry" is not
supported anymore.');
```

```

break;default:console.warn('THREE.ObjectLoader: Unsupported
geometry type "' + o.type + '"');continue}a.uuid=o.uuid,void 0!==o.name&&
(a.name=o.name),!0===a.isBufferGeometry&&void 0!==o.userData&&
(a.userData=o.userData),n[o.uuid]=a}}return n}parseMaterials(t,e){const n={},i=
{};if(void 0!==t){const r=new ic;r.setTextures(e);for(let e=0,s=t.length;e<s;e++)
{const s=t[e];if("MultiMaterial"===s.type){const t=[];for(let
e=0;e<s.materials.length;e++){const i=s.materials[e];void 0===n[i.uuid]&&
(n[i.uuid]=r.parse(i)),t.push(n[i.uuid])}i[s.uuid]=t}else void 0===n[s.uuid]&&
(n[s.uuid]=r.parse(s)),i[s.uuid]=n[s.uuid]}return i}parseAnimations(t){const e=
{};if(void 0!==t)for(let n=0;n<t.length;n++){const
i=t[n],r=n1.parse(i);e[r.uuid]=r}return e}parseImages(t,e){const n=this,i={};let
r;function s(t){if("string"===typeof t){const e=t;return function(t){return
n.manager.itemStart(t),r.load(t,(function(){n.manager.itemEnd(t)}),void 0,
(function(){n.manager.itemError(t),n.manager.itemEnd(t)})))(/^(\//\//)|([a-z] +=
(\//\//)?)/i.test(e)?e:n.resourcePath+e)}return t.data?
{data:yn(t.type,t.data),width:t.width,height:t.height}:null}if(void
0!==t&&t.length>0){const n=new sl(e);r=new
hl(n),r.setCrossOrigin(this.crossOrigin);for(let e=0,n=t.length;e<n;e++){const
n=t[e],r=n.url;if(Array.isArray(r)){i[n.uuid]=[];for(let t=0,e=r.length;t<e;t++)
{const e=s(r[t]);null!==e&&(e instanceof HTMLImageElement?
i[n.uuid].push(e):i[n.uuid].push(new ii(e.data,e.width,e.height)))}}else{const
t=s(n.url);null!==t&&(i[n.uuid]=t)}}return i}parseTextures(t,e){function n(t,e)
{return"number"===typeof t?t:(console.warn("THREE.ObjectLoader.parseTexture:
Constant should be in numeric form.",t),e[t])}const i={};if(void 0!==t)for(let
r=0,s=t.length;r<s;r++){const s=t[r];let a;void
0===s.image&&console.warn('THREE.ObjectLoader: No "image" specified
for',s.uuid),void 0===e[s.image]&&console.warn("THREE.ObjectLoader: Undefined
image",s.image);const o=e[s.image];Array.isArray(o)?(a=new ei(o),6===o.length&&
(a.needsUpdate=!0)):(a=o&&o.data?new ii(o.data,o.width,o.height):new bt(o),o&&
(a.needsUpdate=!0)),a.uuid=s.uuid,void 0!==s.name&&(a.name=s.name),void
0!==s.mapping&&(a.mapping=n(s.mapping,lc)),void
0!==s.offset&&a.offset.fromArray(s.offset),void
0!==s.repeat&&a.repeat.fromArray(s.repeat),void
0!==s.center&&a.center.fromArray(s.center),void 0!==s.rotation&&
(a.rotation=s.rotation),void 0!==s.wrap&&
(a.wrapS=n(s.wrap[0],cc),a.wrapT=n(s.wrap[1],cc)),void 0!==s.format&&
(a.format=s.format),void 0!==s.type&&(a.type=s.type),void 0!==s.encoding&&
(a.encoding=s.encoding),void 0!==s.minFilter&&(a.minFilter=n(s.minFilter,hc)),void
0!==s.magFilter&&(a.magFilter=n(s.magFilter,hc)),void 0!==s.anisotropy&&
(a.anisotropy=s.anisotropy),void 0!==s.flipY&&(a.flipY=s.flipY),void
0!==s.premultiplyAlpha&&(a.premultiplyAlpha=s.premultiplyAlpha),void
0!==s.unpackAlignment&&(a.unpackAlignment=s.unpackAlignment),i[s.uuid]=a}return
i}parseObject(t,e,n,i){let r,s,a;function o(t){return void
0===e[t]&&console.warn("THREE.ObjectLoader: Undefined geometry",t),e[t]}function
l(t){if(void 0!==t){if(Array.isArray(t)){const e=[];for(let i=0,r=t.length;i<r;i++)
{const r=t[i];void 0===n[r]&&console.warn("THREE.ObjectLoader: Undefined
material",r),e.push(n[r])}return e}return void
0===n[t]&&console.warn("THREE.ObjectLoader: Undefined
material",t),n[t]}switch(t.type){case"Scene":r=new Ts,void
0!==t.background&&Number.isInteger(t.background)&&(r.background=new
tn(t.background)),void 0!==t.fog&&("Fog"===t.fog.type?r.fog=new
Ss(t.fog.color,t.fog.near,t.fog.far):"FogExp2"===t.fog.type&&(r.fog=new
Ms(t.fog.color,t.fog.density)));break;case"PerspectiveCamera":r=new
Kn(t.fov,t.aspect,t.near,t.far),void 0!==t.focus&&(r.focus=t.focus),void
0!==t.zoom&&(r.zoom=t.zoom),void 0!==t.filmGauge&&(r.filmGauge=t.filmGauge),void
0!==t.filmOffset&&(r.filmOffset=t.filmOffset),void 0!==t.view&&
(r.view=Object.assign({},t.view));break;case"OrthographicCamera":r=new

```

```

Jl(t.left,t.right,t.top,t.bottom,t.near,t.far),void 0!==t.zoom&&
(r.zoom=t.zoom),void 0!==t.view&&
(r.view=Object.assign({},t.view));break;case"AmbientLight":r=new
$1(t.color,t.intensity);break;case"DirectionalLight":r=new
K1(t.color,t.intensity);break;case"PointLight":r=new
Z1(t.color,t.intensity,t.distance,t.decay);break;case"RectAreaLight":r=new
tc(t.color,t.intensity,t.width,t.height);break;case"SpotLight":r=new
W1(t.color,t.intensity,t.distance,t.angle,t.penumbra,t.decay);break;case"Hemisphere
Light":r=new O1(t.color,tgroundColor,t.intensity);break;case"LightProbe":r=(new
nc).fromJSON(t);break;case"SkinnedMesh":s=o(t.geometry),a=l(t.material),r=new
$s(s,a),void 0!==t.bindMode&&(r.bindMode=t.bindMode),void
0!==t.bindMatrix&&r.bindMatrix.fromArray(t.bindMatrix),void 0!==t.skeleton&&
(r.skeleton=t.skeleton);break;case"Mesh":s=o(t.geometry),a=l(t.material),r=new
Wn(s,a);break;case"InstancedMesh":s=o(t.geometry),a=l(t.material);const
e=t.count,n=t.instanceMatrix,i=t.instanceColor;r=new la(s,a,e),r.instanceMatrix=new
sn(new Float32Array(n.array),16),void 0!==i&&(r.instanceColor=new sn(new
Float32Array(i.array),i.itemSize));break;case"LOD":r=new Xs;break;case"Line":r=new
fa(o(t.geometry),l(t.material));break;case"LineLoop":r=new
xa(o(t.geometry),l(t.material));break;case"LineSegments":r=new
ya(o(t.geometry),l(t.material));break;case"PointCloud":case"Points":r=new
Ta(o(t.geometry),l(t.material));break;case"Sprite":r=new
Vs(l(t.material));break;case"Group":r=new gs;break;case"Bone":r=new
ta;break;default:r=new Ce;if(r.uuid=t.uuid,void 0!==t.name&&(r.name=t.name),void
0!==t.matrix?(r.matrix.fromArray(t.matrix),void 0!==t.matrixAutoUpdate&&
(r.matrixAutoUpdate=t.matrixAutoUpdate),r.matrixAutoUpdate&&r.matrix.decompose(r.po
sition,r.quaternion,r.scale)):(void
0!==t.position&&r.position.fromArray(t.position),void
0!==t.rotation&&r.rotation.fromArray(t.rotation),void
0!==t.quaternion&&r.quaternion.fromArray(t.quaternion),void
0!==t.scale&&r.scale.fromArray(t.scale)),void 0!==t.castShadow&&
(r.castShadow=t.castShadow),void 0!==t.receiveShadow&&
(r.receiveShadow=t.receiveShadow),t.shadow&&(void 0!==t.shadow.bias&&
(r.shadow.bias=t.shadow.bias),void 0!==t.shadow.normalBias&&
(r.shadow.normalBias=t.shadow.normalBias),void 0!==t.shadow.radius&&
(r.shadow.radius=t.shadow.radius),void
0!==t.shadow.mapSize&&r.shadow.mapSize.fromArray(t.shadow.mapSize),void
0!==t.shadow.camera&&(r.shadow.camera=this.parseObject(t.shadow.camera)),void
0!==t.visible&&(r.visible=t.visible),void 0!==t.frustumCulled&&
(r.frustumCulled=t.frustumCulled),void 0!==t.renderOrder&&
(r.renderOrder=t.renderOrder),void 0!==t.userData&&(r.userData=t.userData),void
0!==t.layers&&(r.layers.mask=t.layers),void 0!==t.children){const
s=t.children;for(let t=0;t<s.length;t++)r.add(this.parseObject(s[t],e,n,i))}if(void
0!==t.animations){const e=t.animations;for(let t=0;t<e.length;t++){const
n=e[t];r.animations.push(i[n])}}if("LOD"===t.type){void 0!==t.autoUpdate&&
(r.autoUpdate=t.autoUpdate);const e=t.levels;for(let t=0;t<e.length;t++){const
n=e[t],i=r.getObjectByProperty("uuid",n.object);void
0!==i&&r.addLevel(i,n.distance)}}return r}bindSkeletons(t,e)
{0!==Object.keys(e).length&&t.traverse((function(t){if(!0===t.isSkinnedMesh&&void
0!==t.skeleton){const n=e[t.skeleton];void 0===n?console.warn("THREE.ObjectLoader:
No skeleton found with
UUID:",t.skeleton):t.bind(n,t.bindMatrix)}}))}setTexturePath(t){return
console.warn("THREE.ObjectLoader: .setTexturePath() has been renamed to
.setResourcePath()."),this.setResourcePath(t)},t.ObjectSpaceNormalMap=1,t.Octahedr
onBufferGeometry=_o,t.OctahedronGeometry=_o,t.OneFactor=201,t.OneMinusDstAlphaFacto
r=207,t.OneMinusDstColorFactor=209,t.OneMinusSrcAlphaFactor=205,t.OneMinusSrcColorF
actor=203,t.OrthographicCamera=J1,t.PCFShadowMap=1,t.PCFSoftShadowMap=2,t.PMREMGene
rator=class{constructor(t)
{this._renderer=t,this._pingPongRenderTarget=null,this._blurMaterial=function(t)

```

```
const e=new Float32Array(t),n=new Lt(0,1,0);return new  
Io({name:"SphericalGaussianBlur",defines:{n:t},uniforms:{envMap:  
{value:null},samples:{value:1},weights:{value:e},latitudinal:{value:!1},dTheta:  
{value:0},mipInt:{value:0},poleAxis:{value:n},inputEncoding:  
{value:Nh[3e3]}},outputEncoding:  
{value:Nh[3e3]}}},vertexShader:`\n\n\t\t\tprecision mediump  
float;\n\t\t\t\tprecision mediump int;\n\t\t\t\t\tvarying vec3  
vOutputDirection;\n\t\t\t\t\tuniform sampler2D envMap;\n\t\t\t\t\tuniform int  
samples;\n\t\t\t\t\tuniform float weights[ n ];\n\t\t\t\t\tuniform bool  
latitudinal;\n\t\t\t\t\tuniform float dTheta;\n\t\t\t\t\tuniform float  
mipInt;\n\t\t\t\t\tuniform vec3 poleAxis;\n\t\t\t\t\t\t$tu()\n\t\t\t\t\t\t#define  
ENVMAP_TYPE_CUBE_UV\n\t\t\t\t\t\t#include <cube_uv_reflection_fragment>\n\t\t\t\t\t\tvec3  
getSample( float theta, vec3 axis ) {\n\t\t\t\t\t\t\tfloat cosTheta = cos( theta  
);\n\t\t\t\t\t\t\t// Rodrigues' axis-angle rotation\n\t\t\t\t\t\t\tvec3 sampleDirection =  
vOutputDirection * cosTheta\n\t\t\t\t\t\t\t+ cross( axis, vOutputDirection ) * sin(  
theta )\n\t\t\t\t\t\t\t+ axis * dot( axis, vOutputDirection ) * ( 1.0 - cosTheta  
);\n\t\t\t\t\t\t\treturn bilinearCubeUV( envMap, sampleDirection, mipInt  
);\n\t\t\t\t\t\t}\n\t\t\t\t\t\tvoid main() {\n\t\t\t\t\t\t\tvec3 axis = latitudinal ? poleAxis  
: cross( poleAxis, vOutputDirection );\n\t\t\t\t\t\t\tif ( all( equal( axis, vec3( 0.0  
)) ) ) {\n\t\t\t\t\t\t\t\taxis = vec3( vOutputDirection.z, 0.0, - vOutputDirection.x  
);\n\t\t\t\t\t\t\t}\n\t\t\t\t\t\t\taxis = normalize( axis );\n\t\t\t\t\t\t\tgl_FragColor =  
vec4( 0.0, 0.0, 0.0, 1.0 );\n\t\t\t\t\t\t\tgl_FragColor.rgb += weights[ 0 ] * getSample(  
0.0, axis );\n\t\t\t\t\t\t\tfor ( int i = 1; i < n; i++ ) {\n\t\t\t\t\t\t\t\tif ( i >=  
samples ) {\n\t\t\t\t\t\t\t\t\tbreak;\n\t\t\t\t\t\t\t\t}\n\t\t\t\t\t\t\t\tfloat theta = dTheta  
* float( i );\n\t\t\t\t\t\t\t\tgl_FragColor.rgb += weights[ i ] * getSample( -1.0 *  
theta, axis );\n\t\t\t\t\t\t\t\tgl_FragColor.rgb += weights[ i ] * getSample( theta, axis  
);\n\t\t\t\t\t\t\t\tgl_FragColor = linearToOutputTexel( gl_FragColor  
);\n\t\t\t\t\t\t\t}\n\t\t\t\t\t\t\t`,\n\t\t\t\t\t\t\tblending:0,depthTest:!1,depthWrite:!1)}  
(Ih),this._equirectShader=null,this._cubemapShader=null,this._compileMaterial(this.  
blurMaterial)}}fromScene(t,e=0,n=.1,i=100)  
{kh=this._renderer.getRenderTarget();const r=this._allocateTargets();return  
this._sceneToCubeUV(t,n,i,r),e>0&&this._blur(r,0,0,e),this._applyPMREM(r),this._cle  
anup(r),r}fromEquirectangular(t){return this._fromTexture(t)}fromCubemap(t){return  
this._fromTexture(t)}compileCubemapShader(){null===this._cubemapShader&&  
(this._cubemapShader=Kh(),this._compileMaterial(this._cubemapShader))}compileEquire  
ctangularShader(){null===this._equirectShader&&  
(this._equirectShader=Qh(),this._compileMaterial(this._equirectShader))}dispose()  
{this._blurMaterial.dispose(),null!==this._cubemapShader&&this._cubemapShader.dispo  
se(),null!==this._equirectShader&&this._equirectShader.dispose();for(let  
t=0;t<Oh.length;t++)Oh[t].dispose()}_cleanup(t)  
{this._pingPongRenderTarget.dispose(),this._renderer.setRenderTarget(kh),t.scissorT  
est=!1,Jh(t,0,0,t.width,t.height)}_fromTexture(t)  
{kh=this._renderer.getRenderTarget();const e=this._allocateTargets(t);return  
this._textureToCubeUV(t,e),this._applyPMREM(e),this._cleanup(e),e}_allocateTargets(  
t){const e=  
{magFilter:p,minFilter:p,generateMipmaps:!1,type:x,format:1023,encoding:Xh(t)?  
t.encoding:J,depthBuffer:!1},n=Zh(e);return  
n.depthBuffer=!t,this._pingPongRenderTarget=Zh(e),n}_compileMaterial(t){const e=new  
Wn(Oh[0],t);this._renderer.compile(e,Fh)}_sceneToCubeUV(t,e,n,i){const r=new  
Kn(90,1,e,n),s=[1,-1,1,1,1,1],a=  
[1,1,1,-1,-1,-1],o=this._renderer,l=o.autoClear,c=o.outputEncoding,h=o.toneMapping;  
o.getClearColor(Uh),o.toneMapping=0,o.outputEncoding=X,o.autoClear=!1;let  
u=!1;const d=t.background;if(d){if(d.isColor)  
{Bh.color.copy(d).convertSRGBToLinear(),t.background=null;const  
e=qh(Bh.color);Bh.opacity=e,u=!0}else{Bh.color.copy(Uh).convertSRGBToLinear();cons  
t t=qh(Bh.color);Bh.opacity=t,u=!0}for(let e=0;e<6;e++){const n=e%3;0==n?  
(r.up.set(0,s[e],0),r.lookAt(a[e],0,0)):1==n?  
(r.up.set(0,0,s[e]),r.lookAt(0,a[e],0)):
```

```

(r.up.set(0,s[e],0),r.lookAt(0,0,a[e])),Jh(i,n*Ch,e>2?
Ch:0,Ch,Ch),o.setRenderTarget(i),u&o.render(zh,r),o.render(t,r)}o.toneMapping=h,o.
outputEncoding=c,o.autoClear=1}_textureToCubeUV(t,e){const
n=this._renderer;t.isCubeTexture?null==this._cubemapShader&&
(this._cubemapShader=Kh()):null==this._equirectShader&&
(this._equirectShader=Qh());const i=t.isCubeTexture?
this._cubemapShader:this._equirectShader,r=new
Wn(Oh[0],i),s=i.uniforms;s.envMap.value=t,t.isCubeTexture||s.texelSize.value.set(1/
t.image.width,1/t.image.height),s.inputEncoding.value=Nh[t.encoding],s.outputEncodi
ng.value=Nh[e.texture.encoding],Jh(e,0,0,3*Ch,2*Ch),n.setRenderTarget(e),n.render(r
,Fh)}_applyPMREM(t){const e=this._renderer,n=e.autoClear;e.autoClear=!1;for(let
e=1;e<Dh;e++){const n=Math.sqrt(Gh[e]*Gh[e]-Gh[e-1]*Gh[e-1]),i=jh[(e-
1)%jh.length];this._blur(t,e-1,e,n,i)}e.autoClear=n}_blur(t,e,n,i,r){const
s=this._pingPongRenderTarget;this._halfBlur(t,s,e,n,i,"latitudinal",r),this._halfBl
ur(s,t,n,n,i,"longitudinal",r)}_halfBlur(t,e,n,i,r,s,a){const
o=this._renderer,l=this._blurMaterial;"latitudinal"!==s&&"longitudinal"!==s&&consol
e.error("blur direction must be either latitudinal or longitudinal!");const c=new
Wn(Oh[i],l),h=l.uniforms,u=Hh[n]-1,d=isFinite(r)?
Math.PI/(2*u):2*Math.PI/39,p=r/d,m=isFinite(r)?
1+Math.floor(3*p):1h;m>1h&&console.warn(`sigmaRadians, ${r}, is too large and will
clip, as it requested ${m} samples when the maximum is set to 20`);const f=[];let
g=0;for(let t=0;t<1h;++t){const e=t/p,n=Math.exp(-e*e/2);f.push(n),0==t?g+=n:t<m&&
(g+=2*n)}for(let
t=0;t<f.length;t++)f[t]=f[t]/g;h.envMap.value=t.texture,h.samples.value=m,h.weights
.value=f,h.latitudinal.value="latitudinal"===s,a&&
(h.poleAxis.value=a),h.dTheta.value=d,h.mipInt.value=8-
n,h.inputEncoding.value=Nh[t.texture.encoding],h.outputEncoding.value=Nh[t.texture.
encoding];const v=Hh[i];Jh(e,3*Math.max(0,Ch-2*v),(0===i?0:2*Ch)+2*v*(i>4?i-
8+4:0),3*v,2*v),o.setRenderTarget(e),o.render(c,Fh)}},t.ParametricBufferGeometry=wo
,t.ParametricGeometry=wo,t.Particle=function(t){return console.warn("THREE.Particle
has been renamed to THREE.Sprite."),new Vs(t)},t.ParticleBasicMaterial=function(t)
{return console.warn("THREE.ParticleBasicMaterial has been renamed to
THREE.PointsMaterial."),new _a(t)},t.ParticleSystem=function(t,e){return
console.warn("THREE.ParticleSystem has been renamed to THREE.Points."),new
Ta(t,e)},t.ParticleSystemMaterial=function(t){return
console.warn("THREE.ParticleSystemMaterial has been renamed to
THREE.PointsMaterial."),new
_a(t)},t.Path=B1,t.PerspectiveCamera=Kn,t.Plane=Ne,t.PlaneBufferGeometry=ci,t.Plane
Geometry=ci,t.PlaneHelper=class extends fa{constructor(t,e=1,n=16776960){const
i=n,r=new En;r.setAttribute("position",new
mn([1,-1,1,-1,1,1,-1,-1,1,1,1,1,-1,1,1,-1,-1,1,1,1,1,0,0,1,0,0,0],3)),r.comp
uteBoundingSphere(),super(r,new
ca({color:i,toneMapped:!1})),this.type="PlaneHelper",this.plane=t,this.size=e;const
s=new En;s.setAttribute("position",new
mn([1,1,1,-1,1,1,-1,-1,1,1,1,1,-1,-1,1,1,-1,1],3)),s.computeBoundingSphere(),this.a
dd(new Wn(s,new
en({color:i,opacity:.2,transparent:!0,depthWrite:!1,toneMapped:!1})))}updateMatrixW
orld(t){let e=-this.plane.constant;Math.abs(e)<1e-8&&(e=1e-
8),this.scale.set(.5*this.size,.5*this.size,e),this.children[0].material.side=e<0?
1:0,this.lookAt(this.plane.normal),super.updateMatrixWorld(t)}},t.PointCloud=functi
on(t,e){return console.warn("THREE.PointCloud has been renamed to
THREE.Points."),new Ta(t,e)},t.PointCloudMaterial=function(t){return
console.warn("THREE.PointCloudMaterial has been renamed to
THREE.PointsMaterial."),new _a(t)},t.PointLight=Z1,t.PointLightHelper=class extends
Wn{constructor(t,e,n){super(new So(e,4,2),new
en({wireframe:!0,fog:!1,toneMapped:!1})),this.light=t,this.light.updateMatrixWorld(
),this.color=n,this.type="PointLightHelper",this.matrix=this.light.matrixWorld,this
.matrixAutoUpdate=!1,this.update()}dispose()}

```

```

{this.geometry.dispose(),this.material.dispose()}update(){void 0!==this.color?
this.material.color.set(this.color):this.material.color.copy(this.light.color)}}},t.
Points=Ta,t.PointsMaterial=_a,t.PolarGridHelper=class extends
ya{constructor(t=10,e=16,n=8,i=64,r=4473924,s=8947848){r=new tn(r),s=new
tn(s);const a=[],o=[];for(let n=0;n<=e;n++){const i=n/e*
(2*Math.PI),l=Math.sin(i)*t,c=Math.cos(i)*t;a.push(0,0,0),a.push(l,0,c);const
h=l&n?r:s;o.push(h.r,h.g,h.b),o.push(h.r,h.g,h.b)}for(let e=0;e<=n;e++){const
l=1&e?r:s,c=t-t/n*e;for(let t=0;t<i;t++){let e=t/i*
(2*Math.PI),n=Math.sin(e)*c,r=Math.cos(e)*c;a.push(n,0,r),o.push(l.r,l.g,l.b),e=
(t+1)/i*
(2*Math.PI),n=Math.sin(e)*c,r=Math.cos(e)*c,a.push(n,0,r),o.push(l.r,l.g,l.b)}}cons
t l=new En;l.setAttribute("position",new mn(a,3)),l.setAttribute("color",new
mn(o,3));super(l,new
ca({vertexColors:!0,toneMapped:!1})),this.type="PolarGridHelper"}}},t.PolyhedronBuff
erGeometry=Na,t.PolyhedronGeometry=Na,t.PositionalAudio=class extends
Lc{constructor(t)
{super(t),this.panner=this.context.createPanner(),this.panner.panningModel="HRTF",t
his.panner.connect(this.gain)}getOutput(){return this.panner}getRefDistance()
{return this.panner.refDistance}setRefDistance(t){return
this.panner.refDistance=t,this}getRolloffFactor(){return
this.panner.rolloffFactor}setRolloffFactor(t){return
this.panner.rolloffFactor=t,this}getDistanceModel(){return
this.panner.distanceModel}setDistanceModel(t){return
this.panner.distanceModel=t,this}getMaxDistance(){return
this.panner.maxDistance}setMaxDistance(t){return
this.panner.maxDistance=t,this}setDirectionalCone(t,e,n){return
this.panner.coneInnerAngle=t,this.panner.coneOuterAngle=e,this.panner.coneOuterGain
=n,this}updateMatrixWorld(t)
{if(super.updateMatrixWorld(t),!0===this.hasPlaybackControl&&!1===this.isPlaying)re
turn;this.matrixWorld.decompose(Rc,Cc,Pc),Dc.set(0,0,1).applyQuaternion(Cc);const
e=this.panner;if(e.positionX){const
t=this.context.currentTime+this.listener.timeDelta;e.positionX.linearRampToValueAtT
ime(Rc.x,t),e.positionY.linearRampToValueAtTime(Rc.y,t),e.positionZ.linearRampToVal
ueAtTime(Rc.z,t),e.orientationX.linearRampToValueAtTime(Dc.x,t),e.orientationY.line
arRampToValueAtTime(Dc.y,t),e.orientationZ.linearRampToValueAtTime(Dc.z,t)}else
e.setPosition(Rc.x,Rc.y,Rc.z),e.setOrientation(Dc.x,Dc.y,Dc.z)}}},t.PropertyBinding=
jc,t.PropertyMixer=Nc,t.QuadraticBezierCurve=C1,t.QuadraticBezierCurve3=Pl,t.Quater
nion=At,t.QuaternionKeyframeTrack=$o,t.QuaternionLinearInterpolant=Ko,t.REVISION=e,
t.RGBADepthPacking=3201,t.RGBAFormat=E,t.RGBAIntegerFormat=1033,t.RGBA_ASTC_10x10_F
ormat=37819,t.RGBA_ASTC_10x5_Format=37816,t.RGBA_ASTC_10x6_Format=37817,t.RGBA_ASTC
_10x8_Format=37818,t.RGBA_ASTC_12x10_Format=37820,t.RGBA_ASTC_12x12_Format=37821,t.
RGBA_ASTC_4x4_Format=37808,t.RGBA_ASTC_5x4_Format=37809,t.RGBA_ASTC_5x5_Format=3781
0,t.RGBA_ASTC_6x5_Format=37811,t.RGBA_ASTC_6x6_Format=37812,t.RGBA_ASTC_8x5_Format=
37813,t.RGBA_ASTC_8x6_Format=37814,t.RGBA_ASTC_8x8_Format=37815,t.RGBA_BPTC_Format=
36492,t.RGBA_ETC2_EAC_Format=0,t.RGBA_PVRTC_2BPPV1_Format=z,t.RGBA_PVRTC_4BPPV1_For
mat=B,t.RGBA_S3TC_DXT1_Format=C,t.RGBA_S3TC_DXT3_Format=P,t.RGBA_S3TC_DXT5_Format=D
,t.RGBDEncoding=$,t.RGBEEncoding=J,t.RGBEFormat=1023,t.RGBFormat=T,t.RGBIntegerForm
at=1032,t.RGBM16Encoding=K,t.RGBM7Encoding=Q,t.RGB_ETC1_Format=36196,t.RGB_ETC2_For
mat=F,t.RGB_PVRTC_2BPPV1_Format=N,t.RGB_PVRTC_4BPPV1_Format=I,t.RGB_S3TC_DXT1_Forma
t=R,t.RGFormat=1030,t.RGIntegerFormat=1031,t.RawShaderMaterial=Io,t.Ray=re,t.Raycas
ter=class{constructor(t,e,n=0,i=1/0){this.ray=new
re(t,e),this.near=n,this.far=i,this.camera=null,this.layers=new ge,this.params=
{Mesh:{},Line:{threshold:1},LOD:{},Points:{threshold:1},Sprite:{}}set(t,e)
{this.ray.set(t,e)}setFromCamera(t,e){e&&e.isPerspectiveCamera?
(this.ray.origin.setFromMatrixPosition(e.matrixWorld),this.ray.direction.set(t.x,t.
y,.5).unproject(e).sub(this.ray.origin).normalize(),this.camera=e):e&&e.isOrthograp
hicCamera?(this.ray.origin.set(t.x,t.y,(e.near+e.far)/(e.near-
e.far)).unproject(e),this.ray.direction.set(0,0,-1).transformDirection(e.matrixWorl

```

```

d),this.camera=e):console.error("THREE.Raycaster: Unsupported camera type:
"+e.type)}intersectObject(t,e=!1,n=[]){return
$c(t,this,n,e),n.sort(Kc),n}intersectObjects(t,e=!1,n=[]){for(let
i=0,r=t.length;i<r;i++)$c(t[i],this,n,e);return
n.sort(Kc),n}},t.RectAreaLight=tc,t.RedFormat=1028,t.RedIntegerFormat=1029,t.Reinha
rdToneMapping=2,t.RepeatWrapping=h,t.ReplaceStencilOp=7681,t.ReverseSubtractEquatio
n=102,t.RingBufferGeometry=bo,t.RingGeometry=bo,t.SRGB8_ALPHA8_ASTC_10x10_Format=37
851,t.SRGB8_ALPHA8_ASTC_10x5_Format=37848,t.SRGB8_ALPHA8_ASTC_10x6_Format=37849,t.S
RGB8_ALPHA8_ASTC_10x8_Format=37850,t.SRGB8_ALPHA8_ASTC_12x10_Format=37852,t.SRGB8_A
LPHA8_ASTC_12x12_Format=37853,t.SRGB8_ALPHA8_ASTC_4x4_Format=37840,t.SRGB8_ALPHA8_A
STC_5x4_Format=37841,t.SRGB8_ALPHA8_ASTC_5x5_Format=37842,t.SRGB8_ALPHA8_ASTC_6x5_F
ormat=37843,t.SRGB8_ALPHA8_ASTC_6x6_Format=37844,t.SRGB8_ALPHA8_ASTC_8x5_Format=378
45,t.SRGB8_ALPHA8_ASTC_8x6_Format=37846,t.SRGB8_ALPHA8_ASTC_8x8_Format=37847,t.Scen
e=Ts,t.SceneUtils=eu,t.ShaderChunk=hi,t.ShaderLib=di,t.ShaderMaterial=Jn,t.ShadowMa
terial=Do,t.Shape=z1,t.ShapeBufferGeometry=Mo,t.ShapeGeometry=Mo,t.ShapePath=dc,t.S
hapeUtils=po,t.ShortType=1011,t.Skeleton=ia,t.SkeletonHelper=hh,t.SkinnedMesh=$s,t.
SmoothShading=2,t.Sphere=Jt,t.SphereBufferGeometry=So,t.SphereGeometry=So,t.Spheric
al=class{constructor(t=1,e=0,n=0){return
this.radius=t,this.phi=e,this.theta=n,this}set(t,e,n){return
this.radius=t,this.phi=e,this.theta=n,this}copy(t){return
this.radius=t.radius,this.phi=t.phi,this.theta=t.theta,this}makeSafe(){const t=1e-
6;return this.phi=Math.max(t,Math.min(Math.PI-t,this.phi)),this}setFromVector3(t)
{return this.setFromCartesianCoords(t.x,t.y,t.z)}setFromCartesianCoords(t,e,n)
{return this.radius=Math.sqrt(t*t+e*e+n*n),0===this.radius?
(this.theta=0,this.phi=0):
(this.theta=Math.atan2(t,n),this.phi=Math.acos(ht(e/this.radius,-1,1))),this}clone(
){return(new
this.constructor).copy(this)}}},t.SphericalHarmonics3=ec,t.SplineCurve=Dl,t.SpotLigh
t=Wl,t.SpotLightHelper=class extends Ce{constructor(t,e)
{super(),this.light=t,this.light.updateMatrixWorld(),this.matrix=t.matrixWorld,this
.matrixAutoUpdate=!1,this.color=e;const n=new En,i=
[0,0,0,0,0,1,0,0,0,1,0,1,0,0,0,-1,0,1,0,0,0,0,1,1,0,0,0,0,-1,1];for(let
t=0,e=1,n=32;t<n;t++,e++){const
r=t/n*Math.PI*2,s=e/n*Math.PI*2;i.push(Math.cos(r),Math.sin(r),1,Math.cos(s),Math.s
in(s),1)}n.setAttribute("position",new mn(i,3));const r=new
ca({fog:!1,toneMapped:!1});this.cone=new
ya(n,r),this.add(this.cone),this.update()}dispose()
{this.cone.geometry.dispose(),this.cone.material.dispose()}update()
{this.light.updateMatrixWorld();const t=this.light.distance?
this.light.distance:1e3,e=t*Math.tan(this.light.angle);this.cone.scale.set(e,e,t),a
h.setFromMatrixPosition(this.light.target.matrixWorld),this.cone.lookAt(ah),void
0!==(this.color?
this.cone.material.color.set(this.color):this.cone.material.color.copy(this.light.c
olor))},t.Sprite=Vs,t.SpriteMaterial=Rs,t.SrcAlphaFactor=204,t.SrcAlphaSaturateFact
or=210,t.SrcColorFactor=202,t.StaticCopyUsage=35046,t.StaticDrawUsage=et,t.StaticRe
adUsage=35045,t.StereoCamera=class{constructor()
{this.type="StereoCamera",this.aspect=1,this.eyeSep=.064,this.cameraL=new
Kn,this.cameraL.layers.enable(1),this.cameraL.matrixAutoUpdate=!1,this.cameraR=new
Kn,this.cameraR.layers.enable(2),this.cameraR.matrixAutoUpdate=!1,this._cache=
{focus:null,fov:null,aspect:null,near:null,far:null,zoom:null,eyeSep:null}}update(t)
){const
e=this._cache;if(e.focus!==t.focus||e.fov!==t.fov||e.aspect!==t.aspect*this.aspect|
|e.near!==t.near||e.far!==t.far||e.zoom!==t.zoom||e.eyeSep!==this.eyeSep)
{e.focus=t.focus,e.fov=t.fov,e.aspect=t.aspect*this.aspect,e.near=t.near,e.far=t.fa
r,e.zoom=t.zoom,e.eyeSep=this.eyeSep;const
n=t.projectionMatrix.clone(),i=e.eyeSep/2,r=i*e.near/e.focus,s=e.near*Math.tan(ot*e
.fov*.5)/e.zoom;let a,o;wc.elements[12]=-i,_c.elements[12]=i,a=-
s*e.aspect+r,o=s*e.aspect+r,n.elements[0]=2*e.near/(o-a),n.elements[8]=(o+a)/(o-

```

```

a),this.cameraL.projectionMatrix.copy(n),a=-s*e.aspect-r,o=s*e.aspect-
r,n.elements[0]=2*e.near/(o-a),n.elements[8]=(o+a)/(o-
a),this.cameraR.projectionMatrix.copy(n)}this.cameraL.matrixWorld.copy(t.matrixWorld).multiply(wc),this.cameraR.matrixWorld.copy(t.matrixWorld).multiply(_c)}},t.StreamCopyUsage=35042,t.StreamDrawUsage=35040,t.StreamReadUsage=35041,t.StringKeyframeTrack=tl,t.SubtractEquation=101,t.SubtractiveBlending=3,t.TOUCH={ROTATE:0,PAN:1,DOLLY_PAN:2,DOLLY_ROTATE:3},t.TangentSpaceNormalMap=0,t.TetrahedronBufferGeometry=To,t.TetrahedronGeometry=To,t.TextBufferGeometry=Eo,t.TextGeometry=Eo,t.Texture=bt,t.TextureLoader=pl,t.TorusBufferGeometry=Ao,t.TorusGeometry=Ao,t.TorusKnotBufferGeometry=Lo,t.TorusKnotGeometry=Lo,t.Triangle=je,t.TriangleFanDrawMode=2,t.TriangleStripDrawMode=1,t.TrianglesDrawMode=0,t.TubeBufferGeometry=Ro,t.TubeGeometry=Ro,t.UVMapping=i,t.Uint16Attribute=function(t,e){return console.warn("THREE.Uint16Attribute has been removed. Use new THREE.Uint16BufferAttribute() instead."),new hn(t,e)},t.Uint16BufferAttribute=hn,t.Uint32Attribute=function(t,e){return console.warn("THREE.Uint32Attribute has been removed. Use new THREE.Uint32BufferAttribute() instead."),new dn(t,e)},t.Uint32BufferAttribute=dn,t.Uint8Attribute=function(t,e){return console.warn("THREE.Uint8Attribute has been removed. Use new THREE.Uint8BufferAttribute() instead."),new on(t,e)},t.Uint8BufferAttribute=on,t.Uint8ClampedAttribute=function(t,e){return console.warn("THREE.Uint8ClampedAttribute has been removed. Use new THREE.Uint8ClampedBufferAttribute() instead."),new ln(t,e)},t.Uint8ClampedBufferAttribute=ln,t.Uniform=Zc,t.UniformsLib=ui,t.UniformsUtils=Zn,t.UnsignedByteType=x,t.UnsignedInt248Type=S,t.UnsignedIntType=w,t.UnsignedShort4444Type=1017,t.UnsignedShort5551Type=1018,t.UnsignedShort565Type=1019,t.UnsignedShortType=_,t.VSMShadowMap=3,t.Vector2=vt,t.Vector3=Lt,t.Vector4=St,t.VectorKeyframeTrack=el,t.Vertex=function(t,e,n){return console.warn("THREE.Vertex has been removed. Use THREE.Vector3 instead."),new Lt(t,e,n)},t.VertexColors=2,t.VideoTexture=Aa,t.WebGL1Renderer=bs,t.WebGLCubeRenderTarget=ni,t.WebGLMultisampleRenderTarget=Et,t.WebGLRenderTarget=Tt,t.WebGLRenderTargetCube=function(t,e,n){return console.warn("THREE.WebGLRenderTargetCube( width, height, options ) is now WebGLCubeRenderTarget( size, options )."),new ni(t,n)},t.WebGLRenderer=ws,t.WebGLUtils=ms,t.WireframeGeometry=Co,t.WireframeHelper=function(t,e){return console.warn("THREE.WireframeHelper has been removed. Use THREE.WireframeGeometry instead."),new ya(new Co(t.geometry),new ca({color:void 0!==e?e:16777215}))},t.WrapAroundEnding=W,t.XHRLoader=function(t){return console.warn("THREE.XHRLoader has been renamed to THREE.FileLoader."),new cl(t)},t.ZeroCurvatureEnding=k,t.ZeroFactor=200,t.ZeroSlopeEnding=V,t.ZeroStencilOp=0,t.sRGBEncoding=Y,Object.defineProperty(t,"__esModule",{value:!0}}));

```